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## Groundwater chemistry of the Weaber Plain (Goomig Farmlands): baseline results 2010–13

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# **Groundwater chemistry of the Weaber Plain (Goomig Farmlands): baseline results 2010–13**

**Resource management technical report 392**

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# **Groundwater chemistry of the Weaber Plain (Goomig Farmlands): baseline results 2010–13**

**Resource management technical report 392**

**Adam M Lillicrap, Richard George, Arjen Ryder and Don Bennett**

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Cover: Irrigation supply channel, Ord River Irrigation Area (Photo: RJ George)

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The Weaber Plain hydrological investigations underpinned the state's Ord Stage 2 Irrigation Expansion Project. The resulting irrigated area is now known as the Goomig Farmlands. The project was initiated in April 2010 and included about 450 days of field work and over 2000 days of planning, analysis and consultation. As a result, many people have supported this project.

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## Dedication

We dedicate this report to our colleague Arjen Ryder who, with his wife Yvonne, was killed when Malaysia Airlines Flight MH17 was shot down over the Ukraine in July 2014. Arjen undertook two years of monitoring upon which this report is based and also participated in the 2011 drilling program. He worked with enthusiasm and enjoyed his time in the East Kimberley region, making friends of all he met. We thank him and miss him.

## Summary

The Ord River Irrigation Area (ORIA) is located in the north-east of the Kimberley region of Western Australia, near the town of Kununurra. The irrigation area was established in 1963 and over time developed to the current extent of 14 000 hectares (ha). The Weaber Plain (Goomig Farmlands) area is located north-north-east of the existing irrigation area, 30km from Kununurra, and has been identified as being suitable for irrigated agriculture for many decades. However, it was not until 2009, with state government support, that the 7400ha project commenced, with construction starting in 2010.

State and Australian government environmental approvals required the proponent to install a groundwater monitoring network and develop a groundwater management plan. The groundwater monitoring network on the Weaber and Knox Creek plains consists of 20 high-intensity (data loggers and intensive sampling) and 30 low-intensity bores (lower level monitoring status). Of the 50 bores, two were dry during the baseline, pre-development, monitoring period. Another eight bores were set up as reference bores to act as long-term controls. Four of the reference bores are high-intensity and four are low-intensity.

The environmental approvals required seasonal monitoring of groundwater to establish baseline groundwater chemistry conditions. The monitoring bores were sampled for up to three years and showed a large variation in water type and water quality across the Weaber and Knox Creek plains. The highest salinities — over 3000mg/L total dissolved solids (TDS) (450mS/m) — were generally located on the northern and southern margins of the study area in association with Aquitaine soils and low permeability basement geology.

The groundwater was mainly alkaline and reducing. Sodium was the dominant cation and bicarbonate was the dominant anion for low salinity (<1400mg/L TDS or 200mS/m) groundwater. Chloride and sulfate were the dominant anions for higher salinity groundwater. Groundwater in the northern and southern margins often contained relatively high concentrations of metals, such as zinc, copper, lead, molybdenum and vanadium. Statistical analysis and geochemical modelling showed that geology and soils played important roles in determining the groundwater chemistry. The high concentrations of metals and metalloids are natural background levels likely caused by mineralised ore bodies and vadose zone processes. The Keep River also had naturally high background levels of aluminium, zinc, copper, lead and boron.

Comparing groundwater beneath the Weaber Plain with groundwater beneath the existing developed irrigation areas showed potentially hazardous metals and metalloids are unlikely to change with the development of agriculture. Only nutrients and pesticides are likely to slowly increase in groundwater in the irrigated areas of the Weaber Plain, based on the results from the ORIA. Using aquifer characteristics and available modelling, it would take 100–300 years for chemical hazards to reach the Keep River. Therefore, we conclude groundwater from beneath the Goomig Farmlands poses a relatively low and long-term risk to the receiving environment.

Strategen (2012a) developed a groundwater management plan which specifies options to manage watertable levels beneath the Weaber Plain in the future. If required, groundwater would be pumped into the M2 supply channel for reuse during

the dry season and at higher flows during the wet season, it would be pumped to the Keep River. Baseline data was used to derive the quality of the M2 supply channel for different scenarios using a range of proportions of groundwater to M2 supply water to ensure the final water mixture remained suitable for irrigation. Modelling showed that all of the water quality parameters were below the ANZECC and ARMCANZ (2000) long-term trigger values for irrigation. The modelled salinity of the supply channel water would increase from 26mS/m to around 30–35mS/m, which is suitable even for salt-sensitive crops.

The groundwater beneath the Weaber Plain is unsuitable for direct irrigation on to black soils of the area. However, based on the results of the mixing model and other studies, groundwater can be diluted with channel water and used for irrigation. The Aquitaine soils found on the northern and southern margins of the study area have high natural salt loads and irrigation with unmixed channel water is recommended until mixing waters is found suitable by further analysis.

Baseline groundwater results reported here are also used to recommend future changes to the approved groundwater monitoring regime that align with the *Environmental Protection and Biodiversity Conservation Act 1999* (Commonwealth) consent conditions and the groundwater management plan.

## 1 Introduction

The Ord River Irrigation Area (ORIA) is located in the north-east of the Kimberley region of Western Australia, near the town of Kununurra (Figure 1.1). The irrigation area was established in 1963 with the release of five farms and has developed to the current extent of 14 000ha.

The Weaber Plain (Goomig Farmlands) area is located north-north-east of the existing irrigation area, 30km from Kununurra. The area had been identified as suitable for irrigated agriculture for many decades. After initial environmental assessments, approval was given in 2002 by the Western Australian Government to develop the ORIA Stage 2, which included the Weaber Plain, subject to a number of conditions. After a hiatus, the Ord East Kimberley Expansion Project was recommenced in 2009 with state (\$315m for development of irrigation on the Weaber Plain) and Australian Government support (\$195m for community infrastructure).

The Weaber Plain is the first of several areas proposed to be developed for irrigated agriculture. The annual demand for irrigation water supply from Lake Argyle has been calculated at 80–120 gigalitres (GL) for the Weaber Plain and up to 865GL has been allocated for the entire Ord irrigation area (including 225GL used on the existing ORIA). There could potentially be 50 000ha of irrigated agriculture in the area.

Part of the state government approval conditions for ORIA Stage 2 required further investigations and development of specific management plans, including a groundwater management plan. Investigations to underpin the groundwater management plan (GMP) included better delineation of groundwater systems, installation of monitoring bores and development of a groundwater model for the area. DAFWA, with partners Kellogg Brown and Root Pty Ltd (KBR), started work on the GMP in 2009 (KBR 2010). At the same time, investigations also started to underpin other plans to manage the impact of the proposed development on the Keep River (GHD 2010, Bennett and George 2011, 2014).

The management plans were prepared in two stages. The first stage consisted of a suite of groundwater model simulations using updated groundwater level, regolith and climate data based on Lawrie et al. (2010) and CSIRO (2009). During this work, which was completed in February 2010 by DAFWA and KBR, it was determined that the existing groundwater data was inadequate for substantiating modelled options to manage shallow watertables and salinity (KBR 2010). Some of the data gaps directly related to the proposal to manage groundwater levels by pumping from the palaeochannel beneath the Weaber Plain. Specifically, there was a lack of groundwater chemistry data to determine the suitability, in terms of water quality, of mixing groundwater pumped from the palaeochannel with the irrigation supply.

The second stage of investigations consisted of field programs that commenced in 2010 and which updated the groundwater modelling by KBR (2011). The investigations also had to adapt to the timing of the approval processes, requiring a fast-tracked process of analysis and publication. As a result, the 2010 preliminary groundwater-related investigations were first published in 2011 (George et al. 2011, Lillicrap et al. 2011).

In addition to the state approvals process, in June 2010 the Australian Government determined that the project had the potential to significantly impact on Matters of National Environmental Significance and therefore the project required approval

under the *Environmental Protection and Biodiversity Conservation Act 1999* (EPBC Act). This approval required an additional environmental impact statement and supporting studies. The Weaber Plain proposal was approved by the Australian Government in 2011, subject to additional conditions.

These EPBC conditions were incorporated into a revised groundwater management plan (Strategen 2012a) and a stormwater and groundwater discharge management plan (Strategen 2012b). The EPBC conditions required baseline data on the condition of groundwater and surface water systems, delineation of environmental triggers, assessment tools and specified management actions.

To meet EPBC approval conditions, an additional 23 bores at 16 sites were installed between July and August 2011. The resultant groundwater monitoring network consisted of the bores drilled in 2010 and 2011, as well as 16 existing groundwater bores. Bores were monitored over two to three years to establish baseline conditions and the chemistry results are published in this report.

Based on the experience from ORIA Stage 1, a rise in watertables caused by irrigated agriculture on the Weaber Plain could be a risk to the area's future agricultural development (Lillicrap et al. 2011). One of the proposed options to manage the risk of shallow watertables is pumping and reusing the groundwater. The GMP (Strategen 2012a), supported by groundwater modelling (KBR 2011), proposed pumping groundwater into the Keep River during the wet season and into the main M2 supply channel during the dry season. The option was tested by Lillicrap et al. (2011) based on data obtained mostly in 2010. This report updates the initial modelling of Lillicrap et al. (2011) by using the completed baseline analysis and additional bores drilled in 2011. Results for groundwater reuse and disposal to the M2 supply channel are compared to ANZECC and ARMCANZ (2000) irrigation water quality guidelines.

In summary, this report presents the results and analysis of the groundwater monitoring (2010–13) to provide baseline groundwater chemistry for future reference. We have re-analysed the efficacy of the proposed management option for groundwater reuse and disposal to the M2 supply channel and reviewed the requirement for ongoing high-intensity sampling of some bores and analytes. Based on the results of this baseline analysis, we also suggest changes, which remain within the scope and intent of the approved conditions, to the proposed future groundwater monitoring regime.

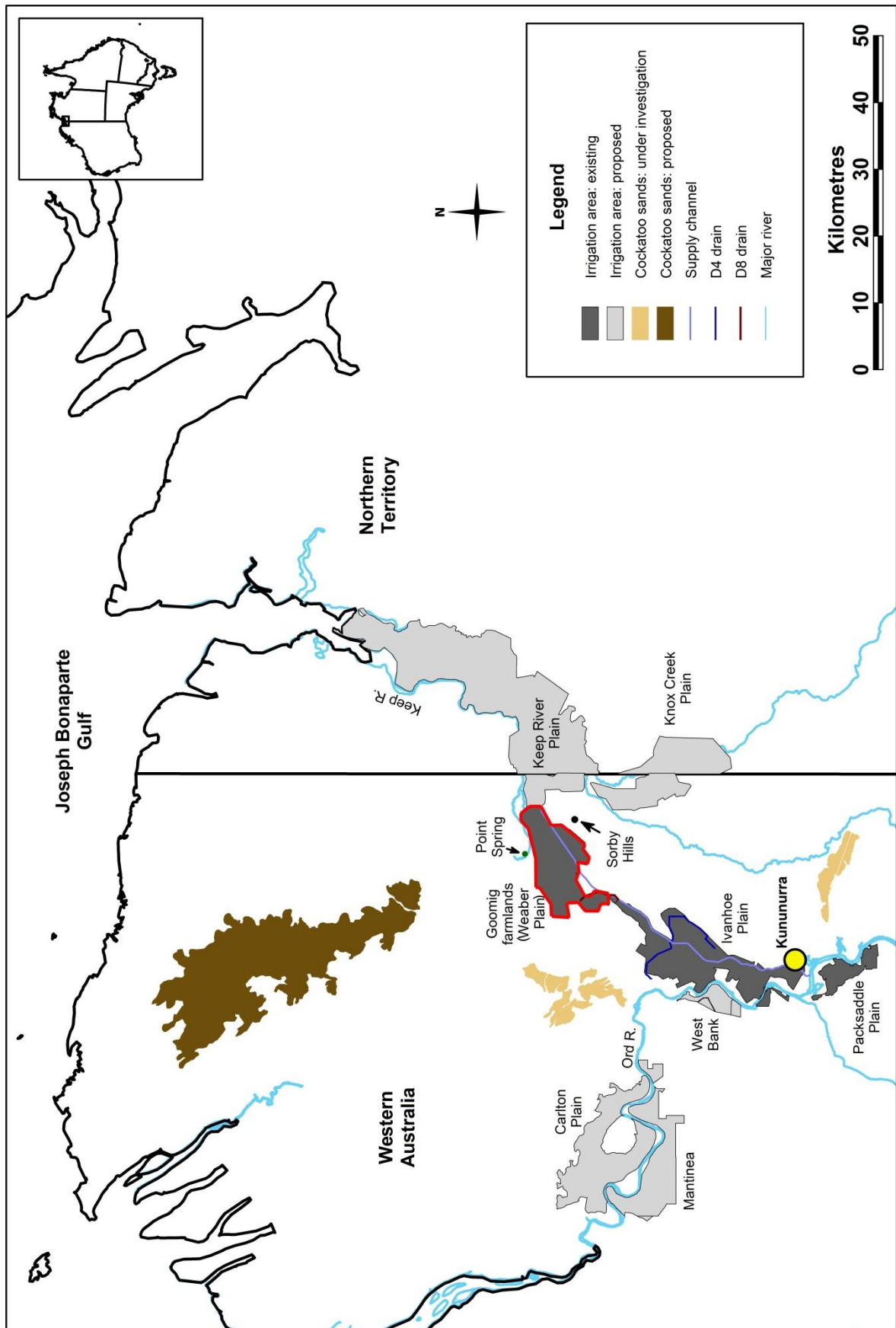


Figure 1.1 Location of the study area

## 2 Methodology

### 2.1 Groundwater chemistry

To meet the EPBC conditions, the groundwater monitoring network of 20 high-intensity bores (instrumented with loggers) and 30 low-intensity bores (non-instrumented) was set up across the proposed farmlands and buffer areas. Another eight bores were set up as reference bores to act as long-term controls (Figure 2.1 and Appendix A). Four of the reference bores are high-intensity monitoring sites and four are low-intensity. The rationale for the design of the monitoring network is outlined in Lillicrap et al. (2011).

In addition to the new bores, existing bores (drilled between 1964 and 1997 on the Weaber and Knox Creek plains) were assessed and used where possible. The existing bores were typically outside the proposed farmland. Some bores that were likely to be affected by infrastructure in strategic sites were redrilled as near as practical to the existing site (these sites have names with the suffix R) and used as either high- or low-intensity bores to capitalise on the longer-term groundwater level and salinity data. While many of the pre-2010 bores will be lost to during development for farmland, some may be retained and used as farm monitoring bores.

All bores were sampled twice a year — at the end of the wet and dry seasons — over at least two years (August 2011, November 2011, May 2012, October 2012, May–June 2013 and October 2013). Bores drilled in 2010 were sampled over a correspondingly longer period. The monitoring is longer than the 18-month baseline period specified in the EPBC conditions. Two of the shallow bores at low-intensity sites 10WP43S and 11WP52S remained dry throughout the baseline period.

The total depth drilled as well as the depth and length of screens and the aquifer screened for each bore, where possible were determined from drill logs (George et al. 2011, George et al. in prep., Humphreys et al. 1995, Laws 1983, Nixon 1997a, Nixon 1997b, Nixon 1997c, Nixon 1997d). The bedrock geology for each bore was determined by intersecting the bore data with regional hydrogeology (O'Boy et al. 2001) in a geographic information system (Appendix A).

#### 2.1.1 Sample collection

All bores were developed by pumping until the casing was free of sediment and pumping rates could be estimated. The majority of bores were sampled with a low-volume electric pump (Proactive Environmental Products Hurricane 12v), and where pumping was not possible because of low hydraulic conductivity, a hand bailer was used. To ensure that aquifer water was sampled rather than stagnant water in the casing, samples were taken from just above the screen in pumped bores when the measured water quality parameters — electrical conductivity (EC), pH, and/or oxidation-reduction potential (ORP) — had stabilised. For wells sampled by bailer, three casing volumes were removed before sampling to ensure aquifer water was sampled. However, this was not always achieved because of the slow recovery rate of water within bores in areas with low hydraulic conductivities.

Water samples for metal analysis were filtered, acidified and stored in acid-washed containers. If there was insufficient time to filter samples for metal analysis, the samples were stored in acid-washed 125mL plastic containers that were completely

filled to exclude air and then kept cool. Samples for nutrient and general chemistry analysis were stored in acid-washed 500mL plastic bottles, completely filled to exclude air and then kept cool. Water samples for atrazine were collected in acid-washed 50mL opaque glass bottles and then kept cool. All water samples collected in the field were refrigerated or frozen and sent to the laboratory within a week of collection.

EC (WTW TetraCon 325), temperature, pH (WTW Sentix 20), ORP (WTW ORP 900) and dissolved oxygen (WTW CelloX 325) were measured in the field. Total alkalinity (Hanna HI 3811) and total acidity (Hach AC-6) were measured either in the field or within 12 hours of collection.

### 2.1.2 Analysis and reporting of samples

All groundwater samples sent to the ChemCentre WA were analysed for:

- major ions (calcium, magnesium, sodium, potassium, bicarbonate/ carbonate, sulfate and chloride)
- minor ions (silicon, fluoride, bromide, boron, dissolved organic carbon)
- metals and metalloids (aluminium, antimony, arsenic, barium, beryllium, bismuth, cadmium, chromium, cobalt, copper, gallium, iron, lanthanum, lead, lithium, manganese, mercury, molybdenum, nickel, selenium, silver, thallium, tin, titanium, uranium, vanadium, zinc)
- nutrients (total nitrogen, oxidised nitrogen or nitrate and nitrite, ammonia, total phosphorus, soluble reactive phosphorus)
- general chemistry (acidity, alkalinity, EC, hardness, pH).

This list of analytes was specified in the EPBC Condition 12B (Appendix B) based on prior analysis (Lillicrap et al. 2011) and as set out in the groundwater management plan (Strategen 2012a).

Atrazine was the only farm chemical found in the chemical analysis of groundwater in the ORIA Stage 1 area (Smith et al. 2007) and therefore baseline analysis of farm chemicals for the Weaber Plain was restricted to atrazine.

Summary statistical analysis was performed on groundwater chemistry and field data as a whole and for individual bores using Data Desk 6.1 (Data Description) and Microsoft Excel. Concentrations in the range parts per million are reported as milligrams per litre (mg/L) and concentrations in the range of parts per billion are reported as micrograms per litre (µg/L). Any chemistry data that was below the laboratory limit of reporting (LOR) was assigned the value of half the LOR, which is consistent with ANZECC and ARMCANZ (2000) guidelines.

Maps showing the spatial distribution in mean concentrations for analytes that had samples greater than 40% above the LOR were created. The calculated means for the species of interest at each bore were interpolated in ArcGIS 10 to create grids using the inverse distance weighting method. The grids were confined to the bore network using a mask.

The temporal changes in TDS and groundwater levels are discussed in George et al. (in press).

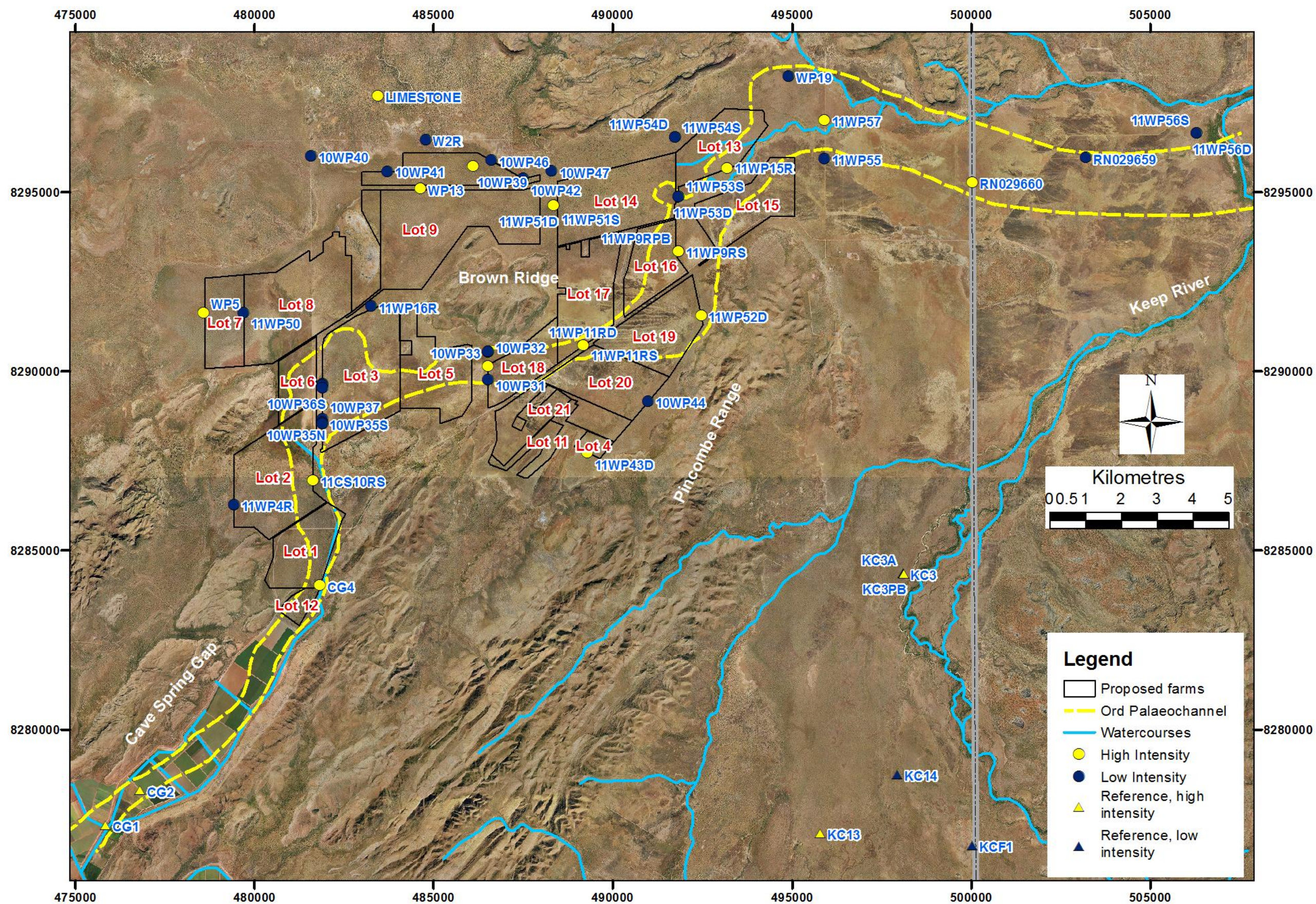


Figure 2.1 Location of monitoring bores on the Weaber and Knox Creek plains

## 2.2 Suitability of groundwater as a resource for agriculture

### 2.2.1 Suitability of groundwater for direct irrigation

The ANZECC and ARMCANZ (2000) irrigation guidelines were used to assess the suitability of groundwater for direct irrigation for all parameters except salinity and sodicity (Chapter 4.2 of this report). For species or parameters that have tolerance classes or that exceeded the ANZECC and ARMCANZ (2000) irrigation guideline long-term trigger values (LTV), distribution maps were created based on the mean concentration grids to show areas of different tolerance or exceeded guideline values.

For salinity and sodicity, the USDA (1954) classification system was used because the ANZECC and ARMCANZ (2000) irrigation guidelines require specific site details, such as soil type, crop type and management practices, which are not all currently available. Following the methodology of Taylor (1996), the USDA salinity classes used in this report were modified, based on Hart (1974).

The USDA (1954) defined the C2 category as medium salinity hazard, with electrical conductivities, as modified by Taylor (1996), between 28 and 800mS/m. We used the C2 category as the test for groundwater suitability based on its use by Ali et al. (2002) in the nearby established ORIA.

### 2.2.2 Mixing of groundwater with the M2 supply channel

The preferred option in the groundwater management plan to manage shallow watertables and excess soil salinity is to pump groundwater from the Ord palaeochannel under the Weaber Plain and discharge the water to either the main M2 supply channel or to the Keep River (Figure 2.2). To establish a representative composition of pumped groundwater to be discharged into the supply channel, the grids developed for each analyte (Chapter 2.1.2), specified in the ANZECC and ARMCANZ (2000) irrigation guidelines, were used. The concentration of these species at the dewatering bore sites proposed (Figure 2.2) in the groundwater management plan (KBR 2011, Strategen 2012a) was estimated from the interpolated grids. It was assumed that each bore was pumped at the same rate. The final mixed groundwater concentration for each species of interest was then estimated and used in subsequent runs of the mixing model outlined below.

The mixing model assumes that all species modelled behave conservatively and that aquifer conditions, such as redox state, remained similar to current levels. However, there are uncertainties:

- pumping groundwater into surface water could result in changes of redox conditions and chemical reactions may alter the composition of pumped groundwater
- it is possible that species, such as manganese, would precipitate and other metal species could be sorped or co-precipitated
- the conservative approach is likely to overestimate concentrations
- recharge from irrigation water could change the redox state of aquifers and result in chemical reactions that affect the chemistry of the aquifers

- there are dissolved salts stored in the regolith profile, based on downhole electromagnetic surveys (George et al. in press) and rising watertables could cause dissolution of these species, which would also alter the aquifer chemistry.

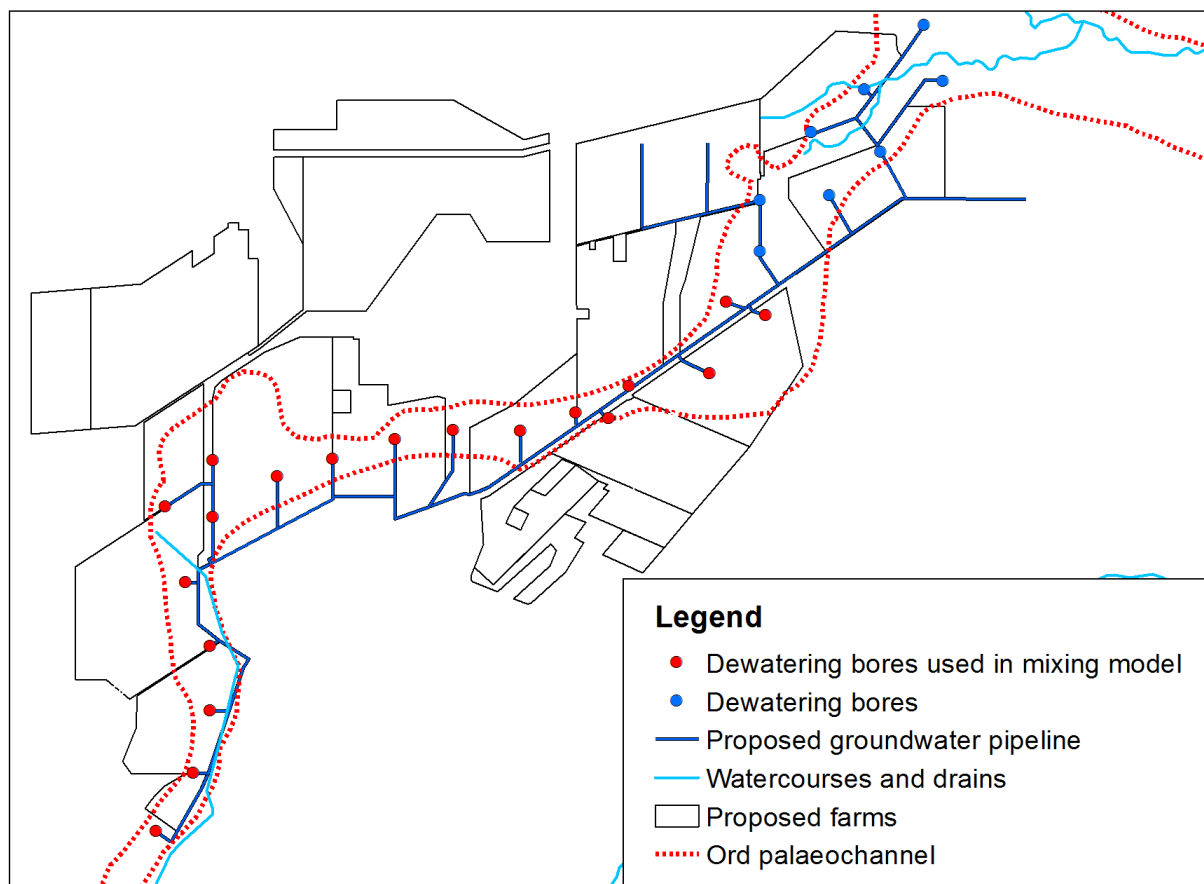


Figure 2.2 Location of Ord palaeochannel and groundwater dewatering bores used in the mixing model

Due to these uncertainties, the mixing model was limited to 20 years from the start of irrigation, rather than the 50 years used in the groundwater modelling (KBR 2011). Over this period, the groundwater model (KBR 2011) forecast that most of the south-west and central part of the model domain (i.e. about 80% of farmland) could have shallow watertables and may require management.

In the water quality model used in this report, two representative groundwater pumping scenarios from KBR (2011) were used: the expected or forecast condition, and the worst case or high groundwater pumping condition. Two supply channel flow conditions — average flow and peak flow — were also modelled, giving four modelling scenarios (Table 2.1). In Scenario A, groundwater is pumped at the forecast flow of 4.4GL over 200 days from the palaeochannel each year during the tropical dry season and discharged into the M2 supply channel flowing at average flow (540 megalitres per day, ML/d). In Scenario B, groundwater is pumped at the forecast flow and discharged into the supply channel flowing at peak flow (769ML/d). In Scenario C, groundwater is pumped at the highest flow of 6.15GL over 200 days from the palaeochannel each year during the tropical dry season and discharged into the supply channel flowing at average flow. In Scenario D, groundwater is pumped at the highest flow and discharged into the supply channel flowing at peak flow.

The future water quality of the M2 supply channel was assumed to be similar to the quality of the M1 supply channel (Lillicrap et al. 2011). The volumetric contribution of M2 water was derived by subtracting the volume of pumped groundwater discharge from the total irrigation supply flow required. The resultant water quality from mixing groundwater into the supply channel was estimated from mixing total volumes and total loads from different water sources as per Lillicrap et al. (2011). The EC was estimated from the TDS of the final water quality using a relationship developed from chemical sampling. The suitability for irrigation was assessed as in Chapter 2.2.1.

Table 2.1 The mixing scenarios used in the water quality model

| Scenario                                           | Description                                                |
|----------------------------------------------------|------------------------------------------------------------|
| A) Forecast groundwater flow, average channel flow | Groundwater pumped at 22ML/d, M2 channel flow at 540ML/d   |
| B) Forecast groundwater flow, peak channel flow    | Groundwater pumped at 22ML/d, M2 channel flow at 769ML/d   |
| C) High groundwater flow, average channel flow     | Groundwater pumped at 30.7ML/d, M2 channel flow at 540ML/d |
| D) High groundwater flow, peak channel flow        | Groundwater pumped at 30.7ML/d, M2 channel flow at 540ML/d |

## 3 Results

### 3.1 Groundwater chemistry

Summary statistics from baseline groundwater chemistry for all bores are shown in Table 3.1. Figures 3.1 to 3.7 show the average concentration of species that had greater than 40% of the samples above the LOR. Appendix C contains the mean concentration of species for each bore and Appendix D contains the summary statistics for species and bores. There was no statistical difference in water quality between the reference and baseline monitoring bores.

Groundwater salinity in the Weaber Plain ranges from 60 to 22 400mg/L TDS, with a mean TDS around 3500mg/L (Table 3.1, Figure 3.1 and Appendices C and D). The highest salinities occurred in the northern (Lots 13, 14, 7 and 8) and southern (Lot 20) margins of the study area.

The groundwater also had a highly variable chemical composition and resultant water types (Figure 3.8 and Appendix A). All bores showed enrichment (higher concentrations than expected) relative to seawater in at least one of the major ions. Sodium was the dominant cation in most of the bores and showed enrichment relative to seawater for bores with chloride concentrations less than 2000mg/L. In bores with chloride concentrations above 2000mg/L, the sodium–chloride ratio was similar to seawater (Figure G1). The sodium concentrations ranged from 5 to 5000mg/L, with a mean of 790mg/L (Table 3.1) and the highest concentrations occurring around the northern and southern margins of the study area.

Calcium was the dominant cation for bores W2R, Limestone, 10WP47, WP13, and 10WP39. Calcium showed enrichment relative to seawater in most bores, and calcium concentrations ranged from about 2 to 1200mg/L, with a mean of 150mg/L (Table 3.1, Figure 3.2 and Appendix A).

Magnesium was the dominant cation for bores KC3, 10WP31, 11WP43D and WP5. Magnesium in most bores was also enriched relative to seawater and had concentrations ranging from 1 to 2200mg/L, with a mean of 170mg/L (Table 3.1, Figure 3.2 and Appendix A).

The dominant anion for groundwater with a TDS less than 1400mg/L was bicarbonate. At higher salinities, the dominant anions were chloride and sulfate. Bicarbonate concentrations ranged from 8 to 8000mg/L, with a mean of 760mg/L. Chloride concentration ranged from 11 to 10 600mg/L, with a mean of 1170mg/L (Table 3.1, Figure 3.1 and Appendix A). In many instances, sulfate was enriched relative to seawater and concentration ranged from 0.5 to 6000mg/L, with a mean of 740mg/L (Table 3.1, Figure 3.2 and Appendix A).

Silicon concentration ranged from 1 to 48mg/L, with a mean around 30mg/L and bromide concentrations ranged from 50 to 30 000µg/L, with a mean of 5700µg/L (Table 3.1). The distribution of bromide was similar to chloride (Figures 3.1 and 3.3). Fluoride concentrations ranged from 60 to 1800µg/L, with a mean of 600µg/L (Table 3.1) and the highest concentrations were mainly found in bores located on or near Antrim Volcanics (Figure 3.3 and Appendices A and C). Boron concentrations were 30–2500µg/L, with a mean of 370µg/L (Table 3.1).

Iron and manganese had the highest concentrations of the minor metals and metalloids, at 2100µg/L (bore 11WP50) and 7700µg/L (bore 10WP44), respectively (Appendix C). The mean concentration of iron was around 120µg/L and the mean

concentration of manganese was around 400µg/L (Table 3.1). Lead and zinc had maximum concentrations around 100µg/L (bore 10WP37) and 200µg/L (bore 11WP54S), respectively, and the mean values were 2µg/L for lead and 20µg/L for zinc (Appendix C and Table 3.1). There are lead–zinc deposits in the study area.

Several bores (10WP44, 11WP9RPB, 11WP9RS, 11WP11, 11WP51, 11WP55) had extremely high concentrations (up to 50 000µg/L) of phosphorus (Appendix C). Upon further investigation, it was found that these bores are at sites with low permeability clays where development chemicals, such as Wellclean (AMC), that contain phosphate compounds had been used during drilling and bore construction. At these sites, development chemicals had contaminated the water samples and the chemicals were not removed by subsequent well pumping. Phosphorus data at these bores was excluded from further analysis.

The mean concentration of total phosphorus for non-contaminated bores was 90µg/L; the mean concentration of soluble reactive phosphorus was 60µg/L; the mean concentration of total nitrogen was 600µg/L and the highest concentration of total nitrogen was around 7100µg/L (bore Limestone). Nitrite was below detection in most bores and the mean nitrate concentration was around 110µg/L. Nitrogen as ammonia was detectable in half the samples collected and had a mean concentration of 180µg/L (Table 3.1). Atrazine was not detected in any of the bores, including reference bores, on the Weaber and Knox Creek plains.

The summary statistics of field measurements for all bores is shown in Table 3.2 and for individual bores in Appendix D. The groundwater pH ranged from moderately acidic (pH around 5) to alkaline (greater than pH 8.0), with most being neutral to alkaline. Groundwater had a mean alkalinity around 530mg/L calcium carbonate (CaCO<sub>3</sub>) and mean acidity around 100mg/L CaCO<sub>3</sub>. For all bores, the alkalinity exceeded the acidity and therefore they are net alkaline. Groundwater beneath the Weaber Plain generally had reducing conditions, with a mean ORP around 1 millivolt (mV). The reducing conditions of groundwater was also reflected in the low dissolved oxygen (DO) levels of groundwater, with an average concentration of 1.5mg/L.

There were slight differences between field and laboratory measured minimum and maximum EC; however, the mean and median values were similar (Tables 3.1 and 3.6). There were greater differences between field and laboratory measured pH, with field pH generally lower than the laboratory pH: the minimum field pH was 4.9 whereas the minimum laboratory pH was 5.5; the mean and median field pH was 7.2 whereas the mean and median laboratory pH was 7.5; the maximum field pH was 8.2 whereas the maximum laboratory pH was 8.8. These differences were also reflected in acidity, with mean field acidity (110mg/L CaCO<sub>3</sub>) being much higher than the mean laboratory acidity (17mg/L CaCO<sub>3</sub>). However, the mean field alkalinity (530mg/L CaCO<sub>3</sub>) varied by less than 10% from the mean laboratory alkalinity (486mg/L CaCO<sub>3</sub>) (Tables 3.1 and 3.6).

Table 3.1 Summary statistics for laboratory analysis of groundwater from all bores

| Description              | Unit                     | Number of samples | Per cent above LOR (%) | Median | Mean | Standard deviation | Min. | Max.   | 20th percentile | 80th percentile |
|--------------------------|--------------------------|-------------------|------------------------|--------|------|--------------------|------|--------|-----------------|-----------------|
| Acidity                  | CaCO <sub>3</sub> - mg/L | 335               | 95.5                   | 13.0   | 16.3 | 12.7               | 3.0  | 96.0   | 6.0             | 23.2            |
| Alkalinity               | CaCO <sub>3</sub> mg/L   | 366               | 99.7                   | 410    | 486  | 357                | 7.0  | 2 020  | 244             | 630             |
| Aluminium                | µg/L                     | 366               | 75.7                   | 17.0   | 22.8 | 25.9               | 5.0  | 240    | 10.2            | 25.0            |
| Antimony                 | µg/L                     | 366               | 27.6                   | 0.2    | 0.3  | 0.4                | 0.1  | 2.9    | 0.1             | 0.4             |
| Arsenic                  | µg/L                     | 366               | 57.1                   | 2.0    | 4.4  | 6.6                | 1.0  | 66.0   | 1.0             | 7.0             |
| Atrazine                 | µg/L                     | 213               | 0                      |        |      |                    |      |        |                 |                 |
| Barium                   | µg/L                     | 366               | 100                    | 55.0   | 71.9 | 60.6               | 8.0  | 370    | 31.0            | 95.0            |
| Beryllium                | µg/L                     | 366               | 1.1                    | 0.2    | 0.6  | 0.8                | 0.1  | 1.8    | 0.2             | 0.8             |
| Bicarbonate              | mg/L                     | 366               | 99.7                   | 482    | 760  | 1089               | 8.0  | 7 900  | 230.0           | 799             |
| Bismuth                  | µg/L                     | 366               | 0                      |        |      |                    |      |        |                 |                 |
| Boron                    | µg/L                     | 366               | 98.1                   | 230    | 372  | 396                | 30.0 | 2 500  | 90.0            | 600             |
| Bromine                  | µg/L                     | 259               | 98.5                   | 700    | 3427 | 5699               | 50.0 | 30 000 | 250             | 4040            |
| Cadmium                  | µg/L                     | 366               | 6.3                    | 0.2    | 0.4  | 0.5                | 0.1  | 2.4    | 0.1             | 0.4             |
| Calcium                  | mg/L                     | 366               | 100                    | 58.6   | 155  | 237                | 1.6  | 1 210  | 26.1            | 195             |
| Carbonate                | mg/L                     | 366               | 1.1                    | 21.0   | 25   | 13.3               | 14.0 | 44.0   | 16.4            | 32.0            |
| Chloride                 | mg/L                     | 366               | 99.7                   | 230    | 1172 | 2041               | 11.0 | 10 600 | 60.8            | 1394            |
| Chromium                 | µg/L                     | 366               | 38.0                   | 2.0    | 2.9  | 3.5                | 0.5  | 23.0   | 1.0             | 3.1             |
| Cobalt                   | µg/L                     | 366               | 49.7                   | 0.7    | 2.5  | 5.0                | 0.1  | 35.0   | 0.2             | 2.5             |
| Copper                   | µg/L                     | 365               | 71.8                   | 0.7    | 2.2  | 4.3                | 0.1  | 36.0   | 0.3             | 2.6             |
| Dissolved organic carbon | mg/L                     | 31                | 29.0                   | 4.1    | 15.9 | 35.4               | 1.4  | 110    | 1.9             | 7.8             |
| Electrical conductivity  | mS/m                     | 366               | 99.7                   | 176    | 508  | 681                | 5.2  | 2 950  | 94.3            | 831             |
| Fluoride                 | µg/L                     | 290               | 90.3                   | 475    | 592  | 397                | 60.0 | 1 800  | 270             | 958             |
| Gallium                  | µg/L                     | 335               | 8.4                    | 0.3    | 0.4  | 0.3                | 0.1  | 1.2    | 0.1             | 0.6             |
| Hardness                 | mg/L CaCO <sub>3</sub>   | 366               | 99.7                   | 389    | 924  | 1519               | 8.0  | 12 000 | 190             | 960             |
| Hydroxide                | mg/L                     | 226               | 0                      |        |      |                    |      |        |                 |                 |
| Iron                     | µg/L                     | 366               | 80.6                   | 21.0   | 124  | 259                | 5.0  | 2 100  | 8.0             | 152             |

| Description                  | Unit | Number of samples | Per cent above LOR (%) | Median | Mean | Standard deviation | Min. | Max.   | 20th percentile | 80th percentile |
|------------------------------|------|-------------------|------------------------|--------|------|--------------------|------|--------|-----------------|-----------------|
| Lanthanum                    | µg/L | 366               | 4.1                    | 0.1    | 0.4  | 0.5                | 0.1  | 1.9    | 0.1             | 0.7             |
| Lead                         | µg/L | 366               | 78.1                   | 0.5    | 2.3  | 8.5                | 0.1  | 97     | 0.2             | 1.5             |
| Lithium                      | µg/L | 364               | 82.1                   | 18.0   | 39.2 | 90.7               | 0.5  | 730    | 3.3             | 39.0            |
| Magnesium                    | mg/L | 366               | 100                    | 43.2   | 170  | 297                | 0.6  | 2 180  | 22.1            | 212             |
| Manganese                    | µg/L | 366               | 95                     | 51.5   | 432  | 1135               | 1.0  | 7 700  | 6.0             | 280             |
| Mercury                      | µg/L | 344               | 1.7                    | 0.3    | 0.5  | 0.6                | 0.1  | 1.6    | 0.1             | 0.4             |
| Molybdenum                   | µg/L | 366               | 44.5                   | 3.0    | 6.8  | 10.4               | 1.0  | 53.0   | 1.4             | 8.0             |
| Nickel                       | µg/L | 366               | 24.6                   | 2.0    | 4.7  | 4.4                | 1.0  | 18.0   | 1.0             | 8.0             |
| Nitrogen, ammonia            | µg/L | 312               | 50.3                   | 5      | 182  | 679                | 5    | 6000   | 5               | 80              |
| Nitrogen, nitrate            | µg/L | 200               | 71.0                   | 50     | 113  | 174                | 5    | 970    | 5               | 170             |
| Nitrogen, nitrite            | µg/L | 113               | 6.2                    | 5      | 6    | 4                  | 5    | 30     | 5               | 5               |
| Nitrogen, total              | µg/L | 353               | 99.2                   | 280    | 593  | 853                | 10   | 7100   | 100             | 950             |
| Oxidised nitrogen            | µg/L | 167               | 76.6                   | 60     | 141  | 229                | 5    | 1300   | 5               | 183             |
| pH                           |      | 366               | 99.7                   | 7.5    | 7.5  | 0.5                | 5.5  | 8.8    | 7.2             | 7.8             |
| Phosphorus, total            | µg/L | 353               | 93.8                   | 60     | 92   | 105                | 5    | 710    | 20              | 140             |
| Phosphorus, soluble reactive | µg/L | 366               | 89.6                   | 40     | 65   | 82                 | 5    | 650    | 10              | 90              |
| Potassium                    | mg/L | 366               | 100                    | 5.4    | 10.1 | 11.4               | 0.3  | 74.5   | 2.6             | 18.1            |
| Selenium                     | µg/L | 366               | 29.5                   | 4.0    | 5.9  | 8.0                | 1.0  | 47.0   | 2.0             | 6.0             |
| Silicon                      | mg/L | 366               | 100                    | 29.0   | 27.6 | 9.5                | 1.1  | 48.0   | 18.0            | 36.0            |
| Silver                       | µg/L | 336               | 14.9                   | 0.6    | 1.6  | 3.0                | 0.0  | 18.0   | 0.2             | 1.5             |
| Sodium                       | mg/L | 366               | 100                    | 256    | 786  | 1161               | 5.4  | 5 180  | 83.5            | 1430            |
| Sulfate, sulfur              | mg/L | 366               | 100                    | 130    | 742  | 1259               | 0.5  | 5 970  | 43.5            | 1100            |
| TDS by summation             | mg/L | 366               | 99.7                   | 990    | 3282 | 4802               | 27   | 19 000 | 520             | 4920            |
| Thallium                     | µg/L | 281               | 9.6                    | 0.2    | 0.6  | 0.7                | 0.1  | 2.2    | 0.1             | 1.1             |
| Tin                          | µg/L | 366               | 7.4                    | 0.2    | 0.3  | 0.2                | 0.1  | 1.2    | 0.1             | 0.4             |
| Titanium                     | µg/L | 335               | 5.7                    | 0.7    | 3.2  | 7.1                | 0.1  | 27.0   | 0.3             | 2.0             |
| Uranium                      | µg/L | 366               | 95.4                   | 6.1    | 19.2 | 29.5               | 0.1  | 150    | 1.7             | 30.4            |
| Vanadium                     | µg/L | 335               | 87.8                   | 17.5   | 29.4 | 38.4               | 0.2  | 250    | 4.2             | 40.4            |
| Zinc                         | µg/L | 366               | 88.0                   | 13     | 20.0 | 21.1               | 1.0  | 200    | 8.0             | 26.0            |

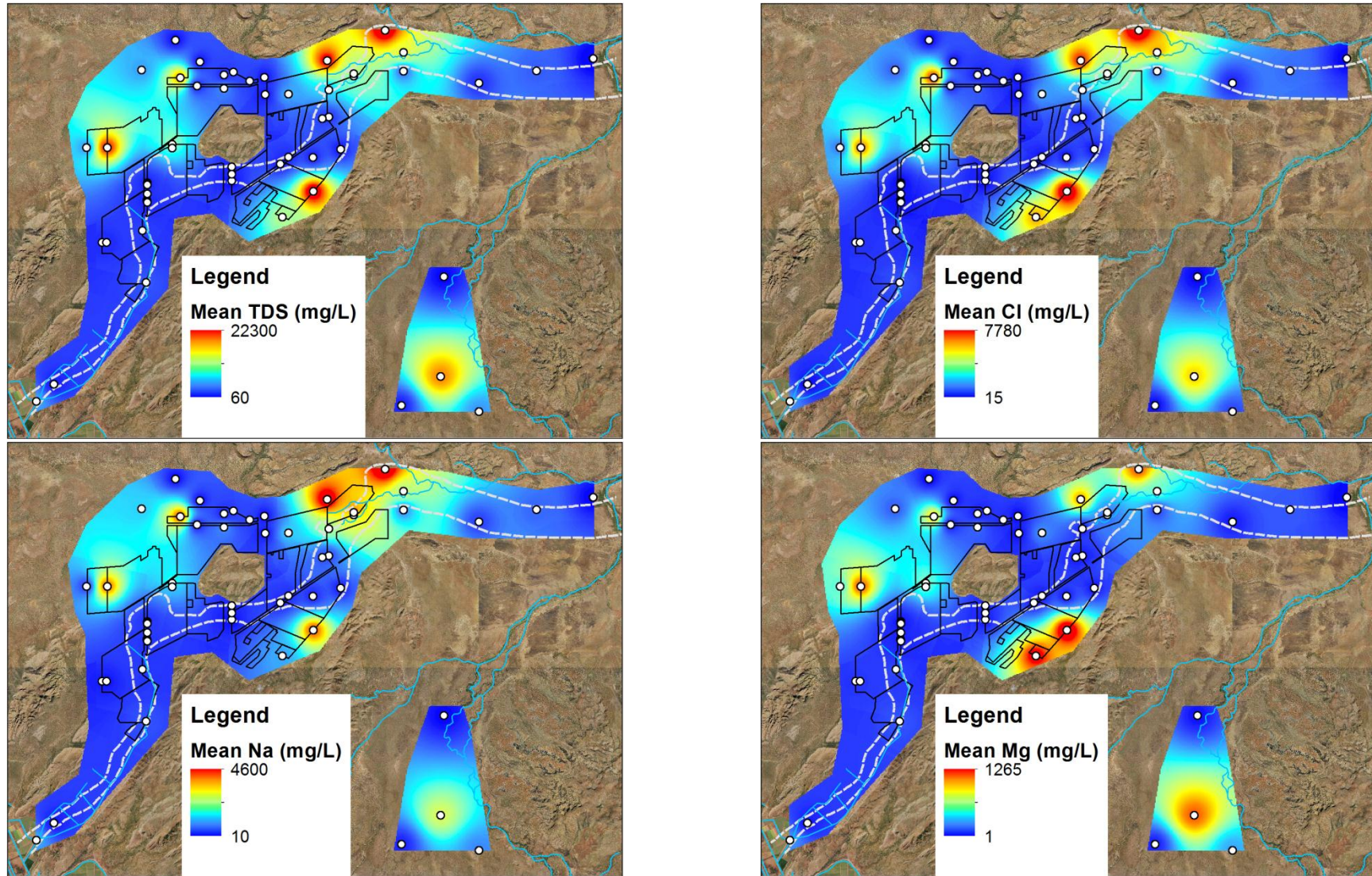


Figure 3.1 Mean groundwater concentration of total dissolved solids (TDS), chloride (Cl), sodium (Na) and magnesium (Mg) across the Weaber and Knox Creek plains

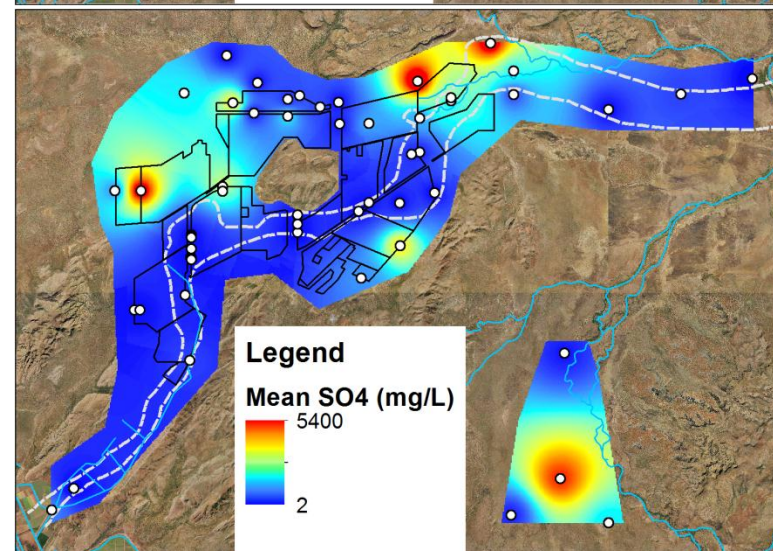
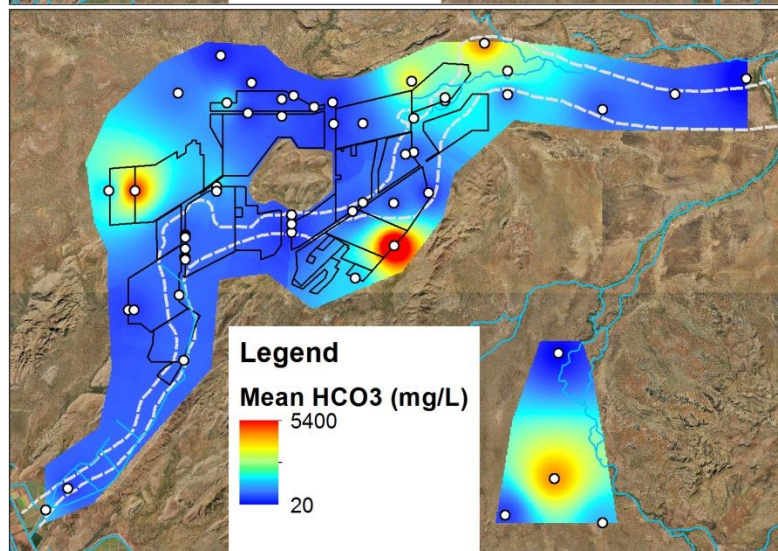
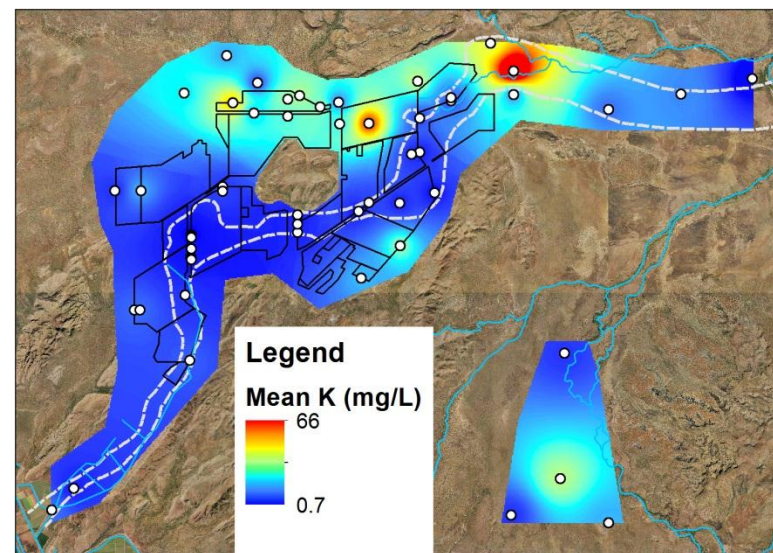
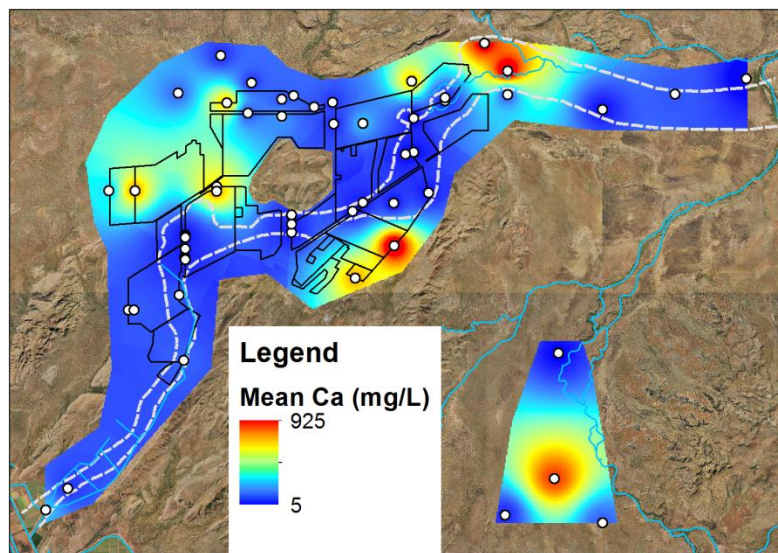


Figure 3.2 Mean groundwater concentration of calcium (Ca), potassium (K), bicarbonate ( $\text{HCO}_3$ ) and sulfate ( $\text{SO}_4$ ) across the Weaber and Knox Creek plains

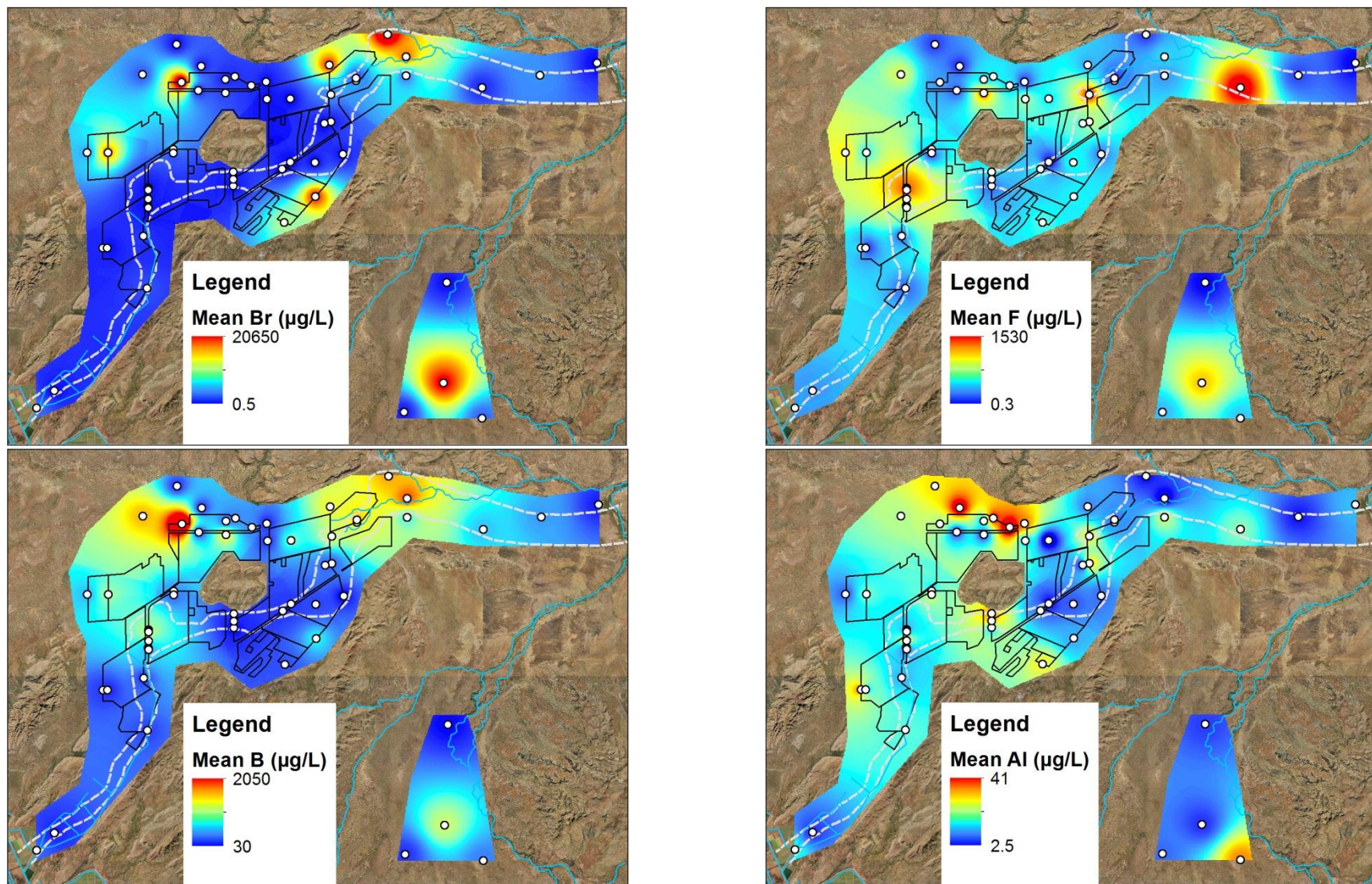


Figure 3.3 Mean groundwater concentration of bromide (Br), fluoride (F), boron (B) and aluminium (Al) across the Weaber and Knox Creek plains

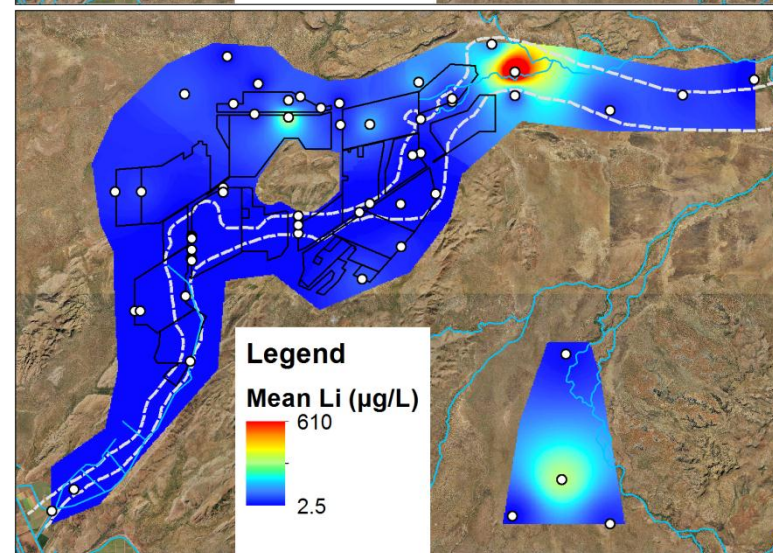
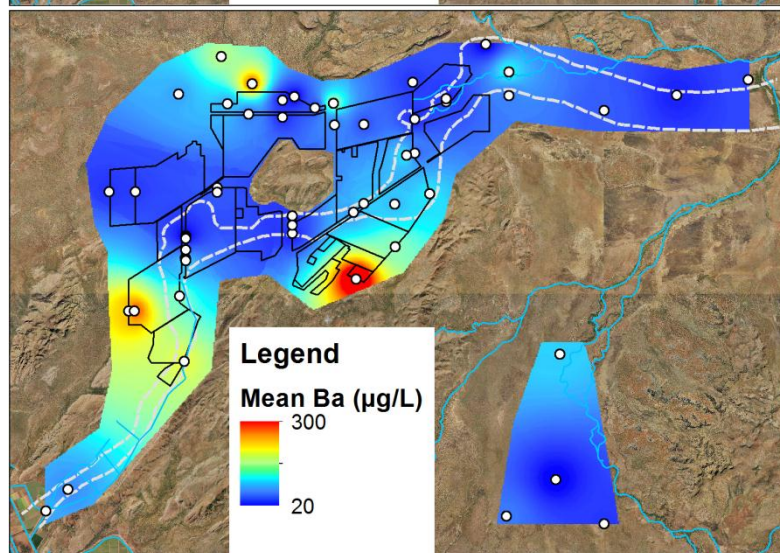
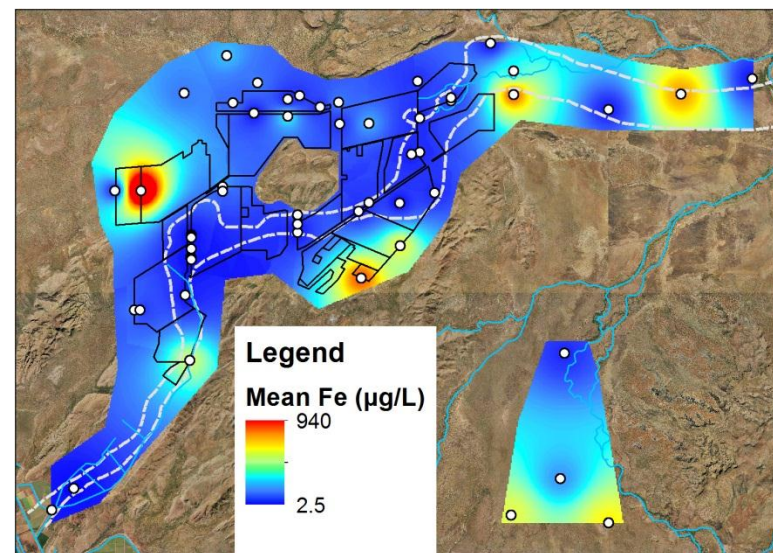
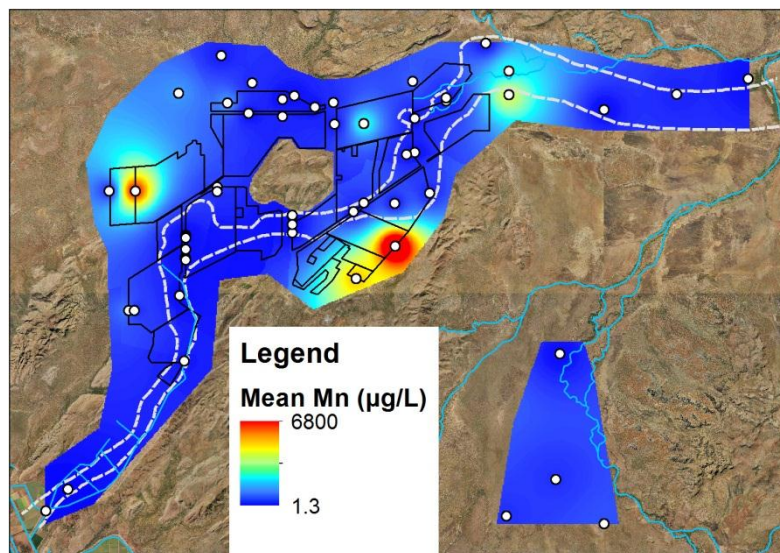


Figure 3.4 Mean groundwater concentration of manganese (Mn), iron (Fe), barium (Ba) and lithium (Li) across the Weaber and Knox Creek plains

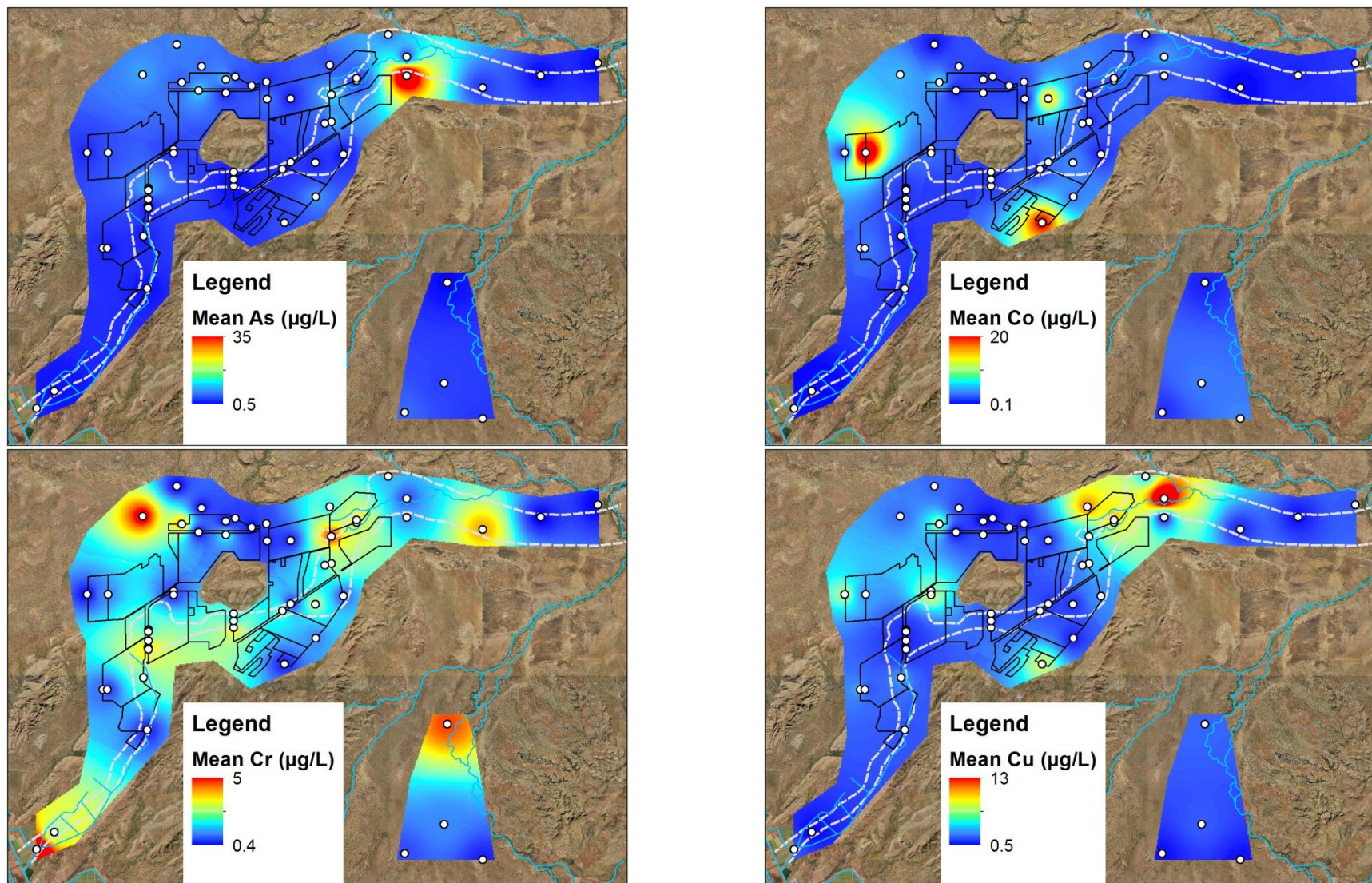


Figure 3.5 Mean groundwater concentration of arsenic (As), cobalt (Co), chromium (Cr) and copper (Cu) across the Weaber and Knox Creek plains

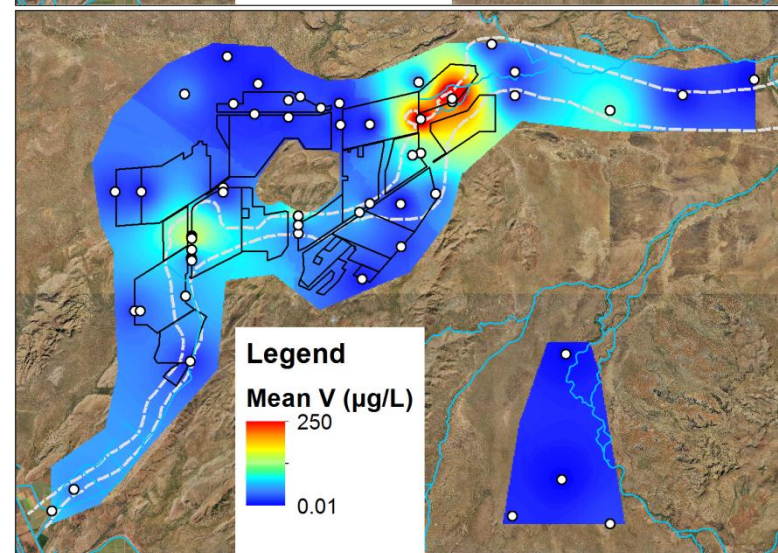
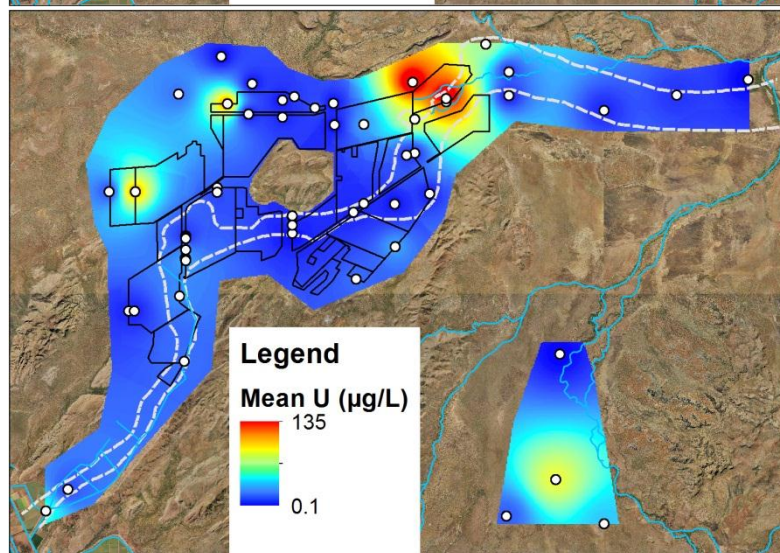
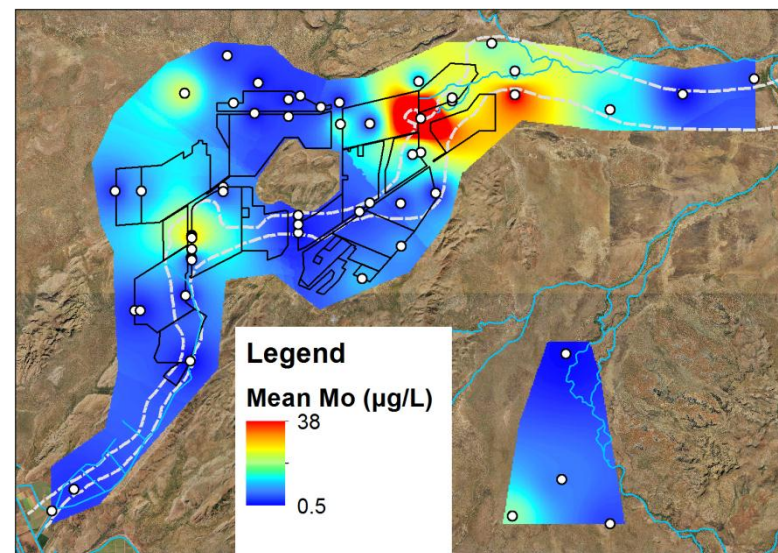
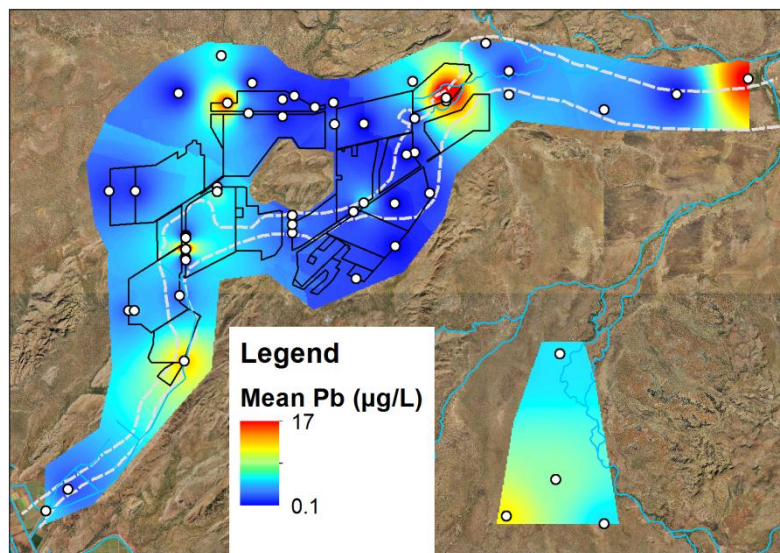


Figure 3.6 Mean groundwater concentration of lead (Pb), molybdenum (Mo), uranium (U) and vanadium (V) across the Weaber and Knox Creek plains

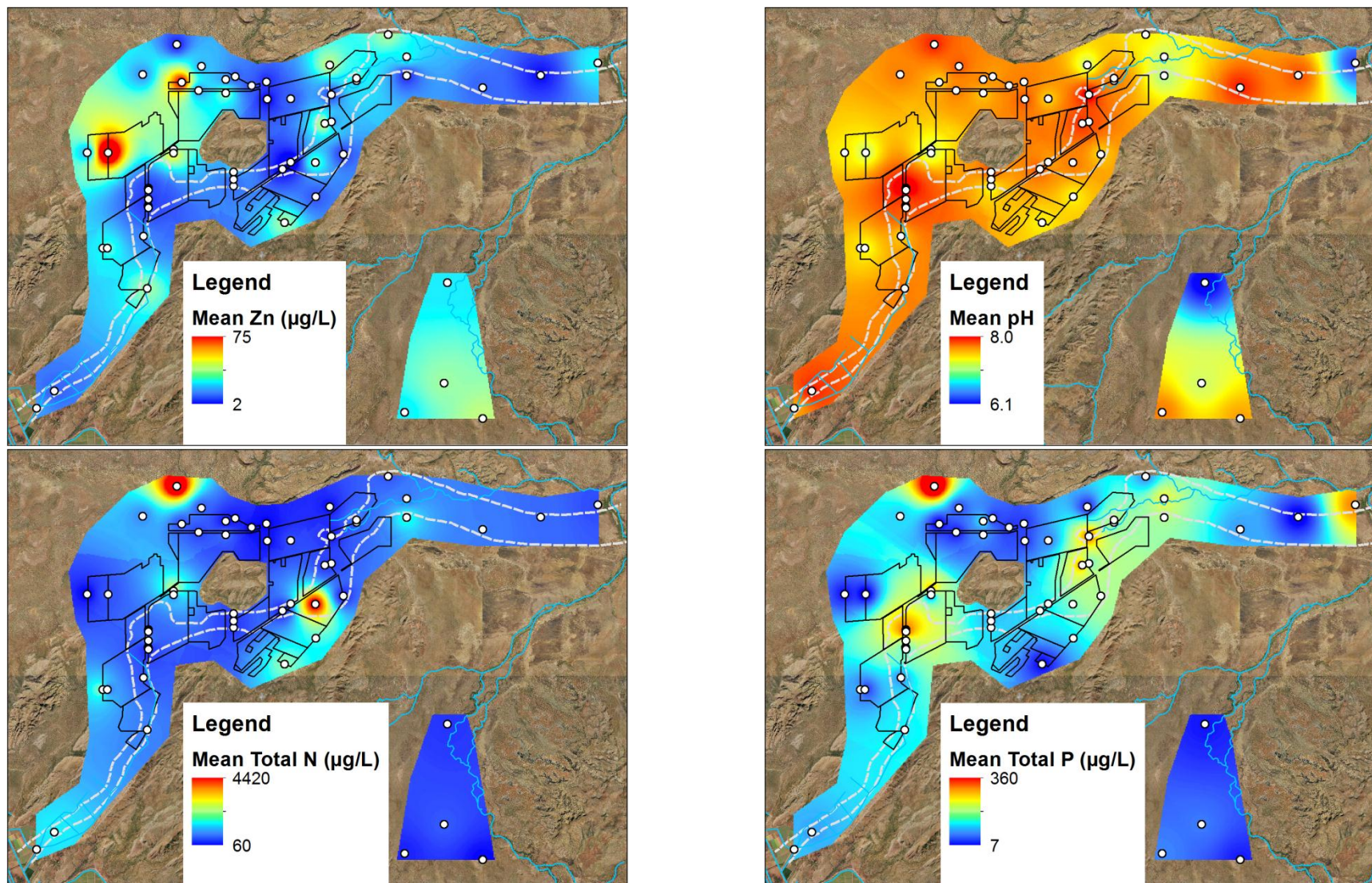
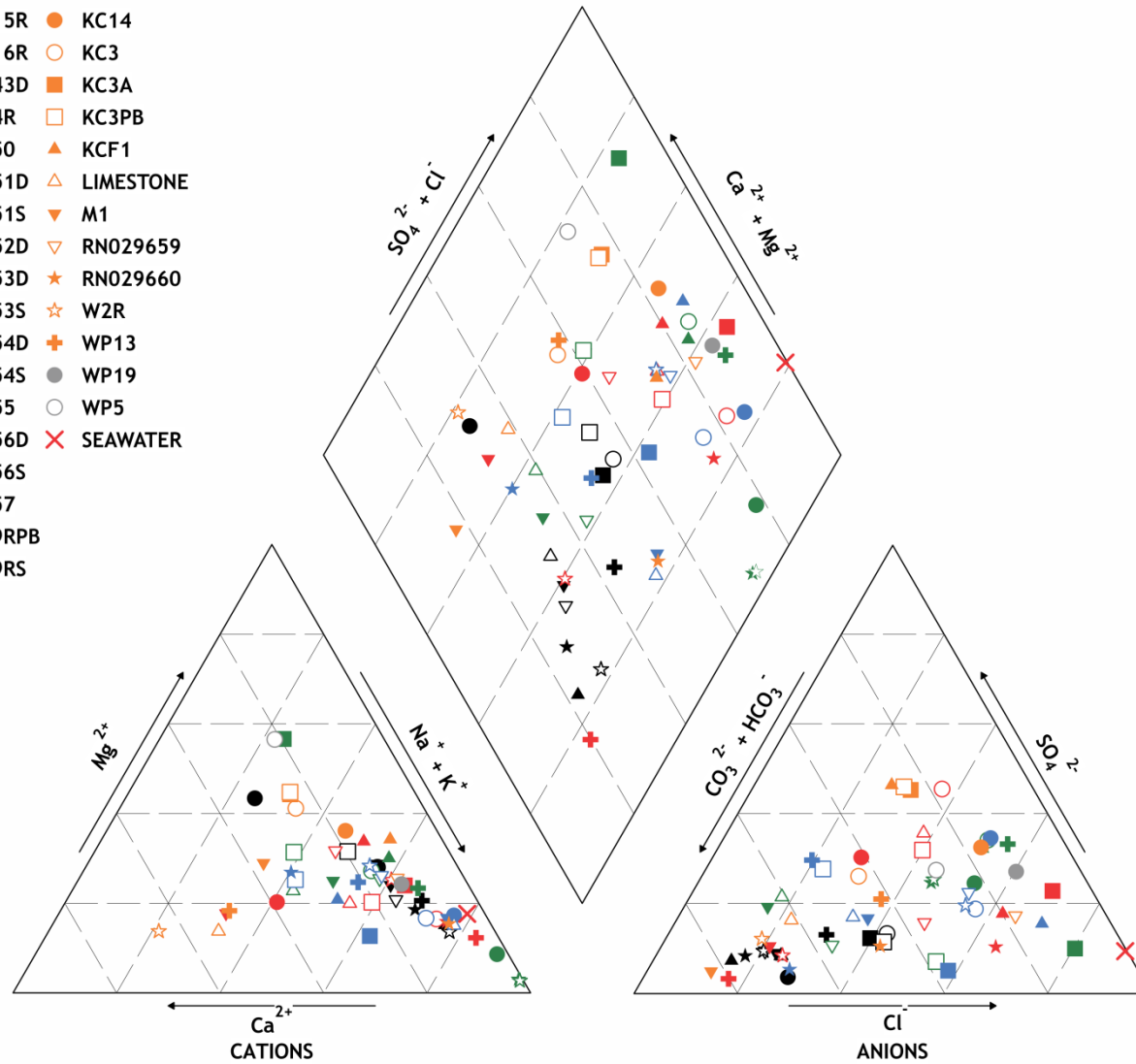


Figure 3.7 Map showing mean groundwater concentration of zinc (Zn), pH, total nitrogen (Total N) and total phosphorus (Total P) across the Weaber and Knox Creek plains

# LEGEND

|            |            |             |
|------------|------------|-------------|
| ● 10WP31   | ● 11WP15R  | ● KC14      |
| ○ 10WP32   | ○ 11WP16R  | ○ KC3       |
| ■ 10WP32PB | ■ 11WP43D  | ■ KC3A      |
| □ 10WP33   | □ 11WP4R   | □ KC3PB     |
| ▲ 10WP35N  | ▲ 11WP50   | ▲ KCF1      |
| △ 10WP35PB | △ 11WP51D  | △ LIMESTONE |
| ▼ 10WP35S  | ▼ 11WP51S  | ▼ M1        |
| ▽ 10WP36N  | ▽ 11WP52D  | ▽ RN029659  |
| ★ 10WP36PB | ★ 11WP53D  | ★ RN029660  |
| ☆ 10WP36S  | ☆ 11WP53S  | ☆ W2R       |
| ⊕ 10WP37   | ⊕ 11WP54D  | ⊕ WP13      |
| ● 10WP39   | ● 11WP54S  | ● WP19      |
| ○ 10WP40   | ○ 11WP55   | ○ WP5       |
| ■ 10WP41   | ■ 11WP56D  | ✕ SEAWATER  |
| □ 10WP42   | □ 11WP56S  |             |
| ▲ 10WP44   | ▲ 11WP57   |             |
| △ 10WP46   | △ 11WP9RPB |             |
| ▼ 10WP47   | ▼ 11WP9RS  |             |
| ▽ 11CS10RD | ▽ CG1      |             |
| ★ 11CS10RS | ★ CG2      |             |
| ☆ 11WP11RD | ☆ CG4      |             |
| ⊕ 11WP11RS | ⊕ KC13     |             |



21 Figure 3.8 Piper diagram for groundwater monitoring bores across the Weaber and Knox Creek plains

Table 3.2 Summary statistics of field measurements of groundwater

| Description                                | Number of samples | Median | Mean | Standard deviation | Min. | Max. | 20th percentile | 80th percentile |
|--------------------------------------------|-------------------|--------|------|--------------------|------|------|-----------------|-----------------|
| EC (mS/m)                                  | 331               | 172    | 504  | 664                | 7.4  | 2750 | 90.1            | 842             |
| pH                                         | 250               | 7.2    | 7.2  | 0.5                | 4.9  | 8.5  | 7               | 7.5             |
| Temperature °C                             | 325               | 29.8   | 30   | 1.4                | 25.6 | 34.9 | 29              | 31              |
| Total acidity<br>mg/L CaCO <sub>3</sub>    | 125               | 80     | 110  | 102                | 0    | 560  | 40              | 160             |
| Total alkalinity<br>mg/L CaCO <sub>3</sub> | 107               | 450    | 530  | 373                | 50   | 1950 | 300             | 676             |
| ORP RAW –<br>Hanna (mV)                    | 56                | 36     | 1    | 128                | -340 | 325  | -137            | 89              |
| ORP RAW –<br>WTW (mV)                      | 146               | 23     | -2   | 144                | -333 | 272  | -156            | 132             |
| DO initial (mg/L)                          | 91                | 1.4    | 1.8  | 1.4                | 0.2  | 6.5  | 0.5             | 3.1             |
| DO final (mg/L)                            | 126               | 0.8    | 1.5  | 1.8                | 0    | 9.4  | 0.2             | 2.7             |
| DO final %                                 | 88                | 11.2   | 20.6 | 23.6               | 1    | 127  | 2.3             | 34.3            |

## 3.2 Suitability of groundwater as a resource for agriculture

### 3.2.1 Suitability of groundwater for direct irrigation

Groundwater chemistry was assessed against modified USDA (1954) salinity and sodicity criteria for the suitability for direct irrigation and is shown in Figure 3.9. Groundwater bores with extremely high salinity (>550mS/m) were not included in Figure 3.9 because they are unsuitable for irrigation. Figure 3.10 shows the irrigation salinity classes of groundwater on the Weaber and Knox Creek plains. Figures 3.11 and 3.12 show the areas where groundwater concentrations have exceeded the ANZECC and ARMCANZ (2000) irrigation LTV and Figure 3.13 shows the plant sensitivity classes for sodium and chloride.

There are no areas within the current farmland lots that would be suitable for direct irrigation by groundwater (<80mS/m) (Figures 3.9 and 3.10). Groundwater in the palaeochannel south-west of Weaber Plain is in the high salinity category, however, most groundwater on the Weaber and Knox Creek plains is in the very high to extreme categories.

Groundwater in areas exceeded the ANZECC and ARMCANZ (2000) irrigation LTV for boron, fluoride, iron, lithium (for citrus crops), manganese, molybdenum, uranium and total phosphorus (Figures 3.11 and 3.12). Groundwater on the Weaber and Knox Creek plains is mostly suitable for plants that are moderately tolerant or tolerant to sodium and chloride (Figure 3.13). The areas that exceeded the irrigation LTV are generally too saline for direct irrigation.

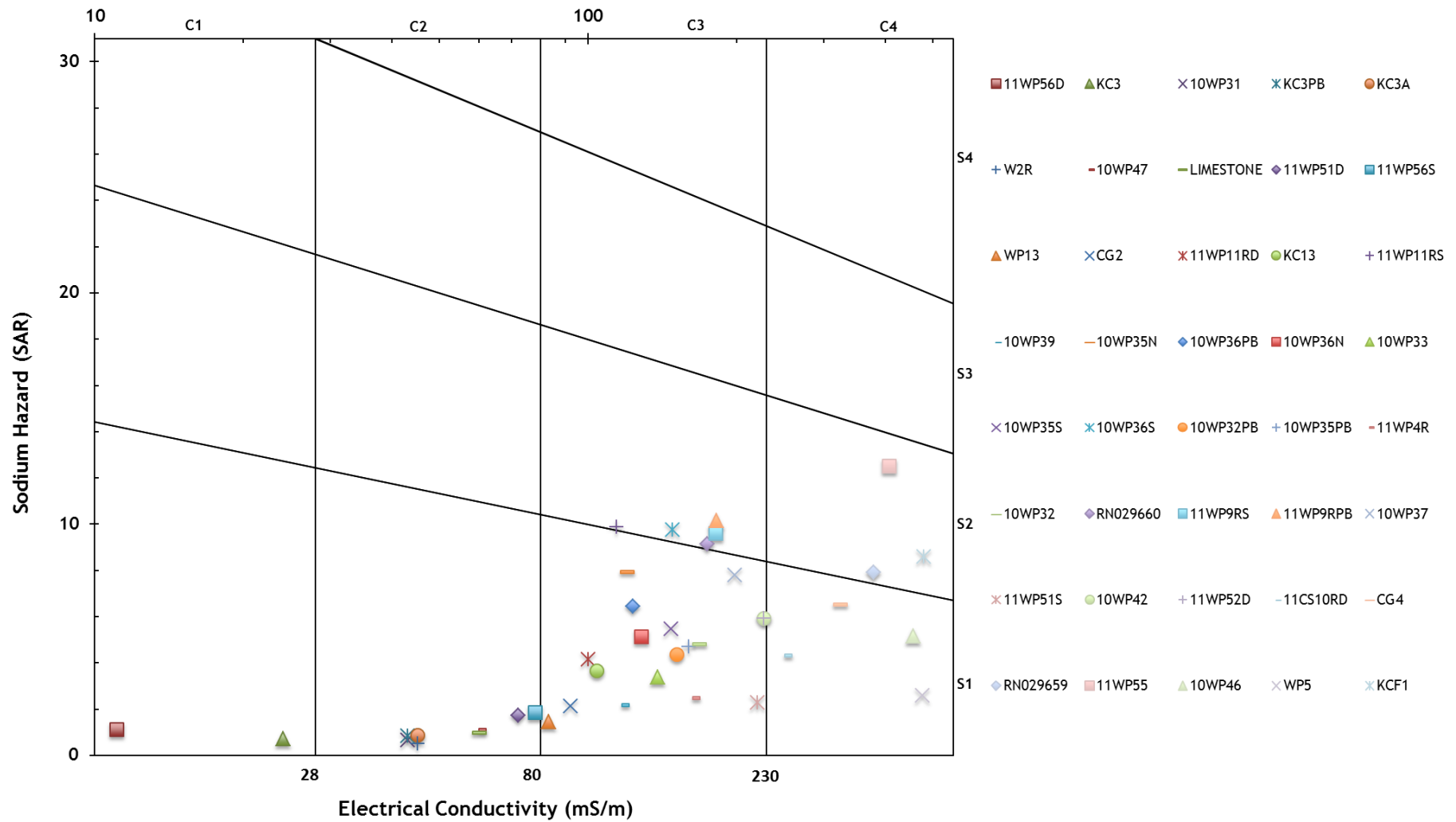


Figure 3.9 Modified Wilcox diagram showing the salinity and sodicity classes for groundwater on the Weaber and Knox Creek plains that are below 550mS/m

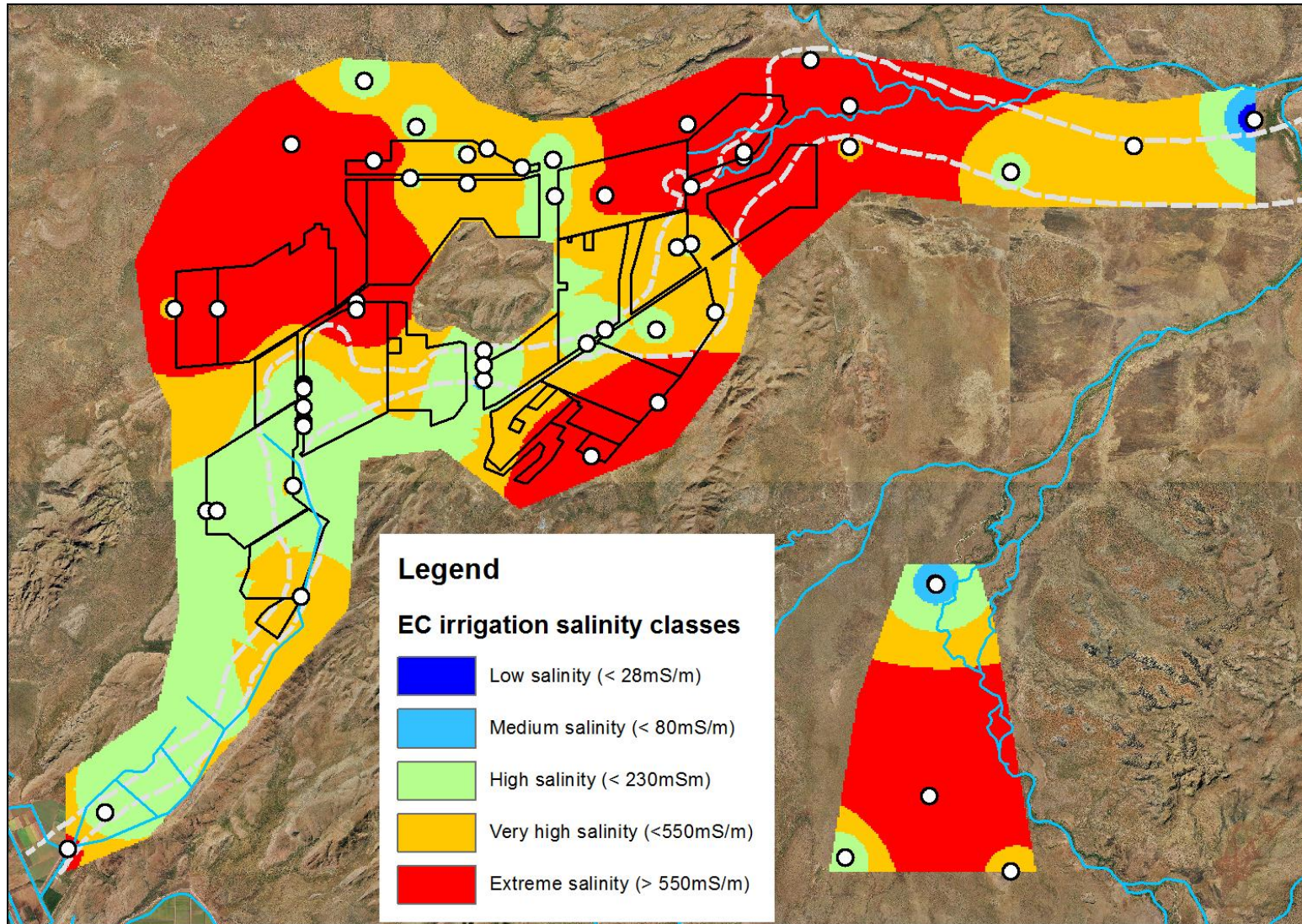


Figure 3.10 Irrigation salinity classes of groundwater on the Weaver and Knox Creek plains

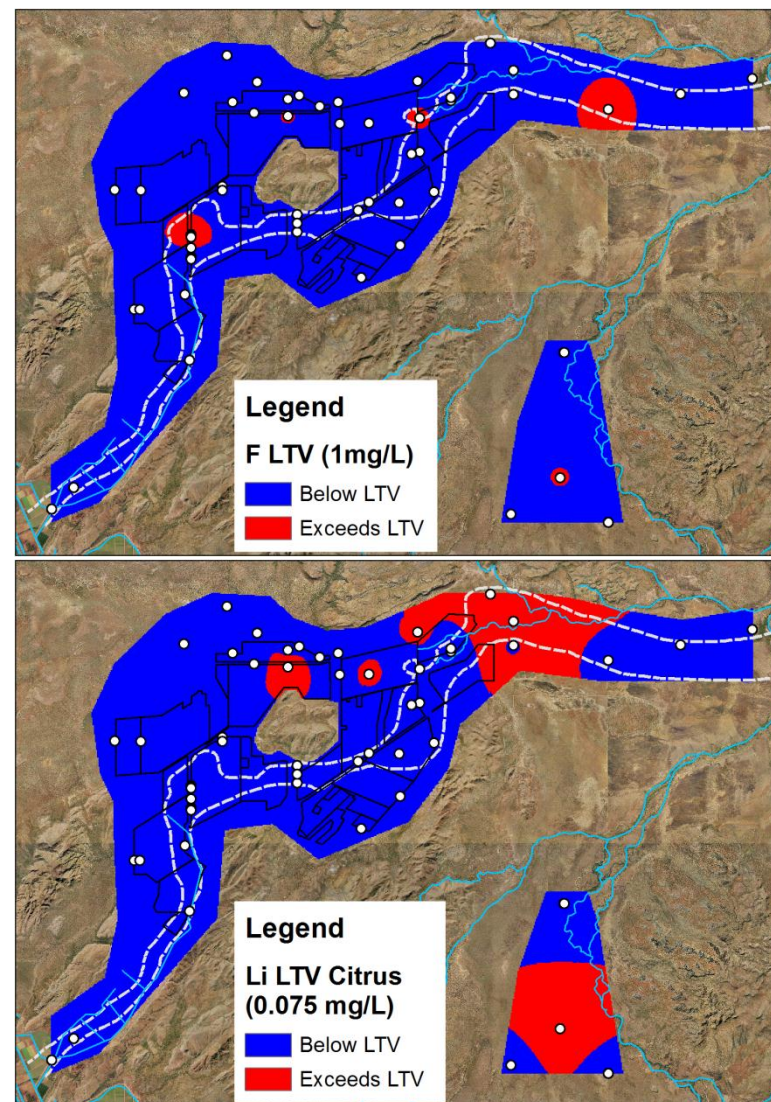
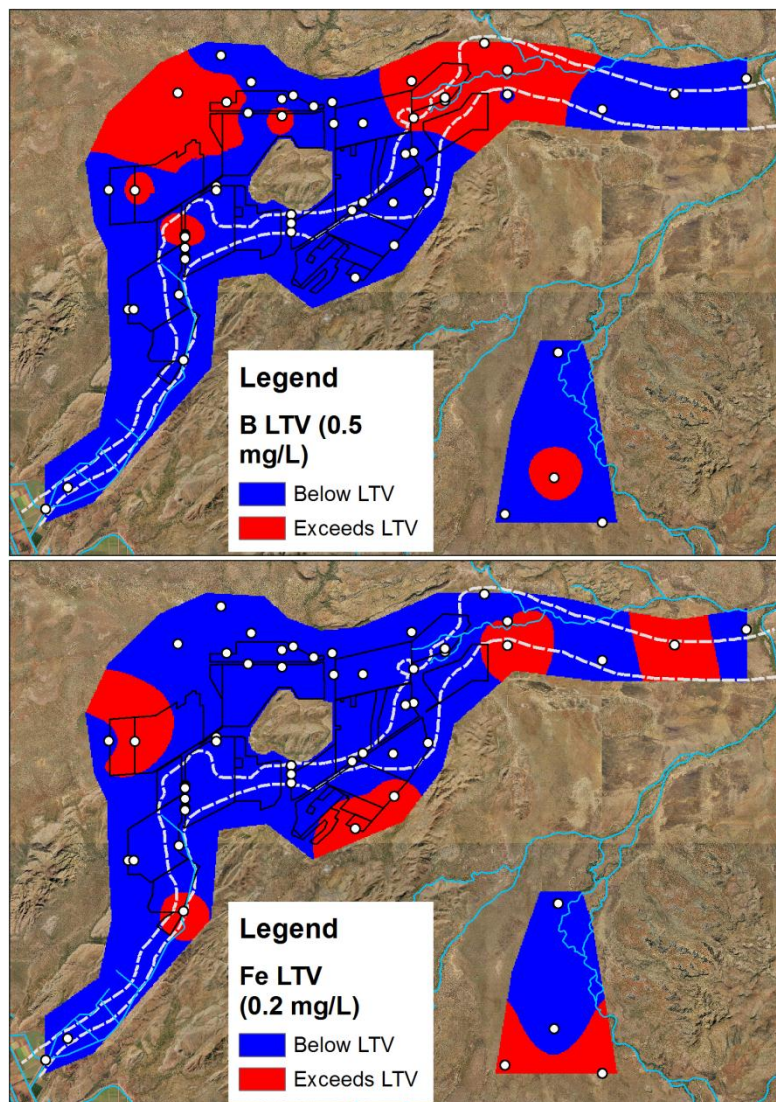


Figure 3.11 Areas where groundwater exceeds the irrigation long-term trigger values (LTV) for boron (B), fluoride (F), iron (Fe) and lithium (Li) for citrus crops across the Weaber and Knox Creek plains

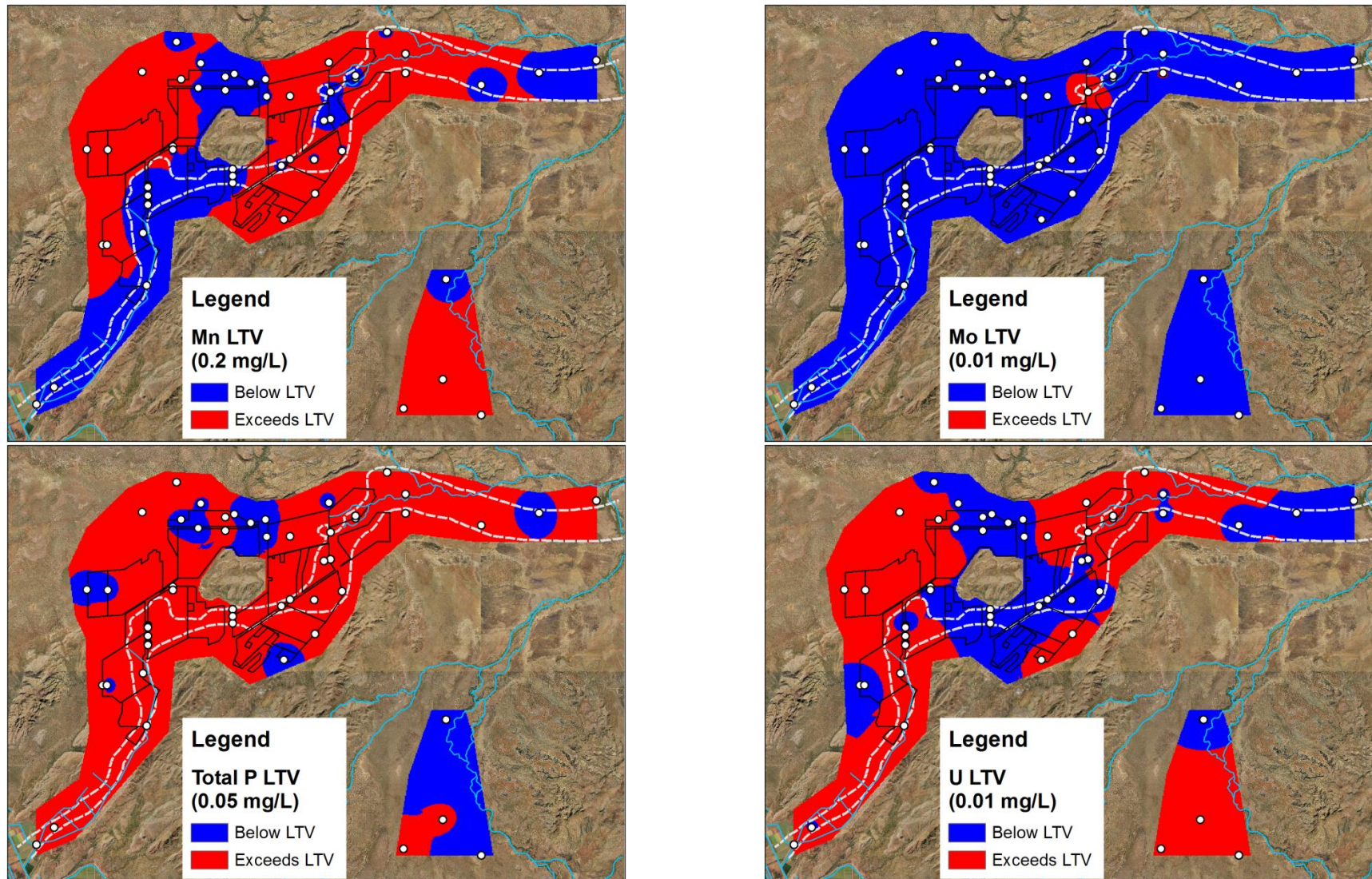


Figure 3.12 Areas where groundwater exceeds the irrigation long-term trigger values (LTV) for manganese (Mn), molybdenum (Mo), total phosphorus (Total P) and uranium (U) across the Weaber and Knox Creek plains

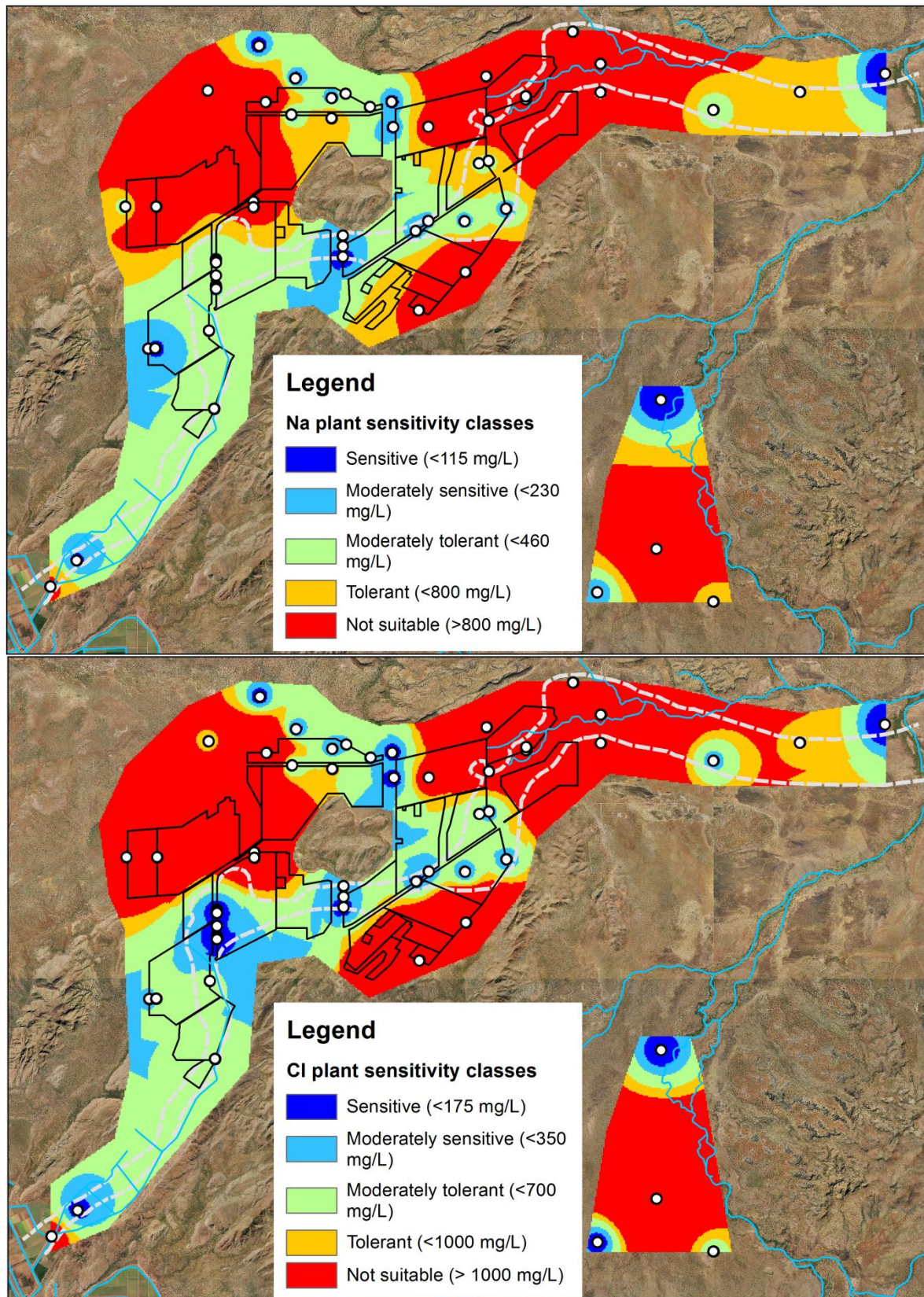


Figure 3.13 Map showing suitability of groundwater for irrigation based on the sensitivity of plants to sodium (Na) and chloride (Cl) across the Weaber and Knox Creek plains

### 3.2.2 Mixing of groundwater with the M2 supply channel

The water quality from the dewatering bores was estimated from groundwater concentration grids (Figures 3.1 to 3.7). The water from the different dewatering bores was equally mixed together and the resultant water quality is shown in Table 3.3. The modelled final groundwater quality forecast to be pumped into the M2 supply channel after 20 years of development was about 1400mg/L TDS (200mS/m). This quality is similar to the salinity (1200mg/L) derived from the groundwater models (KBR 2011) at 50 years.

Table 3.7 shows the results of the four mixing model scenarios. For Scenario A (forecast groundwater flow, average channel flow), the resultant salinity was about 33mS/m. For Scenario B (forecast groundwater flow, peak channel flow), the resultant salinity was 31mS/m. For Scenario C (high groundwater flow, average channel flow), the resultant salinity was 35mS/m. For Scenario D (high groundwater flow, peak channel flow), the resultant salinity was 33mS/m.

For all scenarios, the modelled salinity (EC) was within the USDA (1954) C2 medium category (<80mS/m) and therefore suitable for irrigation. All modelled species were below the ANZECC and ARMCANZ (2000) irrigation LTV (Table 3.7 and Appendix F) and therefore suitable for irrigation.

For all scenarios, sodium and chloride concentrations are suitable for plants sensitive to these ions. Hardness and pH are also at levels suitable for irrigation and should not cause corrosion or fouling problems. For all scenarios, the sodium adsorption ratio and residual sodium carbonate index are low and indicate that irrigation water is not likely to cause soil structural problems.

Table 3.3 Modelled resultant water quality from mixing pumped groundwater with the water in the M2 supply channel and its suitability for irrigation under four scenarios (see Figures 3.11 to 3.13 and Appendix F for long-term trigger values (LTV))

| Analyte            | Unit | Modelled groundwater quality | M1 water quality | Scenario A water quality | Scenario A LTV  | Scenario B water quality | Scenario B LTV  | Scenario C water quality | Scenario C LTV  | Scenario D water quality | Scenario D LTV  |
|--------------------|------|------------------------------|------------------|--------------------------|-----------------|--------------------------|-----------------|--------------------------|-----------------|--------------------------|-----------------|
| Aluminium          | µg/L | 17                           | 16               | 16                       | Below           | 16                       | Below           | 16                       | Below           | 16                       | Below           |
| Arsenic            | µg/L | 2                            | 0.5              | 0.6                      | Below           | 0.6                      | Below           | 0.6                      | Below           | 0.6                      | Below           |
| Boron              | µg/L | 217                          | 30               | 37.36                    | Below           | 35                       | Below           | 40                       | Below           | 38                       | Below           |
| Beryllium          | µg/L | 0.1                          | 0.4              | 0.34                     | Below           | 0.3                      | Below           | 0.3                      | Below           | 0.3                      | Below           |
| Cadmium            | µg/L | 0.07                         | 0.05             | 0.05                     | Below           | 0.05                     | Below           | 0.05                     | Below           | 0.05                     | Below           |
| Chloride           | mg/L | 300                          | 11               | 23                       | Sensitive       | 20                       | Sensitive       | 26                       | Sensitive       | 23                       | Sensitive       |
| Cobalt             | µg/L | 1.0                          | 2.5              | 2.4                      | Below           | 2.5                      | Below           | 2.4                      | Below           | 2.4                      | Below           |
| Chromium           | µg/L | 1.2                          | 1.3              | 1.3                      | Below           | 1.3                      | Below           | 1.3                      | Below           | 1.3                      | Below           |
| Copper             | µg/L | 1.7                          | 1.0              | 1.0                      | Below           | 1.0                      | Below           | 1.0                      | Below           | 1.0                      | Below           |
| EC                 | mS/m | 195                          | 26               | 33                       | Medium salinity | 31                       | Medium salinity | 35                       | Medium salinity | 33                       | Medium salinity |
| TDS                | mg/L | 1400                         | 140              | 230                      |                 | 220                      |                 | 250                      |                 | 230                      |                 |
| Fluoride           | µg/L | 544                          | 190              | 204                      | Below           | 200                      | Below           | 209                      | Below           | 204                      | Below           |
| Hardness           | mg/L | 551                          | 124              | 141                      | Suitable        | 136                      | Suitable        | 147                      | Suitable        | 141                      | Suitable        |
| Iron               | µg/L | 71                           | 19               | 21                       | Below           | 21                       | Below           | 22                       | Below           | 21                       | Below           |
| Mercury            | µg/L | 0.06                         | 0.05             | 0.05                     | Below           | 0.05                     | Below           | 0.05                     | Below           | 0.05                     | Below           |
| Lithium            | µg/L | 5.5                          | 2.5              | 2.6                      | Below           | 2.6                      | Below           | 2.7                      | Below           | 2.6                      | Below           |
| Manganese          | µg/L | 235                          | 4.3              | 13                       | Below           | 11                       | Below           | 16                       | Below           | 14                       | Below           |
| Molybdenum         | µg/L | 2.5                          | 0.5              | 0.6                      | Below           | 0.6                      | Below           | 0.6                      | Below           | 0.6                      | Below           |
| Nitrogen (reduced) | µg/L | 38                           | 5.0              | 6.3                      | Below           | 5.9                      | Below           | 6.8                      | Below           | 6.3                      | Below           |

| Analyte                         | Unit | Modelled groundwater quality | M1 water quality | Scenario A water quality | Scenario A LTV | Scenario B water quality | Scenario B LTV | Scenario C water quality | Scenario C LTV | Scenario D water quality | Scenario D LTV |
|---------------------------------|------|------------------------------|------------------|--------------------------|----------------|--------------------------|----------------|--------------------------|----------------|--------------------------|----------------|
| Nitrogen (oxidised)             | µg/L | 155                          | 12               | 17                       | Below          | 16                       | Below          | 19                       | Below          | 17                       | Below          |
| Nitrogen (total)                | µg/L | 522                          | 193              | 206                      | Below          | 203                      | Below          | 211                      | Below          | 206                      | Below          |
| Sodium                          | mg/L | 278                          | 16               | 27                       | Sensitive      | 24                       | Sensitive      | 30                       | Sensitive      | 27                       | Sensitive      |
| Nickel                          | µg/L | 1.0                          | 0.5              | 0.5                      | Below          | 0.5                      | Below          | 0.5                      | Below          | 0.5                      | Below          |
| Phosphorus (soluble reactive)   | µg/L | 113                          | 8.3              | 12                       | Below          | 11                       | Below          | 14                       | Below          | 13                       | Below          |
| Phosphorus (total)              | µg/L | 150                          | 10               | 15                       | Below          | 14                       | Below          | 17                       | Below          | 16                       | Below          |
| Lead                            | µg/L | 1.5                          | 0.1              | 0.2                      | Below          | 0.1                      | Below          | 0.2                      | Below          | 0.2                      | Below          |
| pH                              |      | 7.6                          | 8.3              | 8.2                      | Suitable       | 8.2                      | Suitable       | 8.2                      | Suitable       | 8.2                      | Suitable       |
| Selenium                        | µg/L | 1.6                          | 0.5              | 0.5                      | Below          | 0.5                      | Below          | 0.6                      | Below          | 0.5                      | Below          |
| Uranium                         | µg/L | 11                           | 0.3              | 0.8                      | Below          | 0.7                      | Below          | 0.9                      | Below          | 0.8                      | Below          |
| Vanadium                        | µg/L | 32                           | 3.8              | 4.9                      | Below          | 4.6                      | Below          | 5.3                      | Below          | 4.9                      | Below          |
| Zinc                            | µg/L | 14                           | 8.8              | 9                        | Below          | 9.0                      | Below          | 9.1                      | Below          | 9.1                      | Below          |
| Sodium ad-sorption ratio        |      | 6.0                          | 0.8              | 1.2                      | Low            | 1.1                      | Low            | 1.4                      | Low            | 1.2                      | Low            |
| Residual sodium carbonate index |      | 0.9                          | 0.5              | 0.5                      | Low            | 0.5                      | Low            | 0.5                      | Low            | 0.5                      | Low            |
| Atrazine                        | µg/L | 0.05                         | 0.05             | 0.05                     | Below          | 0.05                     | Below          | 0.05                     | Below          | 0.05                     | Below          |

## 4 Discussion

### 4.1 Environmental hazards of groundwater

Condition 12 of the Australian Government Ministerial Conditions under the EPBC Act required the installation of high-intensity and low-intensity monitoring sites and reference bores 18 months prior to starting irrigation. Seasonal monitoring of specific analytes was stipulated in the EPBC conditions to provide pre-development or baseline groundwater chemistry. The analytes selected were based on ANZECC and ARMCANZ (2000) risk management and sampling methodologies as detailed in the groundwater management plan

By the end of 2013, 58 bores had been monitored seasonally over 2–3 years. Results of the baseline analysis of these bores are summarised in Table 3.1, Figures 3.1 to 3.7 and Appendix C. The monitoring regime was established to test whether groundwater posed a direct or indirect threat to the environmental assets of the Keep River and the dry season discharge to the irrigation system. Accordingly, a comprehensive suite of analytes was chosen for baseline assessment.

For surface water aquatic environmental assets, the analytes for baseline monitoring should include any substances that are known to harm the target aquatic species (ANZECC & ARMCANZ 2000). The target aquatic species requiring protection in this instance are the EPBC-listed threatened species: freshwater sawfish (*Pristis microdon*) and spartooth shark (*Glyphis glyphis*). While there was some general (e.g. Bartley & Spiers 2010) and local (e.g. Bennett & George 2011, 2014) data about the expected changes to water quality under tropical irrigated agriculture, there was no specific information about which water quality factors are of particular importance or risk to the EPBC-threatened species (Helen Larsen, IRG, 2011, pers. comm., November). Consequently, a precautionary approach was used to select the comprehensive list of analytes that ANZECC and ARMCANZ (2000) categorise as chemical hazards to aquatic biota. The list of analytes in Appendix B encompasses all of the chemical hazards listed in ANZECC and ARMCANZ (2000), including the herbicide atrazine. These were monitored in all 58 bores.

There are two determinants of the direct risk of groundwater on threatened species in the Keep River: the exposure pathways and travel time for a chemical hazard to move from farmland to the Keep River, and the nature of the chemical hazards in the groundwater (Chapter 4.1.1).

The travel time for a chemical hazard to move from the eastern edge of the irrigated farmlands to the Keep River can be estimated from aquifer characteristics and available modelling (KBR 2011, Bennett & George 2011, 2014, Paul et al. 2011). Using this data, travel times for a chemical hazard to reach the Keep River from the eastern edge of the Goomig Farmlands would be 90 to 180 years. If dewatering is implemented, as recommended in the management plans (KBR 2011, Strategen 2012a, b), it would take 100 to 300 years for chemical hazards to reach the river.

#### 4.1.1 Forecasting long-term changes in groundwater quality of Goomig

The second determinant of risk to the Keep River is which chemical hazards are present in groundwater and how they will change with the introduction of irrigated agriculture to the Goomig Farmlands. These changes in water quality can be evaluated by comparing changes in groundwater quality in ORIA Stage 1, which has been irrigated for over 40 years, to groundwater quality in the Weaber Plain.

Smith et al. (2007) looked a broad range of analytes including 29 pesticides plus the herbicide atrazine in 31 monitoring bores in the ORIA Stage 1. When comparing mean groundwater quality in the ORIA Stage 1 to the baseline conditions in Weaber; TDS, metals and metalloids are largely unchanged by agriculture. Only three species changed significantly: the herbicide atrazine (mean concentration of 0.37µg/L) and two nitrogen species (total nitrogen – mean 4200µg/L and total oxidised nitrogen – mean 4035µg/L) (Table 4.1).

After 40 years of irrigation, atrazine was found in only six of 63 samples and in all other bores it was below the detection limit. The data suggests pesticides or herbicides may be slow to accumulate in groundwater beneath the black soil plains.

The speciation of nitrogen in groundwater has changed as a result of irrigated agriculture. Nitrogen in the ORIA Stage 1 was dominated by the oxidised nitrogen fraction (nitrate and nitrite) whereas in the Weaber baseline conditions oxidised nitrogen (mean concentration around 200µg/L) was around 15% of total nitrogen (mean concentration around 1300µg/L). The remainder of the total nitrogen consisted of organic nitrogen. Therefore, under agriculture, nitrogen speciation in the irrigated farmlands of the Weaber Plain is likely to be dominated by oxidised nitrogen that will be more bioavailable.

The specified monitoring regime (discussed below- Chapter 4.3) will detect changes in groundwater chemistry. If chemical hazards exceed baseline conditions, the groundwater management plan and related plans (Strategen 2012a, b) recommend a range of measures to mitigate hazards, ranging from specific actions for chemical use to direct aquifer management.

In summary, on the basis of water quality risk from data derived in the ORIA Stage 1 and travel times, groundwater from beneath the Goomig Farmlands poses a relatively low and long-term risk (100 years) to the receiving environment, the Keep River.

Table 4.1 Baseline chemistry for the Weaber Plain compared to modelled groundwater quality and water quality from the ORIA Stage 1 (Smith et al. 2007) after 40 years of irrigation. The ratio between mean ORIA and mean Weaber baseline concentrations for species is shown

| Analyte                      | Unit                   | Number of samples (Weaber) | Number of samples (ORIA) | Mean (Weaber) | Modelled groundwater | Mean (ORIA) | Ratio |
|------------------------------|------------------------|----------------------------|--------------------------|---------------|----------------------|-------------|-------|
| Alkalinity                   | CaCO <sub>3</sub> mg/L | 366                        | 63                       | 486           | 452                  | 437         | 0.90  |
| Aluminium                    | µg/L                   | 366                        | 8                        | 22.8          | 17                   | 39.9        | 1.75  |
| Atrazine                     | µg/L                   | 213                        | 6                        | LoR           | 0.05                 | 0.37        | n/a   |
| Bicarbonate                  | mg/L                   | 366                        | Calculated               | 760           | 551                  | 533         | 0.65  |
| Calcium                      | mg/L                   | 366                        | 61                       | 155           | 61                   | 100         | 0.65  |
| Chloride                     | mg/L                   | 366                        | 62                       | 1173          | 300                  | 644         | 0.55  |
| EC                           | mS/m                   | 366                        | 63                       | 509           | 195                  | 304         | 0.60  |
| Fluoride                     | µg/L                   | 290                        | 59                       | 593           | 544                  | 842         | 1.42  |
| Iron                         | µg/L                   | 366                        | 11                       | 124           | 71                   | 19.5        | 0.16  |
| Magnesium                    | mg/L                   | 366                        | 63                       | 170           | 61                   | 122         | 0.71  |
| Manganese                    | µg/L                   | 366                        | 33                       | 432           | 235                  | 140         | 0.32  |
| Nitrogen, ammonia            | µg/L                   | 312                        | 38                       | 812           | 39                   | 323         | 0.40  |
| Nitrogen, nitrate            | µg/L                   | 200                        | Calculated               | 169           |                      | 4016        | 23.81 |
| Nitrogen, nitrite            | µg/L                   | 113                        | 6                        | 22.9          |                      | 18.8        | 0.82  |
| Nitrogen, total              | µg/L                   | 353                        | 63                       | 1267          | 522                  | 4161        | 3.28  |
| Oxidised nitrogen            | µg/L                   | 167                        | 58                       | 178           | 154                  | 4035        | 22.70 |
| pH                           |                        | 366                        | 63                       | 7.5           | 7.6                  | 7.52        | 1.00  |
| Phosphorus, total            | µg/L                   | 353                        | 17                       | 801           | 150                  | 305         | 0.38  |
| Phosphorus, soluble reactive | µg/L                   | 366                        | 63                       | 562           | 113                  | 70          | 0.13  |
| Potassium                    | mg/L                   | 366                        | 58                       | 10.1          | 4.1                  | 18.2        | 1.80  |
| Silicon                      | mg/L                   | 366                        | 16                       | 27.6          |                      | 25.7        | 0.93  |
| Sodium                       | mg/L                   | 366                        | 63                       | 786           | 278                  | 459         | 0.58  |
| Sulfate                      | mg/L                   | 366                        | 59                       | 742           | 148                  | 450         | 0.61  |
| TDS by summation             | mg/L                   | 366                        | Calculated               | 3282          | 1403                 | 2085        | 0.61  |

## 4.2 Use of groundwater for irrigation

### 4.2.1 Direct irrigation with groundwater

The baseline monitoring has been used to assess the hazard to irrigated agriculture posed by discharging groundwater into the M2 supply channel during the dry season. In terms of using groundwater for direct irrigation on agricultural crops, the updated baseline chemistry (Figure 3.10 and Appendix F) confirms the conclusions of Lillicrap et al. (2011) that the groundwater is unsuitable for direct irrigation in the proposed irrigated farmlands because the salinity is higher than 80mS/m.

Groundwater in the northern and southern margins of the proposed irrigated areas also have high chloride and sodium concentrations that make them unsuitable for irrigation (Figure 3.13). In these areas, groundwaters also exceed the ANZECC and ARMCANZ (2000) irrigation water quality LTV for total phosphorus.

Similarly, groundwater in the northern and southern margins exceeded the LTV for uranium, manganese, iron, boron and fluoride (Figures 3.11 and 3.12 and Appendix F). These areas are subject to inundation during the wet season and these analytes are species whose solubility is oxidation-reduction (redox) sensitive (Hem 1985, Drever 2002). Therefore, it is likely the concentrations are being controlled by the redox state of infiltrating waters.

The high concentrations of fluoride mainly occur in areas on or near the Antrim Volcanics (Appendix A). Basaltic and similar rocks can contain fluoride minerals (Hem 1985).

Groundwater beneath northern margins of the proposed farmlands and eastwards had relatively high concentrations of metals including zinc, copper, lead, molybdenum and vanadium. The high concentrations of metals and metalloids are natural background levels likely caused by mineralised ore bodies and vadose zone processes (Appendix G). The Keep River also had naturally high background levels of aluminium, zinc, copper, lead and boron (Bennett & George 2014). As the Weaber Plain retained its native vegetation during the baseline monitoring period, the high concentration of ions in the groundwater are naturally occurring.

### 4.2.2 Suitability of groundwater for reuse

The sustainability of pumping groundwater into the irrigation system and the risk to irrigated agriculture was first assessed by Lillicrap et al. (2011) using initial data collected up until 2011 and revised using the complete baseline data (2010–13). The modelling used delivery rates and quantity of pumped groundwater (Chapter 3.3.2 and Table 3.7) to the M2 supply channel as specified in disposal scenarios developed in the GMP (Chapter 3.2.2). Results of the updated mixing model show that for all the KBR (2011) pumping scenarios, including the maximum rate where the irrigation supply (94%) is mixed with groundwater (6%), the resultant water quality remained below the ANZECC and ARMCANZ (2000) LTV for irrigation (Table 3.7). These results are consistent with Lillicrap et al. (2011).

These results compare to a similar study for ORIA Stage 1 where Ali et al. (2002) found that mixing groundwater with water from the supply channel could be suitable for irrigation. Using similar criteria to this study, they found that groundwater with salinities up to 100mS/m required mixing with 70% scheme water before it was suitable for direct irrigation. The groundwater in the south-east of the Weaber Plain,

mainly in the Ord palaeochannel, had the lowest salinities ( $<230\text{mS/m}$ ). Based on the modelling by Ali et al. (2002) groundwater from this area would need to be mixed with around 85% scheme water before it would be suitable for direct irrigation. Therefore the proposed pumping scenarios in the management plans have dilution rates far greater than the minimum required for direct irrigation.

In addition to water quality, soil types affect the suitability of using mixed waters for irrigated agriculture. For example, Aquitaine soils found on the northern and southern margins of the Goomig Farmlands naturally accumulate low solubility salts, such as gypsum and carbonates, because of restricted drainage (Smolinski et al. 2011). Therefore, it is recommended these areas are only irrigated with scheme water.

As described in Chapter 2, the updated mixing model was based on the area of forecast shallow groundwater after 20 years of development (KBR, 2011; 80% of Goomig Farmlands) and baseline groundwater data from bores within that area. It was not derived from the 50-year KBR (2011) forecast pumping areas, owing to uncertainty in the water balance of future farming systems over that length of time. The mixing model determined a 20-year mean TDS of  $1400\text{mg/L}$  for the defined bore field, which is similar to the  $1200\text{mg/L}$  TDS determined by the 50-year forecast groundwater model (KBR 2011).

While the forecast for TDS provides confidence in the expected mass of solutes, changes from the baseline groundwater quality caused by agriculture are likely to occur, as shown by Smith et al. (2007) in ORIA Stage 1. From the comparison of ORIA Stage 1 to the Weaber baseline chemistry (Table 4.1) major ion chemistry is not expected to significantly change with agricultural development and only farm chemicals would increase in groundwater. In the mixing model, the modelled concentration for atrazine was half the limit of detection ( $0.05\mu\text{g/L}$ ) and total nitrogen was  $522\mu\text{g/L}$  which is less than the mean ORIA Stage 1 concentration (Table 4.1). The groundwater concentrations for atrazine and nutrients are probably underestimated in the mixing model, although they are diluted 25–35 times when the groundwater is mixed with the supply channel water. When the mean concentrations of atrazine and nutrients from the ORIA are diluted by the same scale as used in the mixing model, they are below the ANZECC and ARMCANZ (2000) irrigation LTV.

Future changes in water quality will be tracked by ongoing monitoring and there are requirements in the GMP (Strategen 2012a) and the stormwater and groundwater discharge management plan (Strategen 2012b) to update groundwater data and the mixing model every 2–4 years. Water quality parameters used in updating the models will be informed by regular monitoring (Chapter 4.3) and then used to underpin future management responses.

### 4.3 Monitoring groundwater

#### *Future monitoring*

The Ministerial Conditions and the GMP specify a required monitoring frequency during operation of the Goomig Farmlands. They also recommend a list of initial analytes. While some analytes are specified, the list is subject to ongoing determination by the Independent Review Group (IRG). In this chapter we recommend analytes to be monitored and changes to frequency based on the results from the baseline period, in the context of risks to agriculture and the environment.

Currently, the EPBC conditions require seasonal monitoring of various field and laboratory analytes at the high-intensity, low-intensity, reference and farm bores. These monitoring requirements are discussed below in terms of natural and agricultural chemical hazards.

### *Nutrients*

Analysis of the groundwater chemistry in ORIA Stage 1 by Smith et al. (2007) indicated that agricultural nutrients, particularly nitrogen, can enter the groundwater and increase in concentration. For example, the mean baseline concentration of total nitrogen of 1.3mg/L in Weaber Plain baseline is less than the 4.2mg/L recorded in the ORIA. Based on these results, we suggest maintaining the current list of nutrients for analysis. Adding reduced nitrogen (ammonia) and total organic nitrogen to improve the understanding the fate of farm nutrients could further guide improvements to the existing management plans.

### *Metals and metalloids*

The need for ongoing routine monitoring of metals in the proposed irrigated farmlands is not supported by the baseline data. Many of the baseline potentially hazardous metals and metalloids (e.g. boron, manganese, copper and zinc) exceed ANZECC and ARMCANZ (2000) Aquatic Ecosystem and Irrigation trigger values. As detailed in Appendix G, it is likely these values are the direct result of long-term interactions with basement geology and soil-forming processes. Furthermore, based on the comparison of values in the ORIA Stage 1 and baseline values in Goomig, it appears that irrigation practices have little effect on the concentrations of hazardous metal and metalloid species.

Similarly, Bennett and George (2014), who set baseline values for the Keep River, argue that metal and metalloid chemical species in run-off are unlikely to be a major risk factor to an environment. They found there are naturally high background levels of aluminium, boron, zinc and lead which regularly exceed water quality guidelines.

Given these potentially hazardous metals and metalloids are unlikely to change with the development of agriculture, and that major variations can be traced using TDS and conservative elements like chloride, we suggest that the sampling frequency of metals is reduced, as described in Chapter 4.3.1 and Table 4.2, or removed altogether.

### *Pesticides and herbicides*

A risk assessment approach similar to that recommended for metals and metalloids is also appropriate for selecting farm chemical hazards for ongoing monitoring. ANZECC and ARMCANZ (2000) guidelines recommend that any detection of agricultural chemical hazards, such as pesticides and herbicides, could be grounds for investigation. Rather than routinely monitoring a wide array of farm chemicals that may or may not be used, it is more appropriate that chemical usage is monitored and the list of chemicals that are used on the Goomig Farmlands be considered. Those chemicals that are known to pose an eco-toxicological hazard should then be monitored.

For the ORIA Stage 1, there are limited risk assessments of groundwater. There is that which underpins the Smith et al. (2007) review and a related assessment conducted by Oliver and Kookana (2005) who report a risk assessment of off-site impact by farm chemicals that use a 'pesticide impact rating index' for sugar, melon,

hybrid seed and mango crops locally in the ORIA. They concluded that atrazine, chlorpyrifos, chlorothalonil, cypermethrin, diuron, endosulfan, glyphosate, mancozeb, trifluralin and pendimethalin posed a risk, based on usage, site conditions and pesticide properties.

Without a suitable modelling methodology to accurately forecast risk, a list of farm chemicals to be routinely monitored in the Goomig Farmlands groundwater environments should be determined by assessing:

- the chemicals that are actually being used
- the chemicals that are likely to persist, and
- of those used, which are known to be a hazard to biota or agricultural reuse.

New or different chemicals may be used on the Goomig Farmlands and new or different agricultural enterprises may be pursued, so the groundwater data from the ORIA may not always provide a sufficient reference.

#### **4.3.1 Future monitoring – other considerations**

In addition to the nature of analytes to be monitored, consideration is also required of bore monitoring frequency and aspects related to site characteristics. This section looks at the EPBC conditions and makes some recommendations to improve the effectiveness of sampling based on the results of the baseline data evaluated above.

##### *Monitoring frequency*

Monitoring of analytes (discussed above Chapter 4.3) and frequency of sampling for use in the baseline period was determined in the Ministerial Conditions. While these conditions are assumed as the basis for continued use, some discussion of elements of the high-intensity, low-intensity, farm and reference bores is warranted.

High-intensity, low-intensity and farm bores defined by the conditions are primarily used to assess and monitor the impact of development and assist in its management (Figure 2.1). Apart from farm bores, which are to be constructed on-farm by the proponent prior to irrigation (not required to be assessed as part of the baseline monitoring), high-intensity and low-intensity bores are either located immediately adjacent to land to be irrigated or in the areas likely to be affected (as determined by the groundwater model).

During the baseline period, all high-intensity and low-intensity bores were monitored at the same frequency for all analytes. For high-intensity bores, the baseline data indicates little change in major ion chemistry and metals or metalloids (from ORIA) but a probable increase in nutrients and farm chemicals. Therefore, for future monitoring we recommend the high-intensity bores are sampled annually for nutrients (total phosphorus, total nitrogen) and farm chemicals because these substances are most likely to increase with agriculture. We recommend the sampling for major ion chemistry and metals or metalloids as well as nutrient speciation is reduced to a triennial basis. All chemical sampling should be conducted towards the end of the dry season. This proposed chemical sampling is consistent with the risk assessment (low and long-term) in Chapter 4.1.

For low-intensity bores (Condition 12Bii) there was no requirement for major ions and metals/metalloids to be analysed and, as detailed above, we recommend these are only monitored if there is breach of trigger conditions. For nutrients and pesticides or

herbicides, we recommend triennial measurement. This is consistent with the requirement for seasonal measurement at high-intensity bores being able to detect any changes in groundwater quality to trigger management responses and the conclusion of low and long-term risk to the Keep River (Chapter 4.1).

For farm bores, EPBC conditions (12C, D) determine that only EC and pH are required seasonally. In addition, the condition requests consultation with the IRG to determine parameters for ongoing monitoring. Baseline data suggests water levels should be measured seasonally and field measurements, such as EC, pH, chloride using an ion selective electrode and ORP, should be measured annually. Additionally, farm bores should be sampled triennially for key nutrients and farm chemicals (Table 4.2). Sampling would increase in intensity and include other analytes if any of the triggers identified in the management plans are exceeded or there was a breach of conditions.

#### *Reference bores*

Reference bores established under conditions 12I and 12J are currently only required to be monitored for EC, pH and water levels. While these bores were sampled for all parameters during the baseline period, there was no requirement specified for chemical hazards such as metals or metalloids, nutrients and pesticides or herbicides. As a result, without monitoring and/or change to this condition, these bores cannot be used for direct comparison with bores in the farmlands. It is recommended that the reference bores should be monitored for the same parameters as high-intensity bores, with water levels being logged and analysed seasonally for farm chemicals and nutrients.

#### *Sentinel site*

As a means of verifying the risk assessment and travel times for any contaminant to the Keep River, we recommend bore RN029660, located in the palaeochannel down gradient (about 5km) from the proposed farmlands, be used to detect threats to the Keep River. The chemistry analysis at this bore should be done seasonally as per Appendix B, which is more comprehensive than currently stipulated (Table 4.2), as well as for farm chemicals. The proposed analyses would detect any increases in potential contaminants from the farmlands and be used to establish the velocity of any plume.

#### *Dewatering bores*

A monitoring regime for the dewatering bores, has not been stipulated in the EPBC conditions (e.g. Condition 12H) or groundwater management plan. If dewatering bores are installed and operated then we recommend they have baseline chemistry data monitored over the same period as the bores used in the baseline analysis described here (about 18 months). When operating, groundwater discharge to the M2 supply channel should be continuously monitored for EC and chloride with loggers linked to the irrigation monitoring network plus monthly sampling of predetermined chemical hazards as per Table 4.2.

*Summary of discussion on baseline and future monitoring*

- Data collected in May and November 2014 could be included in the baseline dataset, given that irrigation had not yet started over this time.
- High-intensity bores should maintain seasonal monitoring for field parameters; agricultural chemical hazards (total nitrogen, total phosphorus and farm chemicals) should be monitored annually; and the frequency of monitoring general chemistry, major ions and metals/metalloids should be reduced to a triennial basis (Table 4.2).
- Low-intensity bores should be monitored seasonally for water levels and annually for field parameters. Nutrients (total nitrogen, total phosphorus) and farm chemicals should be analysed on a triennial basis (Table 4.2).
- Farm bores should be monitored seasonally for water levels, annually for EC, pH, ORP and chloride, and triennially for total nitrogen, total phosphate and farm chemicals (Table 4.2). Risk-based analysis should be used to inform triennial monitoring of agricultural chemical hazards based on exceedance levels as defined in the GMP.
- Reference bores should be monitored under the same guidelines as high-intensity bores to provide a means of comparative analysis of future trends and responses.
- After irrigation starts, bore RN029660 — a high-intensity bore — can be used specifically to aid risk assessment of the Keep River and inform the various management plans. Therefore, it should be seasonally monitored, as per Appendix B, as well as for farm chemicals.
- If dewatering bores are installed, the chemistry should be monitored as per environmental requirements (Appendix B) and done so monthly for 18 months to inform management models. The groundwater discharge points into the M2 supply channel should be continuously monitored with loggers linked to the irrigation monitoring network for flow, EC and chloride plus monthly determined agricultural chemical hazards.

Table 4.2 Recommended analytes and frequency of groundwater monitoring after irrigation starts

| Description                                   | High-intensity bores* | Low-intensity bores | Farm bores <sup>†</sup> | Reference bores* | Pumped groundwater discharge <sup>‡</sup> |
|-----------------------------------------------|-----------------------|---------------------|-------------------------|------------------|-------------------------------------------|
| <b>Field measurements</b>                     |                       |                     |                         |                  |                                           |
| EC, 25°C                                      | Seasonal              | Annual              | Annual                  | Seasonal         | Continuous                                |
| pH                                            | Seasonal              | Annual              | Annual                  | Seasonal         | Monthly                                   |
| Chloride mg/L (ion selective electrode)       | Seasonal              | Annual              | Annual                  | Seasonal         | Monthly                                   |
| ORP as standard hydrogen electrode            | Seasonal              | Annual              | Annual                  | Seasonal         | Monthly                                   |
| Alkalinity (as CaCO <sub>3</sub> mg/L)        | Seasonal              |                     |                         | Seasonal         | Monthly                                   |
| Acidity (as CaCO <sub>3</sub> mg/L)           | Seasonal              |                     |                         | Seasonal         | Monthly                                   |
| Water level                                   | Loggers               | Seasonal            | Seasonal                | Loggers          | Continuous (flow rate)                    |
| <b>Laboratory measurements</b>                |                       |                     |                         |                  |                                           |
| Aluminium                                     | Triennial             |                     |                         | Triennial        | Annual                                    |
| Alkalinity, total (as CaCO <sub>3</sub> mg/L) | Triennial             |                     |                         | Triennial        | Annual                                    |
| Arsenic                                       | Triennial             |                     |                         | Triennial        | Annual                                    |
| Boron                                         | Triennial             |                     |                         | Triennial        | Annual                                    |
| Beryllium                                     | Triennial             |                     |                         | Triennial        | Annual                                    |
| Calcium                                       | Triennial             |                     |                         | Triennial        | Annual                                    |
| Cadmium                                       | Triennial             |                     |                         | Triennial        | Annual                                    |
| Chloride                                      | Triennial             |                     |                         | Triennial        | Annual                                    |
| Cobalt                                        | Triennial             |                     |                         | Triennial        | Annual                                    |
| Carbonate                                     | Triennial             |                     |                         | Triennial        | Annual                                    |
| Chromium                                      | Triennial             |                     |                         | Triennial        | Annual                                    |
| Copper                                        | Triennial             |                     |                         | Triennial        | Annual                                    |
| EC, 25°C                                      | Seasonal              |                     |                         | Seasonal         | Monthly                                   |
| Fluoride                                      | Triennial             |                     |                         | Triennial        | Annual                                    |
| Iron                                          | Triennial             |                     |                         | Triennial        | Annual                                    |
| Hardness, total (as CaCO <sub>3</sub> mg/L)   | Triennial             |                     |                         | Triennial        | Annual                                    |
| Bicarbonate                                   | Triennial             |                     |                         | Triennial        | Annual                                    |
| Mercury                                       | Triennial             |                     |                         | Triennial        | Annual                                    |
| Potassium                                     | Triennial             |                     |                         | Triennial        | Annual                                    |
| Lithium                                       | Triennial             |                     |                         | Triennial        | Annual                                    |

| Description                          | High-intensity bores* | Low-intensity bores | Farm bores <sup>†</sup> | Reference bores* | Pumped groundwater discharge <sup>‡</sup> |
|--------------------------------------|-----------------------|---------------------|-------------------------|------------------|-------------------------------------------|
| Magnesium                            | Triennial             |                     |                         | Triennial        | Annual                                    |
| Manganese                            | Triennial             |                     |                         | Triennial        | Annual                                    |
| Molybdenum                           | Triennial             |                     |                         | Triennial        | Annual                                    |
| Sodium                               | Triennial             |                     |                         | Triennial        | Annual                                    |
| Nickel                               | Triennial             |                     |                         | Triennial        | Annual                                    |
| Nitrogen, total                      | Annual                | Triennial           | Triennial               | Annual           | Monthly                                   |
| Nitrogen – ammonia                   | Triennial             |                     |                         | Triennial        | Monthly                                   |
| Nitrogen – oxidised nitrogen         | Triennial             |                     |                         | Triennial        | Monthly                                   |
| Nitrogen – total organic             | Triennial             |                     |                         | Triennial        | Annual                                    |
| Lead                                 | Triennial             |                     |                         | Triennial        | Annual                                    |
| pH                                   | Triennial             |                     |                         | Triennial        | Annual                                    |
| Phosphorus, persulphate total        | Annual                | Triennial           | Triennial               | Annual           | Monthly                                   |
| Phosphorus, soluble reactive         | Triennial             |                     |                         | Triennial        |                                           |
| Selenium                             | Triennial             |                     |                         | Triennial        | Annual                                    |
| Silicon                              | Triennial             |                     |                         | Triennial        | Annual                                    |
| Sulfate, sulfur expressed as sulfate | Triennial             |                     |                         | Triennial        | Annual                                    |
| TDS                                  | Triennial             |                     |                         | Triennial        | Annual                                    |
| Uranium                              | Triennial             |                     |                         | Triennial        | Annual                                    |
| Vanadium                             | Triennial             |                     |                         | Triennial        | Annual                                    |
| Zinc                                 | Triennial             |                     |                         | Triennial        | Annual                                    |
| Selected farm chemical hazards       | Annual                | Triennial           | Triennial               | Annual           | Monthly                                   |

\* High-intensity monitoring bores and reference bores are to be monitored at the same frequency and for the same suite of analytes.

<sup>†</sup> Monitoring of farm bores is as specified in the EPBC conditions, with sampling for specified farm chemicals and nutrients as defined in the groundwater management plan for breach of conditions.

<sup>‡</sup> Production bores are to be monitored for chemistry during use as per monthly for 18 months after installation and their frequency reviewed thereafter.

## **Appendices**

**A Groundwater sites sampled**

**B Analytical methods and detection limits**

**C Mean concentration of groundwater chemistry for individual bores**

**D Summary statistics of field measurements**

**E Summary statistics of groundwater chemistry**

**F Suitability of groundwater for irrigation**

**G Salinity processes**

## Appendix A Groundwater sampling sites

Table A1 Location and site characteristics of the groundwater sampling sites

| Sampling site | Monitoring bore category | Easting | Northing | Depth drilled (m) | Screened interval (mBGL) | TDS (mg/L) | Water type (dominant ions)                  | Aquifer screened                               | Basement geology                                         |
|---------------|--------------------------|---------|----------|-------------------|--------------------------|------------|---------------------------------------------|------------------------------------------------|----------------------------------------------------------|
| 10WP31        | Low                      | 486515  | 8289746  | 23.2              | 16.8–22.8                | 230        | Mg-Ca-Na-HCO <sub>3</sub> -Cl               | clayey sands/sands/gravels                     | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP32        | Low                      | 486513  | 8290520  | 24.5              | 17.4–23.4                | 960        | Na-Mg-HCO <sub>3</sub> -Cl                  | sand/silty sand/gravel                         | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP32PB      | Low                      | 486513  | 8290535  | 23                | 9.5–21.5                 | 820        | Na-Mg-HCO <sub>3</sub> -Cl                  | poorly sorted sands/sands and gravels          | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP33        | High                     | 486515  | 8290136  | 31.7              | 19.5–28.5                | 760        | Na-Mg-HCO <sub>3</sub> -Cl                  | coarse sand and medium gravel                  | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP35N       | Low                      | 481886  | 8288653  | 18                | 12.1–18.1                | 700        | Na-HCO <sub>3</sub>                         | coarse sands and gravels                       | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP35PB      | Low                      | 481888  | 8288569  | 35                | 9.4–27.4                 | 930        | Na-Mg-HCO <sub>3</sub> -Cl                  | coarse sands and gravels above siltstone       | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP35S       | Low                      | 481887  | 8288527  | 29.5              | 16.8–25.8                | 850        | Na-Mg-HCO <sub>3</sub> -Cl                  | coarse and fine sands with some medium gravels | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP36N       | Low                      | 481886  | 8289636  | 35.6              | 14.2–26.2                | 700        | Na-Mg-HCO <sub>3</sub> -Cl                  | coarse sand and gravels                        | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP36PB      | Low                      | 481886  | 8289596  | 29                | 14.4–20.4                | 700        | Na-HCO <sub>3</sub>                         | coarse sand and gravels                        | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP36S       | Low                      | 481886  | 8289517  | 18                | 12.1–18.1                | 880        | Na-HCO <sub>3</sub>                         | coarse sand and gravels                        | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP37        | High                     | 481886  | 8289027  | 18                | 10.2–16.2                | 1 150      | Na-Mg-HCO <sub>3</sub> -Cl                  | sand (coarse to fine) and gravels              | Proterozoic Pincombe Formation (siltstone, shale)        |
| 10WP39        | High                     | 486098  | 8295716  | 33                | 6.1–18.1                 | 710        | Ca-Na-HCO <sub>3</sub> -SO <sub>4</sub> -Cl | clay, sand, some gravels and calcarenite       | Carboniferous Burvill Beds (sandstone, shale, limestone) |
| 10WP40        | Low                      | 481573  | 8295995  | 7.3               | 4.1–7.1                  | 4 050      | Na-SO <sub>4</sub> -Cl                      | clay and sands (with quartzite as base)        | Late Devonian Ningbing Limestone                         |

| Sampling site | Monitoring bore category | Easting | Northing | Depth drilled (m) | Screened interval (mBGL) | TDS (mg/L) | Water type (dominant ions)                 | Aquifer screened                              | Basement geology                                                               |
|---------------|--------------------------|---------|----------|-------------------|--------------------------|------------|--------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------|
| 10WP41        | Low                      | 483698  | 8295562  | 7.3               | 3.2–6.2                  | 15 420     | Na-Mg-Cl-SO <sub>4</sub>                   | clay and fine sand (calcareous)               | Carboniferous Milligans Beds (silty shale)                                     |
| 10WP42        | Low                      | 487500  | 8295383  | 35                | 15.3–20.3, 26.3–32.3     | 1 430      | Na-Ca-Cl-SO <sub>4</sub> -HCO <sub>3</sub> | clay sand & gravels                           | Carboniferous Milligans Beds (silty shale)                                     |
| 10WP44        | Low                      | 490993  | 8289155  | 15.7              | 12.7–15.7                | 18 990     | Na-Mg-Cl-HCO <sub>3</sub>                  | clay sand                                     | Cambrian Antrim volcanics (basalt)                                             |
| 10WP46        | Low                      | 486609  | 8295880  | 36                | 12.6–28.6                | 1 440      | Na-Ca-Cl-SO <sub>4</sub> -HCO <sub>3</sub> | sand (fine to coarse)/sandy clay (calcareous) | Burvill Beds (sandstone, shale, limestone) Early Carboniferous                 |
| 10WP47        | Low                      | 488298  | 8295577  | 36                | 28–34                    | 350        | Ca-Na-HCO <sub>3</sub> -Cl                 | sandy clay/coarse sand (calcareous)           | Carboniferous Milligans Beds (silty shale)                                     |
| 11CS10RD      | High                     | 481633  | 8286944  | 20                | 17.3–20.3                | 1 530      | Na-Mg-Ca-Cl-HCO <sub>3</sub>               | gravels                                       | Proterozoic Pincombe Formation (siltstone, shale)                              |
| 11CS10RS      | High                     | 481633  | 8286944  | 10                | 7.1–10.1                 | 4 460      | Na-Cl-HCO <sub>3</sub>                     | silty clay                                    | Proterozoic Pincombe Formation (siltstone, shale)                              |
| 11WP11RD      | High                     | 489177  | 8290712  | 24.5              | 19.6–22.6                | 540        | Na-Mg-HCO <sub>3</sub> -Cl                 | sand/gravels                                  | Cambrian Antrim volcanics (basalt)                                             |
| 11WP11RS      | High                     | 489177  | 8290712  | 15                | 11.1–14.1                | 610        | Na-HCO <sub>3</sub>                        | silty clay (calcareous)                       | Cambrian Antrim volcanics (basalt)                                             |
| 11WP15R       | High                     | 493196  | 8295657  | 30                | 26.8–29.8                | 5 100      | Na-Cl-SO <sub>4</sub> -HCO <sub>3</sub>    | coarse sand/gravels                           | Late Devonian Hargreaves member (sandstone, marl, dolomite) Cockatoo Formation |
| 11WP16R       | Low                      | 483253  | 8291803  | 14.5              | 11.6–14.6                | 9 130      | Na-Mg-Cl-SO <sub>4</sub>                   | silty clay (calcareous)                       | Late Devonian Cockatoo Formation (undifferentiated)                            |
| 11WP43D       | High                     | 489292  | 8287719  | 21                | 17.7–20.7                | 9 570      | Mg-Na-Cl                                   | siltstone                                     | Proterozoic Pincombe Formation (siltstone, shale)                              |
| 11WP4R        | Low                      | 479412  | 8286267  | 24                | 20.8–23.8                | 900        | Na-Mg-Ca-Cl-HCO <sub>3</sub>               | silty clay/gravels                            | Proterozoic King Leopold Sandstone                                             |
| 11WP50        | Low                      | 479695  | 8291619  | 19                | 11.8–14.8                | 17 160     | Na-Mg-Cl-SO <sub>4</sub>                   | silty clay                                    | Carlton Group, predominantly silt sandstones                                   |
| 11WP51D       | High                     | 488354  | 8294621  | 33                | 24.8–27.8                | 430        | Na-Ca-Mg-HCO <sub>3</sub> -SO <sub>4</sub> | sandy clay                                    | Carboniferous Milligans Beds (silty shale)                                     |
| 11WP51S       | High                     | 488354  | 8294621  | 15                | 12.5–15.5                | 540        | Na-Ca-Mg-HCO <sub>3</sub>                  | sandy clay                                    | Carboniferous Milligans Beds (silty)                                           |

| Sampling site | Monitoring bore category | Easting | Northing | Depth drilled (m) | Screened interval (mBGL) | TDS (mg/L) | Water type (dominant ions)                     | Aquifer screened             | Basement geology                                                     |
|---------------|--------------------------|---------|----------|-------------------|--------------------------|------------|------------------------------------------------|------------------------------|----------------------------------------------------------------------|
|               |                          |         |          |                   |                          |            |                                                |                              | shale)                                                               |
| 11WP52D       | High                     | 492473  | 8291543  | 28.5              | 19.7–25.7                | 470        | Na-Mg-HCO <sub>3</sub> -Cl                     | sands, minor gravels         | Proterozoic Pincombe Formation (siltstone, shale)                    |
| 11WP53D       | Low                      | 491839  | 8294861  | 24.5              | 19.6–22.6                | 5 070      | Na-Cl-HCO <sub>3</sub> -SO <sub>4</sub>        | coarse sands/gravels         | Cambrian Antrim volcanics (basalt)                                   |
| 11WP53S       | Low                      | 491839  | 8294861  | 19.2              | 15.6–18.6                | 5 540      | Na-Cl-HCO <sub>3</sub> -SO <sub>4</sub>        | clayey sands                 | Cambrian Antrim volcanics (basalt)                                   |
| 11WP54D       | Low                      | 491750  | 8296530  | 35.2              | 29.3–32.3                | 19 190     | Na-Mg-Cl-SO <sub>4</sub>                       | coarse sands/gravels         | Carboniferous Milligans Beds (silty shale)                           |
| 11WP54S       | Low                      | 491750  | 8296530  | 17.2              | 13.7–16.7                | 12 610     | Na-Cl-SO <sub>4</sub>                          | sand/sandy clay              | Carboniferous Milligans Beds (silty shale)                           |
| 11WP55        | Low                      | 495920  | 8295930  | 22                | 17.6–20.6                | 2 560      | Na-Cl-HCO <sub>3</sub>                         | clay                         | Carboniferous Milligans Beds (silty shale)                           |
| 11WP56D       | High                     | 506320  | 8296634  | 22.5              | 19.1–22.1                | 40         | Na-Ca-Cl-HCO <sub>3</sub>                      | gravels/sands                | Undifferentiated sandstone – Carboniferous, Cambrian, or Proterozoic |
| 11WP56S       | Low                      | 506320  | 8296634  | 9.3               | 5.0–9.0                  | 450        | Na-Ca-Mg-HCO <sub>3</sub> -SO <sub>4</sub> -Cl | clayey sand                  | Undifferentiated sandstone – Carboniferous, Cambrian, or Proterozoic |
| 11WP57        | High                     | 495920  | 8297000  | 22.5              | 17.7–20.7                | 11 090     | Na-Ca-Cl                                       | sandy gravels                | Milligans beds (silty shale) Early Carboniferous                     |
| 11WP9RPB      | Low                      | 491840  | 8293345  | 33.5              | 20.5–26.5                | 1 090      | Na-HCO <sub>3</sub> -Cl                        | coarse sands and gravels     | Cambrian Antrim volcanics (basalt)                                   |
| 11WP9RS       | High                     | 491840  | 8293345  | 20.6              | 16.6–19.6                | 1 080      | Na-HCO <sub>3</sub> -Cl                        | gravels and sands            | Cambrian Antrim volcanics (basalt)                                   |
| CG1           | Reference High           | 475842  | 8277312  | 38                | 29.3–35.3                | 3 870      | Na-Mg-Cl-HCO <sub>3</sub> -SO <sub>4</sub>     | sand/gravels                 | Proterozoic Wyndham Shale                                            |
| CG2           | Reference High           | 476802  | 8278290  | 30                | 19.0–25.0                | 530        | Na-Ca-Mg-HCO <sub>3</sub> -Cl                  | clayey sands/sand to gravels | Proterozoic Wyndham Shale                                            |
| CG4           | High                     | 481824  | 8284010  | 37                | 29.4–35.4                | 2 010      | Na-Mg-Cl-HCO <sub>3</sub>                      | sand and gravel              | Proterozoic Wyndham Shale                                            |
| KC13          | Reference High           | 495809  | 8277098  | 30                | 24.0–30.0                | 660        | Na-Mg-Ca-HCO <sub>3</sub> -SO <sub>4</sub>     | limestone to limey sandstone | Carboniferous Burt Range Formation (limestone)                       |
| KC14          | Reference Low            | 497959  | 8278736  | 32                | 26.0–32.0                | 14 270     | Na-Mg-Cl-SO <sub>4</sub>                       | clay/mudstone/shale          | Carboniferous Milligans Beds (silty                                  |

| Sampling site | Monitoring bore category | Easting | Northing | Depth drilled (m) | Screened interval (mBGL) | TDS (mg/L) | Water type (dominant ions)                    | Aquifer screened              | Basement geology                                                        |
|---------------|--------------------------|---------|----------|-------------------|--------------------------|------------|-----------------------------------------------|-------------------------------|-------------------------------------------------------------------------|
|               |                          |         |          |                   |                          |            |                                               |                               | shale)                                                                  |
| KC3           | Reference Low            | 498123  | 8284349  | 30                | 18.0–24.0                | 140        | Mg-Na-Ca-HCO <sub>3</sub> -Cl-SO <sub>4</sub> | silty clay/sandy clay         | Carboniferous Septimus limestone                                        |
| KC3A          | Reference Low            | 498123  | 8284351  | 36                | 25.2–31.2                | 270        | Mg-Na-Ca-SO <sub>4</sub> -Cl-HCO <sub>3</sub> | sand/clay/gravel              | Carboniferous Septimus limestone                                        |
| KC3PB         | Reference High           | 498133  | 8284345  | 32.7              | 26.0–32.0                | 260        | Mg-Na-Ca-SO <sub>4</sub> -Cl-HCO <sub>3</sub> | sand/clay/gravel              | Carboniferous Septimus limestone                                        |
| KCF1          | Reference Low            | 500051  | 8276746  | 28.5              | 21.0–27.0                | 3 520      | Na-Mg-SO <sub>4</sub> -HCO <sub>3</sub> -Cl   | sand/clay                     | Cambrian Pretlove sandstone                                             |
| LIMESTONE     | High                     | 483441  | 8297674  | ND                | ND                       | 310        | Ca-Na-HCO <sub>3</sub> -Cl                    | ND                            | Carboniferous Milligans Beds (silty shale)                              |
| RN029659      | Low                      | 503209  | 8295952  | 39.7              | 33.0–37.1                | 2 240      | Na-Mg-Cl                                      | sands/gravel                  | Carboniferous Milligans Beds (silty shale)                              |
| RN029660      | High                     | 500048  | 8295266  | 29.7              | 19.4–23.4                | 1 020      | Na-HCO <sub>3</sub> -Cl                       | clay/sand/gravel              | Carboniferous Milligans Beds (silty shale)                              |
| W2R           | Low                      | 484778  | 8296460  | 18                | 12.0–18.0                | 270        | Ca-HCO <sub>3</sub>                           | sand/clay                     | Carboniferous Milligans Beds (silty shale)                              |
| WP13          | High                     | 484628  | 8295088  | 18                | 12.0–18.0                | 590        | Ca-Na-HCO <sub>3</sub> -Cl-SO <sub>4</sub>    | limestone/sandstone/siltstone | Carboniferous Burt Range Formation (limestone)                          |
| WP19          | Low                      | 494914  | 8298225  | 37.5              | 29.3–35.3                | 20 510     | Na-Mg-Cl-SO <sub>4</sub>                      | sand/gravel                   | Carboniferous Milligans Beds (silty shale)                              |
| WP5           | High                     | 478568  | 8291625  | 24                | 18.0–24.0                | 3 440      | Mg-Na-Cl-HCO <sub>3</sub> -SO <sub>4</sub>    | basalt                        | Carlton Group – predominantly silt sandstones/Antrim volcanics (basalt) |

ND No data

## Appendix B Analytical methods and detection limits

Table B1 Analytical methods and detection limits for groundwater samples sent for laboratory measurements

| Analyte                  | Method description                                      | Limit of reporting | Unit                      |
|--------------------------|---------------------------------------------------------|--------------------|---------------------------|
| Acidity                  |                                                         | 2                  | mg/L<br>CaCO <sub>3</sub> |
| Silver                   | Coupled mass spectroscopy                               | 0.0001             | mg/L                      |
| Ion balance              | Calculated                                              | -50                | %                         |
| Aluminium                | Inductively coupled plasma atomic emission spectroscopy | 0.005              | mg/L                      |
| Alkalinity, total        | Titration                                               | 1                  | mg/L<br>CaCO <sub>3</sub> |
| Arsenic                  | Coupled mass spectroscopy                               | 0.001              | mg/L                      |
| Atrazine                 | Liquid chromatography                                   | 0.2 or 0.1         | µg/L                      |
| Boron                    | Inductively coupled plasma atomic emission spectroscopy | 0.02               | mg/L                      |
| Barium                   | Inductively coupled plasma atomic emission spectroscopy | 0.002              | mg/L                      |
| Beryllium                | Coupled mass spectroscopy                               | 0.0001             | mg/L                      |
| Bismuth                  | Coupled mass spectroscopy                               | 0.0001             | mg/L                      |
| Bromide                  | Inductively coupled plasma atomic emission spectroscopy | 0.02               | mg/L                      |
| Calcium                  | Inductively coupled plasma atomic emission spectroscopy | 0.1                | mg/L                      |
| Cadmium                  | Coupled mass spectroscopy                               | 0.0001             | mg/L                      |
| Chloride                 | Colorimetric reaction and discrete analyser             | 1                  | mg/L                      |
| Cobalt                   | Inductively coupled plasma atomic emission spectroscopy | 0.005              | mg/L                      |
| Carbonate                | Titration                                               | 1                  | mg/L                      |
| Chromium                 | Coupled mass spectroscopy                               | 0.0005             | mg/L                      |
| Chromium                 | Inductively coupled plasma atomic emission spectroscopy | 0.001              | mg/L                      |
| Copper                   | Inductively coupled plasma atomic emission spectroscopy | 0.002              | mg/L                      |
| Dissolved organic carbon | Colorimetry                                             | 1                  | mg/L                      |
| Electrical conductivity  | Electronic probe, measured at 25°C                      | 0.2                | mS/m                      |
| Fluoride                 | Selective electricity probe                             | 0.05               | mg/L                      |
| Iron                     | Inductively coupled plasma atomic emission spectroscopy | 0.005              | mg/L                      |
| Gallium                  | Coupled mass spectroscopy                               | 0.0001             | mg/L                      |
| Hardness, total          | Calculated from calcium and magnesium                   | 1                  | mg/L<br>CaCO <sub>3</sub> |
| Bicarbonate              | Titration                                               | 1                  | mg/L                      |
| Mercury                  | Vapour generation and spectroscopy                      | 0.0001             | mg/L                      |
| Mercury                  | Coupled mass spectroscopy                               | 0.0001             | mg/L                      |
| Potassium                | Inductively coupled plasma atomic emission spectroscopy | 0.1                | mg/L                      |
| Lanthanum                | Inductively coupled plasma atomic emission spectroscopy | 0.005              | mg/L                      |
| Lithium                  | Inductively coupled plasma atomic emission spectroscopy | 0.005              | mg/L                      |
| Magnesium                | Inductively coupled plasma atomic emission spectroscopy | 0.1                | mg/L                      |

| Analyte                      | Method description                                                                                                | Limit of reporting | Unit |
|------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------|------|
| Manganese                    | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.001              | mg/L |
| Molybdenum                   | Coupled mass spectroscopy                                                                                         | 0.001              | mg/L |
| Nitrogen, ammonia fraction   | Colorimetric reaction and automated flow injection analysis colorimeter                                           | 0.01               | mg/L |
| Nitrogen, nitrite fraction   | Colorimetric reaction and automated flow injection analysis colorimeter                                           | 0.01               | mg/L |
| Nitrogen, nitrate fraction   | Cadmium reduction, colorimetric reaction and automated flow injection analysis colorimeter                        | 0.01               | mg/L |
| Nitrogen, total oxidised     | Calculation: nitrite + nitrate                                                                                    | 0.01               | mg/L |
| Nitrogen, total              | Persulphate digestion, cadmium reduction, colorimetric reaction and automated flow injection analysis colorimeter | 0.02               | mg/L |
| Sodium                       | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.1                | mg/L |
| Nickel                       | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.001              | mg/L |
| Phosphorus, soluble reactive | Phosphorus from phosphate by colorimetric reaction and automated flow injection analysis colorimeter              | 0.01               | mg/L |
| Phosphorus, total            | Persulphate digestion, colorimetric reaction and automated flow injection analysis colorimeter                    | 0.01               | mg/L |
| Lead                         | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.02               | mg/L |
| Lead                         | Coupled mass spectroscopy                                                                                         | 0.0001             | mg/L |
| pH                           | Electronic probe                                                                                                  | 0.1                |      |
| Antimony                     | Coupled mass spectroscopy                                                                                         | 0.0001             | mg/L |
| Selenium                     | Coupled mass spectroscopy                                                                                         | 0.001              | mg/L |
| Silicon                      | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.05               | mg/L |
| Tin                          | Coupled mass spectroscopy                                                                                         | 0.0001             | mg/L |
| Tin                          | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.02               | mg/L |
| Sulfate                      | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.1                | mg/L |
| TDS by analysis              | Dry mass of evaporated filtrate                                                                                   | 10                 | mg/L |
| TDS by summation             | Summation                                                                                                         | 1                  | mg/L |
| Titanium                     | Coupled mass spectroscopy                                                                                         | 0.0005             | mg/L |
| Thallium                     | Coupled mass spectroscopy                                                                                         | 0.0001             | mg/L |
| Uranium                      | Coupled mass spectroscopy                                                                                         | 0.0001             | mg/L |
| Vanadium                     | Coupled mass spectroscopy                                                                                         | 0.0001             | mg/L |
| Vanadium                     | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.005              | mg/L |
| Zinc                         | Coupled mass spectroscopy                                                                                         | 0.001              | mg/L |
| Zinc                         | Inductively coupled plasma atomic emission spectroscopy                                                           | 0.005              | mg/L |

## Appendix C Mean concentration of groundwater chemistry for individual bores

Table C1 Mean concentrations of analytes for bores 10WP31 to 10WP43

| Analyte                              | 10WP31 | 10WP32 | 10WP32PB | 10WP33 | 10WP35N | 10WP35PB | 10WP35S | 10WP36N | 10WP36PB | 10WP36S | 10WP37 | 10WP38 | 10WP39 | 10WP40 | 10WP41 | 10WP42 | 10WP43 |
|--------------------------------------|--------|--------|----------|--------|---------|----------|---------|---------|----------|---------|--------|--------|--------|--------|--------|--------|--------|
| Acidity (mg/L CaCO <sub>3</sub> )    | 14     | 16     | 14       | 17     | 15      | 19       | 18      | 10      | 12       | 11      | 17     | 18     | 9      | 22     | 38     | 12     | 32     |
| Silver (µg/L)                        | 0.4    | 0.7    | 0.4      | 0.8    | <0.1    | 0.4      | 0.1     | <0.1    | <0.1     | <0.1    | <0.1   | 0.4    | 0.6    | 0.3    | 18.0   | 0.4    | <0.5   |
| Aluminium (µg/L)                     | 14     | 9      | 38       | 44     | 20      | 22       | 17      | 22      | 14       | 23      | 28     | 10     | 16     | 20     | 21     | 43     | 20     |
| Alkalinity (mg/L CaCO <sub>3</sub> ) | 150    | 407    | 411      | 326    | 538     | 612      | 585     | 518     | 513      | 604     | 664    | 376    | 238    | 645    | 508    | 331    | 290    |
| Arsenic (µg/L)                       | <1     | 1.2    | 1.2      | <1     | 1.3     | <1       | 2.0     | 4.3     | 4.2      | 8.9     | 1.3    | <1     | 6.9    | 6.0    | 2.0    | 2.1    | <5     |
| Atrazine (µg/L)                      | <0.01  | <0.01  | <0.01    | <0.01  | <0.01   | <0.01    | <0.01   | <0.01   | <0.01    | <0.01   | <0.01  | <0.01  | <0.01  | <0.01  | <0.01  | <0.01  | <0.01  |
| Boron (µg/L)                         | 33     | 104    | 99       | 73     | 357     | 278      | 293     | 426     | 559      | 896     | 489    | 756    | 157    | 933    | 2071   | 249    | 210    |
| Barium (µg/L)                        | 31     | 60     | 37       | 47     | 40      | 123      | 91      | 23      | 31       | 27      | 52     | 32     | 30     | 79     | 76     | 43     | 290    |
| Beryllium (µg/L)                     | <0.1   | <0.1   | <0.1     | <0.1   | <0.1    | <0.1     | <0.1    | <0.1    | <0.1     | <0.1    | <0.1   | <0.1   | <0.1   | <0.1   | <0.1   | <0.1   | <0.1   |
| Bismuth (µg/L)                       | <0.1   | <0.1   | <0.1     | <0.1   | <0.1    | <0.1     | <0.1    | <0.1    | <0.1     | <0.1    | <0.1   | <0.1   | <0.1   | <0.1   | <0.1   | <0.1   | <0.1   |
| Bromine (µg/L)                       | 152    | 754    | 570      | 608    | 253     | 448      | 394     | 316     | 346      | 434     | 596    | 1 382  | 388    | 2 040  | 20750  | 910    | 12 000 |
| Calcium (mg/L)                       | 26     | 53     | 45       | 53     | 24      | 59       | 49      | 39      | 32       | 30      | 44     | 160    | 90     | 130    | 629    | 95     | 745    |
| Cadmium (µg/L)                       | <0.1   | <0.1   | <0.1     | <0.1   | <0.1    | <0.1     | <0.1    | <0.1    | <0.1     | <0.1    | <0.1   | 0.2    | <0.1   | 0.2    | <0.1   | <0.1   | <0.1   |
| Chloride (mg/L)                      | 42     | 251    | 203      | 205    | 61      | 138      | 116     | 86      | 74       | 102     | 210    | 722    | 123    | 845    | 6616   | 341    | 5 930  |
| Cobalt (µg/L)                        | 0.8    | <0.1   | <0.1     | 0.1    | 0.1     | 0.1      | 0.4     | 0.1     | 0.1      | <0.1    | 0.1    | 1.2    | <0.1   | 3.9    | 6.7    | 0.1    | 21.0   |
| Carbonate (mg/L)                     | <1     | <1     | 24       | <1     | <1      | <1       | <1      | <1      | 44       | <1      | <1     | <1     | <1     | <1     | <1     | <1     | <1     |
| Chromium (µg/L)                      | 2.7    | 3.0    | 1.7      | 1.5    | 23.0    | 1.0      | 6.8     | 3.6     | 0.7      | 6.5     | 9.0    | 3.9    | <1     | 22.0   | 15.0   | <1     | <1     |
| Copper (µg/L)                        | 2.2    | 1.4    | 1.6      | 1.2    | 0.7     | 0.9      | 1.1     | 1.1     | 0.8      | 0.5     | 0.9    | 1.6    | 0.3    | 1.7    | 5.0    | 1.1    | 2.5    |
| Dissolved organic carbon (mg/L)      |        |        |          |        |         | 110.0    |         |         | 4.3      |         |        |        |        |        | 2.6    |        |        |

[illegible]

| Analyte                             | 10WP31 | 10WP32 | 10WP32PB | 10WP33 | 10WP35N | 10WP35PB | 10WP35S | 10WP36N | 10WP36PB | 10WP36S | 10WP37 | 10WP38 | 10WP39 | 10WP40 | 10WP41 | 10WP42 | 10WP43 |
|-------------------------------------|--------|--------|----------|--------|---------|----------|---------|---------|----------|---------|--------|--------|--------|--------|--------|--------|--------|
| Phosphorus, soluble reactive (µg/L) | 64     | 67     | 66       | 57     | 123     | 69       | 73      | 207     | 180      | 199     | 109    | 13     | 14     | 50     | 10     | 21     | <10    |
| Phosphorus, total (µg/L)            | 85     | 88     | 71       | 71     | 132     | 94       | 94      | 208     | 210      | 243     | 128    | 111    | 84     | 87     | 17     | 26     | <10    |
| Lead (µg/L)                         | 0.7    | 0.7    | 0.5      | 3.6    | 0.6     | 0.2      | 0.5     | 0.9     | 0.5      | 1.1     | 16.3   | 0.4    | 0.5    | 0.3    | 9.4    | 0.4    | 0.5    |
| pH                                  | 7.2    | 7.7    | 7.7      | 7.5    | 7.8     | 7.6      | 7.6     | 7.8     | 8.0      | 7.9     | 7.7    | 7.5    | 7.6    | 7.6    | 7.4    | 7.6    | 7.1    |
| Antimony (µg/L)                     | <0.1   | 0.3    | 0.1      | 0.1    | 0.2     | <0.1     | 0.2     | 0.1     | 0.1      | 0.2     | 0.1    | 0.6    | 0.1    | 0.3    | <0.5   | 0.1    | <0.5   |
| Selenium (µg/L)                     | <1     | <1     | 4        | 3      | <1      | <1       | <1      | <1      | <1       | <1      | 6      | <1     | <1     | <2     | <5     | 1      | <5     |
| Silicon (mg/L)                      | 36     | 36     | 33       | 32     | 27      | 29       | 26      | 35      | 35       | 39      | 32     | 14     | 34     | 35     | 16     | 37     | 10     |
| Tin (µg/L)                          | 0.4    | 0.2    | 0.1      | 20     | 20      | 0.1      | 20      | 20      | 20       | 20      | 20     | 20     | 20     | 0.3    | 20     | 0.1    | 20     |
| Sulfate, sulfur (mg/L)              | 7      | 106    | 85       | 74     | 39      | 64       | 60      | 53      | 47       | 63      | 120    | 1 244  | 175    | 1 411  | 2 929  | 370    | 1 130  |
| TDS by summation (mg/L)             | 217    | 946    | 826      | 737    | 707     | 897      | 850     | 707     | 687      | 881     | 1 143  | 3 043  | 679    | 3 857  | 14 571 | 1 371  | 10 000 |
| Titanium (µg/L)                     | <2     | <2     | <2       | <2     | <2      | <2       | 0.2     | <2      | <2       | <2      | <2     | <2     | 0.1    | <2     | <2     | <2     | <2     |
| Thallium (µg/L)                     | <0.1   | 0.5    | 0.2      | <0.1   | <0.1    | <0.1     | <0.1    | <0.1    | 0.2      | 0.1     | <0.1   | 0.2    | 0.3    | <0.2   | 1.4    | 0.3    | <0.5   |
| Uranium (µg/L)                      | 0      | 3      | 4        | 4      | 15      | 25       | 22      | 5       | 5        | 6       | 18     | 3      | 1      | 13     | 81     | 9      | 10     |
| Vanadium (µg/L)                     | 15     | 45     | 63       | 20     | 36      | 22       | 16      | 48      | 41       | 124     | 34     | 8      | 12     | 22     | 8      | 13     | <5     |
| Zinc (µg/L)                         | 13     | 10     | 39       | 13     | 9       | 12       | 16      | 11      | 10       | 13      | 9      | 30     | 17     | 16     | 50     | 14     | 24     |

Table C2 Mean concentrations of analytes for bores 10WP44 to 11WP53D

| Analyte                              | 10WP44 | 10WP46 | 10WP47 | 11CS10RD | 11CS10RS | 11WP11RD | 11WP11RS | 11WP15R | 11WP16R | 11WP43D | 11WP4R | 11WP50 | 11WP51D | 11WP51S | 11WP52D | 11WP53D |
|--------------------------------------|--------|--------|--------|----------|----------|----------|----------|---------|---------|---------|--------|--------|---------|---------|---------|---------|
| Acidity (mg/L CaCO <sub>3</sub> )    | 50     | 14     | 10     | 12       | 15       | 7        | 8        | 22      | 27      | 17      | 11     | 35     | 6       | 14      | 10      | 23      |
| Silver (µg/L)                        | <0.5   | 0.7    | 0.1    | 1.2      | 0.9      | 0.1      | 0.1      | 0.3     | 0.6     | <0.5    | 1.2    | <0.5   | 0.1     | <0.1    | <0.1    | 0.4     |
| Aluminium (µg/L)                     | 17     | 42     | 26     | 19       | 14       | 15       | 77       | 20      | 15      | 21      | 33     | 22     | 16      | 32      | 13      | 37      |
| Alkalinity (mg/L CaCO <sub>3</sub> ) | 835    | 320    | 213    | 442      | 853      | 380      | 528      | 1 203   | 446     | 245     | 306    | 611    | 246     | 315     | 268     | 1 763   |
| Arsenic (µg/L)                       | 7.5    | 1.7    | 1.1    | 1.0      | 3.7      | 1.5      | 4.3      | 2.0     | 3.0     | 1.5     | 3.3    | <5     | 4.2     | 3.0     | 1.0     | 7.3     |
| Atrazine (µg/L)                      | <0.01  | <0.01  | <0.01  | <0.01    | <0.01    | <0.01    | <0.01    | <0.01   | <0.01   | <0.01   | <0.01  | <0.01  | <0.01   | <0.01   | <0.01   | <0.01   |
| Boron (µg/L)                         | 413    | 223    | 73     | 105      | 805      | 95       | 168      | 517     | 330     | 143     | 125    | 558    | 92      | 142     | 40      | 718     |
| Barium (µg/L)                        | 116    | 21     | 134    | 114      | 88       | 71       | 42       | 53      | 48      | 283     | 198    | 39     | 64      | 142     | 98      | 35      |
| Beryllium (µg/L)                     | <0.1   | <0.1   | <0.1   | <0.1     | <0.1     | <0.1     | <0.1     | <0.1    | <0.1    | <0.1    | <0.1   | <0.1   | <0.1    | <0.1    | <0.1    | <0.1    |
| Bismuth (µg/L)                       | <0.1   | <0.1   | <0.1   | <0.1     | <0.1     | <0.1     | <0.1     | <0.1    | <0.1    | <0.1    | <0.1   | <0.1   | <0.1    | <0.1    | <0.1    | <0.1    |
| Bromine (µg/L)                       | 17 000 | 815    | 213    | 1 158    | 4 520    | 283      | 237      | 4 280   | 6 100   | 10 450  | 874    | 13 750 | 152     | 160     | 285     | 3 040   |
| Calcium (mg/L)                       | 929    | 114    | 61     | 112      | 123      | 29       | 11       | 37      | 465     | 639     | 88     | 641    | 50      | 45      | 27      | 14      |
| Cadmium (µg/L)                       | <0.1   | <0.1   | <0.1   | 0.1      | 0.2      | <0.1     | <0.1     | <0.1    | <0.1    | <0.1    | 2.4    | <0.1   | <0.1    | <0.1    | <0.1    | 0.3     |
| Chloride (mg/L)                      | 7 758  | 327    | 45     | 474      | 1 801    | 79       | 57       | 1 588   | 2 960   | 5 330   | 329    | 5 555  | 46      | 53      | 97      | 1277    |
| Cobalt (µg/L)                        | 2.6    | 0.2    | 0.8    | <0.1     | 1.5      | <0.1     | 0.3      | 0.5     | 1.2     | 14.1    | 1.7    | 18.5   | 0.3     | 1.0     | 0.4     | 0.7     |
| Carbonate (mg/L)                     | <1     | <1     | <1     | <1       | <1       | <1       | <1       | <1      | <1      | <1      | <1     | <1     | <1      | <1      | <1      | <1      |
| Chromium (µg/L)                      | <1     | <1     | 2.0    | 1.7      | 1.4      | 1.9      | 2.1      | 4.0     | <1      | <1      | 2.1    | <1     | 1.0     | <1      | 1.4     | 3.2     |
| Copper (µg/L)                        | 2.0    | 0.2    | 0.4    | 2.0      | 4.0      | 1.6      | 1.2      | 3.1     | 7.4     | 9.0     | 2.3    | 2.2    | 1.8     | 1.8     | 1.8     | 2.8     |
| Dissolved organic carbon (mg/L)      |        |        | 1.4    |          |          |          |          |         |         | 4.9     |        |        |         |         |         |         |

| Analyte                            | 10WP44 | 10WP46 | 10WP47 | 11CS10RD | 11CS10RS | 11WP11RD | 11WP11RS | 11WP15R | 11WP16R | 11WP43D | 11WP4R | 11WP50 | 11WP51D | 11WP51S | 11WP52D | 11WP53D |
|------------------------------------|--------|--------|--------|----------|----------|----------|----------|---------|---------|---------|--------|--------|---------|---------|---------|---------|
| Electrical conductivity (mS/m)     | 2 515  | 216    | 60     | 251      | 739      | 100      | 114      | 840     | 1 220   | 1 368   | 163    | 2 085  | 72      | 102     | 90      | 832     |
| Fluoride (µg/L)                    | 435    | 460    | 110    | 194      | 546      | 522      | 1 160    | 362     | 670     | 530     | 538    | 518    | 670     | 566     | 268     | 1 360   |
| Iron (µg/L)                        | 461    | 29     | 89     | 16       | 56       | 12       | 51       | 114     | 165     | 425     | 170    | 853    | 26      | 190     | 19      | 23      |
| Gallium (µg/L)                     | 0.7    | <0.1   | <0.1   | <0.1     | <0.1     | <0.1     | 0.1      | <0.1    | <0.1    | 0.6     | 0.2    | 0.7    | <0.1    | 0.1     | <0.1    | <0.1    |
| Hardness (mg/L CaCO <sub>3</sub> ) | 4 217  | 518    | 218    | 613      | 756      | 289      | 277      | 803     | 2121    | 5 432   | 430    | 3 126  | 239     | 246     | 217     | 843     |
| Bicarbonate (mg/L)                 | 4 250  | 385    | 249    | 608      | 1191     | 371      | 468      | 1117    | 1440    | 1 117   | 392    | 3 094  | 269     | 362     | 283     | 1472    |
| Mercury (µg/L)                     | 0.1    | <0.1   | <0.1   | <0.1     | <0.1     | <0.1     | <0.1     | <0.1    | <0.1    | 0.3     | <0.1   | <0.1   | <0.1    | <0.1    | <0.1    | 1.0     |
| Potassium (mg/L)                   | 19     | 24     | 9      | 5        | 1        | 2        | 1        | 6       | 4       | 13      | 7      | 12     | 18      | 22      | 3       | 6       |
| Lanthanum (µg/L)                   | <5     | <5     | <5     | 0.1      | 0.2      | <5       | 0.6      | <5      | <5      | 1.9     | 0.2    | 1.1    | <5      | 0.1     | <5      | <5      |
| Lithium (µg/L)                     | 26     | 45     | 8      | 4        | 6        | 2        | 1        | 24      | 31      | 55      | 2      | 36     | 22      | 21      | 3       | 19      |
| Magnesium (mg/L)                   | 1 240  | 55     | 13     | 98       | 143      | 31       | 18       | 88      | 448     | 1 130   | 56     | 945    | 20      | 27      | 26      | 32      |
| Manganese (µg/L)                   | 6 550  | 42     | 212    | 5        | 275      | 7        | 26       | 143     | 424     | 3 475   | 1079   | 4 725  | 61      | 502     | 126     | 65      |
| Molybdenum (µg/L)                  | 5      | 1      | <1     | <1       | 16       | 2        | 2        | 5       | <1      | 3       | 1      | <1     | 2       | 6       | 1       | 39      |
| Nitrogen, ammonia (µg/L)           | 12 700 | 50     | 38     | 50       | 170      | <10      | 70       | 85      | 1836    | 398     | 1320   | 60     | 130     | 4114    | 50      | 140     |
| Nitrogen, nitrite (µg/L)           | <10    | <10    | <10    | <10      | <10      | <10      | 20       | 20      | <10     | <10     | 10     | <10    | <10     | <10     | <10     | <10     |
| Nitrogen, nitrate (µg/L)           | 130    | 43     | 10     | 110      | 20       | 117      | 150      | 17      | 33      | 15      | 50     | <10    | 10      | 10      | 67      | 40      |
| Oxidised nitrogen (µg/L)           | 155    | 43     | <10    | 150      | <10      | 130      | 155      | 35      | 60      | 385     | 130    | 10     | 20      | 170     | 73      | 65      |
| Nitrogen, total (µg/L)             | 17 475 | 236    | 189    | 487      | 392      | 318      | 533      | 290     | 2260    | 1 390   | 1873   | 538    | 438     | 3933    | 315     | 303     |
| Sodium (mg/L)                      | 3 523  | 275    | 40     | 270      | 1293     | 137      | 230      | 1730    | 1732    | 899     | 125    | 3 423  | 61      | 87      | 109     | 1915    |
| Nickel (µg/L)                      | 12     | <1     | <1     | <1       | 2        | 2        | 3        | <1      | <1      | 8       | 5      | 10     | <1      | 1       | 11      | <1      |
| Hydroxide (mg/L)                   | <1     | <1     | <1     | <1       | <1       | <1       | <1       | <1      | <1      | <1      | <1     | <1     | <1      | <1      | <1      | <1      |

| Analyte                             | 10WP44 | 10WP46 | 10WP47 | 11CS10RD | 11CS10RS | 11WP11RD | 11WP11RS | 11WP15R | 11WP16R | 11WP43D | 11WP4R | 11WP50 | 11WP51D | 11WP51S | 11WP52D | 11WP53D |
|-------------------------------------|--------|--------|--------|----------|----------|----------|----------|---------|---------|---------|--------|--------|---------|---------|---------|---------|
| Phosphorus, soluble reactive (µg/L) | 1 588  | 21     | 26     | 58       | 38       | 440      | 362      | 82      | 264     | 20      | 57     | 10     | 22      | 446     | 93      | 188     |
| Phosphorus, total (µg/L)            | 1 720  | 25     | 31     | 66       | 58       | 686      | 572      | 105     | 271     | 42      | 76     | 18     | 32      | 417     | 133     | 240     |
| Lead (µg/L)                         | 0.6    | 0.4    | 0.4    | 1.0      | 1.8      | 0.3      | 0.4      | 1.1     | 1.3     | 1.4     | 1.1    | 0.9    | 1.5     | 15.4    | 0.4     | 0.6     |
| pH                                  | 7.3    | 7.5    | 7.6    | 7.4      | 7.5      | 7.7      | 8.0      | 7.6     | 7.3     | 7.2     | 7.4    | 7.1    | 7.7     | 7.6     | 7.4     | 7.9     |
| Antimony (µg/L)                     | 0.4    | 0.1    | 0.1    | <0.1     | 0.2      | 0.1      | 0.2      | 0.3     | <0.1    | 0.6     | 0.5    | 0.6    | 0.3     | 0.4     | 0.1     | 0.4     |
| Selenium (µg/L)                     | <1     | <1     | <1     | 2        | <1       | 1        | <1       | 5       | <1      | <1      | <1     | <1     | <1      | <1      | 1       | 7       |
| Silicon (mg/L)                      | 18     | 33     | 34     | 33       | 40       | 33       | 30       | 23      | 34      | 9       | 17     | 14     | 14      | 17      | 27      | 32      |
| Tin (µg/L)                          | 0.6    | <20    | <20    | 0.1      | <20      | <20      | 0.1      | <20     | <20     | 0.5     | 0.7    | <20    | 0.2     | 0.2     | <20     | <20     |
| Sulfate, sulfur (mg/L)              | 2 975  | 414    | 30     | 208      | 385      | 36       | 14       | 981     | 2658    | 887     | 56     | 4 725  | 75      | 85      | 42      | 942     |
| TDS by summation (mg/L)             | 17 000 | 1400   | 329    | 1452     | 4200     | 547      | 647      | 5050    | 8183    | 9 225   | 855    | 15 750 | 420     | 502     | 468     | 5100    |
| Titanium (µg/L)                     | <2     | <2     | <2     | <2       | <2       | <2       | 2.8      | <2      | <2      | 27.0    | 0.7    | <2     | 0.2     | 0.3     | <2      | 1.1     |
| Thallium(µg/L)                      | <0.2   | <0.2   | <0.2   | <0.2     | <0.2     | <0.2     | 0.1      | 0.9     | <0.2    | <0.2    | <0.2   | 2.0    | 0.2     | 0.1     | <0.2    | <0.2    |
| Uranium (µg/L)                      | 23     | 6      | 1      | 18       | 42       | 3        | 4        | 57      | 16      | 14      | 3      | 73     | 0       | 1       | 2       | 56      |
| Vanadium (µg/L)                     | 8      | 12     | 2      | 25       | 41       | 37       | 127      | 24      | 7       | 4       | 7      | 4      | 1       | 3       | 18      | 138     |
| Zinc (µg/L)                         | 12     | 15     | 14     | 14       | 16       | 9        | 10       | 18      | 18      | 28      | 23     | 69     | 10      | 31      | 10      | 13      |

Table C3 Mean concentrations of analytes for bores 11WP53S to KC3PB

| Analyte                              | 11WP53S | 11WP54D | 11WP54S | 11WP55 | 11WP56D | 11WP56S | 11WP57 | 11WP9RPB | 11WP9RS | CG1   | CG2   | CG4   | KC13  | KC14   | KC3   | KC3A  | KC3PB |
|--------------------------------------|---------|---------|---------|--------|---------|---------|--------|----------|---------|-------|-------|-------|-------|--------|-------|-------|-------|
| Acidity (mg/L CaCO <sub>3</sub> )    | 18      | 35      | 22      | 22     | 11      | 6       | 27     | 10       | 14      | 25    | 10    | 14    | 8     | 27     | 7     | 12    | 13    |
| Silver (µg/L)                        | 1.1     | 2.1     | 1.7     | <0.1   | 0.9     | 0.7     | 1.2    | 0.2      | 0.1     | 0.6   | 0.1   | 0.2   | 0.5   | 9.8    | 5.5   | 4.6   | 5.4   |
| Aluminium (µg/L)                     | 30      | 14      | 29      | 20     | 13      | 20      | 10     | 35       | 21      | 24    | 12    | 19    | 16    | 9      | 27    | 14    | 14    |
| Alkalinity (mg/L CaCO <sub>3</sub> ) | 1 903   | 682     | 1 200   | 437    | 31      | 296     | 380    | 517      | 464     | 737   | 331   | 412   | 315   | 306    | 45    | 28    | 29    |
| Arsenic (µg/L)                       | 7.5     | 1.0     | 3.0     | 34.2   | <1      | 1.0     | 2.0    | 6.1      | 2.0     | <2    | 1.0   | 1.0   | 3.4   | <5     | <1    | <1    | <1    |
| Atrazine (µg/L)                      | <0.01   | <0.01   | <0.01   | <0.01  | <0.01   | <0.01   | <0.01  | <0.01    | <0.01   | <0.02 | <0.01 | <0.01 | <0.01 | <0.01  | <0.01 | <0.01 | <0.01 |
| Boron (µg/L)                         | 842     | 760     | 978     | 440    | 62      | 80      | 1 110  | 170      | 168     | 187   | 66    | 125   | 93    | 672    | 40    | 38    | 42    |
| Barium (µg/L)                        | 49      | 37      | 43      | 77     | 40      | 62      | 94     | 50       | 55      | 75    | 69    | 157   | 59    | 32     | 65    | 86    | 87    |
| Beryllium (µg/L)                     | <0.1    | <0.5    | <0.5    | <0.1   | 0.2     | <0.1    | <0.1   | <0.1     | <0.1    | <0.1  | <0.1  | <0.1  | <0.1  | <0.5   | <0.1  | <0.1  | <0.1  |
| Bismuth (µg/L)                       | <0.1    | <0.5    | <0.5    | <0.1   | <0.1    | <0.1    | <0.1   | <0.1     | <0.1    | <0.1  | <0.1  | <0.1  | <0.1  | <0.5   | <0.1  | <0.1  | <0.1  |
| Bromine (µg/L)                       | 3 400   | 16 600  | 10 940  | 2 600  | 122     | 225     | 14 600 | 602      | 598     | 2 475 | 233   | 1 633 | 356   | 17 000 | 154   | 250   | 262   |
| Calcium (mg/L)                       | 13      | 596     | 248     | 100    | 4       | 47      | 921    | 27       | 31      | 207   | 63    | 114   | 48    | 745    | 11    | 19    | 18    |
| Cadmium (µg/L)                       | 0.3     | <0.5    | 0.8     | 0.1    | <0.1    | <0.1    | 0.4    | <0.1     | <0.1    | <0.1  | <0.2  | 0.1   | 0.1   | 0.3    | 0.4   | 1.1   | 0.4   |
| Chloride (mg/L)                      | 1 402   | 6 570   | 3 855   | 891    | 16      | 64      | 5 142  | 217      | 228     | 1 267 | 91    | 681   | 75    | 4 706  | 26    | 52    | 49    |
| Cobalt (µg/L)                        | 0.5     | 2.5     | 16.5    | 1.2    | 0.7     | 1.3     | 3.0    | 0.2      | 0.3     | <0.2  | <0.1  | 0.3   | 0.5   | 2.3    | 0.2   | <0.1  | 0.2   |
| Carbonate (mg/L)                     | <1      | <1      | <1      | <1     | <1      | <1      | <1     | <1       | <1      | <1    | <1    | <1    | <1    | <1     | <1    | <1    | <1    |
| Chromium (µg/L)                      | 2.2     | 4.0     | 7.5     | 4.0    | <1      | <1      | 0.5    | 2.1      | 1.5     | 4.8   | 1.7   | 1.0   | 1.0   | 2.0    | 0.8   | 1.0   | 3.6   |
| Copper (µg/L)                        | 0.6     | 12.2    | 4.3     | 0.6    | 1.8     | 1.2     | 15.9   | 2.6      | 0.6     | 2.1   | 0.7   | 1.7   | 0.5   | 1.6    | 0.8   | 1.8   | 1.4   |
| Dissolved organic carbon (mg/L)      |         |         |         |        |         |         |        |          |         |       |       | 1.9   |       |        |       |       |       |

| Analyte                            | 11WP53S | 11WP54D | 11WP54S | 11WP55 | 11WP56D | 11WP56S | 11WP57 | 11WP9RPB | 11WP9RS | CG1   | CG2   | CG4  | KC13 | KC14  | KC3  | KC3A | KC3PB |
|------------------------------------|---------|---------|---------|--------|---------|---------|--------|----------|---------|-------|-------|------|------|-------|------|------|-------|
| Electrical conductivity (mS/m)     | 898     | 2 425   | 1 682   | 408    | 9       | 78      | 1 598  | 182      | 181     | 606   | 92    | 324  | 104  | 1 752 | 24   | 45   | 43    |
| Fluoride (µg/L)                    | 1 320   | 338     | 272     | 270    | <50     | 195     | 424    | 774      | 835     | 300   | 292   | 228  | 362  | 1 068 | <50  | <50  | <50   |
| Iron (µg/L)                        | 36      | 28      | 89      | 670    | 11      | 35      | 207    | 26       | 13      | 17    | 7     | 334  | 378  | 131   | 22   | 8    | 24    |
| Gallium (µg/L)                     | <0.2    | <0.2    | <0.5    | 0.3    | <0.1    | <0.1    | 0.4    | <0.1     | <0.1    | <0.2  | <0.1  | <0.1 | <0.1 | <0.5  | <0.1 | <0.1 | <0.1  |
| Hardness (mg/L CaCO <sub>3</sub> ) | 912     | 3 500   | 2 057   | 464    | 35      | 332     | 2 898  | 301      | 314     | 1303  | 306   | 703  | 309  | 3 383 | 62   | 99   | 97    |
| Bicarbonate (mg/L)                 | 1 567   | 2 265   | 1 770   | 665    | 16      | 238     | 1 665  | 541      | 494     | 979   | 381   | 564  | 335  | 2 650 | 64   | 68   | 70    |
| Mercury (µg/L)                     | <0.2    | <0.2    | <0.5    | 0.3    | <0.1    | <0.1    | <0.5   | 0.3      | 0.1     | <0.2  | <0.1  | <0.1 | <0.1 | <0.5  | <0.1 | <0.1 | <0.1  |
| Potassium (mg/L)                   | 6       | 25      | 20      | 11     | 1       | 1       | 67     | 3        | 3       | 8     | 3     | 7    | 4    | 24    | 2    | 4    | 4     |
| Lanthanum (µg/L)                   | <5      | <5      | <5      | <5     | <5      | <5      | <5     | <5       | <5      | <5    | <5    | <5   | <5   | <5    | <5   | 0.1  | 0.1   |
| Lithium (µg/L)                     | 19      | 112     | 25      | 40     | 3       | 1       | 616    | 5        | 5       | 10    | 3     | 7    | 9    | 232   | 11   | 13   | 12    |
| Magnesium (mg/L)                   | 31      | 836     | 421     | 85     | 1       | 22      | 432    | 35       | 40      | 209   | 31    | 117  | 34   | 924   | 11   | 21   | 21    |
| Manganese (µg/L)                   | 143     | 507     | 1 310   | 3105   | 18      | 136     | 1 408  | 37       | 28      | 3     | 3     | 93   | 212  | 230   | 11   | 15   | 13    |
| Molybdenum (µg/L)                  | 40      | <5      | <5      | 11     | <1      | <1      | 12     | 4        | 2       | <2    | 2     | <1   | 5    | <5    | <1   | <1   | <1    |
| Nitrogen, ammonia (µg/L)           | 140     | 30      | 233     | 574    | 55      | <10     | 368    | 70       | 50      | 40    | 60    | 93   | 68   | 86    | 23   | 17   | 15    |
| Nitrogen, nitrite (µg/L)           | <10     | <10     | <10     | <10    | <10     | <10     | <10    | <10      | <10     | <10   | <10   | <10  | <10  | <10   | <10  | <10  | <10   |
| Nitrogen, nitrate (µg/L)           | 20      | <10     | 55      | <10    | 113     | 10      | <10    | 85       | 140     | 770   | 570   | 37   | 10   | <10   | 190  | 157  | 147   |
| Oxidised nitrogen (µg/L)           | 50      | <10     | 35      | 20     | 93      | 60      | <10    | 175      | 113     | 965   | 1 200 | 50   | 75   | <10   | 240  | 180  | 167   |
| Nitrogen, total (µg/L)             | 250     | 242     | 487     | 1142   | 552     | 103     | 635    | 543      | 440     | 1 213 | 1 192 | 677  | 170  | 404   | 362  | 325  | 243   |
| Sodium (mg/L)                      | 2 090   | 4 498   | 3 520   | 682    | 10      | 69      | 2 015  | 335      | 342     | 876   | 86    | 421  | 140  | 2 220 | 16   | 26   | 25    |
| Nickel (µg/L)                      | <2      | 8       | 12      | 3      | 1       | <1      | 4      | 1        | <1      | <2    | <1    | 2    | 2    | 2     | 2    | 2    | 11    |

| Analyte                             | 11WP53S | 11WP54D | 11WP54S | 11WP55 | 11WP56D | 11WP56S | 11WP57 | 11WP9RPB | 11WP9RS | CG1   | CG2  | CG4   | KC13 | KC14   | KC3  | KC3A | KC3PB |
|-------------------------------------|---------|---------|---------|--------|---------|---------|--------|----------|---------|-------|------|-------|------|--------|------|------|-------|
| Hydroxide (mg/L)                    | <1      | <1      | <1      | <1     | <1      | <1      | <1     | <1       | <1      | <1    | <1   | <1    | <1   | <1     | <1   | <1   | <1    |
| Phosphorus, soluble reactive (µg/L) | 175     | 32      | 57      | 22 333 | 30      | 57      | 72     | 1 800    | 434     | 57    | 50   | 35    | 20   | 15     | 13   | 17   | 10    |
| Phosphorus, total (µg/L)            | 227     | 28      | 69      | 29 333 | 157     | 217     | 164    | 4 797    | 612     | 75    | 64   | 87    | 56   | 75     | 21   | 40   | 30    |
| Lead (µg/L)                         | 0.4     | 2.8     | 0.7     | 1.5    | 10.5    | 0.7     | 0.6    | 0.3      | 0.4     | 1.9   | 1.3  | 5.0   | 4.6  | 4.0    | 0.4  | 1.1  | 2.8   |
| pH                                  | 8.0     | 7.1     | 7.4     | 7.1    | 6.1     | 7.1     | 7.1    | 7.7      | 7.7     | 7.5   | 7.8  | 7.6   | 7.6  | 7.1    | 6.7  | 6.3  | 6.1   |
| Antimony (µg/L)                     | 0.5     | 0.6     | <0.5    | 0.7    | 0.3     | 0.4     | <0.5   | 0.2      | <0.1    | <0.2  | <0.2 | 0.3   | 1.3  | <0.5   | 0.2  | 0.2  | 0.2   |
| Selenium (µg/L)                     | 6       | 7       | 5       | <1     | <1      | <1      | <5     | 4        | 5       | 37    | 2    | 1     | <1   | 2      | <1   | <1   | <1    |
| Silicon (mg/L)                      | 30      | 30      | 27      | 29     | 11      | 35      | 26     | 35       | 36      | 35    | 39   | 33    | 10   | 16     | 32   | 22   | 20    |
| Tin (µg/L)                          | 0.2     | <20     | 0.5     | 0.3    | <20     | 0.3     | <20    | <20      | <20     | <20   | 0.3  | <20   | <20  | <20    | <20  | 0.3  | <20   |
| Sulfate, sulfur (mg/L)              | 1 061   | 5 302   | 3 485   | 398    | 2       | 105     | 1 512  | 147      | 139     | 718   | 23   | 331   | 154  | 4 064  | 31   | 103  | 103   |
| TDS by summation (mg/L)             | 5 583   | 17 667  | 12 083  | 2 367  | 55      | 477     | 10 167 | 1 044    | 1 074   | 3 733 | 498  | 1 900 | 640  | 13 000 | 126  | 243  | 240   |
| Titanium (µg/L)                     | 1.2     | <2      | <2      | 1.9    | <2      | 0.5     | <2     | 0.5      | <2      | <2    | <2   | <2    | <2   | <2     | <2   | 0.3  | <2    |
| Thallium (µg/L)                     | <0.2    | <0.5    | <0.5    | <0.1   | <0.1    | <0.1    | 1.2    | <0.1     | <0.1    | <0.2  | <0.1 | <0.1  | <0.1 | 2.2    | <0.1 | <0.1 | 0.1   |
| Uranium (µg/L)                      | 63      | 109     | 128     | 1      | 0       | 2       | 6      | 6        | 6       | 35    | 9    | 14    | 9    | 56     | 0    | 0    | 0     |
| Vanadium (µg/L)                     | 118     | 20      | 27      | 9      | 1       | 10      | 4      | 52       | 40      | 33    | 23   | 9     | 6    | 1      | 3    | 1    | 0     |
| Zinc (µg/L)                         | 17      | 28      | 49      | 15     | 21      | 61      | 22     | 16       | 9       | 24    | 18   | 25    | 21   | 31     | 20   | 27   | 19    |

Table C4 Mean concentrations of analytes for bores KCF1 to WP5

| Analyte                              | KCF1  | LIMESTONE | RN029659 | RN029660 | W2R   | WP13  | WP19   | WP5   |
|--------------------------------------|-------|-----------|----------|----------|-------|-------|--------|-------|
| Acidity (mg/L CaCO <sub>3</sub> )    | 19    | 10        | 10       | 9        | 9     | 5     | 28     | 27    |
| Silver (µg/L)                        | 2.2   | <0.1      | 1.0      | 5.0      | <0.1  | 1.4   | <1     | 1.1   |
| Aluminium (µg/L)                     | 31    | 24        | 15       | 17       | 33    | 15    | 11     | 17    |
| Alkalinity (mg/L CaCO <sub>3</sub> ) | 715   | 163       | 273      | 450      | 160   | 193   | 447    | 541   |
| Arsenic (µg/L)                       | 1.5   | 5.2       | <1       | 1.7      | 3.7   | 6.8   | <10    | 1.7   |
| Atrazine (µg/L)                      | <0.01 | <0.01     | <0.01    | <0.01    | <0.01 | <0.01 | <0.01  | <0.01 |
| Boron (µg/L)                         | 222   | 97        | 278      | 327      | 75    | 103   | 928    | 319   |
| Barium (µg/L)                        | 38    | 174       | 34       | 49       | 232   | 62    | 25     | 37    |
| Beryllium (µg/L)                     | 1.8   | <0.1      | <0.1     | <0.1     | <0.1  | <0.1  | <1     | <0.1  |
| Bismuth (µg/L)                       | <0.1  | <0.1      | <0.1     | <0.1     | <0.1  | <0.1  | <1     | <0.1  |
| Bromine (µg/L)                       | 2140  | 153       | 3 000    | 703      | 150   | 407   | 20 250 | 3 560 |
| Calcium (mg/L)                       | 110   | 58        | 96       | 29       | 58    | 95    | 823    | 241   |
| Cadmium (µg/L)                       | 0.3   | <0.1      | <0.1     | 0.1      | <0.1  | <0.1  | 0.5    | 0.2   |
| Chloride (mg/L)                      | 561   | 41        | 918      | 258      | 31    | 136   | 7 732  | 1 019 |
| Cobalt (µg/L)                        | 2.3   | 0.4       | 0.3      | <0.1     | 1.3   | <0.1  | <1     | 0.5   |
| Carbonate (mg/L)                     | <1    | <1        | 14       | <1       | <1    | <1    | <1     | <1    |
| Chromium (µg/L)                      | 0.8   | <1        | <1       | 3.0      | 0.6   | <1    | <1     | <1    |
| Copper (µg/L)                        | 0.7   | 2.1       | 0.9      | 0.7      | 1.3   | 0.4   | 4.2    | 3.9   |
| Dissolved organic carbon (mg/L)      |       | 1.9       |          |          |       |       |        |       |
| Electrical conductivity (mS/m)       | 480   | 60        | 379      | 174      | 45    | 99    | 2 566  | 476   |
| Fluoride (µg/L)                      | 398   | 197       | 158      | 1 500    | 105   | 185   | 165    | 936   |
| Iron (µg/L)                          | 360   | 197       | 533      | 13       | 105   | 11    | 14     | 20    |
| Gallium (µg/L)                       | 0.1   | 0.1       | <0.1     | <0.1     | 0.1   | <0.1  | <1     | <0.1  |
| Hardness (mg/L CaCO <sub>3</sub> )   | 1095  | 189       | 616      | 264      | 173   | 295   | 3 729  | 1 783 |
| Bicarbonate (mg/L)                   | 976   | 193       | 423      | 493      | 196   | 260   | 2 776  | 1 090 |
| Mercury (µg/L)                       | <0.1  | <0.1      | <0.1     | <0.1     | <0.1  | <0.1  | <1     | <0.1  |
| Potassium (mg/L)                     | 12    | 12        | 10       | 9        | 6     | 15    | 29     | 4     |
| Lanthanum (µg/L)                     | 0.1   | 0.2       | <5       | <5       | 0.1   | <5    | <5     | 0.1   |
| Lithium (µg/L)                       | 22    | 14        | 40       | 30       | 7     | 25    | 92     | 25    |
| Magnesium (mg/L)                     | 229   | 9         | 115      | 33       | 7     | 22    | 946    | 390   |
| Manganese (µg/L)                     | 390   | 96        | 147      | 7        | 147   | 6     | 63     | 210   |
| Molybdenum (µg/L)                    | 1     | 1         | <1       | 3        | <1    | <1    | <10    | 2     |

| Analyte                             | KCF1 | LIMESTONE | RN029659 | RN029660 | W2R  | WP13 | WP19   | WP5  |
|-------------------------------------|------|-----------|----------|----------|------|------|--------|------|
| Nitrogen, ammonia (µg/L)            | 30   | 3 820     | 30       | 30       | 575  | <10  | 43     | 80   |
| Nitrogen, nitrite (µg/L)            | <10  | <10       | <10      | <10      | <10  | <10  | <10    | <10  |
| Nitrogen, nitrate (µg/L)            | 13   | 45        | <10      | 390      | <10  | 75   | 10     | <10  |
| Oxidised nitrogen (µg/L)            | 23   | 95        | 10       | 397      | 15   | 103  | 10     | 13   |
| Nitrogen, total (µg/L)              | 113  | 25 350    | 532      | 740      | 618  | 312  | 486    | 260  |
| Sodium (mg/L)                       | 698  | 34        | 516      | 306      | 18   | 64   | 4 630  | 289  |
| Nickel (µg/L)                       | 2    | <1        | <1       | <1       | 3    | <1   | <10    | 3    |
| Hydroxide (mg/L)                    | <1   | <1        | <1       | <1       | <1   | <1   | <1     | <1   |
| Phosphorus, soluble reactive (µg/L) | 18   | 350       | 15       | 57       | 30   | 10   | 34     | 20   |
| Phosphorus, total (µg/L)            | 23   | 1165      | 21       | 66       | 47   | 20   | 64     | 20   |
| Lead (µg/L)                         | 2.2  | 2.6       | 0.4      | 2.0      | 0.5  | 0.9  | 1.6    | 0.6  |
| pH                                  | 7.5  | 7.7       | 7.7      | 7.8      | 7.7  | 7.6  | 7.2    | 7.3  |
| Antimony (µg/L)                     | 0.2  | 0.2       | 0.2      | 0.2      | 0.1  | 0.3  | <1     | 0.2  |
| Selenium (µg/L)                     | 6    | <1        | <1       | <1       | <1   | <1   | 1      | <1   |
| Silicon (mg/L)                      | 28   | 23        | 14       | 28       | 15   | 35   | 26     | 24   |
| Tin (µg/L)                          | 0.1  | <20       | <20      | 0.1      | 0.2  | <20  | <20    | <20  |
| Sulfate, sulfur (mg/L)              | 1320 | 40        | 328      | 86       | 27   | 103  | 4 686  | 844  |
| TDS by summation (mg/L)             | 3267 | 317       | 2 147    | 977      | 243  | 546  | 19 000 | 3143 |
| Titanium (µg/L)                     | <2   | 0.8       | <2       | <2       | 10.0 | <2   | <2     | <2   |
| Thallium (µg/L)                     | 0.1  | <0.1      | <0.1     | <0.1     | <0.1 | 0.1  | <1     | <0.1 |
| Uranium (µg/L)                      | 25   | 1         | 6        | 3        | 0    | 1    | 45     | 9    |
| Vanadium (µg/L)                     | 10   | 3         | 2        | 46       | 1    | 6    | 11     | 7    |
| Zinc (µg/L)                         | 28   | 12        | 13       | 14       | 23   | 17   | 28     | 17   |

## Appendix D Summary statistics of groundwater chemistry

Table D1 Concentration of acidity (mg/L CaCO<sub>3</sub>)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 11.5   | 13.5 | 11.5               | 3    | 34   | 22.1            | 6                 | 0                  |
| 10WP32   | 17.0   | 15.8 | 5.8                | 5    | 21   | 20.3            | 6                 | 0                  |
| 10WP32PB | 14.0   | 13.7 | 5.9                | 6    | 23   | 18.1            | 6                 | 0                  |
| 10WP33   | 16.5   | 17.3 | 8.4                | 8    | 30   | 24.4            | 6                 | 0                  |
| 10WP35N  | 15.5   | 14.7 | 6.8                | 5    | 22   | 21.3            | 6                 | 0                  |
| 10WP35PB | 20.0   | 19.0 | 6.7                | 9    | 29   | 23.4            | 6                 | 0                  |
| 10WP35S  | 16.0   | 17.7 | 9.2                | 8    | 34   | 24.2            | 6                 | 0                  |
| 10WP36N  | 9.5    | 9.7  | 5.0                | 4    | 16   | 14.6            | 6                 | 0                  |
| 10WP36PB | 8.0    | 10.3 | 8.1                | 1    | 22   | 19.2            | 6                 | 17                 |
| 10WP36S  | 10.5   | 11.0 | 5.6                | 5    | 17   | 17.0            | 6                 | 0                  |
| 10WP37   | 19.5   | 17.2 | 8.2                | 7    | 28   | 23.1            | 6                 | 0                  |
| 10WP39   | 8.0    | 9.3  | 2.2                | 8    | 13   | 11.6            | 6                 | 0                  |
| 10WP40   | 22.0   | 22.3 | 12.3               | 10   | 44   | 30.7            | 6                 | 0                  |
| 10WP41   | 34.0   | 37.7 | 18.8               | 16   | 70   | 53.2            | 6                 | 0                  |
| 10WP42   | 11.5   | 12.3 | 4.8                | 6    | 20   | 16.5            | 6                 | 0                  |
| 10WP44   | 44.5   | 49.5 | 35.9               | 13   | 96   | 84.3            | 4                 | 0                  |
| 10WP46   | 9.5    | 13.5 | 14.8               | 3    | 43   | 20.6            | 6                 | 0                  |
| 10WP47   | 9.0    | 10.0 | 4.5                | 6    | 17   | 14.2            | 6                 | 0                  |
| 11CS10RD | 10.0   | 11.5 | 5.6                | 7    | 22   | 15.7            | 6                 | 0                  |
| 11CS10RS | 14.5   | 15.2 | 8.8                | 4    | 26   | 24.6            | 6                 | 0                  |
| 11WP4R   | 6.5    | 10.7 | 8.4                | 5    | 26   | 18.3            | 6                 | 0                  |
| 11WP9RPB | 10.0   | 10.3 | 6.6                | 3    | 22   | 15.7            | 7                 | 0                  |
| 11WP9RS  | 11.0   | 13.6 | 6.3                | 6    | 22   | 20.0            | 5                 | 0                  |
| 11WP11RD | 6.0    | 6.8  | 3.1                | 4    | 11   | 10.3            | 6                 | 0                  |
| 11WP11RS | 7.5    | 7.7  | 3.7                | 3    | 13   | 10.9            | 6                 | 0                  |
| 11WP15R  | 20.0   | 18.7 | 13.0               | 1    | 33   | 30.2            | 6                 | 17                 |
| 11WP16R  | 28.0   | 26.7 | 13.8               | 7    | 43   | 38.8            | 6                 | 0                  |
| 11WP43D  | 14.0   | 17.0 | 13.6               | 5    | 35   | 30.5            | 4                 | 0                  |
| 11WP50   | 34.5   | 35.0 | 18.1               | 14   | 57   | 51.9            | 4                 | 0                  |
| 11WP51D  | 4.5    | 5.7  | 3.4                | 3    | 12   | 8.5             | 6                 | 0                  |

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP51S <sup>†</sup>  | 6.5    | 9.7  | 9.8                | 1    | 25   | 20.1            | 6                 | 33                 |
| 11WP52D               | 7.0    | 9.5  | 6.3                | 3    | 20   | 15.8            | 6                 | 0                  |
| 11WP53D               | 21.0   | 19.3 | 12.0               | 1    | 35   | 28.7            | 6                 | 17                 |
| 11WP53S               | 15.0   | 15.2 | 11.5               | 1    | 32   | 25.0            | 6                 | 17                 |
| 11WP54D               | 38.5   | 35.2 | 18.0               | 10   | 55   | 52.2            | 6                 | 0                  |
| 11WP54S               | 16.5   | 22.3 | 16.8               | 9    | 55   | 33.3            | 6                 | 0                  |
| 11WP55                | 20.5   | 22.0 | 9.6                | 11   | 39   | 29.2            | 6                 | 0                  |
| 11WP56D               | 9.5    | 10.7 | 8.7                | 3    | 25   | 17.3            | 6                 | 0                  |
| 11WP56S               | 6.0    | 6.3  | 1.5                | 5    | 8    | 7.8             | 3                 | 0                  |
| 11WP57                | 23.0   | 26.7 | 18.4               | 9    | 61   | 39.3            | 6                 | 0                  |
| CG1                   | 28.5   | 25.2 | 6.2                | 15   | 30   | 29.3            | 6                 | 0                  |
| CG2                   | 7.0    | 9.6  | 5.5                | 5    | 17   | 15.5            | 5                 | 0                  |
| CG4                   | 13.0   | 12.0 | 8.7                | 1    | 21   | 20.3            | 6                 | 17                 |
| KC3                   | 5.0    | 7.4  | 3.9                | 4    | 13   | 11.5            | 5                 | 0                  |
| KC3A                  | 10.5   | 12.2 | 9.5                | 3    | 24   | 22.6            | 6                 | 0                  |
| KC3PB                 | 13.0   | 12.8 | 8.9                | 4    | 28   | 18.9            | 6                 | 0                  |
| KC13                  | 5.5    | 6.5  | 4.3                | 1    | 13   | 10.9            | 6                 | 17                 |
| KC14                  | 30.0   | 27.4 | 13.4               | 10   | 46   | 38.5            | 5                 | 0                  |
| KCF1                  | 20.0   | 19.2 | 8.0                | 10   | 30   | 26.5            | 6                 | 0                  |
| LIMESTONE             | 10.5   | 9.8  | 5.0                | 4    | 18   | 13.1            | 6                 | 0                  |
| M1 <sup>†</sup>       | 1.0    | 1.0  | 0.0                | 1    | 1    | 1.0             | 2                 | 100                |
| RN029659 <sup>†</sup> | 5.5    | 7.0  | 6.4                | 1    | 17   | 13.5            | 6                 | 33                 |
| RN029660              | 8.0    | 8.7  | 4.9                | 4    | 14   | 14.0            | 6                 | 0                  |
| W2R                   | 8.5    | 8.0  | 5.3                | 1    | 16   | 11.8            | 6                 | 17                 |
| WP5                   | 28.0   | 26.7 | 12.3               | 8    | 39   | 37.6            | 6                 | 0                  |
| WP13                  | 4.0    | 5.4  | 3.0                | 3    | 10   | 8.5             | 5                 | 0                  |
| WP19                  | 29.0   | 28.3 | 14.7               | 12   | 43   | 41.5            | 4                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D2 Concentration of silver (µg/L)

| Site                   | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>    | 0.05   | 0.11 | 0.14               | 0.05 | 0.40  | 0.16            | 6                 | 83                 |
| 10WP32 <sup>†</sup>    | 0.05   | 0.16 | 0.27               | 0.05 | 0.70  | 0.25            | 6                 | 83                 |
| 10WP32PB <sup>†</sup>  | 0.05   | 0.11 | 0.14               | 0.05 | 0.40  | 0.16            | 6                 | 83                 |
| 10WP33 <sup>†</sup>    | 0.05   | 0.18 | 0.31               | 0.05 | 0.80  | 0.28            | 6                 | 83                 |
| 10WP35N <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05  | 0.05            | 6                 | 100                |
| 10WP35PB <sup>†</sup>  | 0.05   | 0.11 | 0.14               | 0.05 | 0.40  | 0.16            | 6                 | 83                 |
| 10WP35S <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 83                 |
| 10WP36N <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05  | 0.05            | 6                 | 100                |
| 10WP36PB <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05  | 0.05            | 6                 | 100                |
| 10WP36S <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05  | 0.05            | 6                 | 100                |
| 10WP37 <sup>†</sup>    | 0.05   | 0.05 | 0.00               | 0.05 | 0.05  | 0.05            | 6                 | 100                |
| 10WP39 <sup>†</sup>    | 0.05   | 0.14 | 0.22               | 0.05 | 0.60  | 0.22            | 6                 | 83                 |
| 10WP40 <sup>†</sup>    | 0.10   | 0.12 | 0.09               | 0.05 | 0.30  | 0.16            | 6                 | 83                 |
| 10WP41 <sup>†</sup>    | 0.25   | 3.21 | 7.25               | 0.25 | 18.00 | 5.58            | 6                 | 83                 |
| 10WP42 <sup>†</sup>    | 0.05   | 0.11 | 0.14               | 0.05 | 0.40  | 0.16            | 6                 | 83                 |
| 10WP44 <sup>†</sup>    | 0.25   | 0.25 | 0.00               | 0.25 | 0.25  | 0.25            | 4                 | 100                |
| 10WP46 <sup>†</sup>    | 0.05   | 0.19 | 0.26               | 0.05 | 0.70  | 0.39            | 6                 | 83                 |
| 10WP47 <sup>†</sup>    | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 83                 |
| 11CS10 <sup>†</sup> RD | 0.05   | 0.43 | 0.87               | 0.05 | 2.20  | 0.80            | 6                 | 67                 |
| 11CS10 <sup>†</sup> RS | 0.10   | 0.23 | 0.33               | 0.05 | 0.90  | 0.34            | 6                 | 83                 |
| 11WP4R <sup>†</sup>    | 0.05   | 0.24 | 0.47               | 0.05 | 1.20  | 0.40            | 6                 | 83                 |
| 11WP9RP <sup>†</sup> B | 0.05   | 0.09 | 0.07               | 0.05 | 0.20  | 0.20            | 7                 | 71                 |
| 11WP9RS <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.08            | 5                 | 80                 |
| 11WP11RD <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 83                 |
| 11WP11RS               | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 83                 |
| 11WP15R <sup>†</sup>   | 0.10   | 0.13 | 0.08               | 0.10 | 0.30  | 0.16            | 6                 | 83                 |
| 11WP16R <sup>†</sup>   | 0.25   | 0.35 | 0.16               | 0.25 | 0.60  | 0.53            | 6                 | 67                 |
| 11WP43D <sup>†</sup>   | 0.18   | 0.23 | 0.20               | 0.05 | 0.50  | 0.43            | 4                 | 100                |
| 11WP50 <sup>†</sup>    | 0.25   | 0.25 | 0.00               | 0.25 | 0.25  | 0.25            | 4                 | 100                |
| 11WP51D <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 83                 |
| 11WP51S <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 100                |
| 11WP52D <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10  | 0.07            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.15 | 0.12               | 0.10 | 0.40 | 0.19            | 6                 | 83                 |
| 11WP53S <sup>†</sup>   | 0.10   | 0.27 | 0.41               | 0.10 | 1.10 | 0.40            | 6                 | 83                 |
| 11WP54D <sup>†</sup>   | 0.25   | 0.87 | 1.22               | 0.25 | 3.30 | 1.62            | 6                 | 67                 |
| 11WP54S <sup>†</sup>   | 0.25   | 0.49 | 0.59               | 0.25 | 1.70 | 0.69            | 6                 | 83                 |
| 11WP55 <sup>†</sup>    | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 0.05   | 0.19 | 0.35               | 0.05 | 0.90 | 0.31            | 6                 | 83                 |
| 11WP56S <sup>†</sup>   | 0.25   | 0.33 | 0.33               | 0.05 | 0.70 | 0.66            | 3                 | 67                 |
| 11WP57 <sup>†</sup>    | 0.25   | 0.38 | 0.41               | 0.05 | 1.20 | 0.54            | 6                 | 83                 |
| CG1 <sup>†</sup>       | 0.10   | 0.25 | 0.28               | 0.10 | 0.80 | 0.45            | 6                 | 67                 |
| CG2 <sup>†</sup>       | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.08            | 5                 | 80                 |
| CG4 <sup>†</sup>       | 0.05   | 0.08 | 0.06               | 0.05 | 0.20 | 0.10            | 6                 | 83                 |
| KC3 <sup>†</sup>       | 0.05   | 1.14 | 2.44               | 0.05 | 5.50 | 2.78            | 5                 | 80                 |
| KC3A <sup>†</sup>      | 0.05   | 0.81 | 1.86               | 0.05 | 4.60 | 1.42            | 6                 | 83                 |
| KC3PB <sup>†</sup>     | 0.05   | 0.94 | 2.18               | 0.05 | 5.40 | 1.66            | 6                 | 83                 |
| KC13 <sup>†</sup>      | 0.05   | 0.16 | 0.19               | 0.05 | 0.50 | 0.33            | 6                 | 83                 |
| KC14 <sup>††</sup>     | 0.25   | 2.12 | 4.29               | 0.05 | 9.80 | 5.03            | 5                 | 80                 |
| CF1 <sup>†</sup>       | 0.05   | 0.41 | 0.88               | 0.05 | 2.20 | 0.70            | 6                 | 83                 |
| LIMESTONE <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 2                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.21 | 0.39               | 0.05 | 1.00 | 0.34            | 6                 | 83                 |
| RN029660 <sup>†</sup>  | 0.05   | 0.88 | 2.02               | 0.05 | 5.00 | 1.54            | 6                 | 83                 |
| W2R <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.05   | 0.23 | 0.43               | 0.05 | 1.10 | 0.40            | 6                 | 83                 |
| WP13 <sup>†</sup>      | 0.05   | 0.28 | 0.55               | 0.05 | 1.40 | 0.46            | 6                 | 83                 |
| WP19 <sup>†</sup>      | 0.50   | 0.44 | 0.13               | 0.25 | 0.50 | 0.50            | 4                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D3 Concentration of aluminium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31                | 16.0   | 12.5 | 6.1                | 2.5  | 19   | 17.2            | 7                 | 14                 |
| 10WP32                | 7.0    | 7.6  | 4.6                | 2.5  | 17   | 9.8             | 7                 | 14                 |
| 10WP32PB <sup>†</sup> | 6.0    | 22.8 | 43.1               | 2.5  | 120  | 23.7            | 7                 | 43                 |
| 10WP33 <sup>†</sup>   | 15.0   | 26.1 | 42.2               | 2.5  | 120  | 31.8            | 7                 | 43                 |
| 10WP35N <sup>†</sup>  | 18.0   | 15.3 | 9.9                | 2.5  | 26   | 25.1            | 7                 | 29                 |
| 10WP35PB <sup>†</sup> | 2.5    | 9.1  | 10.2               | 2.5  | 25   | 22.6            | 9                 | 67                 |
| 10WP35S <sup>†</sup>  | 12.0   | 10.6 | 8.0                | 2.5  | 20   | 18.2            | 7                 | 43                 |
| 10WP36N <sup>†</sup>  | 17.0   | 16.1 | 14.0               | 2.5  | 38   | 30.8            | 7                 | 29                 |
| 10WP36PB <sup>†</sup> | 2.5    | 6.1  | 6.6                | 2.5  | 21   | 11.0            | 10                | 70                 |
| 10WP36S <sup>†</sup>  | 18.0   | 17.0 | 11.5               | 2.5  | 31   | 30.1            | 7                 | 29                 |
| 10WP37                | 18.0   | 27.6 | 33.1               | 6.0  | 100  | 36.1            | 7                 | 0                  |
| 10WP39 <sup>†</sup>   | 14.0   | 12.4 | 7.1                | 2.5  | 19   | 18.1            | 7                 | 29                 |
| 10WP40                | 20.0   | 19.7 | 6.6                | 7.0  | 26   | 25.1            | 7                 | 0                  |
| 10WP41                | 21.0   | 18.2 | 8.0                | 2.5  | 25   | 24.1            | 7                 | 14                 |
| 10WP42                | 16.0   | 37.4 | 67.6               | 2.5  | 190  | 35.2            | 7                 | 14                 |
| 10WP44                | 15.5   | 13.1 | 7.3                | 2.5  | 19   | 18.1            | 4                 | 25                 |
| 10WP46 <sup>†</sup>   | 12.0   | 30.7 | 57.3               | 2.5  | 160  | 31.3            | 7                 | 29                 |
| 10WP47                | 17.0   | 26.4 | 37.2               | 6.0  | 110  | 27.2            | 7                 | 0                  |
| 11CS10RD <sup>†</sup> | 10.5   | 13.2 | 12.2               | 2.5  | 35   | 23.1            | 6                 | 33                 |
| 11CS10RS              | 12.5   | 11.9 | 7.0                | 2.5  | 20   | 17.9            | 6                 | 17                 |
| 11WP4R                | 15.5   | 27.6 | 36.0               | 2.5  | 100  | 44.0            | 6                 | 17                 |
| 11WP9RPB <sup>†</sup> | 15.0   | 26.0 | 42.1               | 2.5  | 120  | 29.1            | 7                 | 29                 |
| 11WP9RS               | 20.0   | 17.5 | 9.7                | 2.5  | 27   | 25.5            | 5                 | 20                 |
| 11WP11RD              | 13.5   | 12.6 | 6.7                | 2.5  | 21   | 18.2            | 6                 | 17                 |
| 11WP11RS              | 39.0   | 77.3 | 87.1               | 13.0 | 240  | 149.0           | 6                 | 0                  |
| 11WP15R <sup>†</sup>  | 14.0   | 14.2 | 11.4               | 2.5  | 26   | 26.0            | 6                 | 33                 |
| 11WP16R               | 13.0   | 12.8 | 7.9                | 2.5  | 22   | 19.9            | 6                 | 17                 |
| 11WP43D               | 21.0   | 21.3 | 6.7                | 15.0 | 28   | 27.4            | 4                 | 0                  |
| 11WP50                | 19.5   | 17.1 | 10.4               | 2.5  | 27   | 24.9            | 4                 | 25                 |
| 11WP51D               | 12.0   | 14.1 | 9.7                | 2.5  | 31   | 21.9            | 6                 | 17                 |
| 11WP51S               | 29.5   | 27.1 | 18.4               | 2.5  | 44   | 43.3            | 6                 | 17                 |
| 11WP52D               | 12.5   | 11.3 | 6.1                | 2.5  | 18   | 16.6            | 6                 | 17                 |

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>  | 30.5   | 25.5 | 18.9               | 2.5  | 46   | 42.5            | 6                 | 33                 |
| 11WP53S               | 36.0   | 29.7 | 16.2               | 9.0  | 48   | 41.7            | 6                 | 0                  |
| 11WP54D               | 12.0   | 11.9 | 7.3                | 2.5  | 22   | 18.5            | 6                 | 17                 |
| 11WP54S               | 30.5   | 28.5 | 16.2               | 6.0  | 47   | 43.5            | 6                 | 0                  |
| 11WP55                | 20.5   | 20.0 | 9.8                | 8.0  | 36   | 26.9            | 6                 | 0                  |
| 11WP56D               | 12.0   | 12.5 | 4.7                | 5.0  | 18   | 17.3            | 6                 | 0                  |
| 11WP56S               | 21.0   | 19.7 | 2.3                | 17.0 | 21   | 21.0            | 3                 | 0                  |
| 11WP57 <sup>†</sup>   | 4.8    | 6.4  | 5.0                | 2.5  | 15   | 10.8            | 6                 | 50                 |
| G1 <sup>†</sup>       | 15.0   | 16.8 | 15.5               | 2.5  | 44   | 28.6            | 6                 | 33                 |
| CG2 <sup>†</sup>      | 10.0   | 8.4  | 5.8                | 2.5  | 16   | 13.5            | 5                 | 40                 |
| CG4                   | 15.0   | 16.4 | 10.3               | 2.5  | 34   | 23.5            | 6                 | 17                 |
| KC3                   | 18.0   | 27.0 | 17.7               | 13.0 | 55   | 44.5            | 5                 | 0                  |
| KC3A <sup>†</sup>     | 12.5   | 10.0 | 6.0                | 2.5  | 16   | 14.6            | 6                 | 33                 |
| KC3PB <sup>†</sup>    | 5.8    | 8.4  | 7.2                | 2.5  | 19   | 16.2            | 6                 | 50                 |
| KC13 <sup>†</sup>     | 12.5   | 11.5 | 7.7                | 2.5  | 20   | 19.3            | 6                 | 33                 |
| KC14                  | 6.0    | 9.0  | 4.1                | 6.0  | 14   | 13.5            | 5                 | 0                  |
| KCF1                  | 22.5   | 25.9 | 22.3               | 2.5  | 66   | 42.2            | 6                 | 17                 |
| LIMESTONE             | 20.0   | 23.5 | 11.7               | 14.0 | 46   | 30.6            | 6                 | 0                  |
| M1                    | 15.0   | 16.0 | 5.6                | 11.0 | 22   | 21.3            | 3                 | 0                  |
| RN029659 <sup>†</sup> | 8.3    | 8.9  | 7.2                | 2.5  | 18   | 15.2            | 6                 | 50                 |
| RN029660              | 19.0   | 17.3 | 6.0                | 6.0  | 22   | 22.0            | 6                 | 0                  |
| W2R                   | 27.5   | 33.0 | 25.4               | 9.0  | 80   | 49.9            | 6                 | 0                  |
| WP5 <sup>†</sup>      | 8.0    | 10.6 | 8.8                | 2.5  | 21   | 21.0            | 7                 | 43                 |
| WP13 <sup>†</sup>     | 5.8    | 8.6  | 7.7                | 2.5  | 21   | 16.1            | 6                 | 50                 |
| WP19 <sup>†</sup>     | 6.0    | 7.6  | 5.8                | 2.5  | 16   | 13.5            | 5                 | 40                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D4 Concentration of alkalinity (mg/L CaCO<sub>3</sub>)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 150    | 150  | 6                  | 140  | 156  | 156             | 7                 | 0                  |
| 10WP32   | 402    | 407  | 21                 | 385  | 439  | 426             | 7                 | 0                  |
| 10WP32PB | 403    | 411  | 31                 | 375  | 475  | 417             | 7                 | 0                  |
| 10WP33   | 320    | 326  | 19                 | 299  | 350  | 344             | 7                 | 0                  |
| 10WP35N  | 528    | 538  | 93                 | 382  | 638  | 626             | 7                 | 0                  |
| 10WP35PB | 620    | 612  | 55                 | 530  | 682  | 661             | 9                 | 0                  |
| 10WP35S  | 580    | 585  | 65                 | 499  | 662  | 640             | 7                 | 0                  |
| 10WP36N  | 529    | 518  | 29                 | 481  | 561  | 538             | 7                 | 0                  |
| 10WP36PB | 505    | 513  | 62                 | 436  | 607  | 591             | 10                | 0                  |
| 10WP36S  | 630    | 604  | 88                 | 407  | 659  | 649             | 7                 | 0                  |
| 10WP37   | 697    | 664  | 54                 | 590  | 721  | 706             | 7                 | 0                  |
| 10WP39   | 238    | 238  | 7                  | 228  | 246  | 245             | 7                 | 0                  |
| 10WP40   | 655    | 645  | 76                 | 550  | 780  | 686             | 7                 | 0                  |
| 10WP41   | 478    | 508  | 97                 | 390  | 657  | 587             | 7                 | 0                  |
| 10WP42   | 333    | 331  | 27                 | 285  | 358  | 355             | 7                 | 0                  |
| 10WP44   | 819    | 835  | 267                | 573  | 1130 | 1088            | 4                 | 0                  |
| 10WP46   | 285    | 320  | 97                 | 276  | 540  | 320             | 7                 | 0                  |
| 10WP47   | 217    | 213  | 28                 | 163  | 250  | 237             | 7                 | 0                  |
| 11CS10RD | 457    | 442  | 42                 | 368  | 478  | 475             | 6                 | 0                  |
| 11CS10RS | 770    | 853  | 207                | 683  | 1230 | 1035            | 6                 | 0                  |
| 11WP4R   | 300    | 306  | 43                 | 260  | 377  | 344             | 6                 | 0                  |
| 11WP9RPB | 559    | 517  | 124                | 238  | 587  | 572             | 7                 | 0                  |
| 11WP9RS  | 555    | 464  | 147                | 278  | 581  | 578             | 5                 | 0                  |
| 11WP11RD | 378    | 380  | 11                 | 365  | 395  | 392             | 6                 | 0                  |
| 11WP11RS | 539    | 528  | 29                 | 471  | 553  | 546             | 6                 | 0                  |
| 11WP15R  | 1190   | 1203 | 22                 | 1190 | 1240 | 1226            | 6                 | 0                  |
| 11WP16R  | 424    | 446  | 66                 | 385  | 529  | 523             | 6                 | 0                  |
| 11WP43D  | 232    | 245  | 140                | 105  | 411  | 380             | 4                 | 0                  |
| 11WP50   | 614    | 611  | 9                  | 598  | 619  | 618             | 4                 | 0                  |
| 11WP51D  | 247    | 246  | 9                  | 235  | 258  | 252             | 6                 | 0                  |
| 11WP51S  | 395    | 315  | 140                | 99   | 414  | 413             | 6                 | 0                  |
| 11WP52D  | 277    | 268  | 40                 | 192  | 303  | 297             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 1785   | 1763 | 59                 | 1670 | 1810 | 1810            | 6                 | 0                  |
| 11WP53S   | 1910   | 1903 | 91                 | 1760 | 2020 | 1978            | 6                 | 0                  |
| 11WP54D   | 678    | 682  | 25                 | 655  | 715  | 711             | 6                 | 0                  |
| 11WP54S   | 1215   | 1200 | 85                 | 1070 | 1300 | 1272            | 6                 | 0                  |
| 11WP55    | 543    | 437  | 227                | 26   | 605  | 592             | 6                 | 0                  |
| 11WP56D   | 14     | 31   | 43                 | 7    | 117  | 52              | 6                 | 0                  |
| 11WP56S   | 295    | 296  | 61                 | 235  | 357  | 351             | 3                 | 0                  |
| 11WP57    | 345    | 380  | 101                | 310  | 583  | 426             | 6                 | 0                  |
| CG1       | 731    | 737  | 27                 | 709  | 783  | 762             | 6                 | 0                  |
| CG2       | 338    | 331  | 18                 | 309  | 346  | 346             | 5                 | 0                  |
| CG4       | 411    | 412  | 15                 | 395  | 435  | 426             | 6                 | 0                  |
| KC3       | 51     | 45   | 12                 | 27   | 55   | 54              | 5                 | 0                  |
| KC3A      | 28     | 28   | 2                  | 26   | 30   | 29              | 6                 | 0                  |
| KC3PB     | 29     | 29   | 2                  | 25   | 31   | 30              | 6                 | 0                  |
| KC13      | 310    | 315  | 40                 | 260  | 384  | 341             | 6                 | 0                  |
| KC14      | 365    | 306  | 142                | 53   | 384  | 378             | 5                 | 0                  |
| KCF1      | 713    | 715  | 17                 | 695  | 738  | 732             | 6                 | 0                  |
| LIMESTONE | 173    | 163  | 58                 | 79   | 239  | 211             | 6                 | 0                  |
| M1        | 105    | 105  | 0                  | 105  | 105  | 105             | 3                 | 0                  |
| RN029659  | 312    | 273  | 113                | 100  | 377  | 370             | 6                 | 0                  |
| RN029660  | 444    | 450  | 20                 | 428  | 480  | 471             | 6                 | 0                  |
| W2R       | 182    | 160  | 42                 | 81   | 187  | 186             | 6                 | 0                  |
| WP5       | 560    | 541  | 72                 | 381  | 590  | 583             | 7                 | 0                  |
| WP13      | 215    | 193  | 51                 | 102  | 219  | 218             | 5                 | 0                  |
| WP19      | 463    | 447  | 36                 | 384  | 473  | 470             | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D5 Concentration of arsenic (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP32                | 1.0    | 1.1  | 0.4                | 0.5  | 2.0  | 1.1             | 7                 | 14                 |
| 10WP32PB              | 1.0    | 1.1  | 0.4                | 0.5  | 2.0  | 1.1             | 7                 | 14                 |
| 10WP33 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP35N               | 1.0    | 1.2  | 0.6                | 0.5  | 2.0  | 2.0             | 7                 | 14                 |
| 10WP35PB <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.5    | 0.9  | 0.7                | 0.5  | 2.0  | 2.0             | 7                 | 71                 |
| 10WP36N               | 4.0    | 4.3  | 1.3                | 2.0  | 6.0  | 5.1             | 7                 | 0                  |
| 10WP36PB              | 4.5    | 4.2  | 2.3                | 1.0  | 8.0  | 6.0             | 10                | 0                  |
| 10WP36S               | 7.0    | 8.9  | 3.7                | 5.0  | 15.0 | 13.2            | 7                 | 0                  |
| 10WP37                | 1.0    | 1.3  | 0.5                | 1.0  | 2.0  | 2.0             | 7                 | 0                  |
| 10WP39                | 7.0    | 6.9  | 0.7                | 6.0  | 8.0  | 7.1             | 7                 | 0                  |
| 10WP40                | 5.0    | 6.0  | 2.8                | 3.0  | 10.0 | 9.1             | 7                 | 0                  |
| 10WP41 <sup>†</sup>   | 2.5    | 2.8  | 1.0                | 2.0  | 5.0  | 2.8             | 7                 | 86                 |
| 10WP42                | 2.0    | 2.1  | 0.4                | 2.0  | 3.0  | 2.1             | 7                 | 0                  |
| 10WP44 <sup>†</sup>   | 4.3    | 5.0  | 3.1                | 2.5  | 9.0  | 8.1             | 4                 | 50                 |
| 10WP46                | 2.0    | 1.8  | 0.6                | 1.0  | 2.5  | 2.1             | 7                 | 14                 |
| 10WP47                | 1.0    | 1.1  | 0.4                | 1.0  | 2.0  | 1.1             | 7                 | 0                  |
| 11CS10RD <sup>†</sup> | 0.8    | 0.8  | 0.3                | 0.5  | 1.0  | 1.0             | 6                 | 50                 |
| 11CS10RS <sup>†</sup> | 1.0    | 2.3  | 2.8                | 1.0  | 8.0  | 3.8             | 6                 | 50                 |
| 11WP4R <sup>†</sup>   | 2.0    | 2.3  | 1.9                | 0.5  | 5.0  | 4.3             | 6                 | 33                 |
| 11WP9RPB              | 2.0    | 6.1  | 11.1               | 1.0  | 31.0 | 7.6             | 7                 | 0                  |
| 11WP9RS               | 2.0    | 2.0  | 0.7                | 1.0  | 3.0  | 2.5             | 5                 | 0                  |
| 11WP11RD <sup>†</sup> | 1.0    | 1.2  | 0.9                | 0.5  | 3.0  | 1.6             | 6                 | 33                 |
| 11WP11RS              | 3.0    | 4.3  | 3.9                | 1.0  | 12.0 | 6.4             | 6                 | 0                  |
| 11WP15R               | 2.0    | 1.8  | 0.4                | 1.0  | 2.0  | 2.0             | 6                 | 17                 |
| 11WP16R               | 2.5    | 2.6  | 0.2                | 2.5  | 3.0  | 2.7             | 6                 | 83                 |
| 11WP43D <sup>†</sup>  | 1.5    | 2.3  | 1.9                | 1.0  | 5.0  | 4.1             | 4                 | 50                 |
| 11WP50 <sup>†</sup>   | 2.5    | 2.0  | 1.0                | 0.5  | 2.5  | 2.5             | 4                 | 100                |
| 11WP51D               | 3.5    | 4.2  | 1.6                | 3.0  | 7.0  | 5.6             | 6                 | 0                  |
| 11WP51S               | 3.0    | 3.0  | 1.4                | 1.0  | 5.0  | 4.3             | 6                 | 0                  |
| 11WP52D <sup>†</sup>  | 0.5    | 0.7  | 0.3                | 0.5  | 1.0  | 1.0             | 6                 | 83                 |

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D               | 7.5    | 7.3  | 0.8                | 6.0  | 8.0  | 8.0             | 6                 | 0                  |
| 11WP53S               | 7.0    | 7.5  | 0.8                | 7.0  | 9.0  | 8.3             | 6                 | 0                  |
| 11WP54D <sup>†</sup>  | 2.5    | 2.3  | 0.6                | 1.0  | 2.5  | 2.5             | 6                 | 83                 |
| 11WP54S <sup>†</sup>  | 2.5    | 2.6  | 0.2                | 2.5  | 3.0  | 2.7             | 6                 | 83                 |
| 11WP55                | 29.5   | 34.2 | 15.9               | 23.0 | 66.0 | 42.2            | 6                 | 0                  |
| 11WP56D <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP56S <sup>†</sup>  | 1.0    | 1.3  | 1.0                | 0.5  | 2.5  | 2.4             | 3                 | 67                 |
| 11WP57 <sup>†</sup>   | 2.5    | 2.3  | 0.3                | 2.0  | 2.5  | 2.5             | 6                 | 67                 |
| CG1 <sup>†</sup>      | 1.0    | 0.8  | 0.3                | 0.5  | 1.0  | 1.0             | 6                 | 100                |
| CG2 <sup>†</sup>      | 0.5    | 0.6  | 0.2                | 0.5  | 1.0  | 0.8             | 5                 | 80                 |
| CG4 <sup>†</sup>      | 0.5    | 0.6  | 0.2                | 0.5  | 1.0  | 0.7             | 6                 | 83                 |
| KC3 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| KC3A <sup>†</sup>     | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC3PB <sup>†</sup>    | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC13                  | 3.0    | 3.3  | 0.9                | 2.5  | 5.0  | 3.6             | 6                 | 17                 |
| KC14 <sup>†</sup>     | 2.5    | 1.7  | 1.1                | 0.5  | 2.5  | 2.5             | 5                 | 100                |
| KCF1                  | 1.5    | 1.5  | 0.5                | 1.0  | 2.0  | 2.0             | 6                 | 0                  |
| LIMESTONE             | 4.5    | 5.2  | 3.8                | 1.0  | 10.0 | 9.3             | 6                 | 0                  |
| M1 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 3                 | 100                |
| RN029659 <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| RN029660              | 2.0    | 1.7  | 0.5                | 1.0  | 2.0  | 2.0             | 6                 | 0                  |
| W2R                   | 4.0    | 3.7  | 2.1                | 1.0  | 7.0  | 4.9             | 6                 | 0                  |
| WP5 <sup>†</sup>      | 1.0    | 1.1  | 0.9                | 0.5  | 3.0  | 1.2             | 7                 | 57                 |
| WP13                  | 7.0    | 6.8  | 0.4                | 6.0  | 7.0  | 7.0             | 6                 | 0                  |
| WP19 <sup>†</sup>     | 5.0    | 3.6  | 2.0                | 0.5  | 5.0  | 5.0             | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D6 Concentration of boron (µg/L)

| Site                 | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31               | 30     | 30   | 12                 | 10   | 50   | 32              | 7                 | 14                 |
| 10WP32               | 110    | 104  | 33                 | 50   | 160  | 115             | 7                 | 0                  |
| 10WP32PB             | 100    | 99   | 23                 | 60   | 120  | 120             | 7                 | 0                  |
| 10WP33               | 70     | 73   | 19                 | 50   | 110  | 83              | 7                 | 0                  |
| 10WP35N              | 370    | 357  | 55                 | 240  | 400  | 391             | 7                 | 0                  |
| 10WP35PB             | 280    | 278  | 28                 | 240  | 320  | 304             | 9                 | 0                  |
| 10WP35S              | 300    | 293  | 29                 | 250  | 340  | 313             | 7                 | 0                  |
| 10WP36N              | 440    | 426  | 108                | 200  | 530  | 512             | 7                 | 0                  |
| 10WP36PB             | 565    | 559  | 104                | 340  | 720  | 640             | 10                | 0                  |
| 10WP36S              | 890    | 896  | 128                | 690  | 1100 | 983             | 7                 | 0                  |
| 10WP37               | 490    | 489  | 36                 | 440  | 540  | 522             | 7                 | 0                  |
| 10WP39               | 160    | 157  | 23                 | 110  | 180  | 171             | 7                 | 0                  |
| 10WP40               | 870    | 933  | 309                | 600  | 1500 | 1140            | 7                 | 0                  |
| 10WP41               | 2200   | 2071 | 345                | 1500 | 2500 | 2230            | 7                 | 0                  |
| 10WP42               | 240    | 249  | 27                 | 210  | 290  | 272             | 7                 | 0                  |
| 10WP44               | 415    | 413  | 43                 | 370  | 450  | 450             | 4                 | 0                  |
| 10WP46               | 230    | 223  | 42                 | 140  | 270  | 252             | 7                 | 0                  |
| 10WP47               | 70     | 64   | 30                 | 10   | 90   | 90              | 7                 | 14                 |
| 11CS10RD             | 100    | 105  | 18                 | 80   | 130  | 123             | 6                 | 0                  |
| 11CS10RS             | 870    | 805  | 315                | 320  | 1200 | 1060            | 6                 | 0                  |
| 11WP4R               | 125    | 125  | 23                 | 90   | 160  | 139             | 6                 | 0                  |
| 11WP9RPB             | 170    | 170  | 24                 | 140  | 200  | 191             | 7                 | 0                  |
| 11WP9RS              | 170    | 168  | 8                  | 160  | 180  | 175             | 5                 | 0                  |
| 11WP11RD             | 105    | 95   | 24                 | 60   | 120  | 113             | 6                 | 0                  |
| 11WP11RS             | 190    | 168  | 48                 | 90   | 210  | 203             | 6                 | 0                  |
| 11WP15R              | 525    | 517  | 29                 | 460  | 540  | 533             | 6                 | 0                  |
| 11WP16R              | 305    | 330  | 89                 | 230  | 470  | 421             | 6                 | 0                  |
| 11WP43D              | 160    | 143  | 59                 | 60   | 190  | 187             | 4                 | 0                  |
| 11WP50               | 555    | 558  | 32                 | 530  | 590  | 587             | 4                 | 0                  |
| 11WP51D              | 95     | 92   | 25                 | 60   | 120  | 113             | 6                 | 0                  |
| 11WP51S              | 100    | 120  | 89                 | 10   | 230  | 223             | 6                 | 17                 |
| 11WP52D <sup>†</sup> | 35     | 30   | 17                 | 10   | 50   | 43              | 6                 | 33                 |

| Site            | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D         | 720    | 718  | 45                 | 640  | 770  | 756             | 6                 | 0                  |
| 11WP53S         | 835    | 842  | 59                 | 760  | 930  | 895             | 6                 | 0                  |
| 11WP54D         | 765    | 760  | 74                 | 670  | 850  | 829             | 6                 | 0                  |
| 11WP54S         | 990    | 978  | 171                | 710  | 1200 | 1130            | 6                 | 0                  |
| 11WP55          | 440    | 440  | 24                 | 400  | 470  | 463             | 6                 | 0                  |
| 11WP56D         | 60     | 62   | 8                  | 50   | 70   | 70              | 6                 | 0                  |
| 11WP56S         | 80     | 80   | 20                 | 60   | 100  | 98              | 3                 | 0                  |
| 11WP57          | 1150   | 1110 | 132                | 860  | 1200 | 1200            | 6                 | 0                  |
| CG1             | 195    | 187  | 30                 | 130  | 210  | 210             | 6                 | 0                  |
| CG2             | 70     | 66   | 11                 | 50   | 80   | 75              | 5                 | 0                  |
| CG4             | 125    | 125  | 14                 | 110  | 140  | 140             | 6                 | 0                  |
| KC3             | 40     | 40   | 7                  | 30   | 50   | 45              | 5                 | 0                  |
| KC3A            | 35     | 33   | 14                 | 10   | 50   | 43              | 6                 | 17                 |
| KC3PB           | 40     | 37   | 15                 | 10   | 50   | 50              | 6                 | 17                 |
| KC13            | 90     | 93   | 5                  | 90   | 100  | 100             | 6                 | 0                  |
| KC14            | 700    | 672  | 93                 | 540  | 790  | 745             | 5                 | 0                  |
| KCF1            | 210    | 222  | 23                 | 200  | 260  | 246             | 6                 | 0                  |
| LIMESTONE       | 115    | 97   | 34                 | 40   | 120  | 120             | 6                 | 0                  |
| M1 <sup>†</sup> | 10     | 30   | 35                 | 10   | 70   | 64              | 3                 | 67                 |
| RN029659        | 320    | 278  | 123                | 30   | 350  | 350             | 6                 | 0                  |
| RN029660        | 315    | 327  | 37                 | 290  | 380  | 366             | 6                 | 0                  |
| W2R             | 75     | 75   | 15                 | 60   | 100  | 86              | 6                 | 0                  |
| WP5             | 310    | 319  | 21                 | 300  | 360  | 333             | 7                 | 0                  |
| WP13            | 100    | 103  | 14                 | 90   | 120  | 120             | 6                 | 0                  |
| WP19            | 940    | 928  | 33                 | 870  | 950  | 950             | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D7 Concentration of barium (µg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 32     | 31   | 6                  | 22   | 36   | 36              | 7                 | 0                  |
| 10WP32   | 53     | 60   | 20                 | 39   | 88   | 88              | 7                 | 0                  |
| 10WP32PB | 36     | 37   | 15                 | 13   | 53   | 52              | 7                 | 0                  |
| 10WP33   | 49     | 47   | 10                 | 36   | 60   | 57              | 7                 | 0                  |
| 10WP35N  | 37     | 40   | 11                 | 26   | 55   | 50              | 7                 | 0                  |
| 10WP35PB | 130    | 123  | 39                 | 65   | 170  | 157             | 9                 | 0                  |
| 10WP35S  | 89     | 91   | 36                 | 37   | 140  | 122             | 7                 | 0                  |
| 10WP36N  | 19     | 23   | 7                  | 17   | 36   | 28              | 7                 | 0                  |
| 10WP36PB | 27     | 31   | 16                 | 8    | 64   | 42              | 10                | 0                  |
| 10WP36S  | 26     | 27   | 7                  | 18   | 38   | 33              | 7                 | 0                  |
| 10WP37   | 53     | 52   | 14                 | 34   | 66   | 66              | 7                 | 0                  |
| 10WP39   | 28     | 30   | 7                  | 23   | 42   | 35              | 7                 | 0                  |
| 10WP40   | 69     | 79   | 30                 | 51   | 130  | 112             | 7                 | 0                  |
| 10WP41   | 67     | 76   | 29                 | 55   | 140  | 83              | 7                 | 0                  |
| 10WP42   | 45     | 43   | 7                  | 32   | 53   | 47              | 7                 | 0                  |
| 10WP44   | 115    | 116  | 23                 | 93   | 140  | 137             | 4                 | 0                  |
| 10WP46   | 23     | 21   | 4                  | 15   | 26   | 24              | 7                 | 0                  |
| 10WP47   | 130    | 134  | 24                 | 100  | 170  | 152             | 7                 | 0                  |
| 11CS10RD | 107    | 114  | 36                 | 69   | 160  | 153             | 6                 | 0                  |
| 11CS10RS | 100    | 88   | 38                 | 31   | 120  | 120             | 6                 | 0                  |
| 11WP4R   | 190    | 198  | 104                | 80   | 320  | 313             | 6                 | 0                  |
| 11WP9RPB | 53     | 50   | 14                 | 21   | 62   | 61              | 7                 | 0                  |
| 11WP9RS  | 57     | 55   | 9                  | 45   | 65   | 63              | 5                 | 0                  |
| 11WP11RD | 75     | 71   | 15                 | 44   | 87   | 82              | 6                 | 0                  |
| 11WP11RS | 41     | 42   | 12                 | 28   | 55   | 55              | 6                 | 0                  |
| 11WP15R  | 43     | 53   | 31                 | 25   | 110  | 79              | 6                 | 0                  |
| 11WP16R  | 48     | 48   | 15                 | 28   | 67   | 64              | 6                 | 0                  |
| 11WP43D  | 285    | 283  | 91                 | 190  | 370  | 364             | 4                 | 0                  |
| 11WP50   | 40     | 39   | 7                  | 31   | 47   | 46              | 4                 | 0                  |
| 11WP51D  | 65     | 64   | 16                 | 44   | 87   | 77              | 6                 | 0                  |
| 11WP51S  | 112    | 142  | 96                 | 50   | 260  | 260             | 6                 | 0                  |
| 11WP52D  | 93     | 98   | 23                 | 68   | 130  | 123             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 25     | 35   | 24                 | 16   | 81   | 53              | 6                 | 0                  |
| 11WP53S   | 50     | 49   | 15                 | 25   | 70   | 61              | 6                 | 0                  |
| 11WP54D   | 35     | 37   | 11                 | 26   | 58   | 43              | 6                 | 0                  |
| 11WP54S   | 44     | 43   | 17                 | 18   | 67   | 57              | 6                 | 0                  |
| 11WP55    | 68     | 77   | 51                 | 18   | 150  | 129             | 6                 | 0                  |
| 11WP56D   | 22     | 40   | 49                 | 13   | 140  | 60              | 6                 | 0                  |
| 11WP56S   | 55     | 62   | 33                 | 34   | 98   | 94              | 3                 | 0                  |
| 11WP57    | 57     | 94   | 91                 | 49   | 280  | 132             | 6                 | 0                  |
| CG1       | 77     | 75   | 17                 | 44   | 94   | 86              | 6                 | 0                  |
| CG2       | 68     | 69   | 7                  | 60   | 77   | 76              | 5                 | 0                  |
| CG4       | 165    | 157  | 56                 | 53   | 210  | 203             | 6                 | 0                  |
| KC3       | 55     | 65   | 26                 | 44   | 110  | 85              | 5                 | 0                  |
| KC3A      | 91     | 86   | 12                 | 65   | 95   | 94              | 6                 | 0                  |
| KC3PB     | 88     | 87   | 11                 | 71   | 100  | 97              | 6                 | 0                  |
| KC13      | 58     | 59   | 11                 | 46   | 78   | 66              | 6                 | 0                  |
| KC14      | 26     | 32   | 18                 | 21   | 63   | 46              | 5                 | 0                  |
| KCF1      | 38     | 38   | 6                  | 31   | 46   | 45              | 6                 | 0                  |
| LIMESTONE | 130    | 174  | 113                | 69   | 360  | 290             | 6                 | 0                  |
| M1        | 30     | 31   | 1                  | 30   | 32   | 32              | 3                 | 0                  |
| RN029659  | 36     | 34   | 13                 | 11   | 47   | 46              | 6                 | 0                  |
| RN029660  | 43     | 49   | 18                 | 37   | 85   | 57              | 6                 | 0                  |
| W2R       | 240    | 232  | 44                 | 180  | 280  | 273             | 6                 | 0                  |
| WP5       | 33     | 37   | 11                 | 30   | 61   | 40              | 7                 | 0                  |
| WP13      | 59     | 62   | 6                  | 56   | 70   | 69              | 6                 | 0                  |
| WP19      | 25     | 25   | 4                  | 20   | 29   | 29              | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D8 Concentration of beryllium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 0.05   | 0.18 | 0.22               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP33 <sup>†</sup>   | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 0.05   | 0.20 | 0.23               | 0.05 | 0.50 | 0.50            | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.23 | 0.23               | 0.05 | 0.50 | 0.50            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP39 <sup>†</sup>   | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 0.10   | 0.14 | 0.16               | 0.05 | 0.50 | 0.14            | 7                 | 100                |
| 10WP41 <sup>†</sup>   | 0.25   | 0.32 | 0.12               | 0.25 | 0.50 | 0.50            | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.05   | 0.18 | 0.22               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.05   | 0.21 | 0.21               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.05   | 0.18 | 0.22               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.10   | 0.16 | 0.17               | 0.05 | 0.50 | 0.22            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.11 | 0.17               | 0.05 | 0.50 | 0.10            | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.14 | 0.20               | 0.05 | 0.50 | 0.28            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| 11WP15R <sup>†</sup>  | 0.10   | 0.17 | 0.16               | 0.10 | 0.50 | 0.22            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 0.25   | 0.29 | 0.10               | 0.25 | 0.50 | 0.33            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.18   | 0.23 | 0.20               | 0.05 | 0.50 | 0.43            | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| 11WP51S               | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.22            | 6                 | 100                |
| 11WP52D <sup>†</sup>  | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.22            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.17 | 0.16               | 0.10 | 0.50 | 0.22            | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 0.10   | 0.17 | 0.16               | 0.10 | 0.50 | 0.22            | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 0.25   | 0.29 | 0.10               | 0.25 | 0.50 | 0.33            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 0.25   | 0.29 | 0.10               | 0.25 | 0.50 | 0.33            | 6                 | 100                |
| 11WP55 <sup>†</sup>    | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 0.15   | 0.18 | 0.17               | 0.05 | 0.50 | 0.29            | 6                 | 50                 |
| 11WP56S <sup>†</sup>   | 0.25   | 0.27 | 0.23               | 0.05 | 0.50 | 0.48            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.25   | 0.26 | 0.14               | 0.05 | 0.50 | 0.33            | 6                 | 100                |
| CG1 <sup>†</sup>       | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| CG2 <sup>††</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.05   | 0.16 | 0.19               | 0.05 | 0.50 | 0.33            | 6                 | 100                |
| KC14 <sup>†</sup>      | 0.25   | 0.21 | 0.09               | 0.05 | 0.25 | 0.25            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.05   | 0.42 | 0.70               | 0.05 | 1.80 | 0.89            | 6                 | 83                 |
| LIMESTONE <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.50   | 0.35 | 0.26               | 0.05 | 0.50 | 0.50            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| W2R <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.10            | 7                 | 100                |
| WP13 <sup>†</sup>      | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.19            | 6                 | 100                |
| WP19 <sup>†</sup>      | 0.50   | 0.45 | 0.11               | 0.25 | 0.50 | 0.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D9 Concentration of bismuth (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP33 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP39 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 0.10   | 0.09 | 0.02               | 0.05 | 0.10 | 0.10            | 7                 | 100                |
| 10WP41 <sup>†</sup>   | 0.25   | 0.29 | 0.09               | 0.25 | 0.50 | 0.28            | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 0.25   | 0.20 | 0.10               | 0.05 | 0.25 | 0.25            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.05   | 0.08 | 0.08               | 0.05 | 0.25 | 0.07            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.10   | 0.09 | 0.02               | 0.05 | 0.10 | 0.10            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP15R <sup>†</sup>  | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP43D               | 0.18   | 0.23 | 0.20               | 0.05 | 0.50 | 0.43            | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 100                |
| 11WP52D <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP55 <sup>†</sup>    | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.05   | 0.12 | 0.12               | 0.05 | 0.25 | 0.23            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.25   | 0.22 | 0.08               | 0.05 | 0.25 | 0.25            | 6                 | 100                |
| CG1 <sup>†</sup>       | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.05   | 0.08 | 0.08               | 0.05 | 0.25 | 0.11            | 6                 | 100                |
| KC14 <sup>†</sup>      | 0.25   | 0.21 | 0.09               | 0.05 | 0.25 | 0.25            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| LIMESTONE <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| W2R <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.10            | 7                 | 100                |
| WP13 <sup>†</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP19 <sup>†</sup>      | 0.50   | 0.45 | 0.11               | 0.25 | 0.50 | 0.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D10 Concentration of bromide (mg/L)

| Site     | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 10WP31   | 0.13   | 0.15  | 0.05               | 0.10  | 0.20  | 0.20            | 5                 | 0                  |
| 10WP32   | 0.74   | 0.75  | 0.12               | 0.60  | 0.93  | 0.87            | 5                 | 0                  |
| 10WP32PB | 0.57   | 0.57  | 0.14               | 0.40  | 0.78  | 0.69            | 5                 | 0                  |
| 10WP33   | 0.60   | 0.61  | 0.08               | 0.50  | 0.70  | 0.69            | 5                 | 0                  |
| 10WP35N  | 0.22   | 0.20  | 0.12               | 0.01  | 0.30  | 0.30            | 5                 | 20                 |
| 10WP35PB | 0.48   | 0.45  | 0.08               | 0.30  | 0.50  | 0.50            | 5                 | 0                  |
| 10WP35S  | 0.43   | 0.39  | 0.09               | 0.30  | 0.50  | 0.47            | 5                 | 0                  |
| 10WP36N  | 0.40   | 0.32  | 0.16               | 0.05  | 0.43  | 0.42            | 5                 | 0                  |
| 10WP36PB | 0.30   | 0.35  | 0.12               | 0.20  | 0.53  | 0.47            | 5                 | 0                  |
| 10WP36S  | 0.41   | 0.43  | 0.10               | 0.30  | 0.56  | 0.53            | 5                 | 0                  |
| 10WP37   | 0.60   | 0.60  | 0.12               | 0.40  | 0.70  | 0.69            | 5                 | 0                  |
| 10WP39   | 0.40   | 0.39  | 0.08               | 0.30  | 0.50  | 0.46            | 5                 | 0                  |
| 10WP40   | 2.10   | 2.04  | 0.31               | 1.70  | 2.50  | 2.30            | 5                 | 0                  |
| 10WP41   | 21.00  | 21.00 | 8.00               | 11.00 | 30.00 | 27.00           | 4                 | 0                  |
| 10WP42   | 0.92   | 0.91  | 0.08               | 0.80  | 1.00  | 0.98            | 4                 | 0                  |
| 10WP44   | 18.00  | 17.00 | 1.00               | 15.00 | 18.00 | 18.00           | 4                 | 0                  |
| 10WP46   | 0.84   | 0.82  | 0.09               | 0.70  | 0.88  | 0.88            | 4                 | 0                  |
| 10WP47   | 0.21   | 0.21  | 0.07               | 0.14  | 0.30  | 0.27            | 4                 | 0                  |
| 11CS10RD | 1.10   | 1.16  | 0.38               | 0.59  | 1.60  | 1.50            | 5                 | 0                  |
| 11CS10RS | 4.50   | 4.52  | 1.92               | 1.80  | 6.70  | 6.30            | 5                 | 0                  |
| 11WP4R   | 0.89   | 0.87  | 0.13               | 0.70  | 1.00  | 0.99            | 5                 | 0                  |
| 11WP9RPB | 0.58   | 0.60  | 0.08               | 0.50  | 0.70  | 0.70            | 6                 | 0                  |
| 11WP9RS  | 0.60   | 0.60  | 0.18               | 0.40  | 0.80  | 0.77            | 4                 | 0                  |
| 11WP11RD | 0.28   | 0.22  | 0.14               | 0.01  | 0.30  | 0.30            | 4                 | 25                 |
| 11WP11RS | 0.21   | 0.18  | 0.12               | 0.01  | 0.30  | 0.27            | 4                 | 25                 |
| 11WP15R  | 4.20   | 4.28  | 0.28               | 3.90  | 4.60  | 4.55            | 5                 | 0                  |
| 11WP16R  | 7.40   | 6.10  | 3.16               | 0.50  | 8.10  | 7.80            | 5                 | 0                  |
| 11WP43D  | 10.00  | 10.00 | 8.00               | 5.00  | 16.00 | 16.00           | 2                 | 0                  |
| 11WP50   | 14.00  | 14.00 | 1.00               | 12.00 | 15.00 | 15.00           | 4                 | 0                  |
| 11WP51D  | 0.13   | 0.15  | 0.05               | 0.10  | 0.20  | 0.20            | 5                 | 0                  |
| 11WP51S  | 0.18   | 0.16  | 0.07               | 0.08  | 0.24  | 0.22            | 5                 | 0                  |
| 11WP52D  | 0.33   | 0.29  | 0.14               | 0.09  | 0.40  | 0.39            | 4                 | 0                  |

| Site      | Median  | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------|---------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 11WP53D   | 3.30    | 3.04  | 0.71               | 1.80  | 3.50  | 3.50            | 5                 | 0                  |
| 11WP53S   | 3.70    | 3.40  | 0.87               | 2.00  | 4.30  | 4.05            | 5                 | 0                  |
| 11WP54D   | 17.00   | 17.00 | 1.00               | 15.00 | 15.00 | 18.00           | 5                 | 0                  |
| 11WP54S   | 11.00   | 11.00 | 2.00               | 9.00  | 13.00 | 13.00           | 5                 | 0                  |
| 11WP55    | 2.50    | 2.60  | 0.23               | 2.40  | 3.00  | 2.80            | 5                 | 0                  |
| 11WP56D   | 0.07    | 0.12  | 0.10               | 0.07  | 0.30  | 0.20            | 5                 | 0                  |
| 11WP56S   | 0.23    | 0.23  | 0.04               | 0.20  | 0.25  | 0.25            | 2                 | 0                  |
| 11WP57    | 14.00   | 15.00 | 1.00               | 13.00 | 16.00 | 16.00           | 5                 | 0                  |
| CG1       | 2.55    | 2.48  | 0.26               | 2.10  | 2.70  | 2.67            | 4                 | 0                  |
| CG2       | 0.22    | 0.23  | 0.06               | 0.18  | 0.30  | 0.29            | 3                 | 0                  |
| CG4       | 1.60    | 1.63  | 0.06               | 1.60  | 1.70  | 1.69            | 3                 | 0                  |
| KC3       | 0.13    | 0.15  | 0.04               | 0.12  | 0.20  | 0.20            | 5                 | 0                  |
| KC3A      | 0.22    | 0.25  | 0.05               | 0.21  | 0.30  | 0.30            | 5                 | 0                  |
| KC3PB     | 0.21    | 0.26  | 0.09               | 0.19  | 0.40  | 0.35            | 5                 | 0                  |
| KC13      | 0.36    | 0.36  | 0.05               | 0.30  | 0.40  | 0.40            | 5                 | 0                  |
| KC14      | 17.00   | 17.00 | 2.00               | 16.00 | 18.00 | 18.00           | 5                 | 0                  |
| KCF1      | 2.10    | 2.14  | 0.15               | 2.00  | 2.30  | 2.30            | 5                 | 0                  |
| LIMESTONE | 0.20    | 0.15  | 0.08               | 0.06  | 0.20  | 0.20            | 3                 | 0                  |
| M1        | No data |       |                    |       |       |                 |                   |                    |
| RN029659  | 3.05    | 3.00  | 0.47               | 2.40  | 3.50  | 3.41            | 4                 | 0                  |
| RN029660  | 0.71    | 0.70  | 0.10               | 0.60  | 0.80  | 0.79            | 4                 | 0                  |
| W2R       | 0.15    | 0.15  | 0.06               | 0.10  | 0.20  | 0.20            | 4                 | 0                  |
| WP5       | 3.60    | 3.56  | 0.36               | 3.00  | 4.00  | 3.85            | 5                 | 0                  |
| WP13      | 0.40    | 0.41  | 0.01               | 0.40  | 0.42  | 0.42            | 3                 | 0                  |
| WP19      | 20.00   | 20.00 | 3.00               | 17.00 | 25.00 | 24.00           | 4                 | 0                  |

Table D11 Concentration of calcium (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 25     | 26   | 3                  | 23   | 30   | 30              | 7                 | 0                  |
| 10WP32   | 50     | 53   | 14                 | 34   | 71   | 69              | 7                 | 0                  |
| 10WP32PB | 44     | 45   | 12                 | 31   | 62   | 58              | 7                 | 0                  |
| 10WP33   | 49     | 53   | 9                  | 41   | 65   | 63              | 7                 | 0                  |
| 10WP35N  | 23     | 24   | 7                  | 15   | 33   | 32              | 7                 | 0                  |
| 10WP35PB | 63     | 59   | 13                 | 39   | 77   | 68              | 9                 | 0                  |
| 10WP35S  | 46     | 49   | 11                 | 33   | 64   | 59              | 7                 | 0                  |
| 10WP36N  | 35     | 39   | 10                 | 29   | 57   | 49              | 7                 | 0                  |
| 10WP36PB | 26     | 32   | 17                 | 8    | 69   | 45              | 10                | 0                  |
| 10WP36S  | 29     | 30   | 10                 | 16   | 45   | 40              | 7                 | 0                  |
| 10WP37   | 46     | 44   | 10                 | 31   | 56   | 54              | 7                 | 0                  |
| 10WP39   | 90     | 90   | 9                  | 76   | 102  | 99              | 7                 | 0                  |
| 10WP40   | 111    | 130  | 83                 | 48   | 297  | 162             | 7                 | 0                  |
| 10WP41   | 692    | 629  | 178                | 353  | 833  | 746             | 7                 | 0                  |
| 10WP42   | 95     | 95   | 7                  | 87   | 104  | 103             | 7                 | 0                  |
| 10WP44   | 952    | 929  | 75                 | 821  | 992  | 983             | 4                 | 0                  |
| 10WP46   | 114    | 114  | 11                 | 96   | 132  | 122             | 7                 | 0                  |
| 10WP47   | 64     | 61   | 8                  | 45   | 69   | 67              | 7                 | 0                  |
| 11CS10RD | 123    | 112  | 35                 | 67   | 143  | 142             | 6                 | 0                  |
| 11CS10RS | 127    | 123  | 69                 | 20   | 222  | 181             | 6                 | 0                  |
| 11WP4R   | 90     | 88   | 23                 | 48   | 113  | 109             | 6                 | 0                  |
| 11WP9RPB | 27     | 27   | 7                  | 17   | 36   | 35              | 7                 | 0                  |
| 11WP9RS  | 26     | 31   | 10                 | 22   | 47   | 41              | 5                 | 0                  |
| 11WP11RD | 30     | 29   | 3                  | 23   | 32   | 31              | 6                 | 0                  |
| 11WP11RS | 11     | 11   | 2                  | 9    | 13   | 12              | 6                 | 0                  |
| 11WP15R  | 34     | 37   | 4                  | 34   | 43   | 42              | 6                 | 0                  |
| 11WP16R  | 440    | 465  | 99                 | 368  | 588  | 585             | 6                 | 0                  |
| 11WP43D  | 596    | 639  | 474                | 153  | 1210 | 1096            | 4                 | 0                  |
| 11WP50   | 656    | 641  | 43                 | 578  | 676  | 670             | 4                 | 0                  |
| 11WP51D  | 51     | 50   | 4                  | 44   | 55   | 54              | 6                 | 0                  |
| 11WP51S  | 52     | 45   | 18                 | 18   | 64   | 60              | 6                 | 0                  |
| 11WP52D  | 27     | 27   | 4                  | 21   | 32   | 32              | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 12     | 14   | 5                  | 10   | 23   | 20              | 6                 | 0                  |
| 11WP53S   | 12     | 13   | 3                  | 11   | 17   | 16              | 6                 | 0                  |
| 11WP54D   | 584    | 596  | 34                 | 562  | 652  | 630             | 6                 | 0                  |
| 11WP54S   | 258    | 248  | 26                 | 210  | 276  | 266             | 6                 | 0                  |
| 11WP55    | 105    | 100  | 20                 | 63   | 117  | 116             | 6                 | 0                  |
| 11WP56D   | 4      | 4    | 2                  | 2    | 6    | 5               | 6                 | 0                  |
| 11WP56S   | 49     | 47   | 30                 | 16   | 77   | 74              | 3                 | 0                  |
| 11WP57    | 932    | 921  | 81                 | 766  | 992  | 984             | 6                 | 0                  |
| CG1       | 202    | 207  | 25                 | 179  | 254  | 221             | 6                 | 0                  |
| CG2       | 64     | 63   | 5                  | 54   | 67   | 66              | 5                 | 0                  |
| CG4       | 114    | 114  | 12                 | 99   | 128  | 128             | 6                 | 0                  |
| KC3       | 10     | 11   | 1                  | 9    | 12   | 12              | 5                 | 0                  |
| KC3A      | 19     | 19   | 2                  | 17   | 21   | 21              | 6                 | 0                  |
| KC3PB     | 18     | 18   | 3                  | 15   | 22   | 21              | 6                 | 0                  |
| KC13      | 49     | 48   | 11                 | 27   | 58   | 57              | 6                 | 0                  |
| KC14      | 764    | 745  | 48                 | 675  | 795  | 784             | 5                 | 0                  |
| KCF1      | 112    | 110  | 7                  | 99   | 118  | 117             | 6                 | 0                  |
| LIMESTONE | 62     | 58   | 15                 | 36   | 76   | 70              | 6                 | 0                  |
| M1        | 18     | 17   | 1                  | 15   | 18   | 18              | 3                 | 0                  |
| RN029659  | 117    | 96   | 57                 | 10   | 144  | 143             | 6                 | 0                  |
| RN029660  | 28     | 29   | 5                  | 23   | 37   | 32              | 6                 | 0                  |
| W2R       | 61     | 58   | 14                 | 37   | 71   | 70              | 6                 | 0                  |
| WP5       | 242    | 241  | 21                 | 209  | 267  | 260             | 7                 | 0                  |
| WP13      | 102    | 95   | 15                 | 69   | 108  | 106             | 6                 | 0                  |
| WP19      | 820    | 823  | 17                 | 807  | 851  | 838             | 5                 | 0                  |

Table D12 Concentration of cadmium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP33 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP39 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 0.10   | 0.10 | 0.05               | 0.05 | 0.20 | 0.11            | 7                 | 86                 |
| 10WP41 <sup>†</sup>   | 0.25   | 0.29 | 0.09               | 0.25 | 0.50 | 0.28            | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.05   | 0.08 | 0.08               | 0.05 | 0.25 | 0.07            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| 11CS10RS <sup>†</sup> | 0.10   | 0.11 | 0.05               | 0.05 | 0.20 | 0.13            | 6                 | 83                 |
| 11WP4R <sup>†</sup>   | 0.05   | 0.44 | 0.96               | 0.05 | 2.40 | 0.76            | 6                 | 83                 |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP15R <sup>†</sup>  | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.18   | 0.23 | 0.20               | 0.05 | 0.50 | 0.43            | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 100                |
| 11WP52D <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.15 | 0.08               | 0.10 | 0.30 | 0.23            | 6                 | 67                 |
| 11WP53S <sup>†</sup>   | 0.10   | 0.15 | 0.08               | 0.10 | 0.30 | 0.23            | 6                 | 67                 |
| 11WP54D <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 0.25   | 0.34 | 0.22               | 0.25 | 0.80 | 0.42            | 6                 | 83                 |
| 11WP55 <sup>†</sup>    | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| 11WP56D <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.05   | 0.12 | 0.12               | 0.05 | 0.25 | 0.23            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.25   | 0.28 | 0.06               | 0.25 | 0.40 | 0.30            | 6                 | 83                 |
| CG1 <sup>†</sup>       | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| KC3 <sup>†</sup>       | 0.05   | 0.12 | 0.16               | 0.05 | 0.40 | 0.23            | 5                 | 80                 |
| KC3A <sup>†</sup>      | 0.05   | 0.23 | 0.43               | 0.05 | 1.10 | 0.37            | 6                 | 83                 |
| KC3PB <sup>†</sup>     | 0.05   | 0.11 | 0.14               | 0.05 | 0.40 | 0.16            | 6                 | 83                 |
| KC13 <sup>†</sup>      | 0.05   | 0.09 | 0.08               | 0.05 | 0.25 | 0.15            | 6                 | 83                 |
| KC14 <sup>†</sup>      | 0.25   | 0.26 | 0.02               | 0.25 | 0.30 | 0.28            | 5                 | 80                 |
| KCF1 <sup>†</sup>      | 0.05   | 0.09 | 0.10               | 0.05 | 0.30 | 0.13            | 6                 | 83                 |
| LIMESTONE <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| RN029660 <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| W2R <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.05   | 0.09 | 0.06               | 0.05 | 0.20 | 0.11            | 7                 | 86                 |
| WP13 <sup>†</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP19 <sup>†</sup>      | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 5                 | 80                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D13 Concentration of chloride (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31   | 43     | 42   | 6                  | 33   | 49    | 47              | 7                 | 0                  |
| 10WP32   | 238    | 251  | 97                 | 137  | 370   | 344             | 7                 | 0                  |
| 10WP32PB | 214    | 203  | 71                 | 110  | 312   | 266             | 7                 | 0                  |
| 10WP33   | 223    | 205  | 40                 | 158  | 261   | 236             | 7                 | 0                  |
| 10WP35N  | 67     | 61   | 11                 | 46   | 73    | 69              | 7                 | 0                  |
| 10WP35PB | 144    | 138  | 37                 | 96   | 205   | 162             | 9                 | 0                  |
| 10WP35S  | 130    | 116  | 35                 | 58   | 158   | 146             | 7                 | 0                  |
| 10WP36N  | 80     | 86   | 15                 | 73   | 107   | 106             | 7                 | 0                  |
| 10WP36PB | 64     | 74   | 34                 | 22   | 143   | 103             | 10                | 0                  |
| 10WP36S  | 104    | 102  | 25                 | 68   | 137   | 132             | 7                 | 0                  |
| 10WP37   | 225    | 210  | 28                 | 161  | 234   | 233             | 7                 | 0                  |
| 10WP39   | 123    | 123  | 12                 | 110  | 140   | 138             | 7                 | 0                  |
| 10WP40   | 790    | 845  | 243                | 570  | 1310  | 985             | 7                 | 0                  |
| 10WP41   | 7020   | 6616 | 1624               | 4330 | 8960  | 7565            | 7                 | 0                  |
| 10WP42   | 329    | 341  | 30                 | 317  | 395   | 375             | 7                 | 0                  |
| 10WP44   | 7715   | 7758 | 126                | 7660 | 7940  | 7880            | 4                 | 0                  |
| 10WP46   | 329    | 327  | 16                 | 303  | 348   | 342             | 7                 | 0                  |
| 10WP47   | 45     | 45   | 6                  | 35   | 51    | 51              | 7                 | 0                  |
| 11CS10RD | 496    | 474  | 195                | 240  | 706   | 668             | 6                 | 0                  |
| 11CS10RS | 1665   | 1801 | 840                | 853  | 3130  | 2598            | 6                 | 0                  |
| 11WP4R   | 328    | 329  | 20                 | 305  | 351   | 351             | 6                 | 0                  |
| 11WP9RPB | 205    | 217  | 43                 | 157  | 280   | 262             | 7                 | 0                  |
| 11WP9RS  | 224    | 228  | 27                 | 199  | 265   | 256             | 5                 | 0                  |
| 11WP11RD | 81     | 79   | 7                  | 65   | 86    | 83              | 6                 | 0                  |
| 11WP11RS | 55     | 57   | 6                  | 52   | 68    | 62              | 6                 | 0                  |
| 11WP15R  | 1525   | 1588 | 207                | 1420 | 1990  | 1717            | 6                 | 0                  |
| 11WP16R  | 2965   | 2960 | 75                 | 2840 | 3070  | 3014            | 6                 | 0                  |
| 11WP43D  | 4805   | 5330 | 4246               | 1110 | 10600 | 9460            | 4                 | 0                  |
| 11WP50   | 5560   | 5555 | 304                | 5250 | 5850  | 5829            | 4                 | 0                  |
| 11WP51D  | 46     | 46   | 5                  | 41   | 55    | 50              | 6                 | 0                  |
| 11WP51S  | 49     | 53   | 39                 | 15   | 127   | 75              | 6                 | 0                  |
| 11WP52D  | 100    | 97   | 17                 | 75   | 121   | 110             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 1250   | 1277 | 82                 | 1200 | 1410 | 1361            | 6                 | 0                  |
| 11WP53S   | 1380   | 1402 | 116                | 1300 | 1590 | 1499            | 6                 | 0                  |
| 11WP54D   | 6540   | 6570 | 447                | 5880 | 7240 | 6925            | 6                 | 0                  |
| 11WP54S   | 3925   | 3855 | 341                | 3220 | 4180 | 4103            | 6                 | 0                  |
| 11WP55    | 913    | 891  | 69                 | 756  | 953  | 931             | 6                 | 0                  |
| 11WP56D   | 17     | 16   | 3                  | 11   | 18   | 17              | 6                 | 0                  |
| 11WP56S   | 67     | 64   | 43                 | 19   | 105  | 101             | 3                 | 0                  |
| 11WP57    | 5205   | 5142 | 603                | 4050 | 5820 | 5610            | 6                 | 0                  |
| CG1       | 1270   | 1267 | 139                | 1060 | 1420 | 1399            | 6                 | 0                  |
| CG2       | 93     | 91   | 8                  | 82   | 100  | 98              | 5                 | 0                  |
| CG4       | 681    | 681  | 60                 | 585  | 757  | 736             | 6                 | 0                  |
| KC3       | 26     | 26   | 4                  | 20   | 31   | 30              | 5                 | 0                  |
| KC3A      | 51     | 52   | 6                  | 45   | 62   | 56              | 6                 | 0                  |
| KC3PB     | 47     | 49   | 5                  | 44   | 56   | 54              | 6                 | 0                  |
| KC13      | 78     | 75   | 7                  | 63   | 82   | 80              | 6                 | 0                  |
| KC14      | 4720   | 4706 | 109                | 4550 | 4840 | 4800            | 5                 | 0                  |
| KCF1      | 561    | 561  | 28                 | 517  | 590  | 589             | 6                 | 0                  |
| LIMESTONE | 47     | 41   | 17                 | 19   | 59   | 56              | 6                 | 0                  |
| M1        | 9      | 11   | 4                  | 9    | 16   | 15              | 3                 | 0                  |
| RN029659  | 1026   | 918  | 437                | 87   | 1260 | 1260            | 6                 | 0                  |
| RN029660  | 270    | 258  | 38                 | 197  | 297  | 289             | 6                 | 0                  |
| W2R       | 32     | 31   | 7                  | 22   | 39   | 38              | 6                 | 0                  |
| WP5       | 1020   | 1019 | 55                 | 953  | 1110 | 1065            | 7                 | 0                  |
| WP13      | 122    | 136  | 34                 | 107  | 189  | 170             | 5                 | 0                  |
| WP19      | 7810   | 7732 | 291                | 7230 | 7960 | 7930            | 5                 | 0                  |

Table D14 Concentration of cobalt (µg/L)

| Site                  | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.80   | 1.16  | 0.95               | 0.05  | 2.50  | 2.50            | 7                 | 43                 |
| 10WP32 <sup>†</sup>   | 0.05   | 0.75  | 1.20               | 0.05  | 2.50  | 2.50            | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 0.05   | 0.75  | 1.20               | 0.05  | 2.50  | 2.50            | 7                 | 100                |
| 10WP33 <sup>†</sup>   | 0.05   | 0.76  | 1.19               | 0.05  | 2.50  | 2.50            | 7                 | 86                 |
| 10WP35N <sup>†</sup>  | 0.10   | 0.79  | 1.17               | 0.05  | 2.50  | 2.50            | 7                 | 57                 |
| 10WP35PB <sup>†</sup> | 0.10   | 1.14  | 1.29               | 0.05  | 2.50  | 2.50            | 9                 | 89                 |
| 10WP35S <sup>†</sup>  | 0.70   | 1.03  | 1.03               | 0.20  | 2.50  | 2.50            | 7                 | 29                 |
| 10WP36N <sup>†</sup>  | 0.05   | 0.76  | 1.19               | 0.05  | 2.50  | 2.50            | 7                 | 86                 |
| 10WP36PB <sup>†</sup> | 1.30   | 1.28  | 1.29               | 0.05  | 2.50  | 2.50            | 10                | 90                 |
| 10WP36S <sup>†</sup>  | 0.05   | 0.75  | 1.20               | 0.05  | 2.50  | 2.50            | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 0.10   | 0.76  | 1.19               | 0.05  | 2.50  | 2.50            | 7                 | 71                 |
| 10WP39 <sup>†</sup>   | 0.05   | 0.75  | 1.20               | 0.05  | 2.50  | 2.50            | 7                 | 100                |
| 10WP40                | 3.50   | 3.69  | 1.76               | 1.60  | 6.00  | 5.64            | 7                 | 14                 |
| 10WP41 <sup>†</sup>   | 2.90   | 4.61  | 3.43               | 0.50  | 9.60  | 8.70            | 7                 | 43                 |
| 10WP42 <sup>†</sup>   | 0.10   | 0.76  | 1.19               | 0.05  | 2.50  | 2.50            | 7                 | 71                 |
| 10WP44                | 2.85   | 2.63  | 1.28               | 0.90  | 3.90  | 3.69            | 4                 | 0                  |
| 10WP46 <sup>†</sup>   | 0.20   | 0.83  | 1.14               | 0.05  | 2.50  | 2.50            | 7                 | 57                 |
| 10WP47 <sup>†</sup>   | 1.10   | 1.31  | 0.85               | 0.40  | 2.50  | 2.50            | 7                 | 29                 |
| 11CS10RD <sup>†</sup> | 0.05   | 0.46  | 1.00               | 0.05  | 2.50  | 0.79            | 6                 | 100                |
| 11CS10RS              | 1.60   | 1.67  | 0.88               | 0.60  | 2.80  | 2.59            | 6                 | 17                 |
| 11WP4R                | 2.00   | 1.83  | 0.65               | 0.70  | 2.50  | 2.36            | 6                 | 17                 |
| 11WP9RPB <sup>†</sup> | 0.10   | 0.49  | 0.90               | 0.05  | 2.50  | 0.61            | 7                 | 43                 |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.59  | 1.07               | 0.05  | 2.50  | 1.40            | 5                 | 80                 |
| 11WP11RD <sup>†</sup> | 0.05   | 0.46  | 1.00               | 0.05  | 2.50  | 0.79            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.25   | 0.61  | 0.94               | 0.05  | 2.50  | 1.10            | 6                 | 33                 |
| 11WP15R               | 0.65   | 0.87  | 0.83               | 0.30  | 2.50  | 1.31            | 6                 | 17                 |
| 11WP16R <sup>†</sup>  | 1.05   | 1.23  | 0.84               | 0.25  | 2.50  | 2.08            | 6                 | 33                 |
| 11WP43D               | 13.50  | 14.10 | 5.33               | 8.40  | 21.00 | 19.20           | 4                 | 0                  |
| 11WP50                | 17.50  | 18.50 | 5.80               | 13.00 | 26.00 | 24.20           | 4                 | 0                  |
| 11WP51D               | 0.30   | 0.65  | 0.91               | 0.20  | 2.50  | 0.96            | 6                 | 17                 |
| 11WP51S               | 1.15   | 1.28  | 0.88               | 0.30  | 2.50  | 2.15            | 6                 | 17                 |
| 11WP52D <sup>†</sup>  | 0.20   | 0.63  | 0.95               | 0.05  | 2.50  | 1.24            | 6                 | 50                 |

| Site                  | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D               | 0.65   | 1.03  | 0.88               | 0.30 | 2.50  | 1.94            | 6                 | 17                 |
| 11WP53S               | 0.55   | 0.87  | 0.80               | 0.50 | 2.50  | 1.17            | 6                 | 17                 |
| 11WP54D               | 1.90   | 2.52  | 1.34               | 1.80 | 5.20  | 3.31            | 6                 | 17                 |
| 11WP54S               | 14.00  | 16.52 | 10.57              | 4.10 | 35.00 | 25.20           | 6                 | 0                  |
| 11WP55                | 1.25   | 1.45  | 0.68               | 0.60 | 2.50  | 2.15            | 6                 | 17                 |
| 11WP56D               | 0.75   | 0.98  | 0.75               | 0.50 | 2.50  | 1.31            | 6                 | 17                 |
| 11WP56S <sup>†</sup>  | 2.20   | 1.67  | 1.19               | 0.30 | 2.50  | 2.47            | 3                 | 33                 |
| 11WP57                | 2.90   | 2.95  | 0.68               | 2.20 | 4.20  | 3.36            | 6                 | 17                 |
| CG1 <sup>†</sup>      | 0.10   | 0.10  | 0.00               | 0.10 | 0.10  | 0.10            | 6                 | 100                |
| CG2 <sup>†</sup>      | 0.05   | 0.05  | 0.00               | 0.05 | 0.05  | 0.05            | 5                 | 100                |
| CG4                   | 0.30   | 0.65  | 0.91               | 0.20 | 2.50  | 1.03            | 6                 | 17                 |
| KC3 <sup>†</sup>      | 0.05   | 0.09  | 0.07               | 0.05 | 0.20  | 0.15            | 5                 | 60                 |
| KC3A <sup>†</sup>     | 0.05   | 0.46  | 1.00               | 0.05 | 2.50  | 0.79            | 6                 | 100                |
| KC3PB                 | 0.20   | 0.57  | 0.95               | 0.10 | 2.50  | 0.96            | 6                 | 17                 |
| KC13 <sup>†</sup>     | 0.45   | 0.79  | 0.86               | 0.25 | 2.50  | 1.31            | 6                 | 33                 |
| KC14                  | 2.20   | 2.34  | 0.47               | 1.70 | 2.90  | 2.80            | 5                 | 0                  |
| KCF1                  | 2.40   | 2.30  | 0.50               | 1.40 | 2.90  | 2.62            | 6                 | 17                 |
| LIMESTONE             | 0.25   | 0.37  | 0.29               | 0.10 | 0.90  | 0.62            | 6                 | 0                  |
| M1 <sup>†</sup>       | 2.50   | 2.50  | 0.00               | 2.50 | 2.50  | 2.50            | 3                 | 100                |
| RN029659 <sup>†</sup> | 0.40   | 0.64  | 0.92               | 0.05 | 2.50  | 1.03            | 6                 | 33                 |
| RN029660 <sup>†</sup> | 0.05   | 0.05  | 0.00               | 0.05 | 0.05  | 0.05            | 6                 | 100                |
| W2R                   | 0.10   | 1.09  | 2.31               | 0.05 | 5.80  | 2.02            | 6                 | 17                 |
| WP5                   | 0.50   | 0.79  | 0.77               | 0.30 | 2.50  | 0.97            | 7                 | 14                 |
| WP13 <sup>†</sup>     | 0.05   | 0.46  | 1.00               | 0.05 | 2.50  | 0.79            | 6                 | 100                |
| WP19 <sup>†</sup>     | 0.50   | 0.85  | 0.93               | 0.25 | 2.50  | 1.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D15 Concentration of carbonate (mg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 0.5    | 3.9  | 8.9                | 0.5  | 24.0 | 2.9             | 7                 | 86                 |
| 10WP33 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 0.5    | 4.9  | 13.8               | 0.5  | 44.0 | 0.5             | 10                | 90                 |
| 10WP36S <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP39 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP41 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP9RPB <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP15R <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP52D <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP55 <sup>†</sup>    | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| CG1 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC14 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| LIMESTONE <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| M1 <sup>†</sup>        | 0.5    | 2.3  | 3.2                | 0.5  | 6.0  | 5.5             | 3                 | 67                 |
| RN029659 <sup>†</sup>  | 0.5    | 2.8  | 5.5                | 0.5  | 14.0 | 4.6             | 6                 | 83                 |
| RN029660 <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| W2R <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| WP13 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| WP19 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D16 Concentration of chromium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31                | 1.90   | 2.74 | 1.93               | 1.70 | 7.00  | 3.40            | 7                 | 0                  |
| 10WP32                | 1.50   | 3.03 | 2.96               | 0.70 | 9.00  | 5.04            | 7                 | 0                  |
| 10WP32PB              | 1.90   | 1.70 | 0.49               | 1.00 | 2.20  | 2.02            | 7                 | 0                  |
| 10WP33 <sup>†</sup>   | 0.70   | 1.06 | 0.93               | 0.50 | 3.00  | 1.65            | 7                 | 43                 |
| 10WP35N <sup>†</sup>  | 0.25   | 3.57 | 8.57               | 0.25 | 23.00 | 2.75            | 7                 | 86                 |
| 10WP35PB <sup>†</sup> | 0.90   | 0.73 | 0.29               | 0.25 | 1.00  | 1.00            | 9                 | 44                 |
| 10WP35S <sup>†</sup>  | 0.50   | 3.11 | 6.60               | 0.25 | 18.00 | 3.60            | 7                 | 57                 |
| 10WP36N <sup>†</sup>  | 0.50   | 1.72 | 2.84               | 0.25 | 8.00  | 2.60            | 7                 | 57                 |
| 10WP36PB <sup>†</sup> | 0.50   | 0.45 | 0.15               | 0.25 | 0.70  | 0.50            | 10                | 90                 |
| 10WP36S <sup>†</sup>  | 0.25   | 2.07 | 3.99               | 0.25 | 11.00 | 2.90            | 7                 | 71                 |
| 10WP37 <sup>†</sup>   | 0.25   | 1.57 | 3.28               | 0.25 | 9.00  | 1.35            | 7                 | 86                 |
| 10WP39 <sup>†</sup>   | 0.25   | 0.36 | 0.13               | 0.25 | 0.50  | 0.50            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 0.50   | 3.50 | 8.16               | 0.25 | 22.00 | 2.65            | 7                 | 86                 |
| 10WP41 <sup>†</sup>   | 0.50   | 2.79 | 5.40               | 0.50 | 15.00 | 2.63            | 7                 | 86                 |
| 10WP42 <sup>†</sup>   | 0.50   | 0.39 | 0.13               | 0.25 | 0.50  | 0.50            | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 1.25   | 1.06 | 0.38               | 0.50 | 1.25  | 1.25            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.50   | 0.54 | 0.34               | 0.25 | 1.25  | 0.58            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.50   | 0.61 | 0.63               | 0.25 | 2.00  | 0.65            | 7                 | 86                 |
| 11CS10RD              | 1.30   | 1.67 | 0.72               | 1.20 | 3.00  | 2.30            | 6                 | 0                  |
| 11CS10RS <sup>†</sup> | 0.75   | 0.93 | 0.65               | 0.25 | 2.00  | 1.51            | 6                 | 50                 |
| 11WP4R <sup>†</sup>   | 0.38   | 0.89 | 1.08               | 0.25 | 3.00  | 1.67            | 6                 | 67                 |
| 11WP9RPB              | 1.40   | 2.09 | 1.77               | 1.00 | 6.00  | 2.49            | 7                 | 0                  |
| 11WP9RS               | 1.60   | 1.54 | 0.36               | 1.00 | 2.00  | 1.80            | 5                 | 0                  |
| 11WP11RD              | 1.85   | 1.87 | 0.67               | 1.00 | 3.00  | 2.30            | 6                 | 0                  |
| 11WP11RS              | 2.00   | 1.79 | 0.95               | 0.25 | 3.00  | 2.51            | 6                 | 17                 |
| 11WP15R <sup>†</sup>  | 0.50   | 1.67 | 2.62               | 0.50 | 7.00  | 2.80            | 6                 | 67                 |
| 11WP16R <sup>†</sup>  | 0.88   | 0.88 | 0.41               | 0.50 | 1.25  | 1.25            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.50   | 0.69 | 0.38               | 0.50 | 1.25  | 1.03            | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 1.25   | 1.06 | 0.38               | 0.50 | 1.25  | 1.25            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.25   | 0.42 | 0.30               | 0.25 | 1.00  | 0.65            | 6                 | 83                 |
| 11WP51S <sup>†</sup>  | 0.38   | 0.38 | 0.14               | 0.25 | 0.50  | 0.50            | 6                 | 100                |
| 11WP52D <sup>†</sup>  | 0.50   | 0.72 | 0.66               | 0.25 | 2.00  | 1.16            | 6                 | 67                 |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D                | 3.00   | 3.23 | 1.49               | 1.60 | 6.00 | 4.18            | 6                 | 0                  |
| 11WP53S                | 2.30   | 2.22 | 0.65               | 1.20 | 3.00 | 2.72            | 6                 | 0                  |
| 11WP54D <sup>†</sup>   | 1.25   | 1.46 | 1.30               | 0.50 | 4.00 | 2.08            | 6                 | 83                 |
| 11WP54S <sup>†</sup>   | 0.88   | 1.92 | 2.76               | 0.50 | 7.50 | 3.13            | 6                 | 83                 |
| 11WP55 <sup>†</sup>    | 0.25   | 0.92 | 1.51               | 0.25 | 4.00 | 1.55            | 6                 | 83                 |
| 11WP56D <sup>†</sup>   | 0.25   | 0.33 | 0.13               | 0.25 | 0.50 | 0.50            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.50   | 0.67 | 0.52               | 0.25 | 1.25 | 1.18            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.50   | 0.75 | 0.39               | 0.50 | 1.25 | 1.25            | 6                 | 83                 |
| CG1                    | 4.80   | 4.77 | 0.92               | 3.40 | 6.20 | 5.36            | 6                 | 0                  |
| CG2                    | 1.90   | 1.74 | 0.65               | 1.00 | 2.60 | 2.30            | 5                 | 0                  |
| CG4 <sup>†</sup>       | 0.38   | 0.46 | 0.29               | 0.25 | 1.00 | 0.65            | 6                 | 83                 |
| KC3                    | 0.60   | 0.70 | 0.23               | 0.50 | 1.10 | 0.90            | 5                 | 20                 |
| KC3A <sup>†</sup>      | 0.25   | 0.42 | 0.30               | 0.25 | 1.00 | 0.65            | 6                 | 83                 |
| KC3PB                  | 3.00   | 3.62 | 1.93               | 2.00 | 7.20 | 5.17            | 6                 | 0                  |
| KC13 <sup>†</sup>      | 0.38   | 0.58 | 0.44               | 0.25 | 1.25 | 1.08            | 6                 | 83                 |
| KC14 <sup>†</sup>      | 1.25   | 1.05 | 0.69               | 0.25 | 2.00 | 1.63            | 5                 | 80                 |
| KCF1 <sup>†</sup>      | 0.50   | 0.50 | 0.23               | 0.25 | 0.80 | 0.73            | 6                 | 67                 |
| LIMESTONE <sup>†</sup> | 0.38   | 0.38 | 0.14               | 0.25 | 0.50 | 0.50            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.50   | 1.33 | 1.44               | 0.50 | 3.00 | 2.75            | 3                 | 67                 |
| RN029659 <sup>†</sup>  | 0.38   | 0.38 | 0.14               | 0.25 | 0.50 | 0.50            | 6                 | 100                |
| RN029660               | 3.00   | 2.98 | 0.16               | 2.70 | 3.20 | 3.06            | 6                 | 0                  |
| W2R <sup>†</sup>       | 0.38   | 0.39 | 0.16               | 0.25 | 0.60 | 0.53            | 6                 | 83                 |
| WP5 <sup>†</sup>       | 0.50   | 0.39 | 0.13               | 0.25 | 0.50 | 0.50            | 7                 | 100                |
| WP13 <sup>†</sup>      | 0.38   | 0.38 | 0.14               | 0.25 | 0.50 | 0.50            | 6                 | 100                |
| WP19 <sup>†</sup>      | 1.25   | 1.45 | 1.01               | 0.50 | 2.50 | 2.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D17 Concentration of copper (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31                | 1.80   | 1.92 | 1.93               | 0.05 | 5.00  | 4.10            | 7                 | 14                 |
| 10WP32 <sup>†</sup>   | 0.90   | 1.16 | 1.05               | 0.05 | 3.00  | 2.19            | 7                 | 29                 |
| 10WP32PB              | 1.10   | 1.53 | 1.21               | 0.40 | 4.00  | 2.20            | 7                 | 14                 |
| 10WP33 <sup>†</sup>   | 1.00   | 0.98 | 0.71               | 0.05 | 2.10  | 1.65            | 7                 | 43                 |
| 10WP35N               | 0.50   | 0.76 | 0.59               | 0.30 | 2.00  | 1.10            | 7                 | 14                 |
| 10WP35PB <sup>†</sup> | 1.00   | 0.96 | 0.44               | 0.40 | 1.70  | 1.35            | 9                 | 44                 |
| 10WP35S               | 1.00   | 1.06 | 0.91               | 0.30 | 3.00  | 1.29            | 7                 | 14                 |
| 10WP36N               | 0.80   | 1.11 | 1.31               | 0.20 | 4.00  | 1.30            | 7                 | 14                 |
| 10WP36PB <sup>†</sup> | 1.00   | 0.84 | 0.74               | 0.05 | 2.60  | 1.00            | 10                | 60                 |
| 10WP36S <sup>†</sup>  | 0.70   | 0.63 | 0.31               | 0.20 | 1.00  | 1.00            | 7                 | 29                 |
| 10WP37 <sup>†</sup>   | 0.50   | 0.82 | 1.00               | 0.05 | 3.00  | 1.20            | 7                 | 29                 |
| 10WP39 <sup>†</sup>   | 0.40   | 0.49 | 0.38               | 0.10 | 1.00  | 1.00            | 7                 | 29                 |
| 10WP40                | 0.70   | 1.43 | 1.48               | 0.10 | 4.00  | 3.10            | 7                 | 14                 |
| 10WP41 <sup>†</sup>   | 2.10   | 3.74 | 4.38               | 0.25 | 11.00 | 9.20            | 7                 | 29                 |
| 10WP42 <sup>†</sup>   | 0.60   | 1.09 | 1.33               | 0.10 | 4.00  | 1.30            | 7                 | 29                 |
| 10WP44 <sup>†</sup>   | 0.25   | 0.69 | 0.88               | 0.25 | 2.00  | 1.48            | 4                 | 75                 |
| 10WP46 <sup>†</sup>   | 0.50   | 0.56 | 0.43               | 0.10 | 1.00  | 1.00            | 7                 | 57                 |
| 10WP47 <sup>†</sup>   | 0.60   | 0.63 | 0.37               | 0.20 | 1.00  | 1.00            | 7                 | 43                 |
| 11CS10RD              | 0.50   | 1.98 | 3.47               | 0.20 | 9.00  | 3.75            | 6                 | 0                  |
| 11CS10RS              | 0.55   | 3.33 | 6.71               | 0.10 | 17.00 | 6.15            | 6                 | 17                 |
| 11WP4R                | 0.35   | 2.30 | 3.56               | 0.10 | 9.00  | 5.29            | 6                 | 0                  |
| 11WP9RPB              | 0.30   | 2.63 | 5.91               | 0.10 | 16.00 | 2.77            | 7                 | 0                  |
| 11WP9RS               | 0.60   | 0.70 | 0.17               | 0.60 | 1.00  | 0.85            | 5                 | 20                 |
| 11WP11RD              | 0.55   | 1.31 | 1.85               | 0.05 | 5.00  | 2.41            | 6                 | 17                 |
| 11WP11RS              | 0.95   | 1.13 | 0.77               | 0.30 | 2.30  | 1.95            | 6                 | 17                 |
| 11WP15R               | 1.10   | 3.12 | 5.34               | 0.50 | 14.00 | 5.18            | 6                 | 0                  |
| 11WP16R <sup>†</sup>  | 0.25   | 2.63 | 5.57               | 0.25 | 14.00 | 4.76            | 6                 | 67                 |
| 11WP43D <sup>†</sup>  | 2.63   | 4.59 | 6.05               | 0.10 | 13.00 | 10.60           | 4                 | 50                 |
| 11WP50                | 1.85   | 1.69 | 1.18               | 0.25 | 2.80  | 2.71            | 4                 | 25                 |
| 11WP51D               | 0.30   | 1.53 | 3.17               | 0.05 | 8.00  | 2.68            | 6                 | 17                 |
| 11WP51S               | 0.85   | 1.67 | 1.89               | 0.10 | 5.10  | 3.35            | 6                 | 17                 |
| 11WP52D               | 0.65   | 1.80 | 2.62               | 0.20 | 7.00  | 3.43            | 6                 | 0                  |

| Site                   | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D                | 0.65   | 2.35  | 3.82               | 0.10 | 10.00 | 4.54            | 6                 | 17                 |
| 11WP53S                | 0.60   | 0.67  | 0.34               | 0.20 | 1.10  | 1.03            | 6                 | 17                 |
| 11WP54D <sup>†</sup>   | 0.53   | 6.23  | 13.62              | 0.25 | 34.00 | 11.46           | 6                 | 50                 |
| 11WP54S <sup>†</sup>   | 2.05   | 2.97  | 3.07               | 0.25 | 8.00  | 6.04            | 6                 | 33                 |
| 11WP55 <sup>†</sup>    | 0.45   | 0.54  | 0.39               | 0.05 | 1.00  | 1.00            | 6                 | 33                 |
| 11WP56D                | 1.40   | 1.65  | 1.11               | 0.30 | 3.00  | 2.93            | 6                 | 17                 |
| 11WP56S <sup>†</sup>   | 1.00   | 1.13  | 0.61               | 0.60 | 1.80  | 1.72            | 3                 | 33                 |
| 11WP57 <sup>†</sup>    | 0.85   | 10.70 | 16.04              | 0.25 | 36.00 | 29.00           | 6                 | 33                 |
| CG1 <sup>†</sup>       | 1.65   | 1.58  | 0.96               | 0.10 | 2.70  | 2.49            | 6                 | 33                 |
| CG2                    | 0.80   | 0.76  | 0.42               | 0.30 | 1.30  | 1.15            | 5                 | 20                 |
| CG4 <sup>†</sup>       | 0.80   | 1.29  | 1.44               | 0.05 | 4.00  | 2.39            | 6                 | 33                 |
| KC3                    | 0.90   | 0.80  | 0.35               | 0.20 | 1.10  | 1.05            | 5                 | 0                  |
| KC3A                   | 1.05   | 1.70  | 1.76               | 0.10 | 4.10  | 3.82            | 6                 | 17                 |
| KC3PB                  | 0.85   | 1.35  | 1.31               | 0.20 | 3.80  | 2.40            | 6                 | 17                 |
| KC13                   | 0.50   | 0.62  | 0.35               | 0.30 | 1.10  | 1.03            | 6                 | 17                 |
| KC14                   | 1.50   | 1.35  | 1.03               | 0.25 | 2.90  | 2.20            | 5                 | 20                 |
| KCF1                   | 0.70   | 0.72  | 0.38               | 0.30 | 1.20  | 1.10            | 5                 | 20                 |
| LIMESTONE <sup>†</sup> | 0.75   | 1.41  | 2.19               | 0.05 | 5.80  | 2.44            | 6                 | 50                 |
| M1 <sup>†</sup>        | 1.00   | 1.00  | 0.00               | 1.00 | 1.00  | 1.00            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.80   | 0.63  | 0.48               | 0.05 | 1.10  | 1.03            | 6                 | 67                 |
| RN029660 <sup>†</sup>  | 0.85   | 0.80  | 0.38               | 0.20 | 1.30  | 1.09            | 6                 | 33                 |
| W2R <sup>†</sup>       | 0.75   | 1.04  | 1.16               | 0.05 | 3.30  | 1.69            | 6                 | 33                 |
| WP5                    | 5.00   | 3.89  | 2.38               | 0.90 | 7.00  | 5.83            | 7                 | 0                  |
| WP13 <sup>†</sup>      | 0.40   | 0.53  | 0.39               | 0.05 | 1.00  | 1.00            | 6                 | 50                 |
| WP19                   | 1.90   | 3.37  | 3.40               | 0.25 | 9.00  | 6.45            | 5                 | 20                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D18 Concentration of electrical conductivity (mS/m)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 42     | 43   | 3                  | 39   | 47   | 46              | 7                 | 0                  |
| 10WP32   | 174    | 168  | 36                 | 127  | 214  | 202             | 7                 | 0                  |
| 10WP32PB | 150    | 151  | 22                 | 118  | 188  | 166             | 7                 | 0                  |
| 10WP33   | 137    | 138  | 14                 | 116  | 157  | 150             | 7                 | 0                  |
| 10WP35N  | 120    | 120  | 20                 | 86   | 141  | 137             | 7                 | 0                  |
| 10WP35PB | 166    | 160  | 17                 | 133  | 180  | 175             | 9                 | 0                  |
| 10WP35S  | 154    | 147  | 21                 | 112  | 167  | 167             | 7                 | 0                  |
| 10WP36N  | 126    | 128  | 10                 | 116  | 144  | 141             | 7                 | 0                  |
| 10WP36PB | 118    | 123  | 21                 | 87   | 161  | 144             | 10                | 0                  |
| 10WP36S  | 152    | 148  | 26                 | 93   | 174  | 164             | 7                 | 0                  |
| 10WP37   | 193    | 198  | 11                 | 184  | 215  | 207             | 7                 | 0                  |
| 10WP39   | 117    | 117  | 4                  | 111  | 123  | 121             | 7                 | 0                  |
| 10WP40   | 554    | 590  | 195                | 377  | 955  | 682             | 7                 | 0                  |
| 10WP41   | 2310   | 2211 | 490                | 1530 | 2950 | 2491            | 7                 | 0                  |
| 10WP42   | 222    | 227  | 12                 | 214  | 247  | 238             | 7                 | 0                  |
| 10WP44   | 2510   | 2515 | 44                 | 2470 | 2570 | 2558            | 4                 | 0                  |
| 10WP46   | 221    | 216  | 15                 | 187  | 230  | 226             | 7                 | 0                  |
| 10WP47   | 62     | 60   | 10                 | 44   | 73   | 66              | 7                 | 0                  |
| 11CS10RD | 243    | 251  | 73                 | 174  | 350  | 330             | 6                 | 0                  |
| 11CS10RS | 708    | 739  | 219                | 413  | 997  | 976             | 6                 | 0                  |
| 11WP4R   | 162    | 163  | 7                  | 154  | 170  | 170             | 6                 | 0                  |
| 11WP9RPB | 191    | 182  | 41                 | 91   | 212  | 203             | 7                 | 0                  |
| 11WP9RS  | 198    | 181  | 44                 | 102  | 206  | 206             | 5                 | 0                  |
| 11WP11RD | 100    | 100  | 3                  | 94   | 104  | 102             | 6                 | 0                  |
| 11WP11RS | 113    | 114  | 9                  | 100  | 128  | 121             | 6                 | 0                  |
| 11WP15R  | 838    | 840  | 15                 | 824  | 858  | 855             | 6                 | 0                  |
| 11WP16R  | 1220   | 1220 | 38                 | 1180 | 1280 | 1252            | 6                 | 0                  |
| 11WP43D  | 1269   | 1368 | 1071               | 374  | 2560 | 2386            | 4                 | 0                  |
| 11WP50   | 2085   | 2085 | 17                 | 2070 | 2100 | 2100            | 4                 | 0                  |
| 11WP51D  | 75     | 72   | 8                  | 58   | 81   | 78              | 6                 | 0                  |
| 11WP51S  | 99     | 102  | 51                 | 24   | 155  | 145             | 6                 | 0                  |
| 11WP52D  | 90     | 90   | 6                  | 81   | 99   | 95              | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 822    | 832  | 35                 | 805  | 898  | 858             | 6                 | 0                  |
| 11WP53S   | 901    | 898  | 19                 | 872  | 925  | 912             | 6                 | 0                  |
| 11WP54D   | 2440   | 2425 | 60                 | 2320 | 2480 | 2473            | 6                 | 0                  |
| 11WP54S   | 1700   | 1682 | 69                 | 1550 | 1750 | 1722            | 6                 | 0                  |
| 11WP55    | 437    | 408  | 78                 | 249  | 451  | 444             | 6                 | 0                  |
| 11WP56D   | 9      | 9    | 3                  | 5    | 14   | 12              | 6                 | 0                  |
| 11WP56S   | 86     | 78   | 46                 | 28   | 119  | 116             | 3                 | 0                  |
| 11WP57    | 1620   | 1598 | 112                | 1450 | 1730 | 1702            | 6                 | 0                  |
| CG1       | 612    | 606  | 39                 | 536  | 645  | 635             | 6                 | 0                  |
| CG2       | 94     | 92   | 5                  | 86   | 99   | 97              | 5                 | 0                  |
| CG4       | 319    | 324  | 25                 | 296  | 358  | 350             | 6                 | 0                  |
| KC3       | 26     | 24   | 4                  | 18   | 27   | 26              | 5                 | 0                  |
| KC3A      | 44     | 45   | 3                  | 42   | 51   | 46              | 6                 | 0                  |
| KC3PB     | 44     | 43   | 1                  | 41   | 44   | 44              | 6                 | 0                  |
| KC13      | 108    | 104  | 9                  | 86   | 110  | 109             | 6                 | 0                  |
| KC14      | 1750   | 1752 | 23                 | 1730 | 1790 | 1770            | 5                 | 0                  |
| KCF1      | 481    | 480  | 10                 | 466  | 496  | 489             | 6                 | 0                  |
| LIMESTONE | 63     | 60   | 24                 | 29   | 94   | 80              | 6                 | 0                  |
| M1        | 26     | 26   | 1                  | 26   | 28   | 27              | 3                 | 0                  |
| RN029659  | 428    | 379  | 168                | 51   | 498  | 491             | 6                 | 0                  |
| RN029660  | 176    | 174  | 6                  | 166  | 181  | 178             | 6                 | 0                  |
| W2R       | 51     | 45   | 12                 | 24   | 56   | 53              | 6                 | 0                  |
| WP5       | 494    | 476  | 54                 | 355  | 504  | 502             | 7                 | 0                  |
| WP13      | 103    | 83   | 45                 | 64   | 120  | 110             | 6                 | 0                  |
| WP19      | 2620   | 2566 | 168                | 2270 | 2680 | 2670            | 5                 | 0                  |

Table D19 Concentration of fluoride (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 0.20   | 0.19 | 0.02               | 0.16 | 0.21 | 0.21            | 5                 | 0                  |
| 10WP32   | 0.53   | 0.54 | 0.03               | 0.52 | 0.59 | 0.57            | 5                 | 0                  |
| 10WP32PB | 0.55   | 0.54 | 0.03               | 0.48 | 0.56 | 0.56            | 5                 | 0                  |
| 10WP33   | 0.33   | 0.32 | 0.03               | 0.28 | 0.37 | 0.34            | 6                 | 0                  |
| 10WP35N  | 1.20   | 1.18 | 0.08               | 1.10 | 1.30 | 1.25            | 5                 | 0                  |
| 10WP35PB | 0.44   | 0.42 | 0.09               | 0.28 | 0.53 | 0.47            | 6                 | 0                  |
| 10WP35S  | 0.49   | 0.50 | 0.10               | 0.40 | 0.66 | 0.58            | 5                 | 0                  |
| 10WP36N  | 1.20   | 1.26 | 0.19               | 1.00 | 1.50 | 1.45            | 5                 | 0                  |
| 10WP36PB | 1.03   | 0.92 | 0.52               | 0.03 | 1.40 | 1.40            | 6                 | 17                 |
| 10WP36S  | 1.40   | 1.48 | 0.19               | 1.30 | 1.80 | 1.65            | 5                 | 0                  |
| 10WP37   | 1.00   | 1.04 | 0.05               | 1.00 | 1.10 | 1.10            | 5                 | 0                  |
| 10WP39   | 0.42   | 0.40 | 0.04               | 0.35 | 0.45 | 0.44            | 6                 | 0                  |
| 10WP40   | 0.73   | 0.78 | 0.12               | 0.66 | 0.96 | 0.91            | 5                 | 0                  |
| 10WP41   | 0.32   | 0.32 | 0.07               | 0.23 | 0.39 | 0.39            | 5                 | 0                  |
| 10WP42   | 0.49   | 0.46 | 0.06               | 0.37 | 0.51 | 0.51            | 5                 | 0                  |
| 10WP44   | 0.33   | 0.44 | 0.24               | 0.28 | 0.80 | 0.66            | 4                 | 0                  |
| 10WP46   | 0.47   | 0.46 | 0.03               | 0.42 | 0.48 | 0.48            | 5                 | 0                  |
| 10WP47   | 0.11   | 0.11 | 0.03               | 0.07 | 0.14 | 0.13            | 5                 | 0                  |
| 11CS10RD | 0.21   | 0.19 | 0.04               | 0.12 | 0.23 | 0.22            | 5                 | 0                  |
| 11CS10RS | 0.48   | 0.55 | 0.23               | 0.33 | 0.93 | 0.75            | 5                 | 0                  |
| 11WP4R   | 0.53   | 0.54 | 0.02               | 0.52 | 0.57 | 0.56            | 5                 | 0                  |
| 11WP9RPB | 0.79   | 0.65 | 0.32               | 0.03 | 0.85 | 0.84            | 6                 | 17                 |
| 11WP9RS  | 0.87   | 0.84 | 0.06               | 0.74 | 0.87 | 0.87            | 4                 | 0                  |
| 11WP11RD | 0.54   | 0.52 | 0.03               | 0.46 | 0.54 | 0.54            | 5                 | 0                  |
| 11WP11RS | 1.20   | 1.16 | 0.05               | 1.10 | 1.20 | 1.20            | 5                 | 0                  |
| 11WP15R  | 0.38   | 0.36 | 0.03               | 0.32 | 0.38 | 0.38            | 5                 | 0                  |
| 11WP16R  | 0.58   | 0.67 | 0.13               | 0.56 | 0.82 | 0.82            | 5                 | 0                  |
| 11WP43D  | 0.54   | 0.53 | 0.12               | 0.39 | 0.66 | 0.64            | 4                 | 0                  |
| 11WP50   | 0.52   | 0.52 | 0.03               | 0.48 | 0.55 | 0.55            | 4                 | 0                  |
| 11WP51D  | 0.66   | 0.67 | 0.06               | 0.61 | 0.75 | 0.73            | 5                 | 0                  |
| 11WP51S  | 0.52   | 0.57 | 0.34               | 0.16 | 1.10 | 0.82            | 5                 | 0                  |
| 11WP52D  | 0.28   | 0.27 | 0.03               | 0.23 | 0.30 | 0.30            | 5                 | 0                  |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D                | 1.40   | 1.36 | 0.05               | 1.30 | 1.40 | 1.40            | 5                 | 0                  |
| 11WP53S                | 1.30   | 1.32 | 0.04               | 1.30 | 1.40 | 1.35            | 5                 | 0                  |
| 11WP54D                | 0.34   | 0.34 | 0.03               | 0.28 | 0.37 | 0.37            | 5                 | 0                  |
| 11WP54S                | 0.29   | 0.27 | 0.03               | 0.22 | 0.29 | 0.29            | 5                 | 0                  |
| 11WP55                 | 0.27   | 0.27 | 0.09               | 0.16 | 0.36 | 0.36            | 5                 | 0                  |
| 11WP56D <sup>†</sup>   | 0.03   | 0.03 | 0.00               | 0.03 | 0.03 | 0.03            | 5                 | 100                |
| 11WP56S                | 0.20   | 0.20 | 0.04               | 0.17 | 0.22 | 0.22            | 2                 | 0                  |
| 11WP57                 | 0.43   | 0.42 | 0.03               | 0.38 | 0.45 | 0.45            | 5                 | 0                  |
| CG1                    | 0.31   | 0.30 | 0.02               | 0.26 | 0.33 | 0.32            | 6                 | 0                  |
| CG2                    | 0.28   | 0.29 | 0.03               | 0.26 | 0.34 | 0.33            | 5                 | 0                  |
| CG4                    | 0.24   | 0.23 | 0.02               | 0.20 | 0.24 | 0.24            | 5                 | 0                  |
| KC3 <sup>†</sup>       | 0.03   | 0.03 | 0.00               | 0.03 | 0.03 | 0.03            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.03   | 0.03 | 0.00               | 0.03 | 0.03 | 0.03            | 5                 | 100                |
| KC3PB <sup>†</sup>     | 0.03   | 0.03 | 0.00               | 0.03 | 0.03 | 0.03            | 5                 | 100                |
| KC13                   | 0.37   | 0.36 | 0.03               | 0.32 | 0.39 | 0.39            | 5                 | 0                  |
| KC14                   | 1.00   | 1.07 | 0.12               | 0.96 | 1.20 | 1.20            | 5                 | 0                  |
| KCF1                   | 0.41   | 0.40 | 0.03               | 0.34 | 0.43 | 0.42            | 5                 | 0                  |
| LIMESTONE <sup>†</sup> | 0.14   | 0.13 | 0.10               | 0.03 | 0.24 | 0.23            | 5                 | 40                 |
| M1                     | 0.19   | 0.19 | No data            | 0.19 | 0.19 | 0.19            | 1                 | 0                  |
| RN029659               | 0.18   | 0.16 | 0.06               | 0.06 | 0.20 | 0.20            | 5                 | 0                  |
| RN029660               | 1.50   | 1.50 | 0.12               | 1.40 | 1.70 | 1.60            | 5                 | 0                  |
| W2R <sup>†</sup>       | 0.09   | 0.08 | 0.05               | 0.03 | 0.12 | 0.12            | 6                 | 33                 |
| WP5                    | 0.93   | 0.94 | 0.01               | 0.93 | 0.95 | 0.95            | 5                 | 0                  |
| WP13                   | 0.19   | 0.19 | 0.04               | 0.14 | 0.22 | 0.22            | 4                 | 0                  |
| WP19                   | 0.16   | 0.14 | 0.06               | 0.03 | 0.18 | 0.18            | 5                 | 20                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D20 Concentration of iron (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 7      | 8    | 6                  | 3    | 17   | 16              | 7                 | 29                 |
| 10WP32 <sup>†</sup>   | 3      | 3    | 2                  | 3    | 7    | 3               | 7                 | 86                 |
| 10WP32PB <sup>†</sup> | 3      | 66   | 165                | 3    | 440  | 49              | 7                 | 57                 |
| 10WP33 <sup>†</sup>   | 3      | 6    | 5                  | 3    | 12   | 12              | 7                 | 57                 |
| 10WP35N <sup>†</sup>  | 6      | 8    | 8                  | 3    | 25   | 12              | 7                 | 43                 |
| 10WP35PB              | 6      | 7    | 5                  | 3    | 20   | 7               | 9                 | 11                 |
| 10WP35S <sup>†</sup>  | 6      | 6    | 4                  | 3    | 11   | 10              | 7                 | 43                 |
| 10WP36N <sup>†</sup>  | 7      | 11   | 13                 | 3    | 40   | 15              | 7                 | 29                 |
| 10WP36PB <sup>†</sup> | 3      | 4    | 2                  | 3    | 7    | 6               | 10                | 70                 |
| 10WP36S <sup>†</sup>  | 3      | 8    | 10                 | 3    | 31   | 9               | 7                 | 57                 |
| 10WP37                | 7      | 12   | 15                 | 3    | 46   | 14              | 7                 | 14                 |
| 10WP39 <sup>†</sup>   | 6      | 8    | 6                  | 3    | 16   | 13              | 7                 | 43                 |
| 10WP40                | 43     | 101  | 146                | 7    | 410  | 185             | 7                 | 0                  |
| 10WP41                | 15     | 76   | 140                | 3    | 390  | 95              | 7                 | 14                 |
| 10WP42 <sup>†</sup>   | 7      | 22   | 39                 | 3    | 110  | 27              | 7                 | 43                 |
| 10WP44                | 417    | 461  | 494                | 11   | 1000 | 928             | 4                 | 0                  |
| 10WP46                | 12     | 26   | 34                 | 3    | 99   | 42              | 7                 | 14                 |
| 10WP47                | 58     | 89   | 66                 | 17   | 180  | 153             | 7                 | 0                  |
| 11CS10RD <sup>†</sup> | 4      | 9    | 10                 | 3    | 23   | 21              | 6                 | 50                 |
| 11CS10RS              | 31     | 56   | 65                 | 11   | 180  | 101             | 6                 | 0                  |
| 11WP4R                | 116    | 170  | 174                | 10   | 490  | 301             | 6                 | 0                  |
| 11WP9RPB              | 8      | 26   | 38                 | 6    | 110  | 42              | 7                 | 0                  |
| 11WP9RS               | 12     | 13   | 3                  | 9    | 18   | 16              | 5                 | 0                  |
| 11WP11RD              | 13     | 12   | 5                  | 6    | 18   | 17              | 6                 | 0                  |
| 11WP11RS              | 28     | 51   | 69                 | 6    | 190  | 81              | 6                 | 0                  |
| 11WP15R               | 18     | 95   | 133                | 3    | 320  | 236             | 6                 | 17                 |
| 11WP16R               | 165    | 165  | 145                | 14   | 410  | 270             | 6                 | 0                  |
| 11WP43D               | 420    | 425  | 452                | 10   | 850  | 829             | 4                 | 0                  |
| 11WP50                | 645    | 853  | 883                | 23   | 2100 | 1680            | 4                 | 0                  |
| 11WP51D               | 23     | 26   | 14                 | 8    | 43   | 42              | 6                 | 0                  |
| 11WP51S               | 65     | 159  | 220                | 3    | 580  | 328             | 6                 | 17                 |
| 11WP52D               | 10     | 19   | 21                 | 6    | 59   | 35              | 6                 | 0                  |

| Site              | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D           | 19     | 23   | 20                 | 6    | 58   | 39              | 6                 | 0                  |
| 11WP53S           | 42     | 36   | 18                 | 13   | 59   | 50              | 6                 | 0                  |
| 11WP54D           | 28     | 28   | 17                 | 6    | 49   | 43              | 6                 | 0                  |
| 11WP54S           | 39     | 75   | 112                | 3    | 300  | 124             | 6                 | 17                 |
| 11WP55            | 535    | 670  | 604                | 120  | 1400 | 1330            | 6                 | 0                  |
| 11WP56D           | 12     | 11   | 2                  | 7    | 13   | 12              | 6                 | 0                  |
| 11WP56S           | 20     | 35   | 31                 | 15   | 71   | 66              | 3                 | 0                  |
| 11WP57            | 230    | 207  | 128                | 37   | 370  | 314             | 6                 | 0                  |
| CG1 <sup>†</sup>  | 5      | 10   | 10                 | 3    | 26   | 20              | 6                 | 50                 |
| CG2 <sup>†</sup>  | 3      | 3    | 2                  | 3    | 7    | 5               | 5                 | 80                 |
| CG4               | 415    | 334  | 199                | 66   | 560  | 476             | 6                 | 0                  |
| KC3               | 18     | 22   | 11                 | 10   | 40   | 33              | 5                 | 0                  |
| KC3A              | 7      | 7    | 4                  | 3    | 14   | 11              | 6                 | 17                 |
| KC3PB             | 20     | 24   | 16                 | 8    | 54   | 33              | 6                 | 0                  |
| KC13              | 460    | 378  | 229                | 8    | 570  | 563             | 6                 | 0                  |
| KC14              | 73     | 131  | 89                 | 60   | 260  | 225             | 5                 | 0                  |
| KCF1              | 285    | 360  | 335                | 37   | 970  | 613             | 6                 | 0                  |
| LIMESTONE         | 138    | 197  | 173                | 28   | 460  | 383             | 6                 | 0                  |
| M1                | 18     | 19   | 3                  | 17   | 23   | 23              | 3                 | 0                  |
| RN029659          | 450    | 533  | 509                | 31   | 1300 | 999             | 6                 | 0                  |
| RN029660          | 13     | 13   | 6                  | 6    | 19   | 19              | 6                 | 0                  |
| W2R               | 74     | 88   | 95                 | 3    | 270  | 143             | 6                 | 17                 |
| WP5               | 19     | 18   | 10                 | 3    | 35   | 23              | 7                 | 14                 |
| WP13 <sup>†</sup> | 3      | 5    | 4                  | 3    | 12   | 10              | 6                 | 67                 |
| WP19              | 10     | 12   | 8                  | 3    | 23   | 19              | 5                 | 20                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D21 Concentration of gallium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP32PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP33 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP35N <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP35PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP36N <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP36S <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP37 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP39 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP40 <sup>†</sup>   | 0.10   | 0.13 | 0.06               | 0.05 | 0.20 | 0.20            | 6                 | 50                 |
| 10WP41 <sup>†</sup>   | 0.25   | 0.29 | 0.10               | 0.25 | 0.50 | 0.33            | 6                 | 100                |
| 10WP42 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 10WP44                | 0.65   | 0.56 | 0.21               | 0.25 | 0.70 | 0.70            | 4                 | 25                 |
| 10WP46 <sup>†</sup>   | 0.05   | 0.08 | 0.08               | 0.05 | 0.25 | 0.11            | 6                 | 100                |
| 10WP47 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11CS10RD <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.10   | 0.09 | 0.02               | 0.05 | 0.10 | 0.10            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.05   | 0.10 | 0.08               | 0.05 | 0.20 | 0.20            | 6                 | 67                 |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| 11WP15R <sup>†</sup>  | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP43D               | 0.28   | 0.49 | 0.48               | 0.20 | 1.20 | 0.93            | 4                 | 25                 |
| 11WP50                | 0.70   | 0.61 | 0.26               | 0.25 | 0.80 | 0.80            | 4                 | 25                 |
| 11WP51D <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.05   | 0.07 | 0.03               | 0.05 | 0.10 | 0.10            | 6                 | 83                 |
| 11WP52D <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP55                 | 0.35   | 0.28 | 0.15               | 0.10 | 0.40 | 0.40            | 6                 | 0                  |
| 11WP56D <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.05   | 0.12 | 0.12               | 0.05 | 0.25 | 0.23            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.25   | 0.28 | 0.17               | 0.10 | 0.60 | 0.36            | 6                 | 67                 |
| CG1 <sup>†</sup>       | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.05   | 0.08 | 0.08               | 0.05 | 0.25 | 0.11            | 6                 | 100                |
| KC14 <sup>†</sup>      | 0.25   | 0.21 | 0.09               | 0.05 | 0.25 | 0.25            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| LIMESTONE <sup>†</sup> | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 2                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| W2R <sup>†</sup>       | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.07            | 6                 | 83                 |
| WP5 <sup>†</sup>       | 0.05   | 0.07 | 0.03               | 0.05 | 0.10 | 0.10            | 6                 | 100                |
| WP13 <sup>†</sup>      | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP19 <sup>†</sup>      | 0.50   | 0.44 | 0.13               | 0.25 | 0.50 | 0.50            | 4                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D22 Concentration of hardness (mg/L CaCO<sub>3</sub>)

| Site     | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31   | 170    | 163  | 26                 | 130  | 190   | 187             | 7                 | 0                  |
| 10WP32   | 470    | 408  | 108                | 240  | 507   | 492             | 7                 | 0                  |
| 10WP32PB | 400    | 389  | 126                | 240  | 579   | 500             | 7                 | 0                  |
| 10WP33   | 380    | 360  | 62                 | 270  | 420   | 418             | 7                 | 0                  |
| 10WP35N  | 190    | 289  | 247                | 100  | 778   | 497             | 7                 | 0                  |
| 10WP35PB | 420    | 470  | 199                | 250  | 807   | 689             | 9                 | 0                  |
| 10WP35S  | 380    | 442  | 186                | 250  | 775   | 626             | 7                 | 0                  |
| 10WP36N  | 280    | 360  | 187                | 200  | 644   | 612             | 7                 | 0                  |
| 10WP36PB | 175    | 280  | 187                | 140  | 706   | 421             | 10                | 0                  |
| 10WP36S  | 250    | 320  | 243                | 97   | 788   | 526             | 7                 | 0                  |
| 10WP37   | 370    | 472  | 268                | 230  | 858   | 855             | 7                 | 0                  |
| 10WP39   | 320    | 330  | 31                 | 290  | 370   | 361             | 7                 | 0                  |
| 10WP40   | 780    | 960  | 435                | 670  | 1900  | 1090            | 7                 | 0                  |
| 10WP41   | 4600   | 3969 | 1985               | 583  | 6400  | 5500            | 7                 | 0                  |
| 10WP42   | 440    | 456  | 30                 | 430  | 500   | 491             | 7                 | 0                  |
| 10WP44   | 4285   | 4217 | 3689               | 698  | 7600  | 7480            | 4                 | 0                  |
| 10WP46   | 500    | 518  | 68                 | 450  | 659   | 552             | 7                 | 0                  |
| 10WP47   | 210    | 218  | 47                 | 150  | 305   | 238             | 7                 | 0                  |
| 11CS10RD | 519    | 613  | 213                | 430  | 890   | 876             | 6                 | 0                  |
| 11CS10RS | 841    | 756  | 361                | 170  | 1200  | 1033            | 6                 | 0                  |
| 11WP4R   | 415    | 430  | 69                 | 352  | 540   | 498             | 6                 | 0                  |
| 11WP9RPB | 260    | 301  | 191                | 130  | 716   | 334             | 7                 | 0                  |
| 11WP9RS  | 339    | 314  | 86                 | 200  | 402   | 391             | 5                 | 0                  |
| 11WP11RD | 215    | 289  | 143                | 160  | 481   | 467             | 6                 | 0                  |
| 11WP11RS | 110    | 277  | 266                | 94   | 662   | 600             | 6                 | 0                  |
| 11WP15R  | 510    | 803  | 526                | 410  | 1510  | 1468            | 6                 | 0                  |
| 11WP16R  | 2700   | 2121 | 1238               | 479  | 3400  | 2980            | 6                 | 0                  |
| 11WP43D  | 4800   | 5432 | 5673               | 129  | 12000 | 10890           | 4                 | 0                  |
| 11WP50   | 3027   | 3126 | 2746               | 750  | 5700  | 5580            | 4                 | 0                  |
| 11WP51D  | 220    | 239  | 46                 | 190  | 305   | 292             | 6                 | 0                  |
| 11WP51S  | 265    | 246  | 72                 | 120  | 330   | 295             | 6                 | 0                  |
| 11WP52D  | 205    | 217  | 68                 | 140  | 339   | 266             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 210    | 843  | 1033               | 130  | 2210 | 2161            | 6                 | 0                  |
| 11WP53S   | 175    | 912  | 1165               | 140  | 2460 | 2397            | 6                 | 0                  |
| 11WP54D   | 4650   | 3500 | 2063               | 828  | 5200 | 4920            | 6                 | 0                  |
| 11WP54S   | 2200   | 2057 | 436                | 1510 | 2500 | 2430            | 6                 | 0                  |
| 11WP55    | 485    | 464  | 263                | 32   | 730  | 709             | 6                 | 0                  |
| 11WP56D   | 14     | 35   | 53                 | 8    | 142  | 57              | 6                 | 0                  |
| 11WP56S   | 340    | 332  | 108                | 220  | 436  | 426             | 3                 | 0                  |
| 11WP57    | 3750   | 2898 | 1862               | 378  | 4400 | 4400            | 6                 | 0                  |
| CG1       | 1300   | 1303 | 255                | 918  | 1700 | 1490            | 6                 | 0                  |
| CG2       | 290    | 306  | 64                 | 240  | 412  | 356             | 5                 | 0                  |
| CG4       | 695    | 703  | 123                | 497  | 860  | 811             | 6                 | 0                  |
| KC3       | 67     | 62   | 22                 | 33   | 82   | 82              | 5                 | 0                  |
| KC3A      | 125    | 99   | 52                 | 32   | 140  | 140             | 6                 | 0                  |
| KC3PB     | 115    | 97   | 50                 | 35   | 150  | 136             | 6                 | 0                  |
| KC13      | 290    | 309  | 108                | 160  | 468  | 409             | 6                 | 0                  |
| KC14      | 5200   | 3383 | 2864               | 65   | 5800 | 5600            | 5                 | 0                  |
| KCF1      | 1150   | 1095 | 174                | 873  | 1300 | 1230            | 6                 | 0                  |
| LIMESTONE | 200    | 189  | 40                 | 140  | 240  | 219             | 6                 | 0                  |
| M1        | 76     | 75   | 3                  | 72   | 77   | 77              | 3                 | 0                  |
| RN029659  | 625    | 616  | 325                | 140  | 960  | 960             | 6                 | 0                  |
| RN029660  | 210    | 264  | 136                | 170  | 533  | 349             | 6                 | 0                  |
| W2R       | 185    | 173  | 47                 | 98   | 220  | 213             | 6                 | 0                  |
| WP5       | 2300   | 1783 | 827                | 464  | 2400 | 2310            | 7                 | 0                  |
| WP13      | 340    | 295  | 98                 | 125  | 360  | 355             | 5                 | 0                  |
| WP19      | 5800   | 3729 | 2928               | 468  | 5900 | 5900            | 5                 | 0                  |

Table D23 Concentration of bicarbonate (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 180    | 176  | 17                 | 140  | 190  | 190             | 7                 | 0                  |
| 10WP32   | 474    | 453  | 85                 | 280  | 535  | 520             | 7                 | 0                  |
| 10WP32PB | 488    | 418  | 118                | 180  | 500  | 500             | 7                 | 0                  |
| 10WP33   | 385    | 380  | 41                 | 300  | 426  | 419             | 7                 | 0                  |
| 10WP35N  | 604    | 529  | 251                | 130  | 761  | 747             | 7                 | 0                  |
| 10WP35PB | 720    | 659  | 164                | 340  | 831  | 790             | 9                 | 0                  |
| 10WP35S  | 705    | 588  | 240                | 210  | 807  | 781             | 7                 | 0                  |
| 10WP36N  | 598    | 502  | 230                | 140  | 684  | 656             | 7                 | 0                  |
| 10WP36PB | 592    | 533  | 215                | 51   | 740  | 678             | 10                | 0                  |
| 10WP36S  | 756    | 603  | 294                | 140  | 803  | 791             | 7                 | 0                  |
| 10WP37   | 744    | 660  | 234                | 250  | 879  | 853             | 7                 | 0                  |
| 10WP39   | 290    | 300  | 32                 | 278  | 370  | 307             | 7                 | 0                  |
| 10WP40   | 798    | 697  | 234                | 340  | 952  | 836             | 7                 | 0                  |
| 10WP41   | 702    | 1264 | 1696               | 476  | 5100 | 1231            | 7                 | 0                  |
| 10WP42   | 406    | 416  | 55                 | 348  | 520  | 444             | 7                 | 0                  |
| 10WP44   | 4155   | 4250 | 3771               | 790  | 7900 | 7660            | 4                 | 0                  |
| 10WP46   | 347    | 385  | 104                | 337  | 620  | 386             | 7                 | 0                  |
| 10WP47   | 256    | 249  | 29                 | 199  | 287  | 268             | 7                 | 0                  |
| 11CS10RD | 572    | 608  | 100                | 512  | 790  | 692             | 6                 | 0                  |
| 11CS10RS | 1130   | 1191 | 300                | 872  | 1600 | 1530            | 6                 | 0                  |
| 11WP4R   | 367    | 392  | 110                | 270  | 570  | 492             | 6                 | 0                  |
| 11WP9RPB | 672    | 541  | 240                | 170  | 695  | 689             | 7                 | 0                  |
| 11WP9RS  | 677    | 494  | 277                | 190  | 709  | 706             | 5                 | 0                  |
| 11WP11RD | 450    | 371  | 137                | 180  | 475  | 465             | 6                 | 0                  |
| 11WP11RS | 647    | 468  | 293                | 88   | 674  | 665             | 6                 | 0                  |
| 11WP15R  | 1445   | 1117 | 528                | 420  | 1490 | 1462            | 6                 | 0                  |
| 11WP16R  | 594    | 1440 | 1432               | 470  | 3700 | 3070            | 6                 | 0                  |
| 11WP43D  | 438    | 1117 | 1527               | 191  | 3400 | 2530            | 4                 | 0                  |
| 11WP50   | 3074   | 3094 | 2721               | 729  | 5500 | 5470            | 4                 | 0                  |
| 11WP51D  | 294    | 269  | 53                 | 180  | 315  | 307             | 6                 | 0                  |
| 11WP51S  | 482    | 362  | 204                | 70   | 504  | 503             | 6                 | 0                  |
| 11WP52D  | 329    | 283  | 100                | 150  | 370  | 362             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 2065   | 1472 | 1030               | 130  | 2210 | 2203            | 6                 | 0                  |
| 11WP53S   | 2200   | 1567 | 1093               | 160  | 2390 | 2320            | 6                 | 0                  |
| 11WP54D   | 845    | 2265 | 2246               | 799  | 5500 | 5010            | 6                 | 0                  |
| 11WP54S   | 1515   | 1770 | 557                | 1300 | 2700 | 2350            | 6                 | 0                  |
| 11WP55    | 675    | 665  | 67                 | 550  | 738  | 723             | 6                 | 0                  |
| 11WP56D   | 14     | 16   | 9                  | 8    | 29   | 26              | 6                 | 0                  |
| 11WP56S   | 286    | 238  | 151                | 69   | 360  | 353             | 3                 | 0                  |
| 11WP57    | 430    | 1665 | 1925               | 412  | 4200 | 4130            | 6                 | 0                  |
| CG1       | 891    | 979  | 209                | 864  | 1400 | 1089            | 6                 | 0                  |
| CG2       | 383    | 381  | 50                 | 300  | 422  | 422             | 5                 | 0                  |
| CG4       | 509    | 564  | 151                | 482  | 870  | 633             | 6                 | 0                  |
| KC3       | 64     | 64   | 3                  | 59   | 67   | 67              | 5                 | 0                  |
| KC3A      | 36     | 68   | 53                 | 31   | 150  | 129             | 6                 | 0                  |
| KC3PB     | 38     | 70   | 56                 | 31   | 160  | 132             | 6                 | 0                  |
| KC13      | 343    | 335  | 52                 | 250  | 392  | 377             | 6                 | 0                  |
| KC14      | 468    | 2650 | 3018               | 436  | 6200 | 5950            | 5                 | 0                  |
| KCF1      | 877    | 976  | 185                | 848  | 1300 | 1160            | 6                 | 0                  |
| LIMESTONE | 210    | 193  | 77                 | 96   | 292  | 257             | 6                 | 0                  |
| M1        | 128    | 124  | 7                  | 116  | 128  | 128             | 3                 | 0                  |
| RN029659  | 411    | 423  | 298                | 122  | 960  | 610             | 6                 | 0                  |
| RN029660  | 541    | 493  | 145                | 200  | 586  | 575             | 6                 | 0                  |
| W2R       | 221    | 196  | 47                 | 110  | 228  | 227             | 6                 | 0                  |
| WP5       | 702    | 1090 | 692                | 665  | 2200 | 2020            | 7                 | 0                  |
| WP13      | 262    | 260  | 7                  | 250  | 267  | 266             | 5                 | 0                  |
| WP19      | 568    | 2776 | 3041               | 549  | 6400 | 6100            | 5                 | 0                  |

Table D24 Concentration of mercury (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP32PB <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP33 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP35PB <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 7                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP36N <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 7                 | 100                |
| 10WP36S <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP37 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP39 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 0.05   | 0.06 | 0.03               | 0.03 | 0.10 | 0.10            | 6                 | 100                |
| 10WP41 <sup>†</sup>   | 0.25   | 0.25 | 0.15               | 0.03 | 0.50 | 0.33            | 6                 | 100                |
| 10WP42 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 10WP44 <sup>†</sup>   | 0.18   | 0.16 | 0.11               | 0.03 | 0.25 | 0.25            | 4                 | 75                 |
| 10WP46 <sup>†</sup>   | 0.05   | 0.08 | 0.08               | 0.03 | 0.25 | 0.11            | 6                 | 100                |
| 10WP47 <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11CS10RD <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.10   | 0.08 | 0.03               | 0.03 | 0.10 | 0.10            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.08 | 0.10               | 0.03 | 0.30 | 0.08            | 7                 | 86                 |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.06 | 0.03               | 0.03 | 0.10 | 0.08            | 5                 | 80                 |
| 11WP11RD <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11WP15R <sup>†</sup>  | 0.10   | 0.08 | 0.03               | 0.03 | 0.10 | 0.10            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 0.25   | 0.18 | 0.11               | 0.03 | 0.25 | 0.25            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.08   | 0.12 | 0.12               | 0.03 | 0.30 | 0.24            | 4                 | 75                 |
| 11WP50 <sup>†</sup>   | 0.25   | 0.19 | 0.11               | 0.03 | 0.25 | 0.25            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.05   | 0.05 | 0.02               | 0.03 | 0.10 | 0.07            | 6                 | 100                |
| 11WP52D <sup>†</sup>  | 0.05   | 0.05 | 0.02               | 0.03 | 0.10 | 0.07            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.38 | 0.61               | 0.03 | 1.60 | 0.76            | 6                 | 67                 |
| 11WP53S <sup>†</sup>   | 0.10   | 0.08 | 0.03               | 0.03 | 0.10 | 0.10            | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 0.25   | 0.18 | 0.11               | 0.03 | 0.25 | 0.25            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 0.25   | 0.18 | 0.11               | 0.03 | 0.25 | 0.25            | 6                 | 100                |
| 11WP55 <sup>†</sup>    | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.05   | 0.12 | 0.12               | 0.05 | 0.25 | 0.23            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 0.15   | 0.15 | 0.11               | 0.03 | 0.25 | 0.25            | 6                 | 100                |
| CG1 <sup>†</sup>       | 0.08   | 0.07 | 0.04               | 0.03 | 0.10 | 0.10            | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.05   | 0.08 | 0.08               | 0.03 | 0.25 | 0.11            | 6                 | 100                |
| KC14 <sup>†</sup>      | 0.05   | 0.13 | 0.11               | 0.03 | 0.25 | 0.25            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| LIMESTONE <sup>†</sup> | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| W2R <sup>†</sup>       | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.05   | 0.05 | 0.02               | 0.03 | 0.10 | 0.07            | 6                 | 100                |
| WP13 <sup>†</sup>      | 0.05   | 0.05 | 0.01               | 0.03 | 0.05 | 0.05            | 6                 | 100                |
| WP19 <sup>†</sup>      | 0.25   | 0.27 | 0.23               | 0.03 | 0.50 | 0.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D25 Concentration of potassium (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 3.2    | 3.2  | 0.4                | 2.7  | 4.0  | 3.6             | 7                 | 0                  |
| 10WP32   | 5.1    | 4.4  | 1.4                | 2.4  | 5.9  | 5.5             | 7                 | 0                  |
| 10WP32PB | 3.5    | 3.3  | 1.5                | 0.6  | 5.2  | 4.2             | 7                 | 0                  |
| 10WP33   | 3.4    | 3.6  | 0.4                | 3.1  | 4.4  | 3.9             | 7                 | 0                  |
| 10WP35N  | 2.2    | 2.2  | 0.5                | 1.4  | 2.6  | 2.6             | 7                 | 0                  |
| 10WP35PB | 3.3    | 3.1  | 0.6                | 2.2  | 3.8  | 3.8             | 9                 | 0                  |
| 10WP35S  | 3.2    | 3.2  | 0.4                | 2.6  | 3.8  | 3.5             | 7                 | 0                  |
| 10WP36N  | 2.1    | 2.2  | 0.7                | 0.9  | 3.3  | 2.8             | 7                 | 0                  |
| 10WP36PB | 1.9    | 2.1  | 0.6                | 1.3  | 3.2  | 2.7             | 10                | 0                  |
| 10WP36S  | 2.4    | 2.5  | 0.4                | 1.8  | 2.9  | 2.8             | 7                 | 0                  |
| 10WP37   | 2.8    | 2.5  | 0.5                | 1.6  | 3.0  | 2.8             | 7                 | 0                  |
| 10WP39   | 22.3   | 22.2 | 2.3                | 18.4 | 25.8 | 23.6            | 7                 | 0                  |
| 10WP40   | 17.5   | 16.8 | 4.8                | 11.6 | 25.7 | 19.3            | 7                 | 0                  |
| 10WP41   | 36.7   | 36.4 | 4.3                | 29.1 | 41.2 | 40.0            | 7                 | 0                  |
| 10WP42   | 21.3   | 21.3 | 2.0                | 18.8 | 24.1 | 23.4            | 7                 | 0                  |
| 10WP44   | 17.4   | 19.4 | 6.8                | 13.7 | 29.2 | 25.9            | 4                 | 0                  |
| 10WP46   | 24.4   | 24.3 | 2.3                | 20.4 | 28.0 | 25.7            | 7                 | 0                  |
| 10WP47   | 9.2    | 9.4  | 1.6                | 6.3  | 11.0 | 10.9            | 7                 | 0                  |
| 11CS10RD | 4.5    | 4.7  | 0.9                | 3.7  | 5.8  | 5.7             | 6                 | 0                  |
| 11CS10RS | 1.2    | 1.2  | 0.3                | 0.7  | 1.6  | 1.4             | 6                 | 0                  |
| 11WP4R   | 7.6    | 7.5  | 2.2                | 4.3  | 11.0 | 9.0             | 6                 | 0                  |
| 11WP9RPB | 2.7    | 2.7  | 0.4                | 2.1  | 3.2  | 3.0             | 7                 | 0                  |
| 11WP9RS  | 2.5    | 2.5  | 0.4                | 2.1  | 3.0  | 2.9             | 5                 | 0                  |
| 11WP11RD | 2.5    | 2.4  | 0.4                | 2.0  | 3.1  | 2.7             | 6                 | 0                  |
| 11WP11RS | 0.8    | 0.8  | 0.2                | 0.5  | 1.2  | 1.0             | 6                 | 0                  |
| 11WP15R  | 5.6    | 5.5  | 0.4                | 5.0  | 6.0  | 5.9             | 6                 | 0                  |
| 11WP16R  | 4.5    | 4.3  | 1.2                | 2.2  | 5.4  | 5.4             | 6                 | 0                  |
| 11WP43D  | 13.0   | 12.7 | 5.4                | 5.9  | 19.0 | 17.3            | 4                 | 0                  |
| 11WP50   | 11.9   | 12.0 | 0.8                | 11.2 | 13.2 | 12.8            | 4                 | 0                  |
| 11WP51D  | 18.5   | 17.7 | 2.2                | 14.4 | 20.3 | 19.5            | 6                 | 0                  |
| 11WP51S  | 26.8   | 21.8 | 10.0               | 4.2  | 29.7 | 28.2            | 6                 | 0                  |
| 11WP52D  | 2.6    | 2.5  | 0.4                | 1.9  | 2.9  | 2.8             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 5.6    | 5.7  | 0.4                | 5.2  | 6.3  | 6.2             | 6                 | 0                  |
| 11WP53S   | 6.3    | 6.2  | 0.3                | 5.8  | 6.6  | 6.5             | 6                 | 0                  |
| 11WP54D   | 23.5   | 24.7 | 2.1                | 23.1 | 27.8 | 27.2            | 6                 | 0                  |
| 11WP54S   | 20.0   | 20.4 | 1.9                | 18.4 | 23.0 | 22.3            | 6                 | 0                  |
| 11WP55    | 10.9   | 10.6 | 1.5                | 8.6  | 12.6 | 11.8            | 6                 | 0                  |
| 11WP56D   | 0.6    | 0.8  | 0.5                | 0.3  | 1.4  | 1.3             | 6                 | 0                  |
| 11WP56S   | 1.1    | 0.9  | 0.4                | 0.4  | 1.2  | 1.2             | 3                 | 0                  |
| 11WP57    | 68.3   | 66.7 | 5.9                | 58.5 | 74.5 | 70.9            | 6                 | 0                  |
| CG1       | 8.3    | 8.3  | 0.7                | 7.3  | 9.1  | 9.0             | 6                 | 0                  |
| CG2       | 3.5    | 3.4  | 0.6                | 2.7  | 4.2  | 4.0             | 5                 | 0                  |
| CG4       | 5.8    | 6.7  | 2.7                | 5.0  | 12.1 | 7.9             | 6                 | 0                  |
| KC3       | 1.9    | 2.0  | 0.4                | 1.5  | 2.4  | 2.4             | 5                 | 0                  |
| KC3A      | 4.5    | 4.5  | 0.4                | 3.8  | 5.1  | 4.8             | 6                 | 0                  |
| KC3PB     | 4.6    | 4.4  | 0.6                | 3.7  | 5.2  | 4.9             | 6                 | 0                  |
| KC13      | 4.2    | 4.1  | 0.3                | 3.7  | 4.5  | 4.4             | 6                 | 0                  |
| KC14      | 24.7   | 24.3 | 2.2                | 21.5 | 27.2 | 26.2            | 5                 | 0                  |
| KCF1      | 12.2   | 12.1 | 1.0                | 10.4 | 13.4 | 13.0            | 6                 | 0                  |
| LIMESTONE | 10.0   | 11.7 | 7.9                | 5.5  | 27.1 | 16.0            | 6                 | 0                  |
| M1        | 2.4    | 2.4  | 0.1                | 2.3  | 2.5  | 2.5             | 3                 | 0                  |
| RN029659  | 11.7   | 10.2 | 4.0                | 2.1  | 13.2 | 12.2            | 6                 | 0                  |
| RN029660  | 8.5    | 8.5  | 1.0                | 7.4  | 9.9  | 9.5             | 6                 | 0                  |
| W2R       | 6.6    | 6.3  | 2.3                | 3.3  | 8.5  | 8.4             | 6                 | 0                  |
| WP5       | 4.1    | 4.0  | 0.8                | 3.1  | 5.5  | 4.4             | 7                 | 0                  |
| WP13      | 15.2   | 15.1 | 1.4                | 13.4 | 16.6 | 16.5            | 6                 | 0                  |
| WP19      | 28.7   | 28.6 | 2.4                | 26.4 | 32.3 | 30.7            | 5                 | 0                  |

Table D26 Concentration of lanthanum (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP33 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 2.50   | 2.23 | 0.82               | 0.05 | 2.50 | 2.50            | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 2.50   | 2.26 | 0.77               | 0.05 | 2.50 | 2.50            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP39 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 2.50   | 2.16 | 0.91               | 0.10 | 2.50 | 2.50            | 7                 | 100                |
| 10WP41 <sup>†</sup>   | 2.50   | 2.29 | 0.57               | 1.00 | 2.50 | 2.50            | 7                 | 86                 |
| 10WP42 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 2.50   | 1.94 | 1.13               | 0.25 | 2.50 | 2.50            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 2.50   | 2.16 | 0.91               | 0.10 | 2.50 | 2.50            | 7                 | 86                 |
| 11CS10RD <sup>†</sup> | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 83                 |
| 11CS10RS <sup>†</sup> | 2.50   | 2.12 | 0.94               | 0.20 | 2.50 | 2.50            | 6                 | 83                 |
| 11WP4R <sup>†</sup>   | 2.50   | 2.12 | 0.94               | 0.20 | 2.50 | 2.50            | 6                 | 83                 |
| 11WP9RPB <sup>†</sup> | 2.50   | 2.15 | 0.93               | 0.05 | 2.50 | 2.50            | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 2.50   | 2.01 | 1.10               | 0.05 | 2.50 | 2.50            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 2.50   | 2.18 | 0.78               | 0.60 | 2.50 | 2.50            | 6                 | 83                 |
| 11WP15R <sup>†</sup>  | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 2.50   | 2.13 | 0.92               | 0.25 | 2.50 | 2.50            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 2.50   | 2.35 | 0.30               | 1.90 | 2.50 | 2.50            | 4                 | 75                 |
| 11WP50 <sup>†</sup>   | 2.50   | 2.15 | 0.70               | 1.10 | 2.50 | 2.50            | 4                 | 75                 |
| 11WP51D <sup>†</sup>  | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 83                 |
| 11WP52D <sup>†</sup>  | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 2.50   | 2.13 | 0.92               | 0.25 | 2.50 | 2.50            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 2.50   | 2.13 | 0.92               | 0.25 | 2.50 | 2.50            | 6                 | 100                |
| 11WP55 <sup>†</sup>    | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 2.50   | 2.50 | 0.00               | 2.50 | 2.50 | 2.50            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 2.50   | 2.13 | 0.92               | 0.25 | 2.50 | 2.50            | 6                 | 100                |
| CG1 <sup>†</sup>       | 2.50   | 1.70 | 1.24               | 0.10 | 2.50 | 2.50            | 6                 | 100                |
| CG2 <sup>†</sup>       | 2.50   | 2.01 | 1.10               | 0.05 | 2.50 | 2.50            | 5                 | 100                |
| CG4 <sup>†</sup>       | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| KC3 <sup>†</sup>       | 2.50   | 2.01 | 1.10               | 0.05 | 2.50 | 2.50            | 5                 | 100                |
| KC3A <sup>†</sup>      | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 83                 |
| KC3PB <sup>†</sup>     | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 83                 |
| KC13 <sup>†</sup>      | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| KC14 <sup>†</sup>      | 2.50   | 2.05 | 1.01               | 0.25 | 2.50 | 2.50            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 83                 |
| LIMESTONE <sup>†</sup> | 2.50   | 2.12 | 0.94               | 0.20 | 2.50 | 2.50            | 6                 | 83                 |
| M1 <sup>†</sup>        | 2.50   | 2.50 | 0.00               | 2.50 | 2.50 | 2.50            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| W2R <sup>†</sup>       | 2.50   | 2.10 | 0.98               | 0.10 | 2.50 | 2.50            | 6                 | 83                 |
| WP5 <sup>†</sup>       | 2.50   | 2.16 | 0.91               | 0.10 | 2.50 | 2.50            | 7                 | 86                 |
| WP13 <sup>†</sup>      | 2.50   | 2.09 | 1.00               | 0.05 | 2.50 | 2.50            | 6                 | 100                |
| WP19 <sup>†</sup>      | 2.50   | 2.10 | 0.89               | 0.50 | 2.50 | 2.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D27 Concentration of lithium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 2.3    | 2.2  | 0.3                | 1.7  | 2.5  | 2.5             | 7                 | 43                 |
| 10WP32 <sup>†</sup>   | 2.5    | 2.8  | 0.9                | 1.8  | 4.5  | 3.4             | 7                 | 43                 |
| 10WP32PB <sup>†</sup> | 2.5    | 2.3  | 0.8                | 0.5  | 3.0  | 2.8             | 7                 | 43                 |
| 10WP33 <sup>†</sup>   | 2.5    | 2.6  | 0.2                | 2.4  | 3.0  | 2.7             | 7                 | 43                 |
| 10WP35N <sup>†</sup>  | 2.5    | 2.5  | 0.4                | 1.7  | 2.8  | 2.7             | 7                 | 43                 |
| 10WP35PB <sup>†</sup> | 2.5    | 2.7  | 0.4                | 2.5  | 3.6  | 3.1             | 9                 | 56                 |
| 10WP35S <sup>†</sup>  | 2.5    | 2.7  | 0.3                | 2.5  | 3.3  | 2.9             | 7                 | 43                 |
| 10WP36N <sup>†</sup>  | 2.5    | 2.5  | 0.7                | 1.1  | 3.4  | 2.9             | 7                 | 43                 |
| 10WP36PB <sup>†</sup> | 2.5    | 2.5  | 0.6                | 1.6  | 4.1  | 2.5             | 10                | 60                 |
| 10WP36S <sup>†</sup>  | 2.8    | 3.1  | 0.8                | 2.5  | 4.5  | 3.7             | 7                 | 43                 |
| 10WP37 <sup>†</sup>   | 3.6    | 3.5  | 1.1                | 2.5  | 4.9  | 4.7             | 7                 | 43                 |
| 10WP39                | 41.0   | 40.1 | 3.5                | 35.0 | 44.0 | 43.1            | 7                 | 0                  |
| 10WP40                | 50.0   | 52.9 | 12.5               | 38.0 | 73.0 | 62.2            | 7                 | 0                  |
| 10WP41                | 27.0   | 27.9 | 6.5                | 17.0 | 37.0 | 34.3            | 7                 | 0                  |
| 10WP42                | 40.0   | 39.9 | 5.7                | 33.0 | 50.0 | 43.7            | 7                 | 0                  |
| 10WP44                | 25.5   | 26.3 | 2.6                | 24.0 | 30.0 | 28.8            | 4                 | 0                  |
| 10WP46                | 48.0   | 45.4 | 6.0                | 37.0 | 51.0 | 51.0            | 7                 | 0                  |
| 10WP47                | 8.0    | 8.2  | 1.4                | 6.0  | 10.0 | 9.7             | 7                 | 0                  |
| 11CS10RD <sup>†</sup> | 3.5    | 3.5  | 0.9                | 2.5  | 5.0  | 4.2             | 6                 | 33                 |
| 11CS10RS              | 5.3    | 5.4  | 2.4                | 2.5  | 8.2  | 8.0             | 6                 | 17                 |
| 11WP4R <sup>†</sup>   | 2.4    | 2.3  | 0.3                | 1.7  | 2.5  | 2.5             | 6                 | 33                 |
| 11WP9RPB              | 4.6    | 4.4  | 1.1                | 2.5  | 6.0  | 5.3             | 7                 | 14                 |
| 11WP9RS               | 5.2    | 4.6  | 1.4                | 2.5  | 6.0  | 5.6             | 5                 | 20                 |
| 11WP11RD <sup>†</sup> | 2.3    | 2.2  | 0.5                | 1.3  | 2.7  | 2.6             | 6                 | 33                 |
| 11WP11RS <sup>†</sup> | 1.6    | 1.7  | 0.7                | 0.7  | 2.5  | 2.5             | 6                 | 33                 |
| 11WP15R               | 24.5   | 23.5 | 4.5                | 15.0 | 28.0 | 26.6            | 6                 | 0                  |
| 11WP16R               | 31.0   | 31.3 | 8.5                | 19.0 | 41.0 | 40.3            | 6                 | 0                  |
| 11WP43D               | 59.0   | 55.3 | 37.1               | 14.0 | 89.0 | 87.5            | 4                 | 0                  |
| 11WP50                | 36.5   | 36.0 | 3.2                | 32.0 | 39.0 | 38.7            | 4                 | 0                  |
| 11WP51D               | 22.0   | 21.7 | 5.8                | 12.0 | 28.0 | 27.3            | 6                 | 0                  |
| 11WP51S               | 24.0   | 21.0 | 12.6               | 2.2  | 33.0 | 33.0            | 6                 | 0                  |
| 11WP52D <sup>†</sup>  | 2.5    | 2.5  | 0.3                | 2.1  | 2.8  | 2.8             | 6                 | 33                 |

| Site                 | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------------------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 11WP53D              | 20.0   | 19.2  | 4.5                | 11.0  | 24.0  | 22.6            | 6                 | 0                  |
| 11WP53S              | 21.0   | 18.8  | 4.5                | 10.0  | 22.0  | 21.3            | 6                 | 0                  |
| 11WP54D              | 115.0  | 112.2 | 13.9               | 93.0  | 130.0 | 123.0           | 6                 | 0                  |
| 11WP54S              | 25.5   | 24.5  | 3.1                | 19.0  | 27.0  | 27.0            | 6                 | 0                  |
| 11WP55               | 40.5   | 40.2  | 5.6                | 32.0  | 47.0  | 45.6            | 6                 | 0                  |
| 11WP56D <sup>†</sup> | 2.5    | 2.5   | 0.5                | 1.8   | 3.2   | 3.0             | 6                 | 33                 |
| 11WP56S <sup>†</sup> | 1.2    | 1.6   | 0.8                | 1.0   | 2.5   | 2.4             | 3                 | 33                 |
| 11WP57               | 640.0  | 616.0 | 111.3              | 430.0 | 730.0 | 690.0           | 5                 | 0                  |
| CG1                  | 10.0   | 9.6   | 1.6                | 6.8   | 11.0  | 11.0            | 6                 | 0                  |
| CG2 <sup>†</sup>     | 3.0    | 2.8   | 0.3                | 2.5   | 3.2   | 3.1             | 5                 | 40                 |
| CG4                  | 6.5    | 5.9   | 1.7                | 2.5   | 7.0   | 6.7             | 6                 | 17                 |
| KC3                  | 11.0   | 11.1  | 2.0                | 8.8   | 14.0  | 13.0            | 5                 | 0                  |
| KC3A                 | 13.0   | 13.0  | 1.4                | 11.0  | 15.0  | 14.3            | 6                 | 0                  |
| KC3PB                | 12.0   | 12.0  | 1.4                | 9.9   | 14.0  | 13.3            | 6                 | 0                  |
| KC13                 | 8.1    | 8.3   | 4.0                | 2.5   | 13.0  | 12.3            | 6                 | 17                 |
| KC14                 | 240.0  | 232.0 | 25.9               | 190.0 | 260.0 | 250.0           | 5                 | 0                  |
| KCF1                 | 22.0   | 21.7  | 3.3                | 18.0  | 26.0  | 24.6            | 6                 | 0                  |
| LIMESTONE            | 16.5   | 14.1  | 5.9                | 5.4   | 19.0  | 19.0            | 6                 | 0                  |
| M1 <sup>†</sup>      | 2.5    | 2.5   | 0.0                | 2.5   | 2.5   | 2.5             | 3                 | 100                |
| RN029659             | 39.0   | 33.8  | 15.5               | 2.5   | 44.0  | 41.9            | 6                 | 17                 |
| RN029660             | 30.5   | 29.8  | 1.9                | 27.0  | 32.0  | 31.3            | 6                 | 0                  |
| W2R                  | 7.4    | 6.7   | 3.7                | 1.8   | 10.0  | 10.0            | 6                 | 0                  |
| WP5                  | 25.0   | 25.0  | 2.4                | 22.0  | 28.0  | 28.0            | 7                 | 0                  |
| WP13                 | 25.5   | 25.0  | 1.8                | 22.0  | 27.0  | 26.3            | 6                 | 0                  |
| WP19                 | 85.0   | 91.8  | 12.7               | 80.0  | 110.0 | 105.0           | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D28 Concentration of magnesium (mg/L)

| Site     | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31   | 21.7   | 21.9  | 3.2                | 17.7 | 26.5  | 25.2            | 7                 | 0                  |
| 10WP32   | 59.7   | 57.4  | 15.7               | 37.7 | 77.1  | 73.5            | 7                 | 0                  |
| 10WP32PB | 50.8   | 48.6  | 15.2               | 25.2 | 68.1  | 63.0            | 7                 | 0                  |
| 10WP33   | 48.4   | 51.3  | 8.4                | 42.0 | 62.9  | 60.9            | 7                 | 0                  |
| 10WP35N  | 23.8   | 24.6  | 7.0                | 15.7 | 33.5  | 32.2            | 7                 | 0                  |
| 10WP35PB | 60.6   | 57.4  | 13.1               | 37.4 | 75.2  | 67.4            | 9                 | 0                  |
| 10WP35S  | 46.3   | 47.2  | 10.9               | 30.7 | 61.8  | 57.7            | 7                 | 0                  |
| 10WP36N  | 28.9   | 31.6  | 9.4                | 16.5 | 46.1  | 39.4            | 7                 | 0                  |
| 10WP36PB | 22.5   | 27.9  | 14.1               | 7.6  | 55.8  | 41.5            | 10                | 0                  |
| 10WP36S  | 27.2   | 27.6  | 9.2                | 14.2 | 40.2  | 37.9            | 7                 | 0                  |
| 10WP37   | 54.2   | 51.6  | 11.6               | 37.9 | 67.3  | 61.3            | 7                 | 0                  |
| 10WP39   | 28.5   | 28.4  | 3.2                | 24.2 | 33.3  | 31.5            | 7                 | 0                  |
| 10WP40   | 122.0  | 130.8 | 72.5               | 54.6 | 270.0 | 165.6           | 7                 | 0                  |
| 10WP41   | 813    | 742   | 213                | 442  | 1050  | 875             | 7                 | 0                  |
| 10WP42   | 57.5   | 56.4  | 4.8                | 51.0 | 64.1  | 60.2            | 7                 | 0                  |
| 10WP44   | 1205   | 1240  | 121                | 1140 | 1410  | 1359            | 4                 | 0                  |
| 10WP46   | 53.2   | 55.3  | 6.7                | 49.8 | 69.6  | 58.7            | 7                 | 0                  |
| 10WP47   | 12.7   | 13.1  | 2.1                | 9.5  | 15.7  | 15.0            | 7                 | 0                  |
| 11CS10RD | 101.2  | 98.3  | 28.5               | 64.0 | 130.0 | 127.2           | 6                 | 0                  |
| 11CS10RS | 157.5  | 143.4 | 80.0               | 28.3 | 245.0 | 209.3           | 6                 | 0                  |
| 11WP4R   | 55.2   | 55.7  | 11.7               | 37.5 | 69.7  | 67.0            | 6                 | 0                  |
| 11WP9RPB | 34.5   | 34.5  | 8.3                | 22.2 | 46.3  | 41.7            | 7                 | 0                  |
| 11WP9RS  | 33.9   | 40.1  | 13.5               | 31.2 | 63.5  | 51.7            | 5                 | 0                  |
| 11WP11RD | 31.6   | 30.5  | 3.3                | 25.2 | 33.4  | 33.3            | 6                 | 0                  |
| 11WP11RS | 18.0   | 17.9  | 2.1                | 15.1 | 20.3  | 20.0            | 6                 | 0                  |
| 11WP15R  | 86.1   | 88.5  | 9.0                | 80.4 | 105.0 | 95.1            | 6                 | 0                  |
| 11WP16R  | 430    | 448   | 56                 | 406  | 552   | 490             | 6                 | 0                  |
| 11WP43D  | 1061   | 1130  | 884                | 220  | 2180  | 1979            | 4                 | 0                  |
| 11WP50   | 957    | 945   | 30                 | 902  | 966   | 966             | 4                 | 0                  |
| 11WP51D  | 20.5   | 20.1  | 1.8                | 17.3 | 22.2  | 21.6            | 6                 | 0                  |
| 11WP51S  | 31.3   | 26.8  | 13.6               | 6.3  | 42.5  | 37.0            | 6                 | 0                  |
| 11WP52D  | 25.1   | 25.6  | 4.4                | 21.1 | 31.1  | 30.5            | 6                 | 0                  |

| Site      | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 11WP53D   | 29.4   | 32.1  | 9.1                | 25.4  | 49.7  | 37.9            | 6                 | 0                  |
| 11WP53S   | 31.2   | 31.0  | 1.6                | 28.2  | 32.9  | 32.3            | 6                 | 0                  |
| 11WP54D   | 816    | 836   | 63                 | 773   | 942   | 897             | 6                 | 0                  |
| 11WP54S   | 408    | 421   | 43                 | 377   | 491   | 462             | 6                 | 0                  |
| 11WP55    | 87     | 85    | 21                 | 47    | 107   | 103             | 6                 | 0                  |
| 11WP56D   | 1.1    | 1.1   | 0.4                | 0.6   | 1.7   | 1.4             | 6                 | 0                  |
| 11WP56S   | 22.7   | 22.0  | 14.3               | 7.3   | 35.9  | 34.6            | 3                 | 0                  |
| 11WP57    | 445    | 432   | 44                 | 351   | 469   | 464             | 6                 | 0                  |
| CG1       | 204    | 209   | 24                 | 182   | 253   | 224             | 6                 | 0                  |
| CG2       | 32.5   | 31.1  | 3.5                | 24.8  | 33.3  | 33.0            | 5                 | 0                  |
| CG4       | 113.5  | 116.5 | 13.8               | 102.0 | 133.0 | 132.3           | 6                 | 0                  |
| KC3       | 10.2   | 10.8  | 1.6                | 8.8   | 12.5  | 12.5            | 5                 | 0                  |
| KC3A      | 20.9   | 21.1  | 2.0                | 19.0  | 24.1  | 23.0            | 6                 | 0                  |
| KC3PB     | 20.2   | 20.7  | 2.8                | 16.7  | 24.6  | 23.6            | 6                 | 0                  |
| KC13      | 33.2   | 34.0  | 6.8                | 23.7  | 42.8  | 41.0            | 6                 | 0                  |
| KC14      | 919    | 924   | 69                 | 860   | 1030  | 988             | 5                 | 0                  |
| KCF1      | 231    | 229   | 20                 | 206   | 256   | 247             | 6                 | 0                  |
| LIMESTONE | 9.8    | 9.1   | 3.5                | 4.2   | 13.7  | 11.7            | 6                 | 0                  |
| M1        | 7.9    | 8.0   | 0.2                | 7.9   | 8.2   | 8.2             | 3                 | 0                  |
| RN029659  | 136.5  | 114.7 | 55.3               | 6.2   | 151.0 | 148.2           | 6                 | 0                  |
| RN029660  | 31.9   | 33.2  | 5.4                | 27.5  | 42.7  | 37.6            | 6                 | 0                  |
| W2R       | 8.2    | 7.4   | 2.1                | 4.4   | 9.8   | 9.1             | 6                 | 0                  |
| WP5       | 400    | 390   | 28                 | 348   | 421   | 409             | 7                 | 0                  |
| WP13      | 21.2   | 21.6  | 2.4                | 19.5  | 25.7  | 23.4            | 6                 | 0                  |
| WP19      | 932    | 946   | 54                 | 906   | 1040  | 987             | 5                 | 0                  |

Table D29 Concentration of manganese (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31                | 11     | 22   | 31                 | 4    | 89   | 34              | 7                 | 0                  |
| 10WP32 <sup>†</sup>   | 2      | 4    | 3                  | 1    | 9    | 6               | 7                 | 29                 |
| 10WP32PB              | 4      | 22   | 43                 | 2    | 120  | 28              | 7                 | 0                  |
| 10WP33                | 5      | 5    | 3                  | 1    | 10   | 9               | 7                 | 0                  |
| 10WP35N               | 22     | 33   | 35                 | 6    | 110  | 39              | 7                 | 0                  |
| 10WP35PB              | 9      | 14   | 16                 | 1    | 47   | 27              | 9                 | 22                 |
| 10WP35S               | 98     | 99   | 62                 | 2    | 200  | 146             | 7                 | 0                  |
| 10WP36N               | 22     | 31   | 22                 | 11   | 76   | 45              | 7                 | 0                  |
| 10WP36PB <sup>†</sup> | 1      | 5    | 10                 | 1    | 32   | 6               | 10                | 50                 |
| 10WP36S               | 14     | 13   | 5                  | 6    | 17   | 17              | 7                 | 0                  |
| 10WP37                | 11     | 14   | 10                 | 4    | 27   | 25              | 7                 | 0                  |
| 10WP39                | 5      | 9    | 9                  | 1    | 26   | 16              | 7                 | 0                  |
| 10WP40                | 770    | 1077 | 605                | 560  | 2000 | 1820            | 7                 | 0                  |
| 10WP41                | 1100   | 991  | 933                | 12   | 2500 | 1780            | 7                 | 0                  |
| 10WP42                | 12     | 12   | 9                  | 1    | 24   | 23              | 7                 | 14                 |
| 10WP44                | 6350   | 6550 | 835                | 5800 | 7700 | 7370            | 4                 | 0                  |
| 10WP46                | 33     | 36   | 34                 | 1    | 88   | 70              | 7                 | 14                 |
| 10WP47                | 180    | 182  | 113                | 1    | 330  | 303             | 7                 | 14                 |
| 11CS10RD <sup>†</sup> | 4      | 4    | 3                  | 1    | 8    | 6               | 6                 | 33                 |
| 11CS10RS              | 255    | 275  | 220                | 1    | 560  | 490             | 6                 | 0                  |
| 11WP4R                | 1115   | 1079 | 836                | 62   | 2000 | 1930            | 6                 | 0                  |
| 11WP9RPB              | 16     | 37   | 41                 | 2    | 110  | 83              | 7                 | 0                  |
| 11WP9RS               | 18     | 28   | 24                 | 10   | 69   | 48              | 5                 | 0                  |
| 11WP11RD              | 5      | 7    | 6                  | 1    | 18   | 13              | 6                 | 0                  |
| 11WP11RS              | 24     | 26   | 21                 | 1    | 56   | 47              | 6                 | 0                  |
| 11WP15R               | 122    | 143  | 92                 | 34   | 260  | 246             | 6                 | 0                  |
| 11WP16R               | 335    | 424  | 348                | 56   | 930  | 797             | 6                 | 0                  |
| 11WP43D               | 3450   | 3475 | 818                | 2500 | 4500 | 4200            | 4                 | 0                  |
| 11WP50                | 4400   | 4725 | 2114               | 2600 | 7500 | 6780            | 4                 | 0                  |
| 11WP51D               | 55     | 61   | 31                 | 21   | 100  | 94              | 6                 | 0                  |
| 11WP51S               | 460    | 502  | 390                | 2    | 1000 | 888             | 6                 | 0                  |
| 11WP52D               | 89     | 126  | 106                | 26   | 270  | 249             | 6                 | 0                  |

| Site             | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D          | 41     | 65   | 75                 | 1    | 210  | 110             | 6                 | 0                  |
| 11WP53S          | 160    | 143  | 79                 | 5    | 210  | 210             | 6                 | 0                  |
| 11WP54D          | 395    | 507  | 265                | 270  | 920  | 801             | 6                 | 0                  |
| 11WP54S          | 1300   | 1310 | 977                | 110  | 2700 | 2210            | 6                 | 0                  |
| 11WP55           | 3700   | 3105 | 1733               | 930  | 4800 | 4590            | 6                 | 0                  |
| 11WP56D          | 16     | 18   | 8                  | 10   | 34   | 22              | 6                 | 0                  |
| 11WP56S          | 55     | 136  | 187                | 4    | 350  | 321             | 3                 | 0                  |
| 11WP57           | 545    | 1408 | 2059               | 400  | 5600 | 2282            | 6                 | 0                  |
| CG1              | 3      | 2    | 2                  | 1    | 5    | 4               | 6                 | 17                 |
| CG2 <sup>†</sup> | 2      | 2    | 2                  | 1    | 5    | 4               | 5                 | 40                 |
| CG4              | 86     | 93   | 28                 | 67   | 140  | 119             | 6                 | 0                  |
| KC3              | 6      | 11   | 13                 | 3    | 35   | 22              | 5                 | 0                  |
| KC3A             | 7      | 15   | 22                 | 3    | 60   | 24              | 6                 | 0                  |
| KC3PB            | 12     | 13   | 11                 | 3    | 32   | 19              | 6                 | 0                  |
| KC13             | 255    | 212  | 105                | 2    | 270  | 270             | 6                 | 0                  |
| KC14             | 220    | 230  | 44                 | 180  | 280  | 275             | 5                 | 0                  |
| KCF1             | 330    | 390  | 235                | 150  | 770  | 623             | 6                 | 0                  |
| LIMESTONE        | 78     | 96   | 40                 | 58   | 160  | 139             | 6                 | 0                  |
| M1               | 4      | 4    | 2                  | 3    | 6    | 6               | 3                 | 0                  |
| RN029659         | 130    | 147  | 69                 | 67   | 240  | 226             | 6                 | 0                  |
| RN029660         | 6      | 7    | 4                  | 2    | 15   | 9               | 6                 | 0                  |
| W2R              | 55     | 147  | 263                | 4    | 680  | 261             | 6                 | 0                  |
| WP5              | 180    | 210  | 100                | 99   | 410  | 266             | 7                 | 0                  |
| WP13             | 4      | 5    | 5                  | 1    | 12   | 11              | 6                 | 17                 |
| WP19             | 41     | 63   | 79                 | 8    | 200  | 130             | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D30 Concentration of molybdenum (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 1.0    | 0.9  | 0.5                | 0.5  | 2.0  | 1.1             | 7                 | 43                 |
| 10WP32PB <sup>†</sup> | 0.5    | 1.0  | 0.7                | 0.5  | 2.0  | 2.0             | 7                 | 57                 |
| 10WP33 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP35N               | 2.0    | 2.4  | 0.5                | 2.0  | 3.0  | 3.0             | 7                 | 0                  |
| 10WP35PB <sup>†</sup> | 0.5    | 0.6  | 0.2                | 0.5  | 1.0  | 0.9             | 9                 | 78                 |
| 10WP35S <sup>†</sup>  | 1.0    | 1.1  | 0.9                | 0.5  | 3.0  | 1.2             | 7                 | 43                 |
| 10WP36N               | 5.0    | 5.1  | 2.1                | 2.0  | 9.0  | 6.3             | 7                 | 0                  |
| 10WP36PB              | 6.5    | 6.2  | 3.1                | 2.0  | 10.0 | 9.0             | 10                | 0                  |
| 10WP36S               | 8.0    | 10.3 | 3.5                | 8.0  | 17.0 | 13.4            | 7                 | 0                  |
| 10WP37                | 5.0    | 5.6  | 1.1                | 4.0  | 7.0  | 7.0             | 7                 | 0                  |
| 10WP39 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 10WP40                | 5.0    | 5.9  | 1.9                | 4.0  | 9.0  | 7.2             | 7                 | 0                  |
| 10WP41 <sup>†</sup>   | 2.5    | 2.9  | 0.9                | 2.5  | 5.0  | 2.8             | 7                 | 100                |
| 10WP42                | 1.0    | 1.3  | 0.5                | 1.0  | 2.0  | 2.0             | 7                 | 0                  |
| 10WP44 <sup>†</sup>   | 2.5    | 3.1  | 1.3                | 2.5  | 5.0  | 4.3             | 4                 | 75                 |
| 10WP46 <sup>†</sup>   | 1.0    | 1.0  | 0.7                | 0.5  | 2.5  | 1.2             | 7                 | 57                 |
| 10WP47 <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 4.0    | 11.2 | 17.1               | 1.0  | 45.0 | 21.9            | 6                 | 33                 |
| 11WP4R <sup>†</sup>   | 1.0    | 1.0  | 0.5                | 0.5  | 2.0  | 1.3             | 6                 | 33                 |
| 11WP9RPB              | 2.0    | 4.4  | 5.2                | 2.0  | 16.0 | 5.2             | 7                 | 0                  |
| 11WP9RS               | 2.0    | 2.4  | 0.5                | 2.0  | 3.0  | 3.0             | 5                 | 0                  |
| 11WP11RD <sup>†</sup> | 1.0    | 1.2  | 0.7                | 0.5  | 2.0  | 2.0             | 6                 | 33                 |
| 11WP11RS <sup>†</sup> | 0.5    | 0.8  | 0.6                | 0.5  | 2.0  | 1.0             | 6                 | 83                 |
| 11WP15R               | 4.5    | 4.5  | 1.4                | 3.0  | 6.0  | 6.0             | 6                 | 0                  |
| 11WP16R <sup>†</sup>  | 2.5    | 2.5  | 0.0                | 2.5  | 2.5  | 2.5             | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 2.8    | 2.9  | 1.7                | 1.0  | 5.0  | 4.4             | 4                 | 75                 |
| 11WP50 <sup>†</sup>   | 2.5    | 2.5  | 0.0                | 2.5  | 2.5  | 2.5             | 4                 | 100                |
| 11WP51D               | 2.0    | 1.8  | 0.9                | 0.5  | 3.0  | 2.3             | 6                 | 17                 |
| 11WP51S <sup>†</sup>  | 3.0    | 4.2  | 4.0                | 0.5  | 10.0 | 8.6             | 6                 | 33                 |
| 11WP52D <sup>†</sup>  | 0.5    | 0.7  | 0.3                | 0.5  | 1.0  | 1.0             | 6                 | 83                 |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D                | 36.5   | 38.5 | 7.6                | 30.0 | 51.0 | 45.4            | 6                 | 0                  |
| 11WP53S                | 36.0   | 39.7 | 7.8                | 33.0 | 53.0 | 47.4            | 6                 | 0                  |
| 11WP54D <sup>†</sup>   | 2.5    | 2.5  | 0.0                | 2.5  | 2.5  | 2.5             | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 2.5    | 2.5  | 0.0                | 2.5  | 2.5  | 2.5             | 6                 | 100                |
| 11WP55                 | 10.0   | 10.5 | 4.6                | 5.0  | 19.0 | 12.7            | 6                 | 0                  |
| 11WP56D <sup>†</sup>   | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.5    | 1.2  | 1.2                | 0.5  | 2.5  | 2.3             | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 2.5    | 3.8  | 4.1                | 0.5  | 12.0 | 5.4             | 6                 | 83                 |
| CG1 <sup>†</sup>       | 1.0    | 1.0  | 0.0                | 1.0  | 1.0  | 1.0             | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.5    | 0.8  | 0.7                | 0.5  | 2.0  | 1.3             | 5                 | 80                 |
| CG4 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| KC13                   | 2.3    | 4.3  | 3.7                | 2.0  | 11.0 | 7.5             | 6                 | 17                 |
| KC14 <sup>†</sup>      | 2.5    | 2.1  | 0.9                | 0.5  | 2.5  | 2.5             | 5                 | 100                |
| KCF1                   | 1.0    | 1.2  | 0.4                | 1.0  | 2.0  | 1.3             | 6                 | 0                  |
| LIMESTONE <sup>†</sup> | 0.8    | 0.9  | 0.6                | 0.5  | 2.0  | 1.3             | 6                 | 50                 |
| M1 <sup>†</sup>        | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| RN029660               | 3.0    | 3.3  | 0.8                | 3.0  | 5.0  | 3.6             | 6                 | 0                  |
| W2R <sup>†</sup>       | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| WP5 <sup>†</sup>       | 1.0    | 1.1  | 0.6                | 0.5  | 2.0  | 2.0             | 7                 | 57                 |
| WP13 <sup>†</sup>      | 0.5    | 0.5  | 0.0                | 0.5  | 0.5  | 0.5             | 6                 | 100                |
| WP19 <sup>†</sup>      | 5.0    | 4.5  | 1.1                | 2.5  | 5.0  | 5.0             | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D31 Concentration of nitrogen ammonia (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 5      | 17   | 19                 | 5    | 50   | 36              | 6                 | 67                 |
| 10WP32 <sup>†</sup>   | 13     | 14   | 11                 | 5    | 30   | 23              | 6                 | 50                 |
| 10WP32PB <sup>†</sup> | 13     | 21   | 20                 | 5    | 50   | 43              | 6                 | 50                 |
| 10WP33 <sup>†</sup>   | 5      | 20   | 26                 | 5    | 70   | 42              | 6                 | 67                 |
| 10WP35N <sup>†</sup>  | 18     | 41   | 47                 | 5    | 110  | 96              | 6                 | 50                 |
| 10WP35PB <sup>†</sup> | 5      | 5    | 0                  | 5    | 5    | 5               | 8                 | 100                |
| 10WP35S <sup>†</sup>  | 5      | 6    | 2                  | 5    | 10   | 7               | 6                 | 83                 |
| 10WP36N <sup>†</sup>  | 5      | 52   | 103                | 5    | 260  | 99              | 6                 | 67                 |
| 10WP36PB <sup>†</sup> | 5      | 5    | 0                  | 5    | 5    | 5               | 9                 | 100                |
| 10WP36S <sup>†</sup>  | 5      | 5    | 0                  | 5    | 5    | 5               | 6                 | 100                |
| 10WP37 <sup>†</sup>   | 23     | 124  | 182                | 5    | 440  | 307             | 6                 | 50                 |
| 10WP39 <sup>†</sup>   | 5      | 12   | 14                 | 5    | 40   | 19              | 6                 | 67                 |
| 10WP40                | 180    | 228  | 212                | 5    | 590  | 401             | 6                 | 17                 |
| 10WP41 <sup>†</sup>   | 60     | 75   | 81                 | 5    | 220  | 136             | 6                 | 33                 |
| 10WP42 <sup>†</sup>   | 8      | 9    | 6                  | 5    | 20   | 13              | 6                 | 50                 |
| 10WP44                | ND     | ND   | ND                 | ND   | ND   | ND              | 4/0*              | 0                  |
| 10WP46 <sup>†</sup>   | 5      | 20   | 24                 | 5    | 60   | 46              | 6                 | 67                 |
| 10WP47                | 35     | 33   | 24                 | 5    | 70   | 49              | 6                 | 17                 |
| 11CS10RD <sup>†</sup> | 5      | 23   | 33                 | 5    | 80   | 50              | 5                 | 60                 |
| 11CS10RS <sup>†</sup> | 5      | 71   | 111                | 5    | 260  | 170             | 5                 | 60                 |
| 11WP4R                | 1300   | 1320 | 1061               | 60   | 2900 | 2250            | 5                 | 0                  |
| 11WP9RPB <sup>†</sup> | 5      | 27   | 42                 | 5    | 110  | 54              | 6                 | 67                 |
| 11WP9RS <sup>†</sup>  | 5      | 16   | 23                 | 5    | 50   | 37              | 4                 | 75                 |
| 11WP11RD <sup>†</sup> | 5      | 5    | 0                  | 5    | 5    | 5               | 5                 | 100                |
| 11WP11RS <sup>†</sup> | 40     | 44   | 41                 | 5    | 100  | 85              | 5                 | 40                 |
| 11WP15R <sup>†</sup>  | 5      | 37   | 50                 | 5    | 120  | 85              | 5                 | 60                 |
| 11WP16R               | 320    | 1836 | 2343               | 40   | 5000 | 4350            | 5                 | 0                  |
| 11WP43D               | 240    | 398  | 410                | 110  | 1000 | 793             | 4                 | 0                  |
| 11WP50                | 35     | 46   | 45                 | 5    | 110  | 89              | 4                 | 25                 |
| 11WP51D               | 80     | 130  | 131                | 20   | 330  | 260             | 5                 | 0                  |
| 11WP51S               | 60     | 57   | 45                 | 10   | 100  | 96              | 5/3*              | 0                  |
| 11WP52D <sup>†</sup>  | 5      | 14   | 20                 | 5    | 50   | 28              | 5                 | 80                 |

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>  | 5      | 32   | 60                 | 5    | 140  | 73              | 5                 | 80                 |
| 11WP53S <sup>†</sup>  | 10     | 86   | 110                | 5    | 230  | 205             | 5                 | 40                 |
| 11WP54D <sup>†</sup>  | 10     | 20   | 23                 | 5    | 60   | 40              | 5                 | 40                 |
| 11WP54S <sup>†</sup>  | 20     | 142  | 206                | 5    | 480  | 340             | 5                 | 40                 |
| 11WP55                | 190    | 574  | 715                | 30   | 1700 | 1285            | 5                 | 0                  |
| 11WP56D <sup>†</sup>  | 5      | 25   | 33                 | 5    | 80   | 55              | 5                 | 60                 |
| 11WP56S <sup>†</sup>  | 5      | 5    | 0                  | 5    | 5    | 5               | 2                 | 100                |
| 11WP57                | 350    | 368  | 89                 | 250  | 490  | 450             | 5                 | 0                  |
| CG1 <sup>†</sup>      | 5      | 17   | 26                 | 5    | 70   | 28              | 6                 | 67                 |
| CG2 <sup>†</sup>      | 5      | 16   | 25                 | 5    | 60   | 33              | 5                 | 80                 |
| CG4 <sup>†</sup>      | 10     | 58   | 80                 | 5    | 190  | 135             | 5                 | 40                 |
| KC3 <sup>†</sup>      | 10     | 16   | 19                 | 5    | 50   | 30              | 5                 | 40                 |
| KC3A <sup>†</sup>     | 10     | 12   | 10                 | 5    | 30   | 20              | 5                 | 40                 |
| KC3PB <sup>†</sup>    | 5      | 9    | 7                  | 5    | 20   | 15              | 5                 | 60                 |
| KC13                  | 60     | 55   | 32                 | 5    | 90   | 80              | 5                 | 20                 |
| KC14                  | 70     | 86   | 42                 | 40   | 130  | 130             | 5                 | 0                  |
| KCF1 <sup>†</sup>     | 10     | 20   | 28                 | 5    | 70   | 40              | 5                 | 40                 |
| LIMESTONE             | 3500   | 3820 | 1599               | 1700 | 6000 | 5300            | 5                 | 0                  |
| M1 <sup>†</sup>       | 5      | 5    | ND                 | 5    | 5    | 5               | 1                 | 100                |
| RN029659 <sup>†</sup> | 20     | 20   | 15                 | 5    | 40   | 35              | 5                 | 40                 |
| RN029660 <sup>†</sup> | 5      | 15   | 15                 | 5    | 40   | 30              | 5                 | 60                 |
| W2R <sup>†</sup>      | 310    | 385  | 416                | 5    | 900  | 816             | 6                 | 33                 |
| WP5 <sup>†</sup>      | 5      | 18   | 31                 | 5    | 80   | 28              | 6                 | 83                 |
| WP13 <sup>†</sup>     | 5      | 5    | 0                  | 5    | 5    | 5               | 4                 | 100                |
| WP19 <sup>†</sup>     | 10     | 28   | 30                 | 5    | 70   | 60              | 5                 | 40                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

\* Number of samples and number used in statistical analysis

ND No data

Table D32 Concentration of oxidised nitrogen (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31                | 130    | 123  | 58                 | 40   | 190  | 177             | 7                 | 0                  |
| 10WP32                | 185    | 148  | 82                 | 5    | 230  | 217             | 7                 | 14                 |
| 10WP32PB              | 140    | 140  | 51                 | 90   | 205  | 187             | 7                 | 0                  |
| 10WP33                | 150    | 160  | 78                 | 50   | 300  | 206             | 8/7*              | 0                  |
| 10WP35N               | 115    | 119  | 26                 | 90   | 150  | 146             | 7                 | 0                  |
| 10WP35PB              | 90     | 102  | 28                 | 60   | 145  | 133             | 9                 | 0                  |
| 10WP35S               | 80     | 64   | 41                 | 5    | 105  | 101             | 7                 | 14                 |
| 10WP36N               | 300    | 334  | 183                | 90   | 620  | 526             | 7                 | 0                  |
| 10WP36PB              | 360    | 377  | 297                | 5    | 990  | 585             | 10                | 10                 |
| 10WP36S               | 370    | 381  | 116                | 260  | 600  | 443             | 7                 | 0                  |
| 10WP37                | 560    | 567  | 358                | 5    | 1100 | 857             | 7                 | 14                 |
| 10WP39 <sup>†</sup>   | 10     | 8    | 3                  | 5    | 10   | 10              | 7                 | 71                 |
| 10WP40 <sup>†</sup>   | 5      | 60   | 139                | 5    | 375  | 51              | 7                 | 57                 |
| 10WP41                | 20     | 20   | 9                  | 10   | 35   | 26              | 7                 | 0                  |
| 10WP42                | 30     | 32   | 20                 | 10   | 60   | 51              | 7                 | 14                 |
| 10WP44                | 25     | 25   | 21                 | 10   | 40   | 40              | 4/2*              | 25                 |
| 10WP46                | 55     | 44   | 20                 | 10   | 60   | 60              | 7                 | 0                  |
| 10WP47 <sup>†</sup>   | 5      | 7    | 3                  | 5    | 10   | 10              | 7                 | 86                 |
| 11CS10RD              | 120    | 132  | 45                 | 80   | 190  | 183             | 6                 | 0                  |
| 11CS10RS <sup>†</sup> | 5      | 9    | 8                  | 5    | 25   | 15              | 6                 | 83                 |
| 11WP4R                | 60     | 72   | 53                 | 5    | 160  | 118             | 6                 | 17                 |
| 11WP9RPB              | 135    | 101  | 74                 | 5    | 200  | 155             | 7                 | 14                 |
| 11WP9RS               | 135    | 126  | 47                 | 60   | 180  | 168             | 5                 | 0                  |
| 11WP11RD              | 130    | 104  | 59                 | 5    | 155  | 148             | 6                 | 17                 |
| 11WP11RS              | 155    | 132  | 79                 | 5    | 205  | 195             | 6                 | 17                 |
| 11WP15R               | 30     | 25   | 14                 | 5    | 40   | 37              | 6                 | 17                 |
| 11WP16R               | 48     | 48   | 26                 | 15   | 90   | 69              | 6                 | 0                  |
| 11WP43D               | 88     | 201  | 286                | 10   | 620  | 479             | 4                 | 0                  |
| 11WP50 <sup>†</sup>   | 10     | 9    | 3                  | 5    | 10   | 10              | 4                 | 75                 |
| 11WP51D <sup>†</sup>  | 8      | 10   | 6                  | 5    | 20   | 17              | 6                 | 67                 |
| 11WP51S <sup>†</sup>  | 10     | 63   | 85                 | 5    | 200  | 158             | 6                 | 50                 |
| 11WP52D               | 78     | 72   | 14                 | 50   | 85   | 82              | 6                 | 0                  |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D                | 53     | 44   | 26                 | 5    | 70   | 67              | 6                 | 17                 |
| 11WP53S                | 30     | 29   | 20                 | 5    | 60   | 46              | 6                 | 17                 |
| 11WP54D <sup>†</sup>   | 5      | 7    | 3                  | 5    | 10   | 10              | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 28     | 33   | 33                 | 5    | 95   | 57              | 6                 | 33                 |
| 11WP55 <sup>†</sup>    | 8      | 9    | 6                  | 5    | 20   | 13              | 6                 | 83                 |
| 11WP56D                | 128    | 105  | 47                 | 20   | 145  | 135             | 6                 | 0                  |
| 11WP56S <sup>†</sup>   | 10     | 25   | 30                 | 5    | 60   | 55              | 3                 | 33                 |
| 11WP57 <sup>†</sup>    | 5      | 7    | 3                  | 5    | 10   | 10              | 6                 | 100                |
| CG1                    | 913    | 838  | 225                | 390  | 980  | 977             | 6                 | 0                  |
| CG2                    | 850    | 824  | 390                | 405  | 1300 | 1200            | 5                 | 0                  |
| CG4                    | 48     | 45   | 14                 | 20   | 60   | 57              | 6                 | 0                  |
| KC3                    | 210    | 212  | 40                 | 160  | 270  | 248             | 5                 | 0                  |
| KC3A                   | 178    | 170  | 36                 | 100  | 205  | 195             | 6                 | 0                  |
| KC3PB                  | 178    | 158  | 43                 | 80   | 195  | 185             | 6                 | 0                  |
| KC13 <sup>†</sup>      | 10     | 31   | 36                 | 5    | 90   | 69              | 6                 | 50                 |
| KC14 <sup>†</sup>      | 5      | 7    | 3                  | 5    | 10   | 10              | 5                 | 100                |
| KCF1                   | 20     | 20   | 9                  | 10   | 30   | 30              | 6                 | 0                  |
| LIMESTONE <sup>†</sup> | 48     | 50   | 43                 | 5    | 120  | 85              | 6                 | 33                 |
| M1 <sup>†</sup>        | 10     | 12   | 8                  | 5    | 20   | 19              | 3                 | 33                 |
| RN029659 <sup>†</sup>  | 10     | 8    | 3                  | 5    | 10   | 10              | 6                 | 67                 |
| RN029660               | 418    | 395  | 77                 | 270  | 475  | 458             | 6                 | 0                  |
| W2R <sup>†</sup>       | 10     | 10   | 5                  | 5    | 20   | 13              | 6                 | 67                 |
| WP5 <sup>†</sup>       | 10     | 10   | 5                  | 5    | 20   | 11              | 7                 | 57                 |
| WP13                   | 100    | 93   | 30                 | 50   | 130  | 118             | 5                 | 0                  |
| WP19 <sup>†</sup>      | 10     | 9    | 4                  | 5    | 15   | 13              | 5                 | 60                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

\* Number of samples and number used in statistical analysis

Table D33 Concentration of total nitrogen (µg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 185    | 392  | 359                | 140  | 1000 | 769             | 6                 | 0                  |
| 10WP32   | 410    | 522  | 325                | 150  | 970  | 900             | 6                 | 0                  |
| 10WP32PB | 270    | 486  | 458                | 120  | 1300 | 994             | 7                 | 0                  |
| 10WP33   | 190    | 367  | 297                | 140  | 960  | 591             | 7                 | 0                  |
| 10WP35N  | 210    | 348  | 371                | 150  | 1100 | 519             | 6                 | 0                  |
| 10WP35PB | 160    | 273  | 273                | 90   | 890  | 461             | 9                 | 0                  |
| 10WP35S  | 230    | 408  | 540                | 100  | 1500 | 639             | 6                 | 0                  |
| 10WP36N  | 575    | 705  | 557                | 270  | 1800 | 995             | 6                 | 0                  |
| 10WP36PB | 540    | 585  | 265                | 190  | 990  | 880             | 10                | 0                  |
| 10WP36S  | 510    | 617  | 349                | 390  | 1300 | 810             | 6                 | 0                  |
| 10WP37   | 1000   | 1062 | 526                | 560  | 2000 | 1440            | 6                 | 0                  |
| 10WP39   | 100    | 190  | 267                | 10   | 780  | 240             | 7                 | 14                 |
| 10WP40   | 590    | 852  | 726                | 200  | 1800 | 1730            | 6                 | 0                  |
| 10WP41   | 375    | 452  | 240                | 160  | 820  | 701             | 6                 | 0                  |
| 10WP42   | 90     | 146  | 156                | 50   | 490  | 184             | 7                 | 0                  |
| 10WP44   | ND     | ND   | ND                 | ND   | ND   | ND              | 4/0*              | 0                  |
| 10WP46   | 140    | 236  | 292                | 30   | 860  | 383             | 7                 | 0                  |
| 10WP47   | 110    | 189  | 299                | 20   | 860  | 194             | 7                 | 0                  |
| 11CS10RD | 250    | 487  | 434                | 180  | 1200 | 955             | 6                 | 0                  |
| 11CS10RS | 110    | 392  | 499                | 50   | 1200 | 941             | 6                 | 0                  |
| 11WP4R   | 1500   | 1873 | 1686               | 170  | 4800 | 3330            | 6                 | 0                  |
| 11WP9RPB | 170    | 543  | 715                | 70   | 2000 | 1082            | 7                 | 0                  |
| 11WP9RS  | 180    | 440  | 545                | 100  | 1400 | 880             | 5                 | 0                  |
| 11WP11RD | 140    | 318  | 387                | 130  | 1100 | 519             | 6                 | 0                  |
| 11WP11RS | 270    | 533  | 567                | 160  | 1600 | 998             | 6                 | 0                  |
| 11WP15R  | 90     | 290  | 346                | 30   | 760  | 725             | 6                 | 0                  |
| 11WP16R  | 995    | 2260 | 2604               | 230  | 5800 | 5450            | 6                 | 0                  |
| 11WP43D  | 1300   | 1390 | 697                | 660  | 2300 | 2060            | 4                 | 0                  |
| 11WP50   | 565    | 538  | 416                | 70   | 950  | 911             | 4                 | 0                  |
| 11WP51D  | 125    | 438  | 766                | 90   | 2000 | 733             | 6                 | 0                  |
| 11WP51S  | 400    | 425  | 330                | 100  | 800  | 740             | 6/4*              | 0                  |
| 11WP52D  | 175    | 315  | 353                | 70   | 1000 | 566             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 100    | 303  | 410                | 50   | 1100 | 603             | 6                 | 0                  |
| 11WP53S   | 185    | 250  | 232                | 50   | 650  | 440             | 6                 | 0                  |
| 11WP54D   | 75     | 242  | 337                | 20   | 880  | 516             | 6                 | 0                  |
| 11WP54S   | 440    | 487  | 351                | 70   | 960  | 869             | 6                 | 0                  |
| 11WP55    | 930    | 1142 | 864                | 160  | 2400 | 2050            | 6                 | 0                  |
| 11WP56D   | 175    | 552  | 626                | 130  | 1500 | 1290            | 6                 | 0                  |
| 11WP56S   | 80     | 103  | 59                 | 60   | 170  | 161             | 3                 | 0                  |
| 11WP57    | 475    | 635  | 394                | 300  | 1300 | 1034            | 6                 | 0                  |
| CG1       | 1050   | 1213 | 447                | 900  | 2100 | 1470            | 6                 | 0                  |
| CG2       | 1400   | 1192 | 423                | 460  | 1500 | 1450            | 5                 | 0                  |
| CG4       | 385    | 677  | 813                | 60   | 2200 | 1318            | 6                 | 0                  |
| KC3       | 320    | 362  | 206                | 220  | 720  | 520             | 5                 | 0                  |
| KC3A      | 260    | 325  | 172                | 170  | 570  | 521             | 6                 | 0                  |
| KC3PB     | 190    | 243  | 121                | 160  | 480  | 326             | 6                 | 0                  |
| KC13      | 140    | 170  | 155                | 40   | 470  | 253             | 6                 | 0                  |
| KC14      | 180    | 404  | 509                | 80   | 1300 | 810             | 5                 | 0                  |
| KCF1      | 65     | 113  | 100                | 20   | 280  | 217             | 6                 | 0                  |
| LIMESTONE | 3500   | 4420 | 1766               | 2800 | 7100 | 6200            | 6/5*              | 0                  |
| M1        | 190    | 193  | 15                 | 180  | 210  | 208             | 3                 | 0                  |
| RN029659  | 355    | 532  | 564                | 40   | 1300 | 1160            | 6                 | 0                  |
| RN029660  | 475    | 740  | 475                | 390  | 1400 | 1330            | 6                 | 0                  |
| W2R       | 540    | 618  | 619                | 40   | 1600 | 1110            | 6                 | 0                  |
| WP5       | 55     | 218  | 385                | 10   | 1000 | 398             | 6                 | 17                 |
| WP13      | 140    | 312  | 442                | 70   | 1100 | 625             | 5                 | 0                  |
| WP19      | 620    | 486  | 387                | 40   | 950  | 820             | 5                 | 0                  |

† More than one-quarter of the samples are below the LOR

\* Number of samples and number used in statistical analysis

ND No data

Table D34 Concentration of sodium (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 22     | 22   | 4                  | 16   | 28   | 25              | 7                 | 0                  |
| 10WP32   | 238    | 216  | 52                 | 153  | 270  | 269             | 7                 | 0                  |
| 10WP32PB | 204    | 184  | 60                 | 63   | 247  | 215             | 7                 | 0                  |
| 10WP33   | 142    | 149  | 18                 | 123  | 168  | 168             | 7                 | 0                  |
| 10WP35N  | 247    | 232  | 43                 | 163  | 294  | 258             | 7                 | 0                  |
| 10WP35PB | 216    | 215  | 28                 | 180  | 257  | 240             | 9                 | 0                  |
| 10WP35S  | 227    | 225  | 22                 | 192  | 254  | 245             | 7                 | 0                  |
| 10WP36N  | 197    | 182  | 47                 | 79   | 214  | 213             | 7                 | 0                  |
| 10WP36PB | 193    | 193  | 18                 | 170  | 222  | 210             | 10                | 0                  |
| 10WP36S  | 294    | 297  | 29                 | 254  | 340  | 323             | 7                 | 0                  |
| 10WP37   | 340    | 325  | 48                 | 242  | 387  | 355             | 7                 | 0                  |
| 10WP39   | 93     | 95   | 14                 | 78   | 120  | 108             | 7                 | 0                  |
| 10WP40   | 1090   | 1095 | 387                | 638  | 1820 | 1298            | 7                 | 0                  |
| 10WP41   | 3790   | 3710 | 602                | 2910 | 4380 | 4335            | 7                 | 0                  |
| 10WP42   | 301    | 301  | 39                 | 262  | 364  | 330             | 7                 | 0                  |
| 10WP44   | 3505   | 3523 | 92                 | 3430 | 3650 | 3608            | 4                 | 0                  |
| 10WP46   | 268    | 275  | 34                 | 236  | 346  | 285             | 7                 | 0                  |
| 10WP47   | 38     | 40   | 13                 | 21   | 64   | 48              | 7                 | 0                  |
| 11CS10RD | 223    | 270  | 117                | 166  | 452  | 397             | 6                 | 0                  |
| 11CS10RS | 1330   | 1293 | 324                | 726  | 1650 | 1552            | 6                 | 0                  |
| 11WP4R   | 127    | 125  | 26                 | 78   | 158  | 145             | 6                 | 0                  |
| 11WP9RPB | 335    | 335  | 52                 | 259  | 405  | 392             | 7                 | 0                  |
| 11WP9RS  | 330    | 342  | 42                 | 288  | 395  | 383             | 5                 | 0                  |
| 11WP11RD | 141    | 137  | 13                 | 119  | 149  | 148             | 6                 | 0                  |
| 11WP11RS | 233    | 230  | 25                 | 194  | 256  | 255             | 6                 | 0                  |
| 11WP15R  | 1770   | 1730 | 129                | 1520 | 1880 | 1824            | 6                 | 0                  |
| 11WP16R  | 1580   | 1732 | 350                | 1430 | 2220 | 2157            | 6                 | 0                  |
| 11WP43D  | 882    | 899  | 631                | 221  | 1610 | 1493            | 4                 | 0                  |
| 11WP50   | 3395   | 3423 | 109                | 3330 | 3570 | 3531            | 4                 | 0                  |
| 11WP51D  | 64     | 61   | 8                  | 46   | 69   | 67              | 6                 | 0                  |
| 11WP51S  | 66     | 87   | 67                 | 11   | 181  | 164             | 6                 | 0                  |
| 11WP52D  | 112    | 109  | 18                 | 84   | 129  | 126             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 1925   | 1915 | 152                | 1680 | 2110 | 2047            | 6                 | 0                  |
| 11WP53S   | 2090   | 2090 | 65                 | 1990 | 2180 | 2145            | 6                 | 0                  |
| 11WP54D   | 4440   | 4498 | 387                | 4060 | 5180 | 4774            | 6                 | 0                  |
| 11WP54S   | 3380   | 3520 | 407                | 3150 | 4090 | 3971            | 6                 | 0                  |
| 11WP55    | 652    | 682  | 124                | 553  | 917  | 759             | 6                 | 0                  |
| 11WP56D   | 11     | 10   | 3                  | 5    | 14   | 13              | 6                 | 0                  |
| 11WP56S   | 70     | 69   | 47                 | 21   | 115  | 110             | 3                 | 0                  |
| 11WP57    | 2025   | 2015 | 56                 | 1910 | 2070 | 2056            | 6                 | 0                  |
| CG1       | 854    | 876  | 90                 | 793  | 1050 | 928             | 6                 | 0                  |
| CG2       | 84     | 86   | 21                 | 62   | 107  | 107             | 5                 | 0                  |
| CG4       | 427    | 421  | 50                 | 357  | 480  | 466             | 6                 | 0                  |
| KC3       | 15     | 16   | 4                  | 12   | 21   | 19              | 5                 | 0                  |
| KC3A      | 25     | 26   | 4                  | 21   | 31   | 29              | 6                 | 0                  |
| KC3PB     | 25     | 25   | 4                  | 20   | 29   | 28              | 6                 | 0                  |
| KC13      | 137    | 140  | 30                 | 107  | 175  | 171             | 6                 | 0                  |
| KC14      | 2230   | 2220 | 175                | 1940 | 2390 | 2365            | 5                 | 0                  |
| KCF1      | 674    | 698  | 67                 | 618  | 805  | 765             | 6                 | 0                  |
| LIMESTONE | 40     | 34   | 12                 | 15   | 44   | 43              | 6                 | 0                  |
| M1        | 15     | 16   | 3                  | 15   | 20   | 19              | 3                 | 0                  |
| RN029659  | 604    | 516  | 246                | 31   | 705  | 657             | 6                 | 0                  |
| RN029660  | 298    | 306  | 33                 | 272  | 357  | 342             | 6                 | 0                  |
| W2R       | 20     | 18   | 5                  | 10   | 24   | 22              | 6                 | 0                  |
| WP5       | 266    | 289  | 53                 | 229  | 362  | 343             | 7                 | 0                  |
| WP13      | 62     | 64   | 11                 | 55   | 85   | 72              | 6                 | 0                  |
| WP19      | 4670   | 4630 | 173                | 4360 | 4810 | 4770            | 5                 | 0                  |

Table D35 Concentration of nickel (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.50   | 0.57 | 0.19               | 0.50 | 1.00  | 0.55            | 7                 | 86                 |
| 10WP32 <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 7                 | 100                |
| 10WP32PB <sup>†</sup> | 0.50   | 0.71 | 0.57               | 0.50 | 2.00  | 0.65            | 7                 | 86                 |
| 10WP33 <sup>†</sup>   | 0.50   | 0.79 | 0.57               | 0.50 | 2.00  | 1.10            | 7                 | 71                 |
| 10WP35N <sup>†</sup>  | 0.50   | 0.71 | 0.57               | 0.50 | 2.00  | 0.65            | 7                 | 86                 |
| 10WP35PB <sup>†</sup> | 0.50   | 0.67 | 0.50               | 0.50 | 2.00  | 0.50            | 9                 | 89                 |
| 10WP35S <sup>†</sup>  | 0.50   | 1.07 | 1.02               | 0.50 | 3.00  | 2.10            | 7                 | 71                 |
| 10WP36N <sup>†</sup>  | 0.50   | 0.57 | 0.19               | 0.50 | 1.00  | 0.55            | 7                 | 86                 |
| 10WP36PB <sup>†</sup> | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 0.50   | 0.57 | 0.19               | 0.50 | 1.00  | 0.55            | 7                 | 86                 |
| 10WP37 <sup>†</sup>   | 0.50   | 0.57 | 0.19               | 0.50 | 1.00  | 0.55            | 7                 | 86                 |
| 10WP39 <sup>†</sup>   | 0.50   | 0.71 | 0.57               | 0.50 | 2.00  | 0.65            | 7                 | 86                 |
| 10WP40                | 2.00   | 2.71 | 1.80               | 1.00 | 6.00  | 4.20            | 7                 | 14                 |
| 10WP41 <sup>†</sup>   | 6.00   | 5.43 | 2.56               | 2.50 | 10.00 | 6.40            | 7                 | 43                 |
| 10WP42 <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 7                 | 100                |
| 10WP44 <sup>†</sup>   | 3.75   | 7.00 | 7.43               | 2.50 | 18.00 | 14.10           | 4                 | 50                 |
| 10WP46 <sup>†</sup>   | 0.50   | 0.79 | 0.76               | 0.50 | 2.50  | 0.70            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 1.00   | 1.08 | 0.49               | 0.50 | 2.00  | 1.30            | 6                 | 83                 |
| 11WP4R <sup>†</sup>   | 1.50   | 3.17 | 4.86               | 0.50 | 13.00 | 5.30            | 6                 | 33                 |
| 11WP9RPB <sup>†</sup> | 0.50   | 0.57 | 0.19               | 0.50 | 1.00  | 0.55            | 7                 | 86                 |
| 11WP9RS <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.50   | 0.75 | 0.61               | 0.50 | 2.00  | 0.95            | 6                 | 83                 |
| 11WP11RS <sup>†</sup> | 0.50   | 1.17 | 1.08               | 0.50 | 3.00  | 2.30            | 6                 | 67                 |
| 11WP15R <sup>†</sup>  | 1.00   | 1.00 | 0.00               | 1.00 | 1.00  | 1.00            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 2.50   | 2.50 | 0.00               | 2.50 | 2.50  | 2.50            | 6                 | 100                |
| 11WP43D               | 7.00   | 8.00 | 4.32               | 4.00 | 14.00 | 12.20           | 4                 | 0                  |
| 11WP50                | 10.50  | 9.75 | 1.89               | 7.00 | 11.00 | 11.00           | 4                 | 0                  |
| 11WP51D <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50  | 0.50            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.75   | 0.75 | 0.27               | 0.50 | 1.00  | 1.00            | 6                 | 67                 |
| 11WP52D <sup>†</sup>  | 0.50   | 2.33 | 4.25               | 0.50 | 11.00 | 4.00            | 6                 | 83                 |

| Site                   | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 1.00   | 1.00  | 0.00               | 1.00 | 1.00  | 1.00            | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 1.00   | 1.00  | 0.00               | 1.00 | 1.00  | 1.00            | 6                 | 100                |
| 11WP54D <sup>†</sup>   | 2.50   | 3.42  | 2.25               | 2.50 | 8.00  | 4.15            | 6                 | 83                 |
| 11WP54S                | 11.50  | 10.08 | 3.85               | 2.50 | 13.00 | 12.30           | 6                 | 17                 |
| 11WP55 <sup>†</sup>    | 0.75   | 1.58  | 1.53               | 0.50 | 4.00  | 3.30            | 6                 | 50                 |
| 11WP56D <sup>†</sup>   | 0.75   | 0.75  | 0.27               | 0.50 | 1.00  | 1.00            | 6                 | 50                 |
| 11WP56S <sup>†</sup>   | 0.50   | 1.17  | 1.15               | 0.50 | 2.50  | 2.30            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 2.50   | 2.83  | 1.08               | 2.00 | 5.00  | 3.25            | 6                 | 67                 |
| CG1 <sup>†</sup>       | 1.00   | 1.00  | 0.00               | 1.00 | 1.00  | 1.00            | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.50   | 0.75  | 0.61               | 0.50 | 2.00  | 0.95            | 6                 | 83                 |
| KC3 <sup>†</sup>       | 0.50   | 0.80  | 0.67               | 0.50 | 2.00  | 1.25            | 5                 | 80                 |
| KC3A                   | 2.00   | 2.00  | 0.00               | 2.00 | 2.00  | 2.00            | 6                 | 0                  |
| KC3PB                  | 9.50   | 11.33 | 4.63               | 7.00 | 18.00 | 16.60           | 6                 | 0                  |
| KC13 <sup>†</sup>      | 0.75   | 1.17  | 0.88               | 0.50 | 2.50  | 2.15            | 6                 | 67                 |
| KC14 <sup>†</sup>      | 2.50   | 2.40  | 0.22               | 2.00 | 2.50  | 2.50            | 5                 | 80                 |
| KCF1 <sup>†</sup>      | 0.50   | 0.83  | 0.61               | 0.50 | 2.00  | 1.30            | 6                 | 67                 |
| LIMESTONE <sup>†</sup> | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 6                 | 100                |
| W2R <sup>†</sup>       | 0.50   | 0.92  | 1.02               | 0.50 | 3.00  | 1.25            | 6                 | 83                 |
| WP5 <sup>†</sup>       | 1.00   | 1.21  | 1.25               | 0.50 | 4.00  | 1.30            | 7                 | 71                 |
| WP13 <sup>†</sup>      | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 6                 | 100                |
| WP19 <sup>†</sup>      | 5.00   | 4.50  | 1.12               | 2.50 | 5.00  | 5.00            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D36 Concentration of soluble reactive phosphorus (µg/L)

| Site                 | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31               | 70     | 64   | 29                 | 20   | 100  | 91              | 7                 | 0                  |
| 10WP32               | 60     | 67   | 24                 | 30   | 100  | 91              | 7                 | 0                  |
| 10WP32PB             | 70     | 66   | 17                 | 40   | 90   | 81              | 7                 | 0                  |
| 10WP33               | 60     | 57   | 18                 | 30   | 80   | 71              | 7                 | 0                  |
| 10WP35N              | 130    | 123  | 25                 | 80   | 150  | 141             | 7                 | 0                  |
| 10WP35PB             | 70     | 69   | 24                 | 30   | 100  | 94              | 9                 | 0                  |
| 10WP35S              | 70     | 73   | 36                 | 30   | 130  | 103             | 7                 | 0                  |
| 10WP36N              | 200    | 207  | 57                 | 150  | 310  | 256             | 7                 | 0                  |
| 10WP36PB             | 155    | 180  | 109                | 50   | 400  | 255             | 10                | 0                  |
| 10WP36S              | 170    | 199  | 69                 | 120  | 320  | 248             | 7                 | 0                  |
| 10WP37               | 110    | 109  | 46                 | 60   | 200  | 128             | 7                 | 0                  |
| 10WP39               | 10     | 14   | 8                  | 10   | 30   | 21              | 7                 | 0                  |
| 10WP40               | 40     | 50   | 38                 | 10   | 120  | 84              | 7                 | 0                  |
| 10WP41               | 10     | 9    | 2                  | 5    | 10   | 10              | 7                 | 14                 |
| 10WP42               | 20     | 21   | 11                 | 10   | 40   | 31              | 7                 | 0                  |
| 10WP44               | 30     | 30   | ND                 | 30   | 30   | 30              | 4/1*              | 0                  |
| 10WP46               | 20     | 21   | 12                 | 10   | 40   | 31              | 7                 | 0                  |
| 10WP47               | 20     | 26   | 13                 | 10   | 50   | 32              | 7                 | 0                  |
| 11CS10RD             | 65     | 58   | 24                 | 20   | 80   | 80              | 6                 | 0                  |
| 11CS10RS             | 30     | 38   | 21                 | 20   | 80   | 52              | 6                 | 0                  |
| 11WP4R               | 15     | 57   | 83                 | 10   | 220  | 115             | 6                 | 0                  |
| 11WP9RPB             | 70     | 70   | 10                 | 60   | 80   | 79              | 7/3*              | 0                  |
| 11WP9RS              | ND     | ND   | ND                 | ND   | ND   | ND              | 5/0*              | 0                  |
| 11WP11RD             | 80     | 80   | ND                 | 80   | 80   | 80              | 6/1*              | 0                  |
| 11WP11RS             | 100    | 100  | ND                 | 100  | 100  | 100             | 6/1*              | 0                  |
| 11WP15R              | 85     | 82   | 29                 | 40   | 120  | 106             | 6                 | 0                  |
| 11WP16R              | 30     | 224  | 311                | 10   | 650  | 615             | 6                 | 17                 |
| 11WP43D <sup>†</sup> | 5      | 9    | 8                  | 5    | 20   | 16              | 4                 | 75                 |
| 11WP50 <sup>†</sup>  | 5      | 6    | 3                  | 5    | 10   | 9               | 4                 | 75                 |
| 11WP51D              | 15     | 19   | 16                 | 5    | 50   | 29              | 6                 | 17                 |
| 11WP51S              | 40     | 36   | 23                 | 5    | 60   | 54              | 6/4*              | 17                 |
| 11WP52D              | 90     | 93   | 20                 | 70   | 120  | 113             | 6                 | 0                  |

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D               | 235    | 188  | 88                 | 70   | 260  | 253             | 6                 | 0                  |
| 11WP53S               | 205    | 175  | 90                 | 60   | 260  | 253             | 6                 | 0                  |
| 11WP54D               | 30     | 32   | 12                 | 20   | 50   | 43              | 6                 | 0                  |
| 11WP54S               | 45     | 57   | 43                 | 20   | 120  | 99              | 6                 | 0                  |
| 11WP55                | ND     | ND   | ND                 | ND   | ND   | ND              | 6/0*              | 0                  |
| 11WP56D <sup>†</sup>  | 20     | 22   | 18                 | 5    | 50   | 36              | 6                 | 33                 |
| 11WP56S               | 60     | 57   | 25                 | 30   | 80   | 78              | 3                 | 0                  |
| 11WP57                | 35     | 64   | 74                 | 10   | 210  | 112             | 6                 | 17                 |
| CG1                   | 60     | 57   | 24                 | 30   | 90   | 76              | 6                 | 0                  |
| CG2                   | 60     | 50   | 28                 | 20   | 80   | 75              | 5                 | 0                  |
| CG4                   | 35     | 35   | 21                 | 10   | 60   | 53              | 6                 | 0                  |
| KC3                   | 10     | 11   | 5                  | 5    | 20   | 15              | 5                 | 20                 |
| KC3A <sup>†</sup>     | 8      | 11   | 10                 | 5    | 30   | 16              | 6                 | 50                 |
| KC3PB <sup>†</sup>    | 5      | 7    | 3                  | 5    | 10   | 10              | 6                 | 67                 |
| KC13                  | 10     | 18   | 21                 | 5    | 60   | 25              | 6                 | 17                 |
| KC14 <sup>†</sup>     | 10     | 13   | 9                  | 5    | 25   | 23              | 5                 | 60                 |
| KCF1                  | 20     | 18   | 8                  | 10   | 30   | 23              | 6                 | 0                  |
| LIMESTONE             | 40     | 160  | 177                | 30   | 400  | 350             | 6/5*              | 0                  |
| M1 <sup>†</sup>       | 10     | 8    | 3                  | 5    | 10   | 10              | 3                 | 33                 |
| RN029659 <sup>†</sup> | 5      | 8    | 6                  | 5    | 20   | 13              | 6                 | 67                 |
| RN029660 <sup>†</sup> | 60     | 57   | 15                 | 40   | 80   | 66              | 6                 | 0                  |
| W2R <sup>†</sup>      | 15     | 22   | 19                 | 5    | 50   | 43              | 6                 | 33                 |
| WP5                   | 10     | 18   | 11                 | 5    | 30   | 30              | 7                 | 14                 |
| WP13                  | 10     | 10   | 0                  | 10   | 10   | 10              | 5                 | 0                  |
| WP19                  | 40     | 34   | 13                 | 20   | 50   | 45              | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

\* Number of samples and number used in statistical analysis

ND No data

Table D37 Concentration of total phosphorus (µg/L)

| Site                | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|---------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31              | 85     | 85   | 15                 | 60   | 100  | 100             | 6                 | 0                  |
| 10WP32              | 85     | 88   | 28                 | 60   | 130  | 116             | 6                 | 0                  |
| 10WP32PB            | 80     | 71   | 17                 | 40   | 90   | 81              | 7                 | 0                  |
| 10WP33              | 70     | 71   | 12                 | 60   | 90   | 81              | 7                 | 0                  |
| 10WP35N             | 135    | 132  | 12                 | 110  | 140  | 140             | 6                 | 0                  |
| 10WP35PB            | 100    | 94   | 34                 | 60   | 170  | 107             | 9                 | 0                  |
| 10WP35S             | 86     | 94   | 20                 | 80   | 130  | 109             | 6                 | 0                  |
| 10WP36N             | 210    | 208  | 27                 | 170  | 250  | 229             | 6                 | 0                  |
| 10WP36PB            | 200    | 210  | 97                 | 65   | 400  | 280             | 10                | 0                  |
| 10WP36S             | 240    | 243  | 48                 | 170  | 320  | 278             | 6                 | 0                  |
| 10WP37              | 115    | 128  | 39                 | 90   | 200  | 158             | 6                 | 0                  |
| 10WP39              | 20     | 84   | 175                | 10   | 480  | 75              | 7                 | 0                  |
| 10WP40              | 90     | 87   | 42                 | 30   | 140  | 126             | 6                 | 0                  |
| 10WP41 <sup>†</sup> | 10     | 13   | 10                 | 5    | 30   | 22              | 6                 | 33                 |
| 10WP42              | 20     | 26   | 14                 | 10   | 50   | 41              | 7                 | 0                  |
| 10WP44              | ND     | ND   | ND                 | ND   | ND   | ND              | 4/0*              | 0                  |
| 10WP46              | 23     | 25   | 13                 | 10   | 40   | 40              | 7                 | 0                  |
| 10WP47              | 30     | 31   | 10                 | 20   | 50   | 38              | 7                 | 0                  |
| 11CS10RD            | 70     | 66   | 26                 | 20   | 96   | 85              | 6                 | 0                  |
| 11CS10RS            | 40     | 58   | 52                 | 20   | 160  | 87              | 6                 | 0                  |
| 11WP4R              | 48     | 76   | 93                 | 10   | 260  | 127             | 6                 | 0                  |
| 11WP9RPB            | 100    | 97   | 25                 | 70   | 120  | 118             | 7/3*              | 0                  |
| 11WP9RS             | ND     | ND   | ND                 | ND   | ND   | ND              | 5/0*              | 0                  |
| 11WP11RD            | 94     | 94   | ND                 | 94   | 94   | 94              | 6/1*              | 0                  |
| 11WP11RS            | 190    | 190  | ND                 | 190  | 190  | 190             | 6/1*              | 0                  |
| 11WP15R             | 96     | 105  | 20                 | 90   | 140  | 126             | 6                 | 0                  |
| 11WP16R             | 49     | 227  | 309                | 5    | 650  | 615             | 6                 | 17                 |
| 11WP43D             | 30     | 33   | 26                 | 5    | 65   | 58              | 4                 | 25                 |
| 11WP50 <sup>†</sup> | 8      | 12   | 10                 | 5    | 26   | 21              | 4                 | 50                 |
| 11WP51D             | 25     | 32   | 30                 | 10   | 90   | 49              | 6                 | 0                  |
| 11WP51S             | 65     | 83   | 46                 | 50   | 150  | 126             | 6/4*              | 0                  |
| 11WP52D             | 120    | 133  | 35                 | 100  | 190  | 169             | 6                 | 0                  |

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D               | 245    | 240  | 18                 | 210  | 260  | 253             | 6                 | 0                  |
| 11WP53S               | 230    | 227  | 29                 | 180  | 260  | 253             | 6                 | 0                  |
| 11WP54D               | 25     | 28   | 15                 | 10   | 50   | 43              | 6                 | 0                  |
| 11WP54S               | 61     | 69   | 47                 | 20   | 120  | 120             | 6                 | 0                  |
| 11WP55                | ND     | ND   | ND                 | ND   | ND   | ND              | 6/0*              | 0                  |
| 11WP56D               | 145    | 157  | 65                 | 70   | 230  | 230             | 6                 | 0                  |
| 11WP56S               | 150    | 217  | 180                | 80   | 420  | 393             | 3                 | 0                  |
| 11WP57                | 40     | 138  | 266                | 5    | 680  | 239             | 6                 | 17                 |
| CG1                   | 73     | 75   | 9                  | 66   | 90   | 83              | 6                 | 0                  |
| CG2                   | 60     | 64   | 12                 | 48   | 80   | 75              | 5                 | 0                  |
| CG4                   | 57     | 87   | 86                 | 30   | 260  | 134             | 6                 | 0                  |
| KC3                   | 20     | 21   | 11                 | 10   | 34   | 32              | 5                 | 0                  |
| KC3A <sup>†</sup>     | 10     | 28   | 45                 | 5    | 120  | 49              | 6                 | 33                 |
| KC3PB <sup>†</sup>    | 20     | 22   | 18                 | 5    | 49   | 36              | 6                 | 33                 |
| KC13                  | 43     | 56   | 47                 | 20   | 150  | 80              | 6                 | 0                  |
| KC14 <sup>†</sup>     | 10     | 47   | 80                 | 5    | 190  | 108             | 5                 | 40                 |
| KCF1                  | 20     | 23   | 5                  | 20   | 30   | 30              | 6                 | 0                  |
| LIMESTONE             | 310    | 358  | 245                | 50   | 710  | 585             | 6/5*              | 0                  |
| M1                    | 10     | 10   | ND                 | 10   | 10   | 10              | 1                 | 0                  |
| RN029659 <sup>†</sup> | 13     | 13   | 9                  | 5    | 22   | 21              | 6                 | 50                 |
| RN029660              | 62     | 66   | 8                  | 60   | 80   | 73              | 6                 | 0                  |
| W2R                   | 45     | 40   | 20                 | 5    | 60   | 55              | 6                 | 17                 |
| WP5                   | 20     | 20   | 11                 | 10   | 30   | 30              | 6                 | 0                  |
| WP13                  | 20     | 20   | 12                 | 10   | 40   | 30              | 5                 | 0                  |
| WP19                  | 40     | 64   | 56                 | 20   | 160  | 110             | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

\* Number of samples and number used in statistical analysis

ND No data

Table D38 Concentration of lead (µg/L)

| Site                  | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31                | 0.20   | 0.62  | 0.72               | 0.05 | 1.80  | 1.53            | 7                 | 14                 |
| 10WP32                | 0.60   | 0.59  | 0.50               | 0.05 | 1.60  | 0.79            | 7                 | 14                 |
| 10WP32PB <sup>†</sup> | 0.20   | 0.34  | 0.36               | 0.05 | 0.90  | 0.81            | 7                 | 29                 |
| 10WP33 <sup>†</sup>   | 0.20   | 2.56  | 4.73               | 0.05 | 13.00 | 3.91            | 7                 | 29                 |
| 10WP35N <sup>†</sup>  | 0.10   | 0.43  | 0.87               | 0.05 | 2.40  | 0.42            | 7                 | 29                 |
| 10WP35PB <sup>†</sup> | 0.20   | 0.16  | 0.11               | 0.05 | 0.30  | 0.27            | 9                 | 44                 |
| 10WP35S <sup>†</sup>  | 0.20   | 0.40  | 0.59               | 0.05 | 1.70  | 0.53            | 7                 | 29                 |
| 10WP36N <sup>†</sup>  | 0.10   | 0.55  | 1.17               | 0.05 | 3.20  | 0.59            | 7                 | 43                 |
| 10WP36PB <sup>†</sup> | 0.08   | 0.27  | 0.51               | 0.05 | 1.70  | 0.25            | 10                | 50                 |
| 10WP36S <sup>†</sup>  | 0.30   | 0.77  | 1.56               | 0.05 | 4.30  | 0.70            | 7                 | 29                 |
| 10WP37                | 0.10   | 13.99 | 36.60              | 0.05 | 97.00 | 9.97            | 7                 | 14                 |
| 10WP39                | 0.10   | 0.47  | 0.63               | 0.10 | 1.80  | 0.81            | 7                 | 0                  |
| 10WP40 <sup>†</sup>   | 0.20   | 0.22  | 0.12               | 0.05 | 0.40  | 0.31            | 7                 | 29                 |
| 10WP41 <sup>†</sup>   | 1.00   | 6.79  | 15.97              | 0.25 | 43.00 | 5.47            | 7                 | 29                 |
| 10WP42 <sup>†</sup>   | 0.10   | 0.33  | 0.52               | 0.05 | 1.50  | 0.42            | 7                 | 29                 |
| 10WP44 <sup>†</sup>   | 0.25   | 0.29  | 0.23               | 0.05 | 0.60  | 0.50            | 4                 | 75                 |
| 10WP46 <sup>†</sup>   | 0.20   | 0.23  | 0.23               | 0.05 | 0.70  | 0.34            | 7                 | 57                 |
| 10WP47                | 0.20   | 0.32  | 0.27               | 0.05 | 0.70  | 0.61            | 7                 | 14                 |
| 11CS10RD              | 0.80   | 1.03  | 0.79               | 0.50 | 2.60  | 1.48            | 6                 | 0                  |
| 11CS10RS              | 0.25   | 1.53  | 2.08               | 0.10 | 4.60  | 4.04            | 6                 | 17                 |
| 11WP4R <sup>†</sup>   | 0.20   | 0.75  | 0.97               | 0.05 | 2.10  | 1.96            | 6                 | 33                 |
| 11WP9RPB              | 0.20   | 0.26  | 0.08               | 0.20 | 0.40  | 0.31            | 7                 | 0                  |
| 11WP9RS <sup>†</sup>  | 0.30   | 0.28  | 0.24               | 0.05 | 0.60  | 0.50            | 5                 | 40                 |
| 11WP11RD              | 0.20   | 0.28  | 0.23               | 0.10 | 0.70  | 0.49            | 6                 | 0                  |
| 11WP11RS              | 0.40   | 0.40  | 0.24               | 0.10 | 0.80  | 0.59            | 6                 | 0                  |
| 11WP15R               | 0.95   | 1.12  | 0.87               | 0.20 | 2.20  | 2.13            | 6                 | 0                  |
| 11WP16R <sup>†</sup>  | 0.25   | 0.60  | 0.70               | 0.25 | 2.00  | 1.02            | 6                 | 67                 |
| 11WP43D               | 1.35   | 1.05  | 0.64               | 0.10 | 1.40  | 1.40            | 4                 | 25                 |
| 11WP50 <sup>†</sup>   | 0.48   | 0.58  | 0.41               | 0.25 | 1.10  | 0.98            | 4                 | 50                 |
| 11WP51D               | 0.90   | 1.23  | 1.37               | 0.05 | 3.80  | 2.19            | 6                 | 17                 |
| 11WP51S               | 1.00   | 15.37 | 31.89              | 0.10 | 80.00 | 30.93           | 6                 | 0                  |
| 11WP52D               | 0.40   | 0.43  | 0.20               | 0.20 | 0.70  | 0.63            | 6                 | 0                  |

| Site                  | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D               | 0.70   | 0.62  | 0.21               | 0.30 | 0.80  | 0.80            | 6                 | 0                  |
| 11WP53S               | 0.30   | 0.42  | 0.18               | 0.30 | 0.70  | 0.63            | 6                 | 0                  |
| 11WP54D <sup>†</sup>  | 0.50   | 1.95  | 3.17               | 0.25 | 8.30  | 3.82            | 6                 | 33                 |
| 11WP54S <sup>†</sup>  | 0.43   | 0.46  | 0.24               | 0.25 | 0.80  | 0.66            | 6                 | 50                 |
| 11WP55                | 0.50   | 1.45  | 2.32               | 0.10 | 6.10  | 2.74            | 6                 | 0                  |
| 11WP56D               | 7.25   | 10.53 | 11.00              | 0.30 | 26.00 | 23.20           | 6                 | 0                  |
| 11WP56S <sup>†</sup>  | 0.20   | 0.48  | 0.63               | 0.05 | 1.20  | 1.10            | 3                 | 33                 |
| 11WP57 <sup>†</sup>   | 0.55   | 0.50  | 0.21               | 0.25 | 0.70  | 0.70            | 6                 | 33                 |
| CG1                   | 1.00   | 1.92  | 1.80               | 0.40 | 4.60  | 4.04            | 6                 | 0                  |
| CG2                   | 1.20   | 1.30  | 0.97               | 0.10 | 2.80  | 2.10            | 5                 | 0                  |
| CG4                   | 0.60   | 4.95  | 7.01               | 0.20 | 14.00 | 14.00           | 6                 | 0                  |
| KC3                   | 0.20   | 0.31  | 0.39               | 0.05 | 1.00  | 0.60            | 5                 | 20                 |
| KC3A                  | 0.45   | 1.07  | 1.56               | 0.20 | 4.20  | 1.89            | 6                 | 0                  |
| KC3PB                 | 1.70   | 2.33  | 2.27               | 0.05 | 5.80  | 4.68            | 6                 | 17                 |
| KC13                  | 2.85   | 4.57  | 5.33               | 0.20 | 13.00 | 9.71            | 6                 | 0                  |
| KC14                  | 1.30   | 3.25  | 4.92               | 0.25 | 12.00 | 6.80            | 5                 | 20                 |
| KCF1                  | 0.40   | 2.20  | 4.31               | 0.30 | 11.00 | 3.79            | 6                 | 0                  |
| LIMESTONE             | 0.75   | 2.63  | 3.84               | 0.20 | 9.90  | 5.77            | 6                 | 0                  |
| M1 <sup>†</sup>       | 0.05   | 0.10  | 0.09               | 0.05 | 0.20  | 0.19            | 3                 | 67                 |
| RN029659 <sup>†</sup> | 0.10   | 0.30  | 0.49               | 0.05 | 1.30  | 0.53            | 6                 | 33                 |
| RN029660 <sup>†</sup> | 0.15   | 1.37  | 3.01               | 0.05 | 7.50  | 2.46            | 6                 | 33                 |
| W2R <sup>†</sup>      | 0.13   | 0.28  | 0.34               | 0.05 | 0.90  | 0.55            | 6                 | 50                 |
| WP5                   | 0.40   | 0.52  | 0.36               | 0.05 | 1.10  | 0.83            | 7                 | 14                 |
| WP13                  | 0.50   | 0.92  | 0.88               | 0.20 | 2.50  | 1.73            | 6                 | 0                  |
| WP19 <sup>†</sup>     | 0.50   | 1.14  | 0.90               | 0.50 | 2.40  | 2.10            | 5                 | 40                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D39 Concentration of pH (Laboratory)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 7.1    | 7.2  | 0.5                | 6.7  | 8.2  | 7.2             | 7                 | 0                  |
| 10WP32   | 7.6    | 7.7  | 0.2                | 7.3  | 8.0  | 7.8             | 7                 | 0                  |
| 10WP32PB | 7.7    | 7.7  | 0.3                | 7.5  | 8.4  | 7.8             | 7                 | 0                  |
| 10WP33   | 7.3    | 7.5  | 0.3                | 7.3  | 8.2  | 7.7             | 7                 | 0                  |
| 10WP35N  | 7.8    | 7.8  | 0.2                | 7.7  | 8.2  | 7.9             | 7                 | 0                  |
| 10WP35PB | 7.5    | 7.6  | 0.2                | 7.3  | 8.1  | 7.7             | 9                 | 0                  |
| 10WP35S  | 7.6    | 7.6  | 0.1                | 7.5  | 7.7  | 7.7             | 7                 | 0                  |
| 10WP36N  | 7.8    | 7.8  | 0.1                | 7.7  | 8.0  | 7.8             | 7                 | 0                  |
| 10WP36PB | 7.9    | 8.0  | 0.4                | 7.5  | 8.8  | 8.3             | 10                | 0                  |
| 10WP36S  | 7.9    | 7.9  | 0.2                | 7.7  | 8.3  | 8.1             | 7                 | 0                  |
| 10WP37   | 7.7    | 7.7  | 0.1                | 7.5  | 7.9  | 7.8             | 7                 | 0                  |
| 10WP39   | 7.5    | 7.6  | 0.2                | 7.5  | 8.0  | 7.6             | 7                 | 0                  |
| 10WP40   | 7.6    | 7.6  | 0.2                | 7.3  | 7.9  | 7.7             | 7                 | 0                  |
| 10WP41   | 7.3    | 7.4  | 0.2                | 7.2  | 7.7  | 7.4             | 7                 | 0                  |
| 10WP42   | 7.5    | 7.6  | 0.3                | 7.4  | 8.2  | 7.6             | 7                 | 0                  |
| 10WP44   | 7.3    | 7.3  | 0.1                | 7.2  | 7.5  | 7.4             | 4                 | 0                  |
| 10WP46   | 7.5    | 7.5  | 0.4                | 7.0  | 8.2  | 7.6             | 7                 | 0                  |
| 10WP47   | 7.5    | 7.6  | 0.3                | 7.3  | 8.3  | 7.7             | 7                 | 0                  |
| 11CS10RD | 7.4    | 7.4  | 0.1                | 7.3  | 7.5  | 7.5             | 6                 | 0                  |
| 11CS10RS | 7.5    | 7.5  | 0.2                | 7.3  | 7.8  | 7.7             | 6                 | 0                  |
| 11WP4R   | 7.5    | 7.4  | 0.1                | 7.2  | 7.6  | 7.5             | 6                 | 0                  |
| 11WP9RPB | 7.7    | 7.7  | 0.1                | 7.5  | 7.8  | 7.8             | 7                 | 0                  |
| 11WP9RS  | 7.7    | 7.7  | 0.1                | 7.5  | 7.9  | 7.8             | 5                 | 0                  |
| 11WP11RD | 7.7    | 7.7  | 0.1                | 7.5  | 7.8  | 7.7             | 6                 | 0                  |
| 11WP11RS | 8.0    | 8.0  | 0.2                | 7.6  | 8.3  | 8.2             | 6                 | 0                  |
| 11WP15R  | 7.6    | 7.6  | 0.1                | 7.5  | 7.7  | 7.6             | 6                 | 0                  |
| 11WP16R  | 7.3    | 7.3  | 0.1                | 7.1  | 7.4  | 7.3             | 6                 | 0                  |
| 11WP43D  | 7.3    | 7.2  | 0.2                | 7.0  | 7.4  | 7.4             | 4                 | 0                  |
| 11WP50   | 7.1    | 7.1  | 0.1                | 7.0  | 7.3  | 7.2             | 4                 | 0                  |
| 11WP51D  | 7.7    | 7.7  | 0.1                | 7.6  | 7.7  | 7.7             | 6                 | 0                  |
| 11WP51S  | 7.5    | 7.6  | 0.2                | 7.4  | 7.9  | 7.6             | 6                 | 0                  |
| 11WP52D  | 7.3    | 7.4  | 0.3                | 7.1  | 7.9  | 7.6             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 7.9    | 7.9  | 0.0                | 7.8  | 7.9  | 7.9             | 6                 | 0                  |
| 11WP53S   | 8.0    | 8.0  | 0.1                | 7.9  | 8.1  | 8.0             | 6                 | 0                  |
| 11WP54D   | 7.1    | 7.1  | 0.1                | 7.0  | 7.2  | 7.1             | 6                 | 0                  |
| 11WP54S   | 7.5    | 7.4  | 0.1                | 7.2  | 7.6  | 7.5             | 6                 | 0                  |
| 11WP55    | 7.3    | 7.1  | 0.5                | 6.1  | 7.4  | 7.3             | 6                 | 0                  |
| 11WP56D   | 6.2    | 6.1  | 0.5                | 5.5  | 6.9  | 6.6             | 6                 | 0                  |
| 11WP56S   | 7.2    | 7.1  | 0.2                | 6.9  | 7.3  | 7.3             | 3                 | 0                  |
| 11WP57    | 7.0    | 7.1  | 0.4                | 6.8  | 7.8  | 7.3             | 6                 | 0                  |
| CG1       | 7.5    | 7.5  | 0.0                | 7.5  | 7.6  | 7.5             | 6                 | 0                  |
| CG2       | 7.7    | 7.8  | 0.1                | 7.7  | 7.9  | 7.9             | 5                 | 0                  |
| CG4       | 7.6    | 7.6  | 0.2                | 7.4  | 7.8  | 7.8             | 6                 | 0                  |
| KC3       | 6.8    | 6.7  | 0.4                | 6.0  | 7.2  | 7.1             | 5                 | 0                  |
| KC3A      | 6.2    | 6.3  | 0.2                | 6.0  | 6.5  | 6.5             | 6                 | 0                  |
| KC3PB     | 6.2    | 6.1  | 0.2                | 5.8  | 6.3  | 6.2             | 6                 | 0                  |
| KC13      | 7.7    | 7.6  | 0.3                | 7.1  | 7.8  | 7.8             | 6                 | 0                  |
| KC14      | 7.2    | 7.1  | 0.2                | 6.7  | 7.2  | 7.2             | 5                 | 0                  |
| KCF1      | 7.6    | 7.5  | 0.2                | 7.2  | 7.7  | 7.6             | 6                 | 0                  |
| LIMESTONE | 7.7    | 7.7  | 0.2                | 7.5  | 8.1  | 7.9             | 6                 | 0                  |
| M1        | 8.3    | 8.3  | 0.2                | 8.1  | 8.4  | 8.4             | 3                 | 0                  |
| RN029659  | 7.7    | 7.7  | 0.5                | 7.2  | 8.6  | 8.0             | 6                 | 0                  |
| RN029660  | 7.9    | 7.8  | 0.2                | 7.5  | 7.9  | 7.9             | 6                 | 0                  |
| W2R       | 7.7    | 7.7  | 0.2                | 7.5  | 8.0  | 7.8             | 6                 | 0                  |
| WP5       | 7.3    | 7.3  | 0.3                | 6.8  | 7.7  | 7.4             | 7                 | 0                  |
| WP13      | 7.5    | 7.6  | 0.2                | 7.4  | 7.8  | 7.7             | 5                 | 0                  |
| WP19      | 7.2    | 7.2  | 0.1                | 7.2  | 7.3  | 7.3             | 5                 | 0                  |

Table D40 Concentration of antimony ( $\mu\text{g/L}$ )

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 7                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.09 | 0.09               | 0.05 | 0.30 | 0.08            | 7                 | 86                 |
| 10WP32PB <sup>†</sup> | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.10            | 7                 | 71                 |
| 10WP33 <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.06            | 7                 | 86                 |
| 10WP35N <sup>†</sup>  | 0.05   | 0.07 | 0.06               | 0.05 | 0.20 | 0.07            | 7                 | 86                 |
| 10WP35PB <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.08 | 0.06               | 0.05 | 0.20 | 0.11            | 7                 | 71                 |
| 10WP36N <sup>†</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.10            | 7                 | 71                 |
| 10WP36PB <sup>†</sup> | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.05            | 10                | 90                 |
| 10WP36S <sup>†</sup>  | 0.05   | 0.08 | 0.06               | 0.05 | 0.20 | 0.11            | 7                 | 71                 |
| 10WP37 <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.06            | 7                 | 86                 |
| 10WP39 <sup>††</sup>  | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.06            | 7                 | 86                 |
| 10WP40 <sup>†</sup>   | 0.20   | 0.24 | 0.17               | 0.10 | 0.60 | 0.33            | 7                 | 29                 |
| 10WP41 <sup>†</sup>   | 0.25   | 0.29 | 0.09               | 0.25 | 0.50 | 0.28            | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.06            | 7                 | 86                 |
| 10WP44 <sup>†</sup>   | 0.25   | 0.29 | 0.08               | 0.25 | 0.40 | 0.36            | 4                 | 75                 |
| 10WP46 <sup>†</sup>   | 0.05   | 0.09 | 0.07               | 0.05 | 0.25 | 0.12            | 7                 | 86                 |
| 10WP47 <sup>†</sup>   | 0.05   | 0.06 | 0.02               | 0.05 | 0.10 | 0.06            | 7                 | 86                 |
| 11CS10RD <sup>†</sup> | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.15   | 0.17 | 0.08               | 0.10 | 0.30 | 0.23            | 6                 | 50                 |
| 11WP4R <sup>†</sup>   | 0.05   | 0.20 | 0.30               | 0.05 | 0.80 | 0.38            | 6                 | 67                 |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.11 | 0.13               | 0.05 | 0.40 | 0.13            | 7                 | 57                 |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.08   | 0.08 | 0.03               | 0.05 | 0.10 | 0.10            | 6                 | 50                 |
| 11WP11RS <sup>†</sup> | 0.08   | 0.14 | 0.14               | 0.05 | 0.40 | 0.26            | 6                 | 50                 |
| 11WP15R <sup>†</sup>  | 0.10   | 0.13 | 0.08               | 0.10 | 0.30 | 0.16            | 6                 | 83                 |
| 11WP16R <sup>†</sup>  | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.38   | 0.46 | 0.32               | 0.20 | 0.90 | 0.78            | 4                 | 50                 |
| 11WP50 <sup>†</sup>   | 0.25   | 0.34 | 0.18               | 0.25 | 0.60 | 0.50            | 4                 | 75                 |
| 11WP51D <sup>†</sup>  | 0.05   | 0.12 | 0.11               | 0.05 | 0.30 | 0.23            | 6                 | 67                 |
| 11WP51S <sup>†</sup>  | 0.20   | 0.25 | 0.21               | 0.05 | 0.60 | 0.46            | 6                 | 33                 |
| 11WP52D <sup>†</sup>  | 0.05   | 0.07 | 0.03               | 0.05 | 0.10 | 0.10            | 6                 | 83                 |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.10   | 0.15 | 0.12               | 0.10 | 0.40 | 0.19            | 6                 | 83                 |
| 11WP53S <sup>†</sup>   | 0.10   | 0.23 | 0.28               | 0.10 | 0.80 | 0.38            | 6                 | 67                 |
| 11WP54D <sup>†</sup>   | 0.25   | 0.31 | 0.14               | 0.25 | 0.60 | 0.36            | 6                 | 83                 |
| 11WP54S <sup>†</sup>   | 0.25   | 0.25 | 0.00               | 0.25 | 0.25 | 0.25            | 6                 | 100                |
| 11WP55                 | 0.25   | 0.72 | 1.08               | 0.20 | 2.90 | 1.22            | 6                 | 0                  |
| 11WP56D <sup>†</sup>   | 0.05   | 0.12 | 0.11               | 0.05 | 0.30 | 0.23            | 6                 | 67                 |
| 11WP56S <sup>†</sup>   | 0.30   | 0.35 | 0.13               | 0.25 | 0.50 | 0.48            | 3                 | 33                 |
| 11WP57 <sup>†</sup>    | 0.25   | 0.22 | 0.08               | 0.05 | 0.25 | 0.25            | 6                 | 100                |
| CG1 <sup>†</sup>       | 0.10   | 0.10 | 0.00               | 0.10 | 0.10 | 0.10            | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05   | 0.13 | 0.18               | 0.05 | 0.50 | 0.22            | 6                 | 67                 |
| KC3 <sup>†</sup>       | 0.05   | 0.11 | 0.11               | 0.05 | 0.30 | 0.20            | 5                 | 60                 |
| KC3A <sup>†</sup>      | 0.08   | 0.14 | 0.14               | 0.05 | 0.40 | 0.26            | 6                 | 50                 |
| KC3PB <sup>†</sup>     | 0.05   | 0.10 | 0.10               | 0.05 | 0.30 | 0.16            | 6                 | 67                 |
| KC13 <sup>†</sup>      | 0.33   | 0.69 | 1.03               | 0.05 | 2.70 | 1.30            | 6                 | 50                 |
| KC14 <sup>†</sup>      | 0.25   | 0.21 | 0.09               | 0.05 | 0.25 | 0.25            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.08   | 0.11 | 0.10               | 0.05 | 0.30 | 0.16            | 6                 | 50                 |
| LIMESTONE <sup>†</sup> | 0.08   | 0.11 | 0.07               | 0.05 | 0.20 | 0.20            | 6                 | 50                 |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05 | 0.05            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.05   | 0.08 | 0.06               | 0.05 | 0.20 | 0.10            | 6                 | 83                 |
| RN029660 <sup>†</sup>  | 0.05   | 0.08 | 0.06               | 0.05 | 0.20 | 0.13            | 6                 | 67                 |
| W2R <sup>†</sup>       | 0.10   | 0.13 | 0.06               | 0.05 | 0.20 | 0.20            | 6                 | 17                 |
| WP5 <sup>†</sup>       | 0.10   | 0.15 | 0.12               | 0.05 | 0.40 | 0.22            | 7                 | 43                 |
| WP13                   | 0.20   | 0.25 | 0.12               | 0.20 | 0.50 | 0.29            | 6                 | 0                  |
| WP19 <sup>†</sup>      | 0.50   | 0.45 | 0.11               | 0.25 | 0.50 | 0.50            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D41 Concentration of selenium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 10WP32                | 3.00   | 3.43 | 1.27               | 2.00 | 5.00 | 5.00            | 7                 | 0                  |
| 10WP32PB              | 3.00   | 3.07 | 1.69               | 0.50 | 6.00 | 4.20            | 7                 | 14                 |
| 10WP33                | 3.00   | 2.57 | 0.53               | 2.00 | 3.00 | 3.00            | 7                 | 0                  |
| 10WP35N <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 9                 | 100                |
| 10WP35S <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 10WP37                | 6.00   | 5.71 | 1.11               | 4.00 | 7.00 | 7.00            | 7                 | 0                  |
| 10WP39 <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 1.00   | 0.79 | 0.27               | 0.50 | 1.00 | 1.00            | 7                 | 100                |
| 10WP41 <sup>†</sup>   | 2.50   | 2.57 | 1.30               | 0.50 | 5.00 | 2.75            | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.50   | 0.64 | 0.24               | 0.50 | 1.00 | 1.00            | 7                 | 71                 |
| 10WP44 <sup>†</sup>   | 1.50   | 1.50 | 1.15               | 0.50 | 2.50 | 2.50            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.50   | 0.79 | 0.76               | 0.50 | 2.50 | 0.70            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 7                 | 100                |
| 11CS10RD              | 2.00   | 2.17 | 0.41               | 2.00 | 3.00 | 2.30            | 6                 | 0                  |
| 11CS10RS <sup>†</sup> | 1.00   | 0.92 | 0.20               | 0.50 | 1.00 | 1.00            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 6                 | 100                |
| 11WP9RPB              | 4.00   | 4.29 | 0.95               | 3.00 | 6.00 | 5.10            | 7                 | 0                  |
| 11WP9RS               | 5.00   | 5.00 | 0.71               | 4.00 | 6.00 | 5.50            | 5                 | 0                  |
| 11WP11RD <sup>†</sup> | 1.00   | 1.00 | 0.55               | 0.50 | 2.00 | 1.30            | 6                 | 33                 |
| 11WP11RS <sup>†</sup> | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 6                 | 100                |
| 11WP15R               | 4.50   | 4.83 | 0.98               | 4.00 | 6.00 | 6.00            | 6                 | 0                  |
| 11WP16R <sup>†</sup>  | 2.50   | 2.17 | 0.82               | 0.50 | 2.50 | 2.50            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.75   | 1.75 | 2.18               | 0.50 | 5.00 | 3.80            | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 2.50   | 2.00 | 1.00               | 0.50 | 2.50 | 2.50            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.50   | 0.50 | 0.00               | 0.50 | 0.50 | 0.50            | 6                 | 100                |
| 11WP51S <sup>†</sup>  | 0.50   | 0.58 | 0.20               | 0.50 | 1.00 | 0.65            | 6                 | 100                |
| 11WP52D               | 1.00   | 1.17 | 0.41               | 1.00 | 2.00 | 1.30            | 6                 | 17                 |

| Site                   | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 11WP53D                | 6.00   | 6.50  | 0.84               | 6.00  | 8.00  | 7.30            | 6                 | 0                  |
| 11WP53S                | 6.00   | 6.17  | 1.33               | 4.00  | 8.00  | 7.30            | 6                 | 0                  |
| 11WP54D                | 7.00   | 7.00  | 0.89               | 6.00  | 8.00  | 8.00            | 6                 | 0                  |
| 11WP54S <sup>†</sup>   | 2.50   | 2.92  | 1.02               | 2.50  | 5.00  | 3.25            | 6                 | 83                 |
| 11WP55 <sup>†</sup>    | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| 11WP56D <sup>†</sup>   | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.50   | 1.17  | 1.15               | 0.50  | 2.50  | 2.30            | 3                 | 100                |
| 11WP57 <sup>†</sup>    | 2.50   | 1.83  | 1.03               | 0.50  | 2.50  | 2.50            | 6                 | 100                |
| CG1                    | 38.50  | 36.67 | 8.69               | 24.00 | 47.00 | 44.20           | 6                 | 0                  |
| CG2                    | 2.00   | 2.40  | 0.55               | 2.00  | 3.00  | 3.00            | 5                 | 0                  |
| CG4 <sup>†</sup>       | 1.00   | 1.00  | 0.55               | 0.50  | 2.00  | 1.30            | 6                 | 33                 |
| KC3 <sup>†</sup>       | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| KC3PB <sup>†</sup>     | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.50   | 0.83  | 0.82               | 0.50  | 2.50  | 1.10            | 6                 | 100                |
| KC14 <sup>†</sup>      | 2.50   | 2.00  | 0.87               | 0.50  | 2.50  | 2.50            | 5                 | 80                 |
| KCF1                   | 6.50   | 6.17  | 1.17               | 4.00  | 7.00  | 7.00            | 6                 | 0                  |
| LIMESTONE <sup>†</sup> | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| W2R <sup>†</sup>       | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.50   | 0.64  | 0.24               | 0.50  | 1.00  | 1.00            | 7                 | 100                |
| WP13 <sup>†</sup>      | 0.50   | 0.50  | 0.00               | 0.50  | 0.50  | 0.50            | 6                 | 100                |
| WP19 <sup>†</sup>      | 5.00   | 3.70  | 1.86               | 1.00  | 5.00  | 5.00            | 5                 | 80                 |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D42 Concentration of silicon (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 35     | 36   | 2                  | 33   | 40   | 37              | 7                 | 0                  |
| 10WP32   | 39     | 36   | 8                  | 19   | 43   | 41              | 7                 | 0                  |
| 10WP32PB | 39     | 33   | 11                 | 17   | 43   | 41              | 7                 | 0                  |
| 10WP33   | 32     | 32   | 2                  | 28   | 34   | 33              | 7                 | 0                  |
| 10WP35N  | 29     | 27   | 7                  | 11   | 32   | 31              | 7                 | 0                  |
| 10WP35PB | 29     | 29   | 3                  | 23   | 32   | 31              | 9                 | 0                  |
| 10WP35S  | 28     | 26   | 6                  | 13   | 30   | 30              | 7                 | 0                  |
| 10WP36N  | 38     | 35   | 10                 | 15   | 44   | 41              | 7                 | 0                  |
| 10WP36PB | 37     | 35   | 6                  | 20   | 41   | 38              | 10                | 0                  |
| 10WP36S  | 39     | 39   | 3                  | 34   | 44   | 41              | 7                 | 0                  |
| 10WP37   | 31     | 32   | 2                  | 30   | 36   | 33              | 7                 | 0                  |
| 10WP39   | 34     | 34   | 2                  | 31   | 38   | 35              | 7                 | 0                  |
| 10WP40   | 35     | 35   | 5                  | 29   | 40   | 39              | 7                 | 0                  |
| 10WP41   | 16     | 16   | 3                  | 11   | 20   | 18              | 7                 | 0                  |
| 10WP42   | 38     | 37   | 8                  | 21   | 44   | 42              | 7                 | 0                  |
| 10WP44   | 18     | 18   | 1                  | 17   | 19   | 19              | 4                 | 0                  |
| 10WP46   | 36     | 33   | 8                  | 15   | 39   | 37              | 7                 | 0                  |
| 10WP47   | 37     | 34   | 8                  | 23   | 43   | 40              | 7                 | 0                  |
| 11CS10RD | 34     | 33   | 2                  | 29   | 35   | 35              | 6                 | 0                  |
| 11CS10RS | 44     | 40   | 12                 | 18   | 48   | 48              | 6                 | 0                  |
| 11WP4R   | 19     | 17   | 4                  | 10   | 21   | 20              | 6                 | 0                  |
| 11WP9RPB | 35     | 35   | 3                  | 29   | 39   | 37              | 7                 | 0                  |
| 11WP9RS  | 37     | 36   | 2                  | 33   | 38   | 38              | 5                 | 0                  |
| 11WP11RD | 35     | 33   | 6                  | 21   | 37   | 36              | 6                 | 0                  |
| 11WP11RS | 35     | 30   | 10                 | 10   | 36   | 35              | 6                 | 0                  |
| 11WP15R  | 24     | 23   | 2                  | 20   | 25   | 24              | 6                 | 0                  |
| 11WP16R  | 33     | 34   | 8                  | 25   | 45   | 42              | 6                 | 0                  |
| 11WP43D  | 9      | 9    | 2                  | 6    | 11   | 10              | 4                 | 0                  |
| 11WP50   | 14     | 14   | 1                  | 13   | 14   | 14              | 4                 | 0                  |
| 11WP51D  | 15     | 14   | 3                  | 7    | 17   | 16              | 6                 | 0                  |
| 11WP51S  | 19     | 17   | 6                  | 8    | 25   | 22              | 6                 | 0                  |
| 11WP52D  | 28     | 27   | 2                  | 24   | 29   | 28              | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 34     | 32   | 5                  | 24   | 36   | 35              | 6                 | 0                  |
| 11WP53S   | 32     | 30   | 5                  | 20   | 33   | 33              | 6                 | 0                  |
| 11WP54D   | 31     | 30   | 4                  | 24   | 34   | 33              | 6                 | 0                  |
| 11WP54S   | 28     | 27   | 3                  | 23   | 30   | 29              | 6                 | 0                  |
| 11WP55    | 28     | 29   | 3                  | 25   | 33   | 32              | 6                 | 0                  |
| 11WP56D   | 11     | 11   | 1                  | 9    | 13   | 12              | 6                 | 0                  |
| 11WP56S   | 35     | 35   | 5                  | 31   | 40   | 40              | 3                 | 0                  |
| 11WP57    | 28     | 26   | 5                  | 17   | 29   | 28              | 6                 | 0                  |
| CG1       | 37     | 35   | 6                  | 23   | 39   | 38              | 6                 | 0                  |
| CG2       | 39     | 39   | 3                  | 35   | 43   | 42              | 5                 | 0                  |
| CG4       | 32     | 33   | 2                  | 31   | 36   | 35              | 6                 | 0                  |
| KC3       | 32     | 32   | 3                  | 28   | 35   | 34              | 5                 | 0                  |
| KC3A      | 22     | 22   | 2                  | 19   | 23   | 23              | 6                 | 0                  |
| KC3PB     | 21     | 20   | 2                  | 17   | 22   | 21              | 6                 | 0                  |
| KC13      | 10     | 10   | 1                  | 9    | 11   | 10              | 6                 | 0                  |
| KC14      | 16     | 16   | 2                  | 14   | 18   | 18              | 5                 | 0                  |
| KCF1      | 29     | 28   | 3                  | 24   | 31   | 31              | 6                 | 0                  |
| LIMESTONE | 27     | 23   | 8                  | 11   | 29   | 28              | 6                 | 0                  |
| M1        | 6      | 6    | 0                  | 6    | 6    | 6               | 2                 | 0                  |
| RN029659  | 19     | 14   | 10                 | 1    | 21   | 20              | 6                 | 0                  |
| RN029660  | 28     | 28   | 2                  | 24   | 31   | 30              | 6                 | 0                  |
| W2R       | 15     | 15   | 3                  | 11   | 18   | 18              | 6                 | 0                  |
| WP5       | 24     | 24   | 2                  | 21   | 26   | 25              | 7                 | 0                  |
| WP13      | 37     | 35   | 3                  | 29   | 38   | 37              | 6                 | 0                  |
| WP19      | 26     | 26   | 1                  | 25   | 28   | 27              | 5                 | 0                  |

Table D43 Concentration of tin (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 86                 |
| 10WP32 <sup>†</sup>   | 3.00   | 2.72 | 1.59               | 0.05 | 5.00 | 4.10            | 7                 | 71                 |
| 10WP32PB <sup>†</sup> | 3.00   | 2.22 | 1.46               | 0.05 | 4.00 | 3.10            | 7                 | 86                 |
| 10WP33 <sup>†</sup>   | 2.00   | 2.15 | 1.05               | 0.05 | 3.00 | 3.00            | 7                 | 100                |
| 10WP35N <sup>†</sup>  | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP35PB <sup>†</sup> | 0.50   | 0.45 | 0.15               | 0.05 | 0.50 | 0.50            | 9                 | 89                 |
| 10WP35S <sup>†</sup>  | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP36N <sup>†</sup>  | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP36PB <sup>†</sup> | 0.50   | 0.46 | 0.14               | 0.05 | 0.50 | 0.50            | 10                | 100                |
| 10WP36S <sup>†</sup>  | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP37 <sup>†</sup>   | 6.00   | 5.01 | 2.43               | 0.05 | 7.00 | 7.00            | 7                 | 100                |
| 10WP39 <sup>†</sup>   | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 10WP40 <sup>†</sup>   | 1.00   | 0.73 | 0.36               | 0.10 | 1.00 | 1.00            | 7                 | 86                 |
| 10WP41 <sup>†</sup>   | 2.50   | 2.54 | 1.37               | 0.25 | 5.00 | 2.75            | 7                 | 100                |
| 10WP42 <sup>†</sup>   | 0.50   | 0.51 | 0.27               | 0.05 | 1.00 | 0.55            | 7                 | 86                 |
| 10WP44 <sup>†</sup>   | 1.50   | 1.44 | 1.23               | 0.25 | 2.50 | 2.50            | 4                 | 75                 |
| 10WP46 <sup>†</sup>   | 0.50   | 0.72 | 0.80               | 0.05 | 2.50 | 0.70            | 7                 | 100                |
| 10WP47 <sup>†</sup>   | 0.50   | 0.44 | 0.17               | 0.05 | 0.50 | 0.50            | 7                 | 100                |
| 11CS10RD <sup>†</sup> | 2.00   | 1.85 | 0.95               | 0.10 | 3.00 | 2.30            | 6                 | 83                 |
| 11CS10RS <sup>†</sup> | 1.00   | 0.84 | 0.39               | 0.05 | 1.00 | 1.00            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.50   | 0.43 | 0.18               | 0.05 | 0.50 | 0.50            | 6                 | 67                 |
| 11WP9RPB <sup>†</sup> | 4.00   | 3.44 | 1.60               | 0.05 | 5.00 | 4.10            | 7                 | 100                |
| 11WP9RS <sup>†</sup>  | 5.00   | 3.81 | 2.15               | 0.05 | 5.00 | 5.00            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.75   | 0.68 | 0.39               | 0.05 | 1.00 | 1.00            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 0.50   | 0.43 | 0.18               | 0.05 | 0.50 | 0.50            | 6                 | 83                 |
| 11WP15R <sup>†</sup>  | 4.50   | 4.18 | 2.19               | 0.10 | 6.00 | 6.00            | 6                 | 100                |
| 11WP16R               | 2.50   | 2.13 | 0.92               | 0.25 | 2.50 | 2.50            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.75   | 1.69 | 2.23               | 0.25 | 5.00 | 3.80            | 4                 | 75                 |
| 11WP50 <sup>†</sup>   | 2.50   | 1.94 | 1.13               | 0.25 | 2.50 | 2.50            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.50   | 0.43 | 0.18               | 0.05 | 0.50 | 0.50            | 6                 | 83                 |
| 11WP51S <sup>†</sup>  | 0.50   | 0.51 | 0.30               | 0.05 | 1.00 | 0.65            | 6                 | 67                 |
| 11WP52D <sup>†</sup>  | 1.00   | 0.84 | 0.39               | 0.05 | 1.00 | 1.00            | 6                 | 100                |

| Site                   | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 6.00   | 5.52  | 2.77               | 0.10 | 8.00  | 7.30            | 6                 | 100                |
| 11WP53S <sup>†</sup>   | 6.00   | 5.18  | 2.82               | 0.10 | 8.00  | 7.30            | 6                 | 83                 |
| 11WP54D <sup>†</sup>   | 6.50   | 5.71  | 2.78               | 0.25 | 8.00  | 7.30            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 2.50   | 2.13  | 0.92               | 0.25 | 2.50  | 2.50            | 6                 | 83                 |
| 11WP55 <sup>†</sup>    | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 67                 |
| 11WP56D <sup>†</sup>   | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.50   | 1.17  | 1.15               | 0.50 | 2.50  | 2.30            | 3                 | 67                 |
| 11WP57 <sup>†</sup>    | 2.50   | 1.79  | 1.10               | 0.25 | 2.50  | 2.50            | 6                 | 100                |
| CG1 <sup>†</sup>       | 26.50  | 21.70 | 17.67              | 0.10 | 40.00 | 37.90           | 6                 | 100                |
| CG2 <sup>†</sup>       | 2.00   | 2.02  | 1.18               | 0.10 | 3.00  | 3.00            | 5                 | 60                 |
| CG4 <sup>†</sup>       | 0.75   | 0.84  | 0.67               | 0.05 | 2.00  | 1.30            | 6                 | 100                |
| KC3 <sup>†</sup>       | 0.50   | 0.41  | 0.20               | 0.05 | 0.50  | 0.50            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 83                 |
| KC3PB <sup>†</sup>     | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.50   | 0.76  | 0.87               | 0.05 | 2.50  | 1.10            | 6                 | 100                |
| KC14 <sup>†</sup>      | 2.50   | 1.95  | 0.97               | 0.25 | 2.50  | 2.50            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 6.50   | 5.51  | 2.72               | 0.05 | 7.00  | 7.00            | 6                 | 83                 |
| LIMESTONE <sup>†</sup> | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 100                |
| M1 <sup>†</sup>        | 0.50   | 0.50  | 0.00               | 0.50 | 0.50  | 0.50            | 3                 | 100                |
| RN029659 <sup>†</sup>  | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 83                 |
| W2R <sup>†</sup>       | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 83                 |
| WP5 <sup>†</sup>       | 0.50   | 0.58  | 0.33               | 0.05 | 1.00  | 1.00            | 7                 | 100                |
| WP13 <sup>†</sup>      | 0.50   | 0.43  | 0.18               | 0.05 | 0.50  | 0.50            | 6                 | 100                |
| WP19 <sup>†</sup>      | 5.00   | 3.60  | 2.04               | 0.50 | 5.00  | 5.00            | 5                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D44 Concentration of sulfate from sulfur (mg/L)

| Site     | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 10WP31   | 7.9    | 7.1   | 2.1                | 3.3   | 9.5   | 8.5             | 7                 | 0                  |
| 10WP32   | 126    | 106   | 41                 | 56    | 149   | 146             | 7                 | 0                  |
| 10WP32PB | 93     | 85    | 34                 | 19    | 122   | 111             | 7                 | 0                  |
| 10WP33   | 71.6   | 73.9  | 12.6               | 52.7  | 90.9  | 86.7            | 7                 | 0                  |
| 10WP35N  | 37.7   | 38.7  | 7.5                | 29.8  | 52.1  | 44.8            | 7                 | 0                  |
| 10WP35PB | 67.1   | 63.7  | 12.6               | 45.0  | 83.4  | 73.2            | 9                 | 0                  |
| 10WP35S  | 61.9   | 60.5  | 9.0                | 49.0  | 70.0  | 68.8            | 7                 | 0                  |
| 10WP36N  | 50.7   | 52.7  | 15.3               | 28.9  | 75.3  | 67.2            | 7                 | 0                  |
| 10WP36PB | 47.3   | 47.4  | 14.9               | 18.9  | 72.7  | 59.1            | 10                | 0                  |
| 10WP36S  | 61.3   | 63.3  | 9.0                | 50.1  | 74.5  | 72.6            | 7                 | 0                  |
| 10WP37   | 121    | 120   | 12                 | 102   | 141   | 125             | 7                 | 0                  |
| 10WP39   | 182    | 175   | 18                 | 147   | 192   | 189             | 7                 | 0                  |
| 10WP40   | 1330   | 1411  | 758                | 622   | 2820  | 1938            | 7                 | 0                  |
| 10WP41   | 3200   | 2929  | 519                | 2180  | 3490  | 3283            | 7                 | 0                  |
| 10WP42   | 365    | 370   | 38                 | 307   | 417   | 413             | 7                 | 0                  |
| 10WP44   | 3015   | 2975  | 182                | 2720  | 3150  | 3114            | 4                 | 0                  |
| 10WP46   | 424    | 414   | 54                 | 340   | 500   | 456             | 7                 | 0                  |
| 10WP47   | 25.1   | 29.8  | 11.1               | 17.7  | 52.1  | 34.5            | 7                 | 0                  |
| 11CS10RD | 195.5  | 208.3 | 91.3               | 101.0 | 345.0 | 300.2           | 6                 | 0                  |
| 11CS10RS | 400.0  | 385.0 | 150.5              | 189.0 | 580.0 | 517.0           | 6                 | 0                  |
| 11WP4R   | 60.9   | 56.5  | 20.7               | 29.8  | 83.5  | 73.1            | 6                 | 0                  |
| 11WP9RPB | 141.0  | 146.7 | 28.4               | 99.9  | 192.0 | 165.9           | 7                 | 0                  |
| 11WP9RS  | 141.0  | 139.0 | 17.1               | 120.0 | 165.0 | 153.0           | 5                 | 0                  |
| 11WP11RD | 37.3   | 36.3  | 4.3                | 28.4  | 40.7  | 39.4            | 6                 | 0                  |
| 11WP11RS | 10.3   | 14.4  | 10.1               | 9.6   | 34.9  | 18.7            | 6                 | 0                  |
| 11WP15R  | 999    | 981   | 67                 | 876   | 1070  | 1028            | 6                 | 0                  |
| 11WP16R  | 2655   | 2658  | 180                | 2460  | 2910  | 2812            | 6                 | 0                  |
| 11WP43D  | 851    | 887   | 703                | 156   | 1690  | 1555            | 4                 | 0                  |
| 11WP50   | 4715   | 4725  | 421                | 4240  | 5230  | 5119            | 4                 | 0                  |
| 11WP51D  | 71.9   | 75.0  | 11.5               | 64.6  | 97.2  | 82.2            | 6                 | 0                  |
| 11WP51S  | 26.4   | 84.8  | 103.9              | 9.1   | 232.0 | 212.4           | 6                 | 0                  |
| 11WP52D  | 42.5   | 42.0  | 6.5                | 32.2  | 48.7  | 48.7            | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D   | 942    | 942  | 43                 | 879  | 1010  | 973             | 6                 | 0                  |
| 11WP53S   | 1055   | 1061 | 129                | 880  | 1240  | 1177            | 6                 | 0                  |
| 11WP54D   | 5230   | 5302 | 446                | 4810 | 5970  | 5725            | 6                 | 0                  |
| 11WP54S   | 3395   | 3485 | 472                | 2920 | 4160  | 3992            | 6                 | 0                  |
| 11WP55    | 339    | 398  | 150                | 289  | 682   | 515             | 6                 | 0                  |
| 11WP56D   | 1.0    | 1.8  | 1.6                | 0.5  | 4.2   | 3.6             | 6                 | 0                  |
| 11WP56S   | 128    | 105  | 88                 | 8    | 179   | 174             | 3                 | 0                  |
| 11WP57    | 1520   | 1512 | 120                | 1330 | 1630  | 1623            | 6                 | 0                  |
| CG1       | 687    | 718  | 96                 | 609  | 855   | 826             | 6                 | 0                  |
| CG2       | 24.1   | 23.3 | 4.6                | 16.2 | 28.5  | 27.1            | 5                 | 0                  |
| CG4       | 323    | 331  | 36                 | 289  | 382   | 367             | 6                 | 0                  |
| KC3       | 32.0   | 30.5 | 4.7                | 22.2 | 34.0  | 33.4            | 5                 | 0                  |
| KC3A      | 104    | 103  | 8                  | 90   | 111   | 110             | 6                 | 0                  |
| KC3PB     | 103    | 103  | 8                  | 95   | 115   | 110             | 6                 | 0                  |
| KC13      | 153    | 154  | 31                 | 103  | 190   | 184             | 6                 | 0                  |
| KC14      | 4140   | 4064 | 430                | 3380 | 4560  | 4380            | 5                 | 0                  |
| KCF1      | 1365   | 1320 | 166                | 1100 | 1520  | 1457            | 6                 | 0                  |
| LIMESTONE | 52.4   | 40.5 | 21.8               | 9.8  | 57.2  | 56.2            | 6                 | 0                  |
| M1        | 5.6    | 5.9  | 0.6                | 5.5  | 6.5   | 6.4             | 3                 | 0                  |
| RN029659  | 383    | 328  | 198                | 1    | 518   | 487             | 6                 | 0                  |
| RN029660  | 76.7   | 86.0 | 20.8               | 72.2 | 126.0 | 101.4           | 6                 | 0                  |
| W2R       | 28.4   | 27.0 | 8.3                | 14.3 | 35.0  | 34.9            | 6                 | 0                  |
| WP5       | 839    | 844  | 66                 | 750  | 923   | 902             | 7                 | 0                  |
| WP13      | 99     | 103  | 11                 | 91   | 124   | 111             | 6                 | 0                  |
| WP19      | 4600   | 4686 | 400                | 4160 | 5260  | 5040            | 5                 | 0                  |

Table D45 Concentration of titanium (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP32 <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP32PB <sup>†</sup> | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP33 <sup>†</sup>   | 0.25   | 0.47 | 0.42               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP35N <sup>†</sup>  | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP35PB <sup>†</sup> | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP35S <sup>†</sup>  | 0.63   | 0.62 | 0.42               | 0.20 | 1.00  | 1.00            | 6                 | 83                 |
| 10WP36N <sup>†</sup>  | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP36PB <sup>†</sup> | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP36S <sup>†</sup>  | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP37 <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP39 <sup>†</sup>   | 0.63   | 0.60 | 0.44               | 0.10 | 1.00  | 1.00            | 6                 | 83                 |
| 10WP40 <sup>†</sup>   | 0.75   | 0.68 | 0.38               | 0.10 | 1.00  | 1.00            | 6                 | 100                |
| 10WP41 <sup>†</sup>   | 1.00   | 1.17 | 0.74               | 0.25 | 2.50  | 1.63            | 6                 | 100                |
| 10WP42 <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP44 <sup>†</sup>   | 1.00   | 1.06 | 0.13               | 1.00 | 1.25  | 1.18            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 10WP47 <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 11CS10RD <sup>†</sup> | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 11CS10RS <sup>†</sup> | 0.75   | 0.64 | 0.41               | 0.10 | 1.00  | 1.00            | 6                 | 100                |
| 11WP4R <sup>†</sup>   | 0.85   | 0.67 | 0.42               | 0.05 | 1.00  | 1.00            | 6                 | 83                 |
| 11WP9RPB <sup>†</sup> | 0.50   | 0.58 | 0.42               | 0.05 | 1.00  | 1.00            | 7                 | 86                 |
| 11WP9RS <sup>†</sup>  | 1.00   | 0.66 | 0.47               | 0.05 | 1.00  | 1.00            | 5                 | 100                |
| 11WP11RD <sup>†</sup> | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 11WP11RS <sup>†</sup> | 1.00   | 1.02 | 0.97               | 0.05 | 2.80  | 1.54            | 6                 | 83                 |
| 11WP15R <sup>†</sup>  | 0.75   | 0.68 | 0.38               | 0.10 | 1.00  | 1.00            | 6                 | 100                |
| 11WP16R <sup>†</sup>  | 1.00   | 0.96 | 0.37               | 0.25 | 1.25  | 1.25            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 1.13   | 7.56 | 12.96              | 1.00 | 27.00 | 19.28           | 4                 | 75                 |
| 11WP50 <sup>†</sup>   | 1.00   | 1.06 | 0.13               | 1.00 | 1.25  | 1.18            | 4                 | 100                |
| 11WP51D <sup>†</sup>  | 0.63   | 0.62 | 0.42               | 0.20 | 1.00  | 1.00            | 6                 | 83                 |
| 11WP51S <sup>†</sup>  | 0.65   | 0.63 | 0.40               | 0.25 | 1.00  | 1.00            | 6                 | 83                 |
| 11WP52D <sup>†</sup>  | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 1.00   | 0.78 | 0.40               | 0.10 | 1.10  | 1.03            | 6                 | 83                 |
| 11WP53S                | 1.00   | 0.80 | 0.41               | 0.10 | 1.20  | 1.06            | 6                 | 83                 |
| 11WP54D <sup>†</sup>   | 1.00   | 0.96 | 0.37               | 0.25 | 1.25  | 1.25            | 6                 | 100                |
| 11WP54S <sup>†</sup>   | 1.00   | 0.96 | 0.37               | 0.25 | 1.25  | 1.25            | 6                 | 100                |
| 11WP55 <sup>†</sup>    | 1.00   | 1.14 | 0.86               | 0.05 | 2.70  | 1.58            | 6                 | 67                 |
| 11WP56D <sup>†</sup>   | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| 11WP56S <sup>†</sup>   | 0.50   | 0.52 | 0.48               | 0.05 | 1.00  | 0.95            | 3                 | 67                 |
| 11WP57 <sup>†</sup>    | 1.00   | 0.96 | 0.37               | 0.25 | 1.25  | 1.25            | 6                 | 100                |
| CG1 <sup>†</sup>       | 0.75   | 0.75 | 0.27               | 0.50 | 1.00  | 1.00            | 6                 | 100                |
| CG2 <sup>†</sup>       | 1.00   | 0.70 | 0.41               | 0.25 | 1.00  | 1.00            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| KC3 <sup>†</sup>       | 1.00   | 0.70 | 0.41               | 0.25 | 1.00  | 1.00            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.65   | 0.63 | 0.40               | 0.25 | 1.00  | 1.00            | 6                 | 83                 |
| KC3PB <sup>†</sup>     | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| KC13 <sup>†</sup>      | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| KC14 <sup>†</sup>      | 1.00   | 1.10 | 0.14               | 1.00 | 1.25  | 1.25            | 5                 | 100                |
| KCF1 <sup>†</sup>      | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| LIMESTONE <sup>†</sup> | 1.00   | 0.92 | 0.44               | 0.30 | 1.60  | 1.18            | 6                 | 50                 |
| M1 <sup>†</sup>        | 0.05   | 0.05 | 0.00               | 0.05 | 0.05  | 0.05            | 2                 | 100                |
| RN029659 <sup>†</sup>  | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| RN029660 <sup>†</sup>  | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| W2R <sup>†</sup>       | 1.00   | 3.86 | 7.42               | 0.25 | 19.00 | 6.40            | 6                 | 67                 |
| WP5 <sup>†</sup>       | 0.75   | 0.64 | 0.41               | 0.10 | 1.00  | 1.00            | 6                 | 100                |
| WP13 <sup>†</sup>      | 0.63   | 0.59 | 0.45               | 0.05 | 1.00  | 1.00            | 6                 | 100                |
| WP19 <sup>†</sup>      | 1.00   | 1.38 | 0.75               | 1.00 | 2.50  | 2.05            | 4                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D46 Concentration of thallium (µg/L)

| Site                  | Median | Mean  | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|-------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP32 <sup>†</sup>   | 0.05   | 0.23  | 0.28               | 0.05 | 0.7  | 0.5             | 5                 | 60                 |
| 10WP32PB <sup>†</sup> | 0.05   | 0.08  | 0.07               | 0.05 | 0.2  | 0.125           | 5                 | 80                 |
| 10WP33 <sup>†</sup>   | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP35N <sup>†</sup>  | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP35PB <sup>†</sup> | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP36N <sup>†</sup>  | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.08  | 0.07               | 0.05 | 0.2  | 0.125           | 5                 | 80                 |
| 10WP36S <sup>†</sup>  | 0.05   | 0.06  | 0.02               | 0.05 | 0.1  | 0.075           | 5                 | 80                 |
| 10WP37 <sup>†</sup>   | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 10WP39                | 0.2    | 0.21  | 0.12               | 0.05 | 0.4  | 0.3             | 5                 | 20                 |
| 10WP40 <sup>†</sup>   | 0.1    | 0.08  | 0.03               | 0.05 | 0.1  | 0.1             | 5                 | 100                |
| 10WP41 <sup>†</sup>   | 0.25   | 0.53  | 0.50               | 0.25 | 1.4  | 0.95            | 5                 | 80                 |
| 10WP42 <sup>†</sup>   | 0.05   | 0.1   | 0.11               | 0.05 | 0.3  | 0.175           | 5                 | 80                 |
| 10WP44 <sup>†</sup>   | 0.25   | 0.2   | 0.10               | 0.05 | 0.25 | 0.25            | 4                 | 100                |
| 10WP46 <sup>†</sup>   | 0.05   | 0.09  | 0.09               | 0.05 | 0.25 | 0.15            | 5                 | 100                |
| 10WP47 <sup>†</sup>   | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11CS10RD <sup>†</sup> | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11CS10RS <sup>†</sup> | 0.1    | 0.09  | 0.02               | 0.05 | 0.1  | 0.1             | 5                 | 100                |
| 11WP4R <sup>†</sup>   | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP9RPB <sup>†</sup> | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 4                 | 100                |
| 11WP11RD <sup>†</sup> | 0.05   | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP11RS <sup>†</sup> | 0.05   | 0.06  | 0.02               | 0.05 | 0.1  | 0.075           | 5                 | 80                 |
| 11WP15R <sup>†</sup>  | 0.1    | 0.26  | 0.36               | 0.1  | 0.9  | 0.5             | 5                 | 80                 |
| 11WP16R <sup>†</sup>  | 0.25   | 0.25  | 0.00               | 0.25 | 0.25 | 0.25            | 5                 | 100                |
| 11WP43D <sup>†</sup>  | 0.175  | 0.225 | 0.20               | 0.05 | 0.5  | 0.425           | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 0.25   | 0.688 | 0.88               | 0.25 | 2    | 1.475           | 4                 | 75                 |
| 11WP51D <sup>†</sup>  | 0.05   | 0.08  | 0.07               | 0.05 | 0.2  | 0.125           | 5                 | 80                 |
| 11WP51S <sup>†</sup>  | 0.05   | 0.07  | 0.03               | 0.05 | 0.1  | 0.1             | 5                 | 80                 |
| 11WP52D <sup>†</sup>  | 0.05   | 0.06  | 0.02               | 0.05 | 0.1  | 0.075           | 5                 | 100                |

| Site                   | Median  | Mean  | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|---------|-------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.1     | 0.1   | 0.00               | 0.1  | 0.1  | 0.1             | 5                 | 100                |
| 11WP53S <sup>†</sup>   | 0.1     | 0.1   | 0.00               | 0.1  | 0.1  | 0.1             | 5                 | 100                |
| 11WP54D <sup>†</sup>   | 0.25    | 0.25  | 0.00               | 0.25 | 0.25 | 0.25            | 5                 | 100                |
| 11WP54S <sup>†</sup>   | 0.25    | 0.25  | 0.00               | 0.25 | 0.25 | 0.25            | 5                 | 100                |
| 11WP55 <sup>†</sup>    | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP56D <sup>†</sup>   | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| 11WP56S <sup>†</sup>   | 0.15    | 0.15  | 0.14               | 0.05 | 0.25 | 0.25            | 2                 | 100                |
| 11WP57                 | 1.1     | 1.22  | 0.66               | 0.5  | 1.9  | 1.9             | 5                 | 0                  |
| CG1 <sup>†</sup>       | 0.1     | 0.1   | 0.00               | 0.1  | 0.1  | 0.1             | 6                 | 100                |
| CG2 <sup>†</sup>       | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| CG4 <sup>†</sup>       | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| KC3 <sup>†</sup>       | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| KC3PB <sup>†</sup>     | 0.05    | 0.06  | 0.22               | 0.05 | 0.1  | 0.075           | 5                 | 80                 |
| KC13 <sup>†</sup>      | 0.05    | 0.09  | 0.09               | 0.05 | 0.25 | 0.15            | 5                 | 100                |
| KC14 <sup>†</sup>      | 0.25    | 0.6   | 0.90               | 0.05 | 2.2  | 1.225           | 5                 | 80                 |
| KCF1 <sup>†</sup>      | 0.05    | 0.07  | 0.03               | 0.05 | 0.1  | 0.1             | 5                 | 60                 |
| LIMESTONE <sup>†</sup> | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| M1                     | No data |       |                    |      |      |                 |                   |                    |
| RN029659 <sup>†</sup>  | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| RN029660 <sup>†</sup>  | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 5                 | 100                |
| W2R <sup>†</sup>       | 0.05    | 0.05  | 0.00               | 0.05 | 0.05 | 0.05            | 6                 | 100                |
| WP5 <sup>†</sup>       | 0.05    | 0.06  | 0.02               | 0.05 | 0.1  | 0.075           | 5                 | 100                |
| WP13 <sup>†</sup>      | 0.05    | 0.06  | 0.02               | 0.05 | 0.1  | 0.075           | 5                 | 80                 |
| WP19 <sup>†</sup>      | 0.5     | 0.438 | 0.13               | 0.25 | 0.5  | 0.5             | 4                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D47 Concentration of uranium (µg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31   | 0.3    | 0.3  | 0.1                | 0.1  | 0.4   | 0.3             | 7                 | 0                  |
| 10WP32   | 3.6    | 3.3  | 1.1                | 1.6  | 4.7   | 4.2             | 7                 | 0                  |
| 10WP32PB | 3.8    | 4.0  | 0.6                | 3.4  | 4.6   | 4.6             | 7                 | 0                  |
| 10WP33   | 4.6    | 4.5  | 1.4                | 2.0  | 6.3   | 5.8             | 7                 | 0                  |
| 10WP35N  | 15.0   | 15.5 | 6.8                | 4.3  | 22.0  | 22.0            | 7                 | 0                  |
| 10WP35PB | 30.0   | 25.3 | 9.4                | 9.7  | 35.0  | 33.0            | 9                 | 0                  |
| 10WP35S  | 24.0   | 22.2 | 10.0               | 5.6  | 33.0  | 32.1            | 7                 | 0                  |
| 10WP36N  | 5.5    | 5.4  | 1.4                | 3.1  | 7.5   | 6.4             | 7                 | 0                  |
| 10WP36PB | 4.5    | 4.9  | 2.0                | 2.5  | 9.0   | 6.5             | 10                | 0                  |
| 10WP36S  | 6.3    | 5.8  | 2.2                | 1.4  | 7.9   | 7.6             | 7                 | 0                  |
| 10WP37   | 20.0   | 18.5 | 7.0                | 4.3  | 24.0  | 24.0            | 7                 | 0                  |
| 10WP39   | 0.7    | 0.7  | 0.1                | 0.6  | 1.0   | 0.8             | 7                 | 0                  |
| 10WP40   | 9.1    | 13.4 | 11.5               | 4.9  | 38.0  | 18.2            | 7                 | 0                  |
| 10WP41   | 84.0   | 81.4 | 22.5               | 48.0 | 110.0 | 99.2            | 7                 | 0                  |
| 10WP42   | 9.1    | 9.4  | 1.3                | 7.5  | 11.0  | 11.0            | 7                 | 0                  |
| 10WP44   | 23.0   | 22.8 | 6.9                | 14.0 | 31.0  | 28.6            | 4                 | 0                  |
| 10WP46   | 6.5    | 6.5  | 0.6                | 5.6  | 7.3   | 7.0             | 7                 | 0                  |
| 10WP47   | 0.9    | 0.9  | 0.4                | 0.3  | 1.8   | 1.1             | 7                 | 0                  |
| 11CS10RD | 19.0   | 18.0 | 3.6                | 12.0 | 22.0  | 20.6            | 6                 | 0                  |
| 11CS10RS | 23.0   | 42.1 | 47.1               | 7.3  | 130.0 | 81.0            | 6                 | 0                  |
| 11WP4R   | 3.5    | 3.4  | 0.9                | 2.2  | 4.7   | 4.2             | 6                 | 0                  |
| 11WP9RPB | 5.8    | 5.6  | 0.7                | 4.5  | 6.6   | 6.0             | 7                 | 0                  |
| 11WP9RS  | 6.1    | 6.0  | 0.6                | 5.2  | 6.8   | 6.6             | 5                 | 0                  |
| 11WP11RD | 3.0    | 2.9  | 0.7                | 1.7  | 3.6   | 3.5             | 6                 | 0                  |
| 11WP11RS | 4.0    | 4.0  | 1.3                | 1.9  | 5.9   | 5.1             | 6                 | 0                  |
| 11WP15R  | 57.5   | 56.5 | 2.1                | 53.0 | 58.0  | 58.0            | 6                 | 0                  |
| 11WP16R  | 15.0   | 15.7 | 3.2                | 12.0 | 20.0  | 19.3            | 6                 | 0                  |
| 11WP43D  | 8.3    | 13.5 | 15.8               | 1.5  | 36.0  | 29.1            | 4                 | 0                  |
| 11WP50   | 73.0   | 73.0 | 1.6                | 71.0 | 75.0  | 74.4            | 4                 | 0                  |
| 11WP51D  | 0.5    | 0.4  | 0.1                | 0.2  | 0.5   | 0.5             | 6                 | 0                  |
| 11WP51S  | 1.1    | 1.2  | 0.9                | 0.2  | 2.2   | 2.2             | 6                 | 0                  |
| 11WP52D  | 2.3    | 2.3  | 0.8                | 1.2  | 3.4   | 3.1             | 6                 | 0                  |

| Site                   | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|-------|--------------------|-------|-------|-----------------|-------------------|--------------------|
| 11WP53D                | 56.0   | 55.7  | 3.5                | 50.0  | 61.0  | 57.5            | 6                 | 0                  |
| 11WP53S                | 62.0   | 63.0  | 3.0                | 60.0  | 68.0  | 65.9            | 6                 | 0                  |
| 11WP54D                | 110.0  | 109.2 | 10.2               | 95.0  | 120.0 | 120.0           | 6                 | 0                  |
| 11WP54S                | 125.0  | 128.3 | 11.7               | 120.0 | 150.0 | 136.0           | 6                 | 0                  |
| 11WP55                 | 1.4    | 1.4   | 0.8                | 0.4   | 2.2   | 2.1             | 6                 | 0                  |
| 11WP56D <sup>†</sup>   | 0.2    | 0.1   | 0.1                | 0.1   | 0.2   | 0.2             | 6                 | 33                 |
| 11WP56S <sup>†</sup>   | 1.3    | 1.3   | 1.1                | 0.3   | 2.4   | 2.3             | 3                 | 33                 |
| 11WP57                 | 4.6    | 6.0   | 3.9                | 4.0   | 14.0  | 7.5             | 6                 | 0                  |
| CG1                    | 35.5   | 34.5  | 6.5                | 23.0  | 41.0  | 40.3            | 6                 | 0                  |
| CG2                    | 8.9    | 8.6   | 1.1                | 7.2   | 9.9   | 9.5             | 5                 | 0                  |
| CG4                    | 13.5   | 13.5  | 1.0                | 12.0  | 15.0  | 14.3            | 6                 | 0                  |
| KC3 <sup>†</sup>       | 0.1    | 0.2   | 0.2                | 0.1   | 0.5   | 0.4             | 5                 | 40                 |
| KC3A <sup>†</sup>      | 0.1    | 0.1   | 0.0                | 0.1   | 0.1   | 0.1             | 6                 | 83                 |
| KC3PB <sup>†</sup>     | 0.1    | 0.1   | 0.1                | 0.1   | 0.2   | 0.1             | 6                 | 67                 |
| KC13                   | 4.6    | 9.0   | 11.7               | 1.5   | 32.0  | 16.2            | 6                 | 0                  |
| KC14                   | 56.0   | 56.4  | 2.3                | 54.0  | 60.0  | 58.5            | 5                 | 0                  |
| KCF1                   | 25.0   | 24.5  | 1.8                | 22.0  | 27.0  | 25.6            | 6                 | 0                  |
| LIMESTONE <sup>†</sup> | 0.4    | 0.4   | 0.4                | 0.1   | 1.2   | 0.7             | 6                 | 33                 |
| M1                     | 0.3    | 0.3   | 0.1                | 0.3   | 0.4   | 0.4             | 3                 | 0                  |
| RN029659               | 5.4    | 4.9   | 4.1                | 0.1   | 9.6   | 9.2             | 6                 | 17                 |
| RN029660               | 3.5    | 3.4   | 0.3                | 2.9   | 3.8   | 3.7             | 6                 | 0                  |
| W2R                    | 0.2    | 0.4   | 0.5                | 0.1   | 1.4   | 0.6             | 6                 | 0                  |
| WP5                    | 9.3    | 8.9   | 2.6                | 3.4   | 11.0  | 11.0            | 7                 | 0                  |
| WP13                   | 1.0    | 1.0   | 0.2                | 0.8   | 1.3   | 1.1             | 6                 | 0                  |
| WP19                   | 45.0   | 45.0  | 2.7                | 41.0  | 48.0  | 47.5            | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D48 Concentration of vanadium (µg/L)

| Site                 | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|----------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 10WP31               | 15.5   | 15.3  | 1.2                | 14.0 | 17.0  | 16.3            | 6                 | 0                  |
| 10WP32               | 42.5   | 45.3  | 16.2               | 21.0 | 65.0  | 62.9            | 6                 | 0                  |
| 10WP32PB             | 67.0   | 63.0  | 12.6               | 41.0 | 75.0  | 72.9            | 6                 | 0                  |
| 10WP33               | 19.5   | 19.5  | 1.0                | 18.0 | 21.0  | 20.3            | 6                 | 0                  |
| 10WP35N              | 38.0   | 36.0  | 6.7                | 25.0 | 42.0  | 41.3            | 6                 | 0                  |
| 10WP35PB             | 22.5   | 22.2  | 2.1                | 19.0 | 24.0  | 24.0            | 6                 | 0                  |
| 10WP35S              | 18.0   | 16.5  | 5.3                | 5.9  | 20.0  | 20.0            | 6                 | 0                  |
| 10WP36N              | 52.0   | 47.8  | 13.5               | 25.0 | 63.0  | 57.4            | 6                 | 0                  |
| 10WP36PB             | 38.5   | 41.0  | 25.2               | 16.0 | 82.0  | 61.7            | 6                 | 0                  |
| 10WP36S              | 120.0  | 123.5 | 34.4               | 85.0 | 180.0 | 152.0           | 6                 | 0                  |
| 10WP37               | 35.5   | 34.2  | 2.9                | 30.0 | 37.0  | 36.3            | 6                 | 0                  |
| 10WP39               | 12.0   | 12.0  | 1.5                | 10.0 | 14.0  | 13.3            | 6                 | 0                  |
| 10WP40               | 14.2   | 18.9  | 19.6               | 2.5  | 54.0  | 33.7            | 6                 | 17                 |
| 10WP41               | 6.6    | 6.8   | 3.1                | 2.5  | 11.0  | 9.6             | 6                 | 17                 |
| 10WP42               | 12.0   | 12.7  | 2.0                | 11.0 | 16.0  | 14.6            | 6                 | 0                  |
| 10WP44               | 7.9    | 7.9   | 4.2                | 3.9  | 12.0  | 11.7            | 4                 | 0                  |
| 10WP46               | 12.0   | 11.8  | 1.3                | 10.0 | 13.0  | 13.0            | 6                 | 0                  |
| 10WP47 <sup>†</sup>  | 2.5    | 2.1   | 0.9                | 0.8  | 3.0   | 2.7             | 6                 | 50                 |
| 11CS10RD             | 25.0   | 24.8  | 1.6                | 23.0 | 27.0  | 26.3            | 6                 | 0                  |
| 11CS10RS             | 21.0   | 41.5  | 50.5               | 7.8  | 140.0 | 76.3            | 6                 | 0                  |
| 11WP4R               | 5.6    | 6.1   | 3.4                | 2.5  | 11.0  | 9.6             | 6                 | 17                 |
| 11WP9RPB             | 40.0   | 51.9  | 30.4               | 36.0 | 120.0 | 57.0            | 7                 | 0                  |
| 11WP9RS              | 41.0   | 40.0  | 4.1                | 34.0 | 44.0  | 43.5            | 5                 | 0                  |
| 11WP11RD             | 36.0   | 36.7  | 9.5                | 22.0 | 50.0  | 45.1            | 6                 | 0                  |
| 11WP11RS             | 130.0  | 126.7 | 49.3               | 40.0 | 190.0 | 162.0           | 6                 | 0                  |
| 11WP15R              | 23.0   | 23.7  | 4.2                | 19.0 | 30.0  | 27.9            | 6                 | 0                  |
| 11WP16R              | 4.2    | 6.4   | 4.4                | 2.5  | 13.0  | 11.6            | 6                 | 17                 |
| 11WP43D              | 3.2    | 3.6   | 1.3                | 2.5  | 5.5   | 4.9             | 4                 | 25                 |
| 11WP50               | 3.8    | 3.6   | 0.8                | 2.5  | 4.3   | 4.2             | 4                 | 25                 |
| 11WP51D <sup>†</sup> | 1.6    | 1.6   | 0.8                | 0.3  | 2.5   | 2.5             | 6                 | 33                 |
| 11WP51S <sup>†</sup> | 2.5    | 2.5   | 0.5                | 2.0  | 3.4   | 2.8             | 6                 | 33                 |
| 11WP52D              | 19.0   | 17.7  | 3.5                | 12.0 | 21.0  | 20.3            | 6                 | 0                  |

| Site                   | Median | Mean  | Standard deviation | Min. | Max.  | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|-------|--------------------|------|-------|-----------------|-------------------|--------------------|
| 11WP53D                | 150.0  | 138.3 | 36.0               | 70.0 | 170.0 | 163.0           | 6                 | 0                  |
| 11WP53S                | 115.0  | 118.2 | 30.3               | 69.0 | 150.0 | 150.0           | 6                 | 0                  |
| 11WP54D                | 20.0   | 19.5  | 3.3                | 14.0 | 23.0  | 22.3            | 6                 | 0                  |
| 11WP54S                | 30.0   | 26.7  | 9.7                | 14.0 | 39.0  | 33.4            | 6                 | 0                  |
| 11WP55                 | 2.5    | 7.6   | 13.5               | 0.6  | 35.0  | 12.8            | 6                 | 17                 |
| 11WP56D <sup>†</sup>   | 0.6    | 1.2   | 1.0                | 0.4  | 2.5   | 2.5             | 6                 | 33                 |
| 11WP56S                | 9.7    | 9.9   | 1.0                | 9.0  | 11.0  | 10.9            | 3                 | 0                  |
| 11WP57 <sup>†</sup>    | 3.8    | 3.5   | 0.8                | 2.5  | 4.2   | 4.2             | 6                 | 33                 |
| CG1                    | 34.5   | 33.2  | 5.3                | 23.0 | 38.0  | 36.6            | 6                 | 0                  |
| CG2                    | 22.0   | 22.6  | 3.0                | 20.0 | 27.0  | 25.5            | 5                 | 0                  |
| CG4 <sup>†</sup>       | 8.1    | 7.0   | 3.7                | 2.5  | 11.0  | 10.3            | 6                 | 33                 |
| KC3                    | 2.6    | 2.8   | 0.4                | 2.4  | 3.3   | 3.2             | 5                 | 20                 |
| KC3A <sup>†</sup>      | 0.9    | 1.3   | 0.9                | 0.6  | 2.5   | 2.5             | 6                 | 33                 |
| KC3PB <sup>†</sup>     | 0.6    | 1.1   | 1.1                | 0.2  | 2.5   | 2.5             | 6                 | 33                 |
| KC13 <sup>†</sup>      | 2.0    | 4.1   | 6.9                | 0.1  | 18.0  | 7.2             | 6                 | 33                 |
| KC14 <sup>†</sup>      | 0.3    | 0.8   | 1.0                | 0.3  | 2.5   | 1.5             | 5                 | 80                 |
| KCF1                   | 9.5    | 9.5   | 1.5                | 7.0  | 11.0  | 11.0            | 6                 | 0                  |
| LIMESTONE <sup>†</sup> | 2.5    | 2.5   | 2.2                | 0.4  | 6.6   | 3.7             | 6                 | 50                 |
| M1 <sup>†</sup>        | 3.8    | 3.8   | 1.8                | 2.5  | 5.0   | 5.0             | 2                 | 50                 |
| RN029659 <sup>†</sup>  | 2.5    | 2.4   | 0.3                | 2.0  | 2.7   | 2.6             | 6                 | 50                 |
| RN029660               | 47.0   | 46.2  | 7.5                | 37.0 | 56.0  | 53.2            | 6                 | 0                  |
| W2R <sup>†</sup>       | 2.2    | 1.8   | 0.9                | 0.4  | 2.5   | 2.5             | 6                 | 33                 |
| WP5                    | 6.9    | 6.9   | 0.7                | 6.0  | 8.0   | 7.5             | 6                 | 0                  |
| WP13                   | 6.0    | 5.9   | 1.0                | 4.7  | 7.0   | 7.0             | 6                 | 0                  |
| WP19                   | 10.8   | 10.7  | 1.6                | 9.0  | 12.0  | 12.0            | 4                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D49 Concentration of zinc (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31                | 11     | 13   | 6                  | 7    | 25   | 15              | 7                 | 0                  |
| 10WP32 <sup>†</sup>   | 9      | 8    | 4                  | 3    | 12   | 12              | 7                 | 29                 |
| 10WP32PB <sup>†</sup> | 8      | 28   | 40                 | 3    | 98   | 76              | 7                 | 29                 |
| 10WP33                | 12     | 11   | 6                  | 3    | 21   | 16              | 7                 | 14                 |
| 10WP35N               | 8      | 8    | 5                  | 1    | 13   | 13              | 7                 | 14                 |
| 10WP35PB              | 12     | 11   | 7                  | 3    | 25   | 15              | 9                 | 11                 |
| 10WP35S               | 14     | 14   | 6                  | 3    | 22   | 18              | 7                 | 14                 |
| 10WP36N               | 12     | 11   | 4                  | 4    | 18   | 14              | 7                 | 0                  |
| 10WP36PB <sup>†</sup> | 6      | 7    | 6                  | 3    | 20   | 12              | 10                | 30                 |
| 10WP36S               | 12     | 13   | 7                  | 5    | 28   | 16              | 7                 | 0                  |
| 10WP37                | 8      | 8    | 4                  | 3    | 15   | 11              | 7                 | 14                 |
| 10WP39                | 11     | 15   | 13                 | 3    | 37   | 30              | 7                 | 14                 |
| 10WP40                | 13     | 16   | 13                 | 3    | 44   | 20              | 7                 | 0                  |
| 10WP41                | 26     | 50   | 47                 | 7    | 140  | 89              | 7                 | 0                  |
| 10WP42                | 11     | 12   | 8                  | 3    | 27   | 16              | 7                 | 14                 |
| 10WP44                | 13     | 12   | 2                  | 9    | 13   | 13              | 4                 | 0                  |
| 10WP46 <sup>†</sup>   | 9      | 11   | 10                 | 3    | 31   | 16              | 7                 | 29                 |
| 10WP47 <sup>†</sup>   | 6      | 9    | 10                 | 3    | 30   | 15              | 7                 | 43                 |
| 11CS10RD              | 13     | 14   | 8                  | 5    | 26   | 20              | 6                 | 0                  |
| 11CS10RS              | 11     | 16   | 13                 | 9    | 41   | 22              | 6                 | 0                  |
| 11WP4R                | 13     | 19   | 18                 | 3    | 49   | 39              | 6                 | 17                 |
| 11WP9RPB              | 13     | 14   | 9                  | 3    | 28   | 22              | 7                 | 14                 |
| 11WP9RS               | 6      | 7    | 5                  | 3    | 14   | 12              | 5                 | 20                 |
| 11WP11RD              | 6      | 9    | 6                  | 5    | 19   | 14              | 6                 | 0                  |
| 11WP11RS              | 10     | 9    | 6                  | 2    | 15   | 14              | 6                 | 17                 |
| 11WP15R               | 15     | 16   | 8                  | 3    | 27   | 24              | 6                 | 17                 |
| 11WP16R               | 16     | 18   | 7                  | 12   | 30   | 23              | 6                 | 0                  |
| 11WP43D               | 18     | 28   | 27                 | 10   | 67   | 54              | 4                 | 0                  |
| 11WP50                | 54     | 69   | 49                 | 27   | 140  | 115             | 4                 | 0                  |
| 11WP51D <sup>†</sup>  | 4      | 8    | 8                  | 3    | 21   | 15              | 6                 | 33                 |
| 11WP51S <sup>†</sup>  | 22     | 22   | 18                 | 3    | 41   | 40              | 6                 | 33                 |
| 11WP52D               | 9      | 10   | 3                  | 7    | 16   | 12              | 6                 | 0                  |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D                | 12     | 11   | 5                  | 3    | 19   | 16              | 6                 | 17                 |
| 11WP53S <sup>†</sup>   | 11     | 12   | 10                 | 3    | 25   | 23              | 6                 | 33                 |
| 11WP54D                | 20     | 28   | 17                 | 14   | 55   | 46              | 6                 | 0                  |
| 11WP54S                | 20     | 49   | 75                 | 5    | 200  | 82              | 6                 | 0                  |
| 11WP55                 | 6      | 13   | 14                 | 2    | 32   | 29              | 6                 | 17                 |
| 11WP56D                | 15     | 21   | 15                 | 8    | 49   | 32              | 6                 | 0                  |
| 11WP56S <sup>†</sup>   | 21     | 41   | 52                 | 3    | 100  | 92              | 3                 | 33                 |
| 11WP57                 | 17     | 22   | 18                 | 10   | 57   | 31              | 6                 | 0                  |
| CG1                    | 20     | 24   | 17                 | 9    | 56   | 36              | 6                 | 0                  |
| CG2                    | 14     | 18   | 13                 | 8    | 40   | 28              | 5                 | 0                  |
| CG4                    | 23     | 25   | 17                 | 4    | 50   | 42              | 6                 | 0                  |
| KC3                    | 16     | 20   | 13                 | 8    | 42   | 32              | 5                 | 0                  |
| KC3A                   | 21     | 27   | 24                 | 8    | 74   | 41              | 6                 | 0                  |
| KC3PB                  | 16     | 19   | 14                 | 8    | 45   | 28              | 6                 | 0                  |
| KC13                   | 23     | 21   | 9                  | 8    | 29   | 28              | 6                 | 0                  |
| KC14                   | 20     | 31   | 23                 | 8    | 64   | 55              | 5                 | 0                  |
| KCF1                   | 14     | 28   | 34                 | 8    | 97   | 47              | 6                 | 0                  |
| LIMESTONE <sup>†</sup> | 9      | 9    | 6                  | 3    | 16   | 15              | 6                 | 33                 |
| M1 <sup>†</sup>        | 11     | 9    | 6                  | 3    | 13   | 13              | 3                 | 33                 |
| RN029659 <sup>†</sup>  | 4      | 8    | 8                  | 3    | 24   | 14              | 6                 | 50                 |
| RN029660               | 10     | 12   | 9                  | 3    | 26   | 22              | 6                 | 17                 |
| W2R <sup>†</sup>       | 6      | 16   | 20                 | 3    | 52   | 34              | 6                 | 33                 |
| WP5                    | 17     | 17   | 11                 | 5    | 33   | 29              | 7                 | 0                  |
| WP13                   | 11     | 15   | 13                 | 3    | 39   | 24              | 6                 | 17                 |
| WP19                   | 28     | 28   | 9                  | 14   | 37   | 36              | 5                 | 0                  |

<sup>†</sup> More than one-quarter of the samples are below the LOR

Table D50 Concentration of total dissolved solids (mg/L)

| Site     | Media<br>n | Mean   | Standard<br>deviation | Min.   | Max.   | 80th<br>percentile | Number<br>of<br>samples | Per cent<br>below<br>LOR |
|----------|------------|--------|-----------------------|--------|--------|--------------------|-------------------------|--------------------------|
| 10WP31   | 332        | 335    | 27                    | 295    | 369    | 360                | 7                       | 0                        |
| 10WP32   | 1 234      | 1 178  | 265                   | 828    | 1 488  | 1 487              | 7                       | 0                        |
| 10WP32PB | 1 062      | 1 024  | 287                   | 448    | 1 358  | 1 246              | 7                       | 0                        |
| 10WP33   | 893        | 947    | 104                   | 851    | 1 103  | 1 073              | 7                       | 0                        |
| 10WP35N  | 967        | 938    | 270                   | 449    | 1 225  | 1 189              | 7                       | 0                        |
| 10WP35PB | 1 230      | 1 225  | 204                   | 876    | 1 455  | 1 427              | 9                       | 0                        |
| 10WP35S  | 1 241      | 1 115  | 303                   | 641    | 1 448  | 1 370              | 7                       | 0                        |
| 10WP36N  | 1 015      | 932    | 303                   | 398    | 1 191  | 1 177              | 7                       | 0                        |
| 10WP36PB | 968        | 949    | 269                   | 359    | 1 309  | 1 155              | 10                      | 0                        |
| 10WP36S  | 1 308      | 1 164  | 334                   | 600    | 1 434  | 1 421              | 7                       | 0                        |
| 10WP37   | 1 508      | 1 445  | 272                   | 914    | 1 722  | 1 699              | 7                       | 0                        |
| 10WP39   | 840        | 868    | 66                    | 807    | 1 000  | 919                | 7                       | 0                        |
| 10WP40   | 4 193      | 4 360  | 1 735                 | 2 332  | 7 524  | 5 303              | 7                       | 0                        |
| 10WP41   | 16 829     | 15 942 | 3 667                 | 11 124 | 20 734 | 19 427             | 7                       | 0                        |
| 10WP42   | 1 574      | 1 638  | 154                   | 1474   | 1 932  | 1 729              | 7                       | 0                        |
| 10WP44   | 20 603     | 20 712 | 3 823                 | 17 083 | 24 559 | 24 217             | 4                       | 0                        |
| 10WP46   | 1 569      | 1628   | 209                   | 1 443  | 2 083  | 1 666              | 7                       | 0                        |
| 10WP47   | 485        | 481    | 65                    | 357    | 561    | 529                | 7                       | 0                        |
| 11CS10RD | 1 898      | 1 808  | 485                   | 1 228  | 2 400  | 2 235              | 6                       | 0                        |
| 11CS10RS | 4 711      | 4 977  | 1 526                 | 2 847  | 7 218  | 6 468              | 6                       | 0                        |
| 11WP4R   | 1 048      | 1 071  | 180                   | 823    | 1 345  | 1 229              | 6                       | 0                        |
| 11WP9RPB | 1 477      | 1 340  | 329                   | 777    | 1 619  | 1 579              | 7                       | 0                        |
| 11WP9RS  | 1 474      | 1 313  | 357                   | 925    | 1 666  | 1 617              | 5                       | 0                        |
| 11WP11RD | 781        | 718    | 143                   | 512    | 848    | 828                | 6                       | 0                        |
| 11WP11RS | 1 028      | 829    | 332                   | 398    | 1 072  | 1 051              | 6                       | 0                        |
| 11WP15R  | 5 891      | 5 570  | 610                   | 4 765  | 6 094  | 6 004              | 6                       | 0                        |
| 11WP16R  | 9 157      | 9 741  | 1 525                 | 8 241  | 12 309 | 11 228             | 6                       | 0                        |
| 11WP43D  | 10 104     | 10 023 | 6 617                 | 2 064  | 17 822 | 16 074             | 4                       | 0                        |
| 11WP50   | 18 408     | 18 409 | 3 144                 | 15 331 | 21 490 | 21 259             | 4                       | 0                        |
| 11WP51D  | 583        | 555    | 68                    | 429    | 607    | 603                | 6                       | 0                        |
| 11WP51S  | 772        | 698    | 406                   | 142    | 1 166  | 1 081              | 6                       | 0                        |
| 11WP52D  | 645        | 613    | 130                   | 422    | 745    | 735                | 6                       | 0                        |

| Site      | Media<br>n | Mean   | Standard<br>deviation | Min.   | Max.   | 80th<br>percentile | Number<br>of<br>samples | Per cent<br>below<br>LOR |
|-----------|------------|--------|-----------------------|--------|--------|--------------------|-------------------------|--------------------------|
| 11WP53D   | 6 168      | 5 690  | 1 087                 | 4 202  | 6 705  | 6 540              | 6                       | 0                        |
| 11WP53S   | 6 708      | 6 200  | 987                   | 4 871  | 7 023  | 6 927              | 6                       | 0                        |
| 11WP54D   | 19 254     | 20 122 | 2 666                 | 17 375 | 24 156 | 23 061             | 6                       | 0                        |
| 11WP54S   | 13 222     | 13 346 | 1 469                 | 11 489 | 15 611 | 14 594             | 6                       | 0                        |
| 11WP55    | 2 817      | 2 859  | 238                   | 2 572  | 3 246  | 3 067              | 6                       | 0                        |
| 11WP56D   | 57         | 61     | 14                    | 43     | 78     | 77                 | 6                       | 0                        |
| 11WP56S   | 655        | 581    | 369                   | 181    | 908    | 883                | 3                       | 0                        |
| 11WP57    | 11 071     | 11 779 | 2 173                 | 9 354  | 14 599 | 14 388             | 6                       | 0                        |
| CG1       | 4 249      | 4 300  | 420                   | 3 760  | 4 772  | 4 763              | 6                       | 0                        |
| CG2       | 699        | 717    | 45                    | 670    | 773    | 765                | 5                       | 0                        |
| CG4       | 2 207      | 2 268  | 268                   | 2 017  | 2 726  | 2 480              | 6                       | 0                        |
| KC3       | 198        | 192    | 16                    | 168    | 208    | 204                | 5                       | 0                        |
| KC3A      | 296        | 314    | 61                    | 256    | 417    | 373                | 6                       | 0                        |
| KC3PB     | 278        | 310    | 72                    | 254    | 431    | 386                | 6                       | 0                        |
| KC13      | 827        | 799    | 112                   | 654    | 951    | 885                | 6                       | 0                        |
| KC14      | 13 327     | 15 350 | 3 472                 | 12 341 | 19 571 | 19 117             | 5                       | 0                        |
| KCF1      | 3 911      | 3 935  | 378                   | 3 494  | 4 524  | 4 279              | 6                       | 0                        |
| LIMESTONE | 432        | 411    | 132                   | 211    | 562    | 523                | 6                       | 0                        |
| M1        | 189        | 189    | 4                     | 186    | 192    | 192                | 3/2*                    | 0                        |
| RN029659  | 2 715      | 2 423  | 1 173                 | 295    | 3 519  | 3 263              | 6                       | 0                        |
| RN029660  | 1 255      | 1 242  | 151                   | 973    | 1 408  | 1 366              | 6                       | 0                        |
| W2R       | 398        | 360    | 85                    | 215    | 427    | 422                | 6                       | 0                        |
| WP5       | 3 617      | 3 902  | 687                   | 3 276  | 5 019  | 4 781              | 7                       | 0                        |
| WP13      | 713        | 728    | 75                    | 645    | 839    | 800                | 6/5*                    | 0                        |
| WP19      | 19 680     | 21 648 | 3 102                 | 19045  | 25 512 | 25 015             | 5                       | 0                        |

\* Number of samples and number used in statistical analysis

Table D51 Concentration of sodium adsorption ratio

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31   | 0.7    | 0.8  | 0.1                | 0.6  | 1.1  | 0.8             | 7                 | 0                  |
| 10WP32   | 5.3    | 4.9  | 0.6                | 4.0  | 5.4  | 5.4             | 7                 | 0                  |
| 10WP32PB | 4.7    | 4.5  | 1.2                | 2.0  | 5.7  | 5.2             | 7                 | 0                  |
| 10WP33   | 3.6    | 3.5  | 0.2                | 3.1  | 3.8  | 3.6             | 7                 | 0                  |
| 10WP35N  | 8.4    | 8.0  | 0.9                | 6.2  | 8.8  | 8.7             | 7                 | 0                  |
| 10WP35PB | 5.0    | 4.8  | 0.3                | 4.4  | 5.3  | 5.0             | 9                 | 0                  |
| 10WP35S  | 5.4    | 5.6  | 0.6                | 4.9  | 6.8  | 5.8             | 7                 | 0                  |
| 10WP36N  | 5.6    | 5.2  | 1.1                | 2.9  | 6.3  | 6.0             | 7                 | 0                  |
| 10WP36PB | 6.5    | 6.5  | 1.6                | 4.2  | 10.3 | 7.1             | 10                | 0                  |
| 10WP36S  | 9.3    | 9.8  | 2.4                | 7.5  | 15.0 | 10.7            | 7                 | 0                  |
| 10WP37   | 7.9    | 7.9  | 0.8                | 6.7  | 9.2  | 8.3             | 7                 | 0                  |
| 10WP39   | 2.1    | 2.2  | 0.3                | 2.0  | 2.7  | 2.4             | 7                 | 0                  |
| 10WP40   | 16.6   | 16.6 | 1.1                | 15.0 | 18.4 | 17.2            | 7                 | 0                  |
| 10WP41   | 23.8   | 24.0 | 1.4                | 22.3 | 26.3 | 25.2            | 7                 | 0                  |
| 10WP42   | 5.9    | 6.0  | 0.6                | 5.5  | 6.9  | 6.5             | 7                 | 0                  |
| 10WP44   | 17.9   | 17.8 | 0.5                | 17.2 | 18.2 | 18.2            | 4                 | 0                  |
| 10WP46   | 5.2    | 5.3  | 0.4                | 4.7  | 6.1  | 5.5             | 7                 | 0                  |
| 10WP47   | 1.2    | 1.2  | 0.3                | 0.7  | 1.8  | 1.4             | 7                 | 0                  |
| 11CS10RD | 3.8    | 4.4  | 1.3                | 3.3  | 6.6  | 5.8             | 6                 | 0                  |
| 11CS10RS | 17.5   | 22.3 | 12.9               | 13.8 | 48.2 | 29.2            | 6                 | 0                  |
| 11WP4R   | 2.6    | 2.6  | 0.3                | 2.1  | 2.9  | 2.7             | 6                 | 0                  |
| 11WP9RPB | 9.5    | 10.2 | 2.3                | 8.6  | 15.2 | 10.7            | 7                 | 0                  |
| 11WP9RS  | 10.2   | 9.7  | 0.7                | 8.8  | 10.3 | 10.2            | 5                 | 0                  |
| 11WP11RD | 4.3    | 4.3  | 0.2                | 3.9  | 4.5  | 4.4             | 6                 | 0                  |
| 11WP11RS | 9.7    | 10.0 | 0.9                | 9.0  | 11.5 | 10.9            | 6                 | 0                  |
| 11WP15R  | 35.3   | 35.3 | 2.1                | 31.8 | 38.4 | 36.8            | 6                 | 0                  |
| 11WP16R  | 12.5   | 13.9 | 3.4                | 10.7 | 18.4 | 18.2            | 6                 | 0                  |
| 11WP43D  | 4.9    | 4.7  | 1.7                | 2.7  | 6.4  | 6.2             | 4                 | 0                  |
| 11WP50   | 20.1   | 20.1 | 0.5                | 19.6 | 20.6 | 20.6            | 4                 | 0                  |
| 11WP51D  | 1.9    | 1.8  | 0.2                | 1.5  | 2.0  | 2.0             | 6                 | 0                  |
| 11WP51S  | 1.8    | 2.4  | 1.6                | 0.6  | 4.8  | 4.0             | 6                 | 0                  |
| 11WP52D  | 3.7    | 3.6  | 0.5                | 2.9  | 4.3  | 4.0             | 6                 | 0                  |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D   | 66.0   | 65.3 | 4.9                | 56.7 | 70.0 | 69.6            | 6                 | 0                  |
| 11WP53S   | 72.6   | 71.8 | 4.4                | 66.3 | 76.4 | 75.9            | 6                 | 0                  |
| 11WP54D   | 27.3   | 27.9 | 2.0                | 25.4 | 31.4 | 29.6            | 6                 | 0                  |
| 11WP54S   | 32.0   | 31.6 | 2.8                | 28.5 | 34.2 | 34.2            | 6                 | 0                  |
| 11WP55    | 10.9   | 12.6 | 4.3                | 10.2 | 21.3 | 14.7            | 6                 | 0                  |
| 11WP56D   | 1.3    | 1.2  | 0.4                | 0.8  | 1.7  | 1.5             | 6                 | 0                  |
| 11WP56S   | 2.1    | 2.0  | 0.8                | 1.1  | 2.7  | 2.7             | 3                 | 0                  |
| 11WP57    | 13.5   | 13.8 | 0.9                | 13.0 | 15.5 | 14.3            | 6                 | 0                  |
| CG1       | 10.1   | 10.3 | 0.5                | 9.9  | 11.2 | 10.6            | 6                 | 0                  |
| CG2       | 2.1    | 2.2  | 0.6                | 1.6  | 3.0  | 2.9             | 5                 | 0                  |
| CG4       | 6.7    | 6.6  | 0.5                | 6.0  | 7.1  | 7.1             | 6                 | 0                  |
| KC3       | 0.8    | 0.8  | 0.1                | 0.7  | 1.0  | 0.9             | 5                 | 0                  |
| KC3A      | 0.9    | 1.0  | 0.1                | 0.8  | 1.1  | 1.0             | 6                 | 0                  |
| KC3PB     | 1.0    | 0.9  | 0.1                | 0.8  | 1.0  | 1.0             | 6                 | 0                  |
| KC13      | 3.9    | 3.8  | 0.5                | 3.1  | 4.3  | 4.2             | 6                 | 0                  |
| KC14      | 13.2   | 12.8 | 0.7                | 11.7 | 13.3 | 13.3            | 5                 | 0                  |
| KCF1      | 8.6    | 8.7  | 0.5                | 8.1  | 9.5  | 9.2             | 6                 | 0                  |
| LIMESTONE | 1.2    | 1.1  | 0.3                | 0.6  | 1.3  | 1.3             | 6                 | 0                  |
| M1        | 0.7    | 0.8  | 0.2                | 0.7  | 1.0  | 1.0             | 3                 | 0                  |
| RN029659  | 8.7    | 8.0  | 3.5                | 1.1  | 11.3 | 10.3            | 6                 | 0                  |
| RN029660  | 9.1    | 9.2  | 0.4                | 8.9  | 9.8  | 9.6             | 6                 | 0                  |
| W2R       | 0.6    | 0.6  | 0.1                | 0.4  | 0.8  | 0.7             | 6                 | 0                  |
| WP5       | 2.4    | 2.7  | 0.4                | 2.2  | 3.3  | 3.1             | 7                 | 0                  |
| WP13      | 1.5    | 1.5  | 0.2                | 1.3  | 1.9  | 1.6             | 6                 | 0                  |
| WP19      | 26.0   | 26.1 | 1.0                | 24.9 | 27.2 | 27.1            | 5                 | 0                  |

Table D52 Concentration of atrazine (µg/L)

| Site                  | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|-----------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 10WP31 <sup>†</sup>   | 0.03   | 0.04 | 0.05               | 0.01 | 0.10 | 0.09            | 4                 | 100                |
| 10WP32 <sup>†</sup>   | 0.10   | 0.07 | 0.04               | 0.01 | 0.10 | 0.10            | 5                 | 100                |
| 10WP32PB <sup>†</sup> | 0.08   | 0.08 | 0.04               | 0.05 | 0.10 | 0.10            | 2                 | 100                |
| 10WP33 <sup>†</sup>   | 0.01   | 0.01 | 0.02               | 0.01 | 0.05 | 0.03            | 5                 | 100                |
| 10WP35N <sup>†</sup>  | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.07            | 6                 | 100                |
| 10WP35PB <sup>†</sup> | 0.01   | 0.02 | 0.02               | 0.01 | 0.05 | 0.04            | 4                 | 100                |
| 10WP35S <sup>†</sup>  | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| 10WP36N <sup>†</sup>  | 0.03   | 0.04 | 0.05               | 0.01 | 0.10 | 0.09            | 4                 | 100                |
| 10WP36PB <sup>†</sup> | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 7                 | 100                |
| 10WP36S <sup>†</sup>  | 0.03   | 0.03 | 0.03               | 0.01 | 0.05 | 0.05            | 2                 | 100                |
| 10WP37 <sup>†</sup>   | 0.08   | 0.06 | 0.05               | 0.01 | 0.10 | 0.10            | 4                 | 100                |
| 10WP39 <sup>†</sup>   | 0.03   | 0.04 | 0.05               | 0.01 | 0.10 | 0.09            | 4                 | 100                |
| 10WP40 <sup>†</sup>   | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.08            | 5                 | 100                |
| 10WP41 <sup>†</sup>   | 0.01   | 0.02 | 0.02               | 0.01 | 0.05 | 0.04            | 4                 | 100                |
| 10WP42 <sup>†</sup>   | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| 10WP44 <sup>†</sup>   | 0.05   | 0.05 | ND                 | 0.05 | 0.05 | 0.05            | 1                 | 100                |
| 10WP46 <sup>†</sup>   | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| 10WP47 <sup>†</sup>   | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.08            | 5                 | 100                |
| 11CS10RD <sup>†</sup> | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.08            | 5                 | 100                |
| 11CS10RS <sup>†</sup> | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| 11WP4R <sup>†</sup>   | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| 11WP9RPB <sup>†</sup> | 0.08   | 0.06 | 0.05               | 0.01 | 0.10 | 0.10            | 4                 | 100                |
| 11WP9RS <sup>†</sup>  | 0.03   | 0.03 | 0.03               | 0.01 | 0.05 | 0.05            | 2                 | 100                |
| 11WP11RD <sup>†</sup> | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| 11WP11RS <sup>†</sup> | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.08            | 5                 | 100                |
| 11WP15R <sup>†</sup>  | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| 11WP16R <sup>†</sup>  | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.07            | 6                 | 100                |
| 11WP43D <sup>†</sup>  | 0.03   | 0.04 | 0.05               | 0.01 | 0.10 | 0.09            | 4                 | 100                |
| 11WP50 <sup>†</sup>   | 0.05   | 0.05 | ND                 | 0.05 | 0.05 | 0.05            | 1                 | 100                |
| 11WP51D <sup>†</sup>  | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 5                 | 100                |
| 11WP51S <sup>†</sup>  | 0.08   | 0.06 | 0.05               | 0.01 | 0.10 | 0.10            | 4                 | 100                |
| 11WP52D <sup>†</sup>  | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |

| Site                   | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples | Per cent below LOR |
|------------------------|--------|------|--------------------|------|------|-----------------|-------------------|--------------------|
| 11WP53D <sup>†</sup>   | 0.03   | 0.04 | 0.05               | 0.01 | 0.10 | 0.09            | 4                 | 100                |
| 11WP53S <sup>†</sup>   | 0.03   | 0.03 | 0.03               | 0.01 | 0.05 | 0.05            | 2                 | 100                |
| 11WP54D <sup>†</sup>   | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| 11WP54S <sup>†</sup>   | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| 11WP55 <sup>†</sup>    | 0.01   | 0.02 | 0.02               | 0.01 | 0.05 | 0.04            | 4                 | 100                |
| 11WP56D <sup>†</sup>   | 0.01   | 0.02 | 0.02               | 0.01 | 0.05 | 0.04            | 4                 | 100                |
| 11WP56S <sup>†</sup>   | 0.01   | 0.01 | ND                 | 0.01 | 0.01 | 0.01            | 1                 | 100                |
| 11WP57 <sup>†</sup>    | 0.10   | 0.09 | 0.03               | 0.05 | 0.10 | 0.10            | 4                 | 100                |
| CG1 <sup>†</sup>       | 0.03   | 0.03 | 0.03               | 0.01 | 0.05 | 0.05            | 4                 | 100                |
| CG2 <sup>†</sup>       | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| CG4 <sup>†</sup>       | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| KC3 <sup>†</sup>       | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.08            | 5                 | 100                |
| KC3A <sup>†</sup>      | 0.05   | 0.05 | ND                 | 0.05 | 0.05 | 0.05            | 1                 | 100                |
| KC3PB <sup>†</sup>     | 0.08   | 0.06 | 0.05               | 0.01 | 0.10 | 0.10            | 4                 | 100                |
| KC13 <sup>†</sup>      | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| KC14 <sup>†</sup>      | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| KCF1 <sup>†</sup>      | 0.01   | 0.03 | 0.04               | 0.01 | 0.10 | 0.08            | 5                 | 100                |
| LIMESTONE <sup>†</sup> | 0.03   | 0.04 | 0.05               | 0.01 | 0.10 | 0.10            | 6                 | 100                |
| RN029659 <sup>†</sup>  | 0.01   | 0.02 | 0.02               | 0.01 | 0.05 | 0.04            | 4                 | 100                |
| RN029660 <sup>†</sup>  | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |
| W2R <sup>†</sup>       | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| WP5 <sup>†</sup>       | 0.05   | 0.05 | 0.05               | 0.01 | 0.10 | 0.10            | 3                 | 100                |
| WP13 <sup>†</sup>      | 0.01   | 0.02 | 0.02               | 0.01 | 0.05 | 0.04            | 4                 | 100                |
| WP19 <sup>†</sup>      | 0.01   | 0.02 | 0.03               | 0.01 | 0.05 | 0.05            | 3                 | 100                |

<sup>†</sup> More than one-quarter of the samples are below the LOR

ND No data

## Appendix E Summary statistics of field measurements

Table E1 Concentration of electrical conductivity (mS/m)

| Site     | Median | Mean  | Standard deviation | Min.  | Max.   | 80th percentile | Number of samples |
|----------|--------|-------|--------------------|-------|--------|-----------------|-------------------|
| 10WP31   | 43.0   | 42.1  | 3.6                | 37.2  | 46.3   | 45.8            | 9                 |
| 10WP32   | 176.8  | 170.3 | 34.8               | 122.1 | 211.0  | 205.9           | 8                 |
| 10WP32PB | 154.0  | 152.2 | 24.9               | 123.5 | 184.0  | 175.9           | 5                 |
| 10WP33   | 126.6  | 132.8 | 14.7               | 114.8 | 153.5  | 151.2           | 8                 |
| 10WP35N  | 122.3  | 125.7 | 18.1               | 108.7 | 151.0  | 144.3           | 8                 |
| 10WP35PB | 171.5  | 171.5 | 5.1                | 164.5 | 179.0  | 176.1           | 6                 |
| 10WP35S  | 152.9  | 149.3 | 18.8               | 115.0 | 172.6  | 164.6           | 8                 |
| 10WP36N  | 136.1  | 133.8 | 8.6                | 120.0 | 142.9  | 142.3           | 8                 |
| 10WP36PB | 131.0  | 128.8 | 27.5               | 90.1  | 159.6  | 154.5           | 6                 |
| 10WP36S  | 151.2  | 155.3 | 10.4               | 144.9 | 171.6  | 167.6           | 8                 |
| 10WP37   | 210.0  | 200.3 | 13.7               | 179.6 | 213.0  | 211.0           | 9                 |
| 10WP39   | 120.1  | 293.6 | 429.3              | 111.6 | 1170.0 | 436.3           | 6                 |
| 10WP40   | 612.0  | 601.0 | 166.2              | 360.0 | 943.0  | 649.4           | 9                 |
| 10WP41   | 2370   | 2151  | 396                | 1496  | 2520   | 2394            | 7                 |
| 10WP42   | 238.0  | 192.6 | 96.0               | 22.0  | 244.0  | 243.5           | 5                 |
| 10WP44   | 2340   | 2266  | 337                | 1636  | 2530   | 2523            | 6                 |
| 10WP46   | 227    | 629   | 1016               | 23    | 2440   | 1338            | 5                 |
| 10WP47   | 58.1   | 55.0  | 6.2                | 46.4  | 60.6   | 60.1            | 5                 |
| 11CS10RD | 248.0  | 243.3 | 76.6               | 167.0 | 356.0  | 323.6           | 7                 |
| 11CS10RS | 662.5  | 703.3 | 211.4              | 407.0 | 1044.0 | 879.5           | 6                 |
| 11WP11RD | 102.6  | 101.4 | 4.6                | 92.6  | 106.4  | 104.0           | 6                 |
| 11WP11RS | 118.8  | 128.3 | 21.3               | 113.6 | 169.5  | 144.2           | 6                 |
| 11WP15R  | 850.0  | 852.3 | 19.9               | 833.0 | 890.0  | 863.4           | 6                 |
| 11WP16R  | 1176   | 1188  | 62                 | 1110  | 1264   | 1258.2          | 8                 |
| 11WP43D  | 931    | 1084  | 788                | 384   | 1937   | 1836            | 3                 |
| 11WP4R   | 165.8  | 162.3 | 9.1                | 151.0 | 172.4  | 169.2           | 6                 |
| 11WP50   | 2145   | 2128  | 47                 | 2060  | 2160   | 2160            | 4                 |
| 11WP51D  | 77.6   | 78.7  | 6.7                | 70.0  | 93.3   | 82.6            | 9                 |
| 11WP51S  | 97.0   | 114.4 | 67.2               | 25.4  | 199.4  | 188.4           | 7                 |
| 11WP52D  | 93.7   | 92.1  | 5.1                | 85.4  | 97.2   | 96.6            | 6                 |
| 11WP53D  | 833    | 830   | 34                 | 791   | 884    | 855             | 6                 |

| Site      | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples |
|-----------|--------|-------|--------------------|-------|-------|-----------------|-------------------|
| 11WP53S   | 889    | 891   | 45                 | 827   | 943   | 938             | 6                 |
| 11WP54D   | 2455   | 2066  | 899                | 237   | 2490  | 2483            | 6                 |
| 11WP54S   | 1739   | 1680  | 113                | 1494  | 1760  | 1759            | 5                 |
| 11WP55    | 440.0  | 438.1 | 6.8                | 430.0 | 449.0 | 442.7           | 7                 |
| 11WP56D   | 10.0   | 9.8   | 2.3                | 7.0   | 12.2  | 12.0            | 6                 |
| 11WP56S   | 76.8   | 69.9  | 62.1               | 0.0   | 126.0 | 123.7           | 4                 |
| 11WP57    | 1650   | 1531  | 290                | 794   | 1705  | 1674            | 9                 |
| 11WP9RPB  | 201.0  | 201.4 | 7.7                | 190.1 | 209.0 | 208.5           | 5                 |
| 11WP9RS   | 202.0  | 200.1 | 5.7                | 190.1 | 205.0 | 204.3           | 6                 |
| CG1       | 590.0  | 581.4 | 74.4               | 457.0 | 645.0 | 638.5           | 5                 |
| CG2       | 91.9   | 91.9  | 5.9                | 85.7  | 98.0  | 97.3            | 4                 |
| CG4       | 310.0  | 317.9 | 28.9               | 289.0 | 354.0 | 351.3           | 7                 |
| KC13      | 101.8  | 100.1 | 12.1               | 82.8  | 113.1 | 110.8           | 8                 |
| KC14      | 1754   | 1745  | 29                 | 1705  | 1769  | 1768            | 4                 |
| KC3       | 26.4   | 28.5  | 7.0                | 22.6  | 44.6  | 29.7            | 8                 |
| KC3A      | 41.3   | 43.1  | 2.7                | 41.1  | 49.1  | 45.0            | 9                 |
| KC3PB     | 44.8   | 43.8  | 2.1                | 40.7  | 45.6  | 45.4            | 5                 |
| KCF1      | 481.0  | 477.8 | 18.7               | 456.0 | 497.0 | 495.5           | 5                 |
| LIMESTONE | 50.9   | 51.0  | 19.5               | 30.0  | 72.1  | 69.1            | 4                 |
| M1        | 25.3   | 25.2  | 1.1                | 24.1  | 26.2  | 26.1            | 3                 |
| RN029659  | 400.0  | 356.7 | 177.1              | 50.4  | 485.0 | 477.5           | 5                 |
| RN029660  | 179.6  | 177.6 | 5.3                | 170.0 | 181.4 | 181.3           | 4                 |
| W2R       | 49.8   | 47.8  | 8.5                | 37.0  | 54.7  | 54.6            | 4                 |
| WP13      | 107.2  | 105.9 | 7.4                | 97.5  | 119.5 | 109.4           | 8                 |
| WP19      | 2690   | 2688  | 45                 | 2640  | 2750  | 2730            | 5                 |
| WP5       | 505.0  | 499.4 | 11.5               | 486.0 | 510.0 | 509.0           | 5                 |

Table E2 Concentration of pH (field)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 7.0    | 8.0  | 1.8                | 6.5  | 10.5 | 10.4            | 9                 |
| 10WP32   | 7.5    | 8.3  | 1.5                | 7.2  | 10.7 | 10.4            | 8                 |
| 10WP32PB | 7.3    | 7.4  | 0.1                | 7.2  | 7.6  | 7.5             | 5                 |
| 10WP33   | 7.2    | 8.4  | 1.9                | 6.8  | 10.8 | 10.7            | 8                 |
| 10WP35N  | 7.6    | 8.4  | 1.6                | 7.4  | 11.1 | 10.4            | 8                 |
| 10WP35PB | 7.2    | 7.2  | 0.2                | 7.0  | 7.5  | 7.5             | 6                 |
| 10WP35S  | 7.5    | 8.1  | 1.2                | 7.1  | 10.2 | 9.7             | 8                 |
| 10WP36N  | 7.6    | 8.2  | 1.3                | 7.2  | 10.4 | 10.0            | 8                 |
| 10WP36PB | 7.5    | 7.7  | 0.6                | 7.1  | 8.9  | 8.0             | 6                 |
| 10WP36S  | 7.8    | 8.4  | 1.4                | 7.3  | 10.7 | 10.3            | 8                 |
| 10WP37   | 7.6    | 8.5  | 1.5                | 7.3  | 10.9 | 10.3            | 9                 |
| 10WP39   | 7.2    | 7.2  | 0.1                | 7.0  | 7.4  | 7.4             | 6                 |
| 10WP40   | 7.5    | 8.1  | 1.2                | 7.2  | 9.8  | 9.7             | 9                 |
| 10WP41   | 7.1    | 7.4  | 1.0                | 6.9  | 9.6  | 7.4             | 7                 |
| 10WP42   | 7.2    | 7.2  | 0.1                | 7.0  | 7.4  | 7.3             | 5                 |
| 10WP44   | 7.2    | 7.1  | 0.2                | 6.9  | 7.3  | 7.3             | 6                 |
| 10WP46   | 7.1    | 7.2  | 0.1                | 7.1  | 7.3  | 7.3             | 5                 |
| 10WP47   | 7.1    | 7.2  | 0.1                | 7.1  | 7.3  | 7.3             | 5                 |
| 11CS10RD | 7.1    | 7.9  | 1.4                | 6.9  | 10.1 | 9.9             | 7                 |
| 11CS10RS | 7.1    | 7.8  | 1.4                | 7.0  | 10.6 | 8.6             | 6                 |
| 11WP11RD | 7.4    | 7.7  | 0.8                | 7.1  | 9.3  | 8.1             | 6                 |
| 11WP11RS | 7.8    | 8.0  | 0.6                | 7.7  | 9.2  | 8.2             | 6                 |
| 11WP15R  | 7.4    | 7.4  | 0.1                | 7.2  | 7.6  | 7.5             | 6                 |
| 11WP16R  | 7.1    | 8.0  | 1.5                | 6.8  | 10.0 | 9.8             | 8                 |
| 11WP43D  | 7.1    | 7.1  | 0.3                | 6.8  | 7.4  | 7.4             | 3                 |
| 11WP4R   | 7.2    | 7.5  | 0.7                | 7.0  | 9.0  | 7.9             | 6                 |
| 11WP50   | 6.8    | 6.8  | 0.1                | 6.6  | 6.9  | 6.9             | 4                 |
| 11WP51D  | 7.7    | 8.9  | 1.8                | 7.2  | 11.3 | 10.9            | 9                 |
| 11WP51S  | 7.5    | 8.4  | 1.8                | 7.1  | 11.0 | 10.9            | 7                 |
| 11WP52D  | 7.1    | 7.3  | 0.7                | 6.8  | 8.6  | 7.6             | 6                 |
| 11WP53D  | 7.7    | 7.9  | 0.5                | 7.5  | 8.7  | 8.1             | 6                 |
| 11WP53S  | 7.8    | 8.1  | 0.9                | 7.6  | 9.9  | 8.6             | 6                 |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 11WP54D   | 6.9    | 7.2  | 0.9                | 6.6  | 9.0  | 7.6             | 6                 |
| 11WP54S   | 7.1    | 7.1  | 0.2                | 6.7  | 7.4  | 7.3             | 5                 |
| 11WP55    | 7.1    | 7.8  | 1.3                | 6.9  | 9.9  | 9.4             | 7                 |
| 11WP56D   | 5.6    | 5.8  | 1.2                | 4.7  | 7.9  | 6.7             | 6                 |
| 11WP56S   | 7.2    | 7.4  | 0.8                | 6.7  | 8.3  | 8.2             | 3                 |
| 11WP57    | 7.2    | 7.6  | 1.1                | 6.5  | 8.9  | 8.8             | 9                 |
| 11WP9RPB  | 7.5    | 7.5  | 0.3                | 7.2  | 8.0  | 7.7             | 5                 |
| 11WP9RS   | 7.4    | 7.8  | 1.0                | 7.1  | 9.8  | 8.2             | 6                 |
| CG1       | 7.1    | 7.1  | 0.1                | 7.0  | 7.2  | 7.1             | 5                 |
| CG2       | 7.4    | 7.3  | 0.1                | 7.2  | 7.4  | 7.4             | 4                 |
| CG4       | 7.2    | 8.0  | 1.4                | 6.9  | 10.0 | 9.8             | 7                 |
| KC13      | 7.5    | 7.7  | 0.8                | 6.6  | 9.3  | 8.2             | 8                 |
| KC14      | 6.7    | 6.7  | 0.1                | 6.6  | 6.9  | 6.8             | 4                 |
| KC3       | 6.3    | 7.2  | 1.5                | 5.6  | 9.1  | 9.0             | 8                 |
| KC3A      | 6.4    | 6.3  | 0.6                | 5.3  | 7.0  | 6.9             | 9                 |
| KC3PB     | 5.5    | 5.6  | 0.1                | 5.4  | 5.7  | 5.7             | 5                 |
| KCF1      | 7.1    | 7.1  | 0.1                | 7.1  | 7.3  | 7.2             | 5                 |
| LIMESTONE | 7.4    | 7.4  | 0.1                | 7.3  | 7.5  | 7.5             | 4                 |
| M1        | 8.5    | 8.5  | 0.0                | 8.4  | 8.5  | 8.5             | 3                 |
| RN029659  | 7.6    | 7.7  | 0.6                | 7.3  | 8.7  | 8.2             | 5                 |
| RN029660  | 7.5    | 7.5  | 0.1                | 7.4  | 7.7  | 7.6             | 4                 |
| W2R       | 7.3    | 7.3  | 0.2                | 7.2  | 7.5  | 7.5             | 4                 |
| WP13      | 7.4    | 8.5  | 1.7                | 7.1  | 10.6 | 10.5            | 8                 |
| WP19      | 6.9    | 6.8  | 0.1                | 6.7  | 7.0  | 6.9             | 5                 |
| WP5       | 7.0    | 6.9  | 0.2                | 6.7  | 7.1  | 7.0             | 5                 |

Table E3 Concentration of temperature (°C)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 30.8   | 30.9 | 0.9                | 29.5 | 33   | 31.5            | 9                 |
| 10WP32   | 29.6   | 29.7 | 1.2                | 28.1 | 31   | 30.9            | 8                 |
| 10WP32PB | 30.8   | 30.7 | 0.4                | 30.3 | 31   | 31.0            | 5                 |
| 10WP33   | 30.0   | 30.1 | 0.8                | 29.0 | 32   | 30.7            | 8                 |
| 10WP35N  | 28.0   | 28.7 | 1.5                | 27.2 | 32   | 30.0            | 8                 |
| 10WP35PB | 29.2   | 29.0 | 0.4                | 28.4 | 29   | 29.3            | 5                 |
| 10WP35S  | 28.4   | 28.7 | 0.8                | 27.7 | 30   | 29.7            | 8                 |
| 10WP36N  | 29.1   | 28.8 | 1.0                | 26.7 | 30   | 29.4            | 8                 |
| 10WP36PB | 29.6   | 30.0 | 0.8                | 29.3 | 31   | 30.8            | 5                 |
| 10WP36S  | 29.6   | 29.2 | 1.4                | 26.3 | 31   | 30.0            | 8                 |
| 10WP37   | 29.5   | 29.6 | 1.7                | 26.9 | 33   | 30.6            | 9                 |
| 10WP39   | 30.2   | 30.4 | 0.9                | 29.3 | 32   | 31.1            | 5                 |
| 10WP40   | 30.3   | 30.3 | 1.4                | 28.3 | 32   | 31.6            | 8                 |
| 10WP41   | 29.6   | 29.2 | 1.2                | 27.6 | 31   | 30.4            | 7                 |
| 10WP42   | 29.4   | 29.5 | 0.7                | 28.7 | 31   | 30.2            | 5                 |
| 10WP44   | 30.1   | 30.1 | 1.5                | 28.6 | 33   | 31.2            | 6                 |
| 10WP46   | 30.8   | 30.5 | 0.5                | 29.6 | 31   | 30.9            | 5                 |
| 10WP47   | 28.6   | 28.9 | 0.6                | 28.4 | 30   | 29.5            | 5                 |
| 11CS10RD | 31.1   | 31.0 | 0.8                | 29.5 | 32   | 31.6            | 7                 |
| 11CS10RS | 31.0   | 30.8 | 0.6                | 30.0 | 31   | 31.2            | 6                 |
| 11WP11RD | 30.8   | 30.8 | 1.3                | 29.2 | 33   | 32.2            | 6                 |
| 11WP11RS | 30.7   | 30.3 | 1.6                | 27.8 | 32   | 31.5            | 6                 |
| 11WP15R  | 30.8   | 30.7 | 0.5                | 30.0 | 31   | 31.1            | 6                 |
| 11WP16R  | 29.4   | 29.5 | 1.5                | 26.7 | 32   | 30.8            | 8                 |
| 11WP43D  | 30.0   | 29.8 | 0.5                | 29.3 | 30   | 30.2            | 3                 |
| 11WP4R   | 32.2   | 32.1 | 1.8                | 29.9 | 35   | 33.9            | 6                 |
| 11WP50   | 29.6   | 29.5 | 0.3                | 29.0 | 30   | 29.7            | 4                 |
| 11WP51D  | 29.3   | 29.4 | 0.9                | 28.3 | 31   | 30.3            | 9                 |
| 11WP51S  | 29.7   | 29.8 | 1.0                | 28.5 | 32   | 30.7            | 7                 |
| 11WP52D  | 31.4   | 31.4 | 1.6                | 29.0 | 34   | 32.4            | 6                 |
| 11WP53D  | 29.3   | 29.3 | 0.5                | 28.7 | 30   | 29.7            | 6                 |
| 11WP53S  | 29.4   | 29.4 | 0.7                | 28.3 | 30   | 30.0            | 6                 |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 11WP54D   | 29.0   | 29.0 | 0.6                | 28.3 | 30   | 29.6            | 6                 |
| 11WP54S   | 28.9   | 28.8 | 0.9                | 27.3 | 30   | 29.5            | 5                 |
| 11WP55    | 30.4   | 30.1 | 2.6                | 26.7 | 35   | 31.2            | 7                 |
| 11WP56D   | 31.7   | 31.9 | 0.8                | 30.8 | 33   | 32.6            | 6                 |
| 11WP56S   | 31.6   | 31.0 | 1.1                | 29.7 | 32   | 31.7            | 3                 |
| 11WP57    | 29.8   | 30.2 | 0.7                | 29.5 | 32   | 30.8            | 9                 |
| 11WP9RPB  | 30.5   | 30.6 | 0.6                | 29.9 | 32   | 31.1            | 5                 |
| 11WP9RS   | 30.4   | 30.5 | 0.7                | 29.6 | 32   | 30.9            | 6                 |
| CG1       | 30.0   | 29.6 | 0.8                | 28.5 | 30   | 30.3            | 5                 |
| CG2       | 29.9   | 29.9 | 0.5                | 29.4 | 31   | 30.4            | 4                 |
| CG4       | 30.5   | 30.3 | 0.8                | 29.4 | 31   | 30.8            | 7                 |
| KC13      | 30.1   | 30.2 | 0.9                | 29.2 | 32   | 30.8            | 8                 |
| KC14      | 31.8   | 32.2 | 1.0                | 31.4 | 34   | 33.2            | 4                 |
| KC3       | 30.8   | 30.4 | 0.9                | 29.0 | 31   | 31.1            | 8                 |
| KC3A      | 30.8   | 30.8 | 0.3                | 30.3 | 31   | 31.0            | 9                 |
| KC3PB     | 31.2   | 31.2 | 0.2                | 30.9 | 32   | 31.4            | 5                 |
| KCF1      | 31.4   | 32.4 | 2.2                | 31.0 | 36   | 34.2            | 5                 |
| LIMESTONE | 34.0   | 34.0 | 0.3                | 33.6 | 34   | 34.3            | 4                 |
| M1        | 26.8   | 27.7 | 2.7                | 25.6 | 31   | 30.4            | 3                 |
| RN029659  | 30.3   | 30.3 | 0.4                | 29.8 | 31   | 30.7            | 5                 |
| RN029660  | 30.7   | 30.7 | 0.5                | 30.1 | 31   | 31.1            | 4                 |
| W2R       | 33.9   | 33.8 | 1.0                | 32.4 | 35   | 34.7            | 4                 |
| WP13      | 28.9   | 29.0 | 0.9                | 27.9 | 30   | 30.1            | 8                 |
| WP19      | 30.0   | 30.0 | 0.5                | 29.6 | 31   | 30.5            | 5                 |
| WP5       | 29.4   | 29.3 | 0.4                | 28.9 | 30   | 29.6            | 4                 |

Table E4 Concentration of total acidity (mg/L CaCO<sub>3</sub>)

| Site     | Median  | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|---------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 60      | 73   | 42                 | 40   | 120  | No data         | 3                 |
| 10WP32   | 40      | 67   | 46                 | 40   | 120  | No data         | 3                 |
| 10WP32PB | 100     | 100  | 57                 | 60   | 140  | No data         | 2                 |
| 10WP33   | 110     | 110  | 71                 | 60   | 160  | No data         | 2                 |
| 10WP35N  | 40      | 67   | 46                 | 40   | 120  | No data         | 3                 |
| 10WP35PB | 100     | 120  | 35                 | 100  | 160  | No data         | 3                 |
| 10WP35S  | 60      | 87   | 64                 | 40   | 160  | No data         | 3                 |
| 10WP36N  | 60      | 80   | 53                 | 40   | 140  | No data         | 3                 |
| 10WP36PB | 60      | 60   | No data            | 60   | 60   | No data         | 1                 |
| 10WP36S  | 40      | 67   | 46                 | 40   | 120  | No data         | 3                 |
| 10WP37   | 60      | 87   | 64                 | 40   | 160  | No data         | 3                 |
| 10WP39   | 80      | 80   | 40                 | 40   | 120  | No data         | 3                 |
| 10WP40   | 100     | 100  | 40                 | 60   | 140  | No data         | 3                 |
| 10WP41   | 120     | 87   | 76                 | 0    | 140  | No data         | 3                 |
| 10WP42   | 30      | 30   | 42                 | 0    | 60   | No data         | 2                 |
| 10WP44   | No data |      |                    |      |      |                 |                   |
| 10WP46   | 30      | 30   | 42                 | 0    | 60   | No data         | 2                 |
| 10WP47   | 30      | 30   | 42                 | 0    | 60   | No data         | 2                 |
| 11CS10RD | 140     | 140  | 113                | 60   | 220  | No data         | 2                 |
| 11CS10RS | 220     | 220  | 170                | 100  | 340  | No data         | 2                 |
| 11WP11RD | 90      | 90   | 42                 | 60   | 120  | No data         | 2                 |
| 11WP11RS | 80      | 80   | 28                 | 60   | 100  | No data         | 2                 |
| 11WP15R  | 190     | 190  | 127                | 100  | 280  | No data         | 2                 |
| 11WP16R  | 150     | 150  | 71                 | 100  | 200  | No data         | 2                 |
| 11WP43D  | 0       | 0    | No data            | 0    | 0    | No data         | 1                 |
| 11WP4R   | 90      | 90   | 71                 | 40   | 140  | No data         | 2                 |
| 11WP50   | 400     | 400  | No data            | 400  | 400  | No data         | 1                 |
| 11WP51D  | 70      | 70   | 42                 | 40   | 100  | No data         | 2                 |
| 11WP51S  | 90      | 90   | 42                 | 60   | 120  | No data         | 2                 |
| 11WP52D  | 110     | 110  | 71                 | 60   | 160  | No data         | 2                 |
| 11WP53D  | 150     | 150  | 127                | 60   | 240  | No data         | 2                 |
| 11WP53S  | 130     | 130  | 127                | 40   | 220  | No data         | 2                 |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 11WP54D   | 340    | 340  | 311                | 120  | 560  | No data         | 2                 |
| 11WP54S   | 330    | 330  | 297                | 120  | 540  | No data         | 2                 |
| 11WP55    | 250    | 250  | 127                | 160  | 340  | No data         | 2                 |
| 11WP56D   | 130    | 130  | 99                 | 60   | 200  | No data         | 2                 |
| 11WP56S   | 100    | 100  | 85                 | 40   | 160  | No data         | 2                 |
| 11WP57    | 210    | 210  | 156                | 100  | 320  | No data         | 2                 |
| 11WP9RPB  | 80     | 80   | 57                 | 40   | 120  | No data         | 2                 |
| 11WP9RS   | 140    | 140  | No data            | 140  | 140  | No data         | 1                 |
| CG1       | 0      | 0    | No data            | 0    | 0    | No data         | 1                 |
| CG2       | 0      | 0    | No data            | 0    | 0    | No data         | 1                 |
| CG4       | 20     | 20   | 28                 | 0    | 40   | No data         | 2                 |
| KC13      | 80     | 80   | 57                 | 40   | 120  | No data         | 2                 |
| KC14      | 340    | 340  | No data            | 340  | 340  | No data         | 1                 |
| KC3       | 170    | 170  | 42                 | 140  | 200  | No data         | 2                 |
| KC3A      | 40     | 40   | No data            | 40   | 40   | No data         | 1                 |
| KC3PB     | 160    | 160  | 57                 | 120  | 200  | No data         | 2                 |
| KCF1      | 190    | 190  | 42                 | 160  | 220  | No data         | 2                 |
| LIMESTONE | 0      | 0    | No data            | 0    | 0    | No data         | 1                 |
| M1        | 20     | 20   | 0                  | 20   | 20   | No data         | 3                 |
| RN029659  | 20     | 20   | 28                 | 0    | 40   | No data         | 2                 |
| RN029660  | 0      | 0    | No data            | 0    | 0    | No data         | 1                 |
| W2R       | 0      | 0    | No data            | 0    | 0    | No data         | 1                 |
| WP13      | 20     | 20   | 28                 | 0    | 40   | No data         | 2                 |
| WP19      | 230    | 230  | 71                 | 180  | 280  | No data         | 2                 |
| WP5       | 250    | 250  | 184                | 120  | 380  | No data         | 2                 |

Table E5 Concentration of total alkalinity (mg/L CaCO<sub>3</sub>)

| Site     | Median  | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|---------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 156     | 161  | 17                 | 147  | 180  | No data         | 3                 |
| 10WP32   | 423     | 448  | 45                 | 420  | 500  | No data         | 3                 |
| 10WP32PB | 458     | 458  | 53                 | 420  | 495  | No data         | 2                 |
| 10WP33   | 357     | 357  | 47                 | 324  | 390  | No data         | 2                 |
| 10WP35N  | 550     | 581  | 85                 | 516  | 678  | No data         | 3                 |
| 10WP35PB | 630     | 614  | 39                 | 570  | 642  | No data         | 3                 |
| 10WP35S  | 600     | 589  | 119                | 465  | 702  | No data         | 3                 |
| 10WP36N  | 492     | 500  | 34                 | 470  | 537  | No data         | 3                 |
| 10WP36PB | 450     | 450  | 12                 | 438  | 462  | No data         | 3                 |
| 10WP36S  | 660     | 647  | 42                 | 600  | 680  | No data         | 3                 |
| 10WP37   | 680     | 664  | 81                 | 576  | 735  | No data         | 3                 |
| 10WP39   | 238     | 238  | 25                 | 220  | 255  | No data         | 2                 |
| 10WP40   | 707     | 707  | 52                 | 670  | 744  | No data         | 2                 |
| 10WP41   | 550     | 561  | 117                | 450  | 684  | No data         | 3                 |
| 10WP42   | 340     | 340  | 57                 | 300  | 380  | No data         | 2                 |
| 10WP44   | No data |      |                    |      |      |                 |                   |
| 10WP46   | 320     | 320  | 14                 | 310  | 330  | No data         | 2                 |
| 10WP47   | 155     | 155  | 92                 | 90   | 220  | No data         | 2                 |
| 11CS10RD | 439     | 439  | 54                 | 400  | 477  | No data         | 2                 |
| 11CS10RS | 898     | 898  | 116                | 816  | 980  | No data         | 2                 |
| 11WP11RD | 380     | 380  | 28                 | 360  | 400  | No data         | 2                 |
| 11WP11RS | 565     | 565  | 78                 | 510  | 620  | No data         | 2                 |
| 11WP15R  | 1233    | 1233 | 81                 | 1176 | 1290 | No data         | 2                 |
| 11WP16R  | 485     | 485  | 91                 | 420  | 549  | No data         | 2                 |
| 11WP43D  | 160     | 160  | No data            | 160  | 160  | No data         | 1                 |
| 11WP4R   | 318     | 318  | 4                  | 315  | 320  | No data         | 2                 |
| 11WP50   | No data |      |                    |      |      |                 |                   |
| 11WP51D  | 320     | 320  | No data            | 320  | 320  | No data         | 1                 |
| 11WP51S  | 380     | 380  | No data            | 380  | 380  | No data         | 1                 |
| 11WP52D  | 300     | 300  | No data            | 300  | 300  | No data         | 1                 |
| 11WP53D  | 1803    | 1803 | 38                 | 1776 | 1830 | No data         | 2                 |
| 11WP53S  | 1875    | 1875 | 106                | 1800 | 1950 | No data         | 2                 |

| Site      | Median  | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|---------|------|--------------------|------|------|-----------------|-------------------|
| 11WP54D   | 705     | 705  | 50                 | 669  | 740  | No data         | 2                 |
| 11WP54S   | 1050    | 1050 | No data            | 1050 | 1050 | No data         | 1                 |
| 11WP55    | 680     | 680  | No data            | 680  | 680  | No data         | 1                 |
| 11WP56D   | 50      | 50   | No data            | 50   | 50   | No data         | 1                 |
| 11WP56S   | 360     | 360  | No data            | 360  | 360  | No data         | 1                 |
| 11WP57    | 450     | 450  | No data            | 450  | 450  | No data         | 1                 |
| 11WP9RPB  | 585     | 585  | 8                  | 579  | 590  | No data         | 2                 |
| 11WP9RS   | 591     | 591  | No data            | 591  | 591  | No data         | 1                 |
| CG1       | 450     | 450  | No data            | 450  | 450  | No data         | 1                 |
| CG2       | 450     | 450  | No data            | 450  | 450  | No data         | 1                 |
| CG4       | 585     | 585  | 191                | 450  | 720  | No data         | 2                 |
| KC13      | 270     | 270  | No data            | 270  | 270  | No data         | 1                 |
| KC14      | No data |      |                    |      |      |                 |                   |
| KC3       | No data |      |                    |      |      |                 |                   |
| KC3A      | 90      | 90   | No data            | 90   | 90   | No data         | 1                 |
| KC3PB     | 80      | 80   | No data            | 80   | 80   | No data         | 1                 |
| KCF1      | 800     | 800  | No data            | 800  | 800  | No data         | 1                 |
| LIMESTONE | 300     | 300  | No data            | 300  | 300  | No data         | 1                 |
| M1        | 130     | 131  | 18                 | 114  | 150  | No data         | 3                 |
| RN029659  | 240     | 240  | 85                 | 180  | 300  | No data         | 2                 |
| RN029660  | 375     | 375  | No data            | 375  | 375  | No data         | 1                 |
| W2R       | 150     | 150  | No data            | 150  | 150  | No data         | 1                 |
| WP13      | 210     | 210  | 42                 | 180  | 240  | No data         | 2                 |
| WP19      | No data |      |                    |      |      |                 |                   |
| WP5       | 600     | 600  | No data            | 600  | 600  | No data         | 1                 |

Table E6 Concentration of oxidation–reduction potential, standard hydrogen electrode (mV) – Hanna meter

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 166    | 166  | No data            | 166  | 166  | No data         | 1                 |
| 10WP32   | 126    | 126  | No data            | 126  | 126  | No data         | 1                 |
| 10WP32PB | 110    | 110  | No data            | 110  | 110  | No data         | 1                 |
| 10WP33   | 133    | 133  | No data            | 133  | 133  | No data         | 1                 |
| 10WP35N  | 37     | 37   | No data            | 37   | 37   | No data         | 1                 |
| 10WP35PB | 41     | 41   | No data            | 41   | 41   | No data         | 1                 |
| 10WP35S  | -11    | -11  | No data            | -11  | -11  | No data         | 1                 |
| 10WP36N  | 11     | 11   | No data            | 11   | 11   | No data         | 1                 |
| 10WP36PB | 53     | 53   | No data            | 53   | 53   | No data         | 1                 |
| 10WP36S  | 78     | 78   | No data            | 78   | 78   | No data         | 1                 |
| 10WP37   | -228   | -228 | No data            | -228 | -228 | No data         | 1                 |
| 10WP39   | 80     | 80   | No data            | 80   | 80   | No data         | 1                 |
| 10WP40   | -148   | -148 | No data            | -148 | -148 | No data         | 1                 |
| 10WP41   | 89     | 89   | No data            | 89   | 89   | No data         | 1                 |
| 10WP42   | -152   | -152 | No data            | -152 | -152 | No data         | 1                 |
| 10WP44   | -340   | -340 | No data            | -340 | -340 | No data         | 1                 |
| 10WP46   | 45     | 45   | No data            | 45   | 45   | No data         | 1                 |
| 10WP47   | -123   | -123 | No data            | -123 | -123 | No data         | 1                 |
| 11CS10RD | 68     | 68   | No data            | 68   | 68   | No data         | 1                 |
| 11CS10RS | -162   | -162 | No data            | -162 | -162 | No data         | 1                 |
| 11WP11RD | 68     | 68   | No data            | 68   | 68   | No data         | 1                 |
| 11WP11RS | 27     | 27   | No data            | 27   | 27   | No data         | 1                 |
| 11WP15R  | 32     | 32   | No data            | 32   | 32   | No data         | 1                 |
| 11WP16R  | 325    | 325  | No data            | 325  | 325  | No data         | 1                 |
| 11WP43D  | -169   | -169 | No data            | -169 | -169 | No data         | 1                 |
| 11WP4R   | -205   | -205 | No data            | -205 | -205 | No data         | 1                 |
| 11WP50   | 16     | 16   | No data            | 16   | 16   | No data         | 1                 |
| 11WP51D  | -137   | -137 | No data            | -137 | -137 | No data         | 1                 |
| 11WP51S  | -122   | -122 | No data            | -122 | -122 | No data         | 1                 |
| 11WP52D  | 35     | 35   | No data            | 35   | 35   | No data         | 1                 |
| 11WP53D  | 80     | 80   | No data            | 80   | 80   | No data         | 1                 |

| Site      | Median  | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|---------|------|--------------------|------|------|-----------------|-------------------|
| 11WP53S   | -83     | -83  | No data            | -83  | -83  | No data         | 1                 |
| 11WP54D   | -89     | -89  | No data            | -89  | -89  | No data         | 1                 |
| 11WP54S   | 46      | 46   | No data            | 46   | 46   | No data         | 1                 |
| 11WP55    | -170    | -170 | No data            | -170 | -170 | No data         | 1                 |
| 11WP56D   | 146     | 146  | No data            | 146  | 146  | No data         | 1                 |
| 11WP56S   | -40     | -40  | No data            | -40  | -40  | No data         | 1                 |
| 11WP57    | -167    | -167 | No data            | -167 | -167 | No data         | 1                 |
| 11WP9RPB  | 89      | 89   | No data            | 89   | 89   | No data         | 1                 |
| 11WP9RS   | 40      | 40   | No data            | 40   | 40   | No data         | 1                 |
| CG1       | -5      | -5   | No data            | -5   | -5   | No data         | 1                 |
| CG2       | No data |      |                    |      |      |                 |                   |
| CG4       | 82      | 82   | No data            | 82   | 82   | No data         | 1                 |
| KC13      | -77     | -77  | No data            | -77  | -77  | No data         | 1                 |
| KC14      | 73      | 73   | No data            | 73   | 73   | No data         | 1                 |
| KC3       | 159     | 159  | 44                 | 128  | 190  | No data         | 2                 |
| KC3A      | No data |      |                    |      |      |                 |                   |
| KC3PB     | 187     | 187  | No data            | 187  | 187  | No data         | 1                 |
| KCF1      | -59     | -59  | No data            | -59  | -59  | No data         | 1                 |
| LIMESTONE | -156    | -156 | No data            | -156 | -156 | No data         | 1                 |
| M1        | No data |      |                    |      |      |                 |                   |
| RN029659  | -209    | -209 | No data            | -209 | -209 | No data         | 1                 |
| RN029660  | 70      | 70   | No data            | 70   | 70   | No data         | 1                 |
| W2R       | 12      | 12   | No data            | 12   | 12   | No data         | 1                 |
| WP13      | 69      | 69   | No data            | 69   | 69   | No data         | 1                 |
| WP19      | 79      | 79   | No data            | 79   | 79   | No data         | 1                 |
| WP5       | 106     | 106  | No data            | 106  | 106  | No data         | 1                 |

Table E7 Concentration of oxidation–reduction potential, standard hydrogen electrode (mV) – WTW meter

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 335    | 341  | 28                 | 317  | 377  | 369             | 4                 |
| 10WP32   | 348    | 348  | 28                 | 318  | 378  | 374             | 4                 |
| 10WP32PB | 334    | 336  | 22                 | 315  | 359  | 357             | 3                 |
| 10WP33   | 335    | 332  | 7                  | 323  | 336  | 336             | 3                 |
| 10WP35N  | 54     | 95   | 117                | 8    | 265  | 210             | 4                 |
| 10WP35PB | 312    | 300  | 35                 | 249  | 328  | 324             | 4                 |
| 10WP35S  | 313    | 286  | 111                | 130  | 389  | 370             | 4                 |
| 10WP36N  | 190    | 176  | 166                | -13  | 339  | 324             | 4                 |
| 10WP36PB | 216    | 208  | 140                | 45   | 355  | 335             | 4                 |
| 10WP36S  | 124    | 157  | 107                | 67   | 312  | 256             | 4                 |
| 10WP37   | 98     | 91   | 117                | -31  | 201  | 195             | 4                 |
| 10WP39   | 273    | 264  | 27                 | 225  | 286  | 284             | 4                 |
| 10WP40   | 97     | 156  | 146                | 58   | 372  | 294             | 4                 |
| 10WP41   | 219    | 215  | 103                | 80   | 318  | 314             | 5                 |
| 10WP42   | 136    | 122  | 92                 | 1    | 215  | 200             | 4                 |
| 10WP44   | -64    | -50  | 38                 | -80  | -7   | -13             | 3                 |
| 10WP46   | 76     | 115  | 121                | 24   | 285  | 236             | 4                 |
| 10WP47   | 85     | 78   | 26                 | 42   | 97   | 97              | 4                 |
| 11CS10RD | 206    | 230  | 59                 | 187  | 298  | 289             | 3                 |
| 11CS10RS | 29     | 63   | 60                 | 27   | 132  | 122             | 3                 |
| 11WP11RD | 342    | 281  | 138                | 123  | 378  | 374             | 3                 |
| 11WP11RS | 220    | 225  | 63                 | 165  | 290  | 283             | 3                 |
| 11WP15R  | 23     | 66   | 101                | -7   | 181  | 166             | 3                 |
| 11WP16R  | 1      | 31   | 74                 | -23  | 115  | 104             | 3                 |
| 11WP43D  | 104    | 101  | 51                 | 48   | 150  | 146             | 3                 |
| 11WP4R   | 12     | 13   | 10                 | 4    | 24   | 22              | 3                 |
| 11WP50   | 138    | 121  | 39                 | 77   | 148  | 147             | 3                 |
| 11WP51D  | 14     | 15   | 18                 | -1   | 34   | 32              | 3                 |
| 11WP51S  | -6     | 19   | 67                 | -32  | 95   | 85              | 3                 |
| 11WP52D  | 339    | 310  | 84                 | 215  | 376  | 372             | 3                 |
| 11WP53D  | -2     | 44   | 111                | -37  | 171  | 153             | 3                 |

| Site      | Median  | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|---------|------|--------------------|------|------|-----------------|-------------------|
| 11WP53S   | 1       | 20   | 50                 | -19  | 77   | 69              | 3                 |
| 11WP54D   | 145     | 143  | 87                 | 55   | 229  | 220             | 3                 |
| 11WP54S   | 192     | 168  | 107                | 51   | 261  | 254             | 3                 |
| 11WP55    | 48      | 74   | 55                 | 37   | 138  | 129             | 3                 |
| 11WP56D   | 437     | 449  | 22                 | 434  | 474  | 470             | 3                 |
| 11WP56S   | No data |      |                    |      |      |                 |                   |
| 11WP57    | 38      | 36   | 5                  | 30   | 40   | 40              | 3                 |
| 11WP9RPB  | 294     | 305  | 20                 | 293  | 328  | 325             | 3                 |
| 11WP9RS   | 276     | 276  | 44                 | 232  | 319  | 315             | 3                 |
| CG1       | 307     | 303  | 40                 | 235  | 336  | 333             | 5                 |
| CG2       | 319     | 303  | 62                 | 217  | 355  | 351             | 4                 |
| CG4       | 152     | 193  | 94                 | 137  | 334  | 283             | 4                 |
| KC13      | 59      | 65   | 12                 | 57   | 79   | 77              | 3                 |
| KC14      | 200     | 174  | 69                 | 96   | 226  | 223             | 3                 |
| KC3       | 396     | 377  | 48                 | 322  | 412  | 411             | 3                 |
| KC3A      | 454     | 460  | 21                 | 442  | 483  | 480             | 3                 |
| KC3PB     | 425     | 430  | 9                  | 424  | 440  | 438             | 3                 |
| KCF1      | 133     | 148  | 107                | 49   | 261  | 248             | 3                 |
| LIMESTONE | -8      | -10  | 52                 | -74  | 48   | 36              | 4                 |
| M1        | 379     | 379  | No data            | 379  | 379  | No data         | 1                 |
| RN029659  | -16     | -42  | 59                 | -130 | -6   | -6              | 4                 |
| RN029660  | 324     | 327  | 44                 | 280  | 379  | 369             | 4                 |
| W2R       | 88      | 114  | 127                | 1    | 281  | 240             | 4                 |
| WP13      | 337     | 336  | 44                 | 282  | 389  | 373             | 4                 |
| WP19      | 243     | 242  | 42                 | 191  | 293  | 281             | 4                 |
| WP5       | 284     | 282  | 52                 | 220  | 340  | 329             | 4                 |

Table E8 Concentration of dissolved oxygen (mg/L)

| Site     | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 10WP31   | 2.86   | 3.68 | 2.21               | 2.09 | 6.90 | 5.83            | 4                 |
| 10WP32   | 3.32   | 3.31 | 0.92               | 2.18 | 4.41 | 4.14            | 4                 |
| 10WP32PB | 4.20   | 3.94 | 1.53               | 2.29 | 5.32 | 5.21            | 3                 |
| 10WP33   | 2.62   | 2.75 | 0.26               | 2.58 | 3.05 | 3.01            | 3                 |
| 10WP35N  | 0.31   | 0.94 | 1.01               | 0.09 | 2.18 | 2.04            | 5                 |
| 10WP35PB | 0.24   | 0.79 | 0.94               | 0.05 | 2.20 | 1.75            | 5                 |
| 10WP35S  | 0.18   | 0.89 | 1.07               | 0.09 | 2.36 | 2.03            | 5                 |
| 10WP36N  | 0.17   | 0.90 | 1.07               | 0.05 | 2.12 | 2.07            | 5                 |
| 10WP36PB | 0.82   | 1.19 | 1.08               | 0.15 | 2.50 | 2.34            | 5                 |
| 10WP36S  | 0.25   | 0.87 | 1.01               | 0.04 | 2.14 | 1.97            | 5                 |
| 10WP37   | 0.22   | 1.03 | 1.22               | 0.08 | 2.52 | 2.36            | 5                 |
| 10WP39   | 0.36   | 0.82 | 0.95               | 0.19 | 2.49 | 1.61            | 5                 |
| 10WP40   | 2.28   | 2.71 | 1.20               | 1.41 | 4.30 | 3.95            | 5                 |
| 10WP41   | 2.32   | 2.19 | 1.36               | 0.35 | 4.00 | 3.36            | 5                 |
| 10WP42   | 0.35   | 0.85 | 1.09               | 0.22 | 2.49 | 1.85            | 4                 |
| 10WP44   | 0.38   | 1.19 | 1.47               | 0.30 | 2.89 | 2.64            | 3                 |
| 10WP46   | 0.73   | 1.10 | 0.93               | 0.46 | 2.47 | 1.99            | 4                 |
| 10WP47   | 0.22   | 0.72 | 1.06               | 0.15 | 2.31 | 1.68            | 4                 |
| 11CS10RD | 0.64   | 1.32 | 1.59               | 0.30 | 3.69 | 2.78            | 4                 |
| 11CS10RS | 2.21   | 2.68 | 2.89               | 0.08 | 6.20 | 5.50            | 4                 |
| 11WP11RD | 2.84   | 2.86 | 0.18               | 2.70 | 3.08 | 3.04            | 4                 |
| 11WP11RS | 1.90   | 1.99 | 0.66               | 1.35 | 2.81 | 2.63            | 4                 |
| 11WP15R  | 0.32   | 0.83 | 1.21               | 0.06 | 2.62 | 1.98            | 4                 |
| 11WP16R  | 0.15   | 0.79 | 1.17               | 0.08 | 2.14 | 1.94            | 3                 |
| 11WP43D  | 1.55   | 1.75 | 1.03               | 0.83 | 2.86 | 2.73            | 3                 |
| 11WP4R   | 1.10   | 1.20 | 0.93               | 0.41 | 2.20 | 2.07            | 4                 |
| 11WP50   | 0.85   | 1.07 | 0.89               | 0.35 | 2.21 | 1.95            | 4                 |
| 11WP51D  | 0.49   | 0.89 | 1.00               | 0.22 | 2.35 | 1.86            | 4                 |
| 11WP51S  | 0.63   | 1.01 | 1.09               | 0.17 | 2.60 | 2.06            | 4                 |
| 11WP52D  | 1.85   | 1.97 | 0.85               | 1.21 | 2.96 | 2.79            | 4                 |
| 11WP53D  | 0.26   | 0.70 | 1.00               | 0.08 | 2.19 | 1.64            | 4                 |
| 11WP53S  | 0.25   | 0.72 | 1.06               | 0.09 | 2.30 | 1.72            | 4                 |

| Site      | Median | Mean | Standard deviation | Min. | Max. | 80th percentile | Number of samples |
|-----------|--------|------|--------------------|------|------|-----------------|-------------------|
| 11WP54D   | 0.53   | 0.85 | 0.98               | 0.08 | 2.24 | 1.80            | 4                 |
| 11WP54S   | 2.00   | 2.06 | 0.68               | 1.31 | 2.92 | 2.70            | 4                 |
| 11WP55    | 0.38   | 0.83 | 1.15               | 0.04 | 2.54 | 1.93            | 4                 |
| 11WP56D   | 3.77   | 3.65 | 1.00               | 2.35 | 4.73 | 4.51            | 4                 |
| 11WP56S   | 0.36   | 0.36 | No data            | 0.36 | 0.36 | No data         | 1                 |
| 11WP57    | 0.28   | 0.80 | 1.20               | 0.07 | 2.58 | 1.94            | 4                 |
| 11WP9RPB  | 1.93   | 1.90 | 0.50               | 1.38 | 2.38 | 2.34            | 3                 |
| 11WP9RS   | 1.66   | 1.69 | 0.57               | 1.16 | 2.29 | 2.22            | 4                 |
| CG1       | 2.38   | 2.43 | 1.27               | 0.94 | 4.01 | 3.59            | 4                 |
| CG2       | 2.07   | 2.07 | 0.17               | 1.95 | 2.19 | 2.19            | 2                 |
| CG4       | 0.62   | 1.20 | 1.49               | 0.14 | 3.41 | 2.60            | 4                 |
| KC13      | 1.24   | 1.25 | 1.01               | 0.30 | 2.21 | 2.16            | 4                 |
| KC14      | 1.60   | 1.50 | 1.18               | 0.12 | 2.68 | 2.55            | 4                 |
| KC3       | 3.99   | 3.79 | 1.39               | 2.21 | 5.56 | 5.07            | 5                 |
| KC3A      | 3.11   | 3.04 | 0.71               | 2.29 | 3.71 | 3.65            | 3                 |
| KC3PB     | 2.97   | 4.41 | 3.38               | 2.26 | 9.42 | 7.61            | 4                 |
| KCF1      | 1.38   | 1.61 | 1.18               | 0.56 | 3.10 | 2.77            | 4                 |
| LIMESTONE | 0.12   | 0.64 | 1.07               | 0.07 | 2.25 | 1.61            | 4                 |
| M1        | 5.89   | 5.89 | No data            | 5.89 | 5.89 | No data         | 1                 |
| RN029659  | 0.18   | 0.71 | 1.08               | 0.13 | 2.33 | 1.69            | 4                 |
| RN029660  | 4.72   | 4.51 | 1.93               | 2.45 | 6.16 | 6.16            | 4                 |
| W2R       | 0.16   | 0.67 | 1.06               | 0.12 | 2.26 | 1.63            | 4                 |
| WP13      | 0.89   | 1.20 | 0.71               | 0.76 | 2.25 | 1.85            | 4                 |
| WP19      | 0.27   | 0.81 | 0.97               | 0.15 | 2.43 | 1.72            | 5                 |
| WP5       | 0.21   | 2.06 | 3.21               | 0.06 | 7.56 | 4.92            | 5                 |

Table E9 Concentration of dissolved oxygen (%)

| Site     | Median | Mean  | Standard deviation | Min.  | Max.  | 80th percentile | Number of samples |
|----------|--------|-------|--------------------|-------|-------|-----------------|-------------------|
| 10WP31   | 38.95  | 49.18 | 28.73              | 28.10 | 90.70 | 77.29           | 4                 |
| 10WP32   | 44.35  | 43.50 | 11.46              | 29.00 | 56.30 | 53.63           | 4                 |
| 10WP32PB | 55.90  | 53.97 | 21.96              | 31.10 | 74.90 | 73.00           | 3                 |
| 10WP33   | 34.50  | 36.60 | 4.17               | 33.90 | 41.40 | 40.71           | 3                 |
| 10WP35N  | 14.10  | 14.93 | 13.28              | 2.90  | 28.60 | 27.22           | 4                 |
| 10WP35PB | 10.45  | 12.95 | 12.84              | 1.90  | 29.00 | 25.58           | 4                 |
| 10WP35S  | 12.00  | 14.83 | 15.59              | 1.70  | 33.60 | 30.03           | 4                 |
| 10WP36N  | 13.75  | 14.30 | 14.18              | 1.90  | 27.80 | 27.05           | 4                 |
| 10WP36PB | 6.70   | 10.98 | 12.07              | 2.00  | 28.50 | 22.71           | 4                 |
| 10WP36S  | 12.90  | 13.93 | 13.43              | 1.60  | 28.30 | 26.53           | 4                 |
| 10WP37   | 16.05  | 16.25 | 15.96              | 1.90  | 31.00 | 30.43           | 4                 |
| 10WP39   | 7.65   | 12.85 | 14.30              | 2.40  | 33.70 | 26.71           | 4                 |
| 10WP40   | 28.55  | 33.38 | 16.33              | 19.50 | 56.90 | 48.98           | 4                 |
| 10WP41   | 31.00  | 28.10 | 17.58              | 4.00  | 51.70 | 42.60           | 5                 |
| 10WP42   | 4.45   | 11.23 | 14.60              | 2.90  | 33.10 | 24.55           | 4                 |
| 10WP44   | 5.10   | 17.53 | 24.36              | 1.90  | 45.60 | 41.55           | 3                 |
| 10WP46   | 9.70   | 14.78 | 12.81              | 6.00  | 33.70 | 27.01           | 4                 |
| 10WP47   | 2.85   | 9.63  | 14.12              | 2.00  | 30.80 | 22.46           | 4                 |
| 11CS10RD | 8.70   | 21.20 | 25.74              | 4.10  | 50.80 | 46.59           | 3                 |
| 11CS10RS | 20.00  | 28.63 | 26.43              | 7.60  | 58.30 | 54.47           | 3                 |
| 11WP11RD | 40.40  | 39.30 | 2.72               | 36.20 | 41.30 | 41.21           | 3                 |
| 11WP11RS | 29.10  | 29.63 | 8.71               | 21.20 | 38.60 | 37.65           | 3                 |
| 11WP15R  | 6.50   | 14.83 | 18.38              | 2.10  | 35.90 | 32.96           | 3                 |
| 11WP16R  | 2.40   | 10.60 | 15.34              | 1.10  | 28.30 | 25.71           | 3                 |
| 11WP43D  | 20.90  | 25.83 | 17.53              | 11.30 | 45.30 | 42.86           | 3                 |
| 11WP4R   | 24.80  | 20.20 | 12.93              | 5.60  | 30.20 | 29.66           | 3                 |
| 11WP50   | 4.90   | 13.07 | 14.58              | 4.40  | 29.90 | 27.40           | 3                 |
| 11WP51D  | 3.40   | 12.60 | 16.46              | 2.80  | 31.60 | 28.78           | 3                 |
| 11WP51S  | 5.80   | 14.70 | 18.62              | 2.20  | 36.10 | 33.07           | 3                 |
| 11WP52D  | 32.10  | 30.47 | 11.83              | 17.90 | 41.40 | 40.47           | 3                 |
| 11WP53D  | 4.40   | 11.73 | 14.56              | 2.30  | 28.50 | 26.09           | 3                 |
| 11WP53S  | 5.00   | 12.17 | 15.63              | 1.40  | 30.10 | 27.59           | 3                 |

| Site      | Median  | Mean  | Standard deviation | Min.  | Max.   | 80th percentile | Number of samples |
|-----------|---------|-------|--------------------|-------|--------|-----------------|-------------------|
| 11WP54D   | 9.30    | 14.23 | 13.78              | 3.60  | 29.80  | 27.75           | 3                 |
| 11WP54S   | 24.10   | 23.43 | 5.83               | 17.30 | 28.90  | 28.42           | 3                 |
| 11WP55    | 6.90    | 15.40 | 18.02              | 3.20  | 36.10  | 33.18           | 3                 |
| 11WP56D   | 55.10   | 50.87 | 16.46              | 32.70 | 64.80  | 63.83           | 3                 |
| 11WP56S   | No data |       |                    |       |        |                 |                   |
| 11WP57    | 6.00    | 14.57 | 19.05              | 1.30  | 36.40  | 33.36           | 3                 |
| 11WP9RPB  | 26.00   | 25.47 | 6.82               | 18.40 | 32.00  | 31.40           | 3                 |
| 11WP9RS   | 27.90   | 25.03 | 7.15               | 16.90 | 30.30  | 30.06           | 3                 |
| CG1       | 30.00   | 31.60 | 16.92              | 12.70 | 53.70  | 47.19           | 4                 |
| CG2       | 27.10   | 27.10 | 2.26               | 25.50 | 28.70  | 28.70           | 2                 |
| CG4       | 6.50    | 15.43 | 20.96              | 2.00  | 46.70  | 34.76           | 4                 |
| KC13      | 27.00   | 21.10 | 13.09              | 6.10  | 30.20  | 29.88           | 3                 |
| KC14      | 31.20   | 26.80 | 12.02              | 13.20 | 36.00  | 35.52           | 3                 |
| KC3       | 61.20   | 55.23 | 23.33              | 29.50 | 75.00  | 73.62           | 3                 |
| KC3A      | 41.90   | 41.20 | 10.07              | 30.80 | 50.90  | 50.00           | 3                 |
| KC3PB     | 45.10   | 67.47 | 51.99              | 30.40 | 126.90 | 118.72          | 3                 |
| KCF1      | 27.70   | 26.83 | 15.52              | 10.90 | 41.90  | 40.48           | 3                 |
| LIMESTONE | 1.70    | 8.88  | 14.82              | 1.00  | 31.10  | 22.34           | 4                 |
| M1        | 80.00   | 80.00 | No data            | 80.00 | 80.00  | No data         | 1                 |
| RN029659  | 2.45    | 9.48  | 14.76              | 1.40  | 31.60  | 22.96           | 4                 |
| RN029660  | 57.80   | 50.65 | 38.80              | 4.00  | 83.00  | 82.76           | 4                 |
| W2R       | 2.30    | 9.33  | 14.52              | 1.60  | 31.10  | 22.49           | 4                 |
| WP13      | 11.45   | 15.85 | 9.88               | 9.90  | 30.60  | 25.08           | 4                 |
| WP19      | 8.65    | 13.15 | 13.89              | 2.70  | 32.60  | 26.93           | 4                 |
| WP5       | 2.70    | 11.70 | 15.85              | 2.40  | 30.00  | 27.27           | 3                 |

## Appendix F Suitability of groundwater for irrigation

Table F1 Key for the tables in Appendix F and the irrigation water quality trigger values from ANZECC and ARM CANZ (2000)

| Analyte                            | Parameter                      | Value              |
|------------------------------------|--------------------------------|--------------------|
| Electrical conductivity, EC (mS/m) | Low salinity                   | <28                |
|                                    | Medium salinity                | 80                 |
|                                    | High salinity                  | 230                |
|                                    | Very high salinity             | 550                |
|                                    | Extremely high salinity        | >550               |
| Sodium adsorption ratio            | Low                            | See section 2.2.1  |
|                                    | Medium                         |                    |
|                                    | High                           |                    |
|                                    | Very high                      |                    |
| Chloride, Cl (mg/L)                | Sensitive (S)                  | <175               |
|                                    | Moderately sensitive (MS)      | 350                |
|                                    | Moderately tolerant (MT)       | 700                |
|                                    | Tolerant (T)                   | 1000               |
|                                    | Not suitable (N)               | >1000              |
| Sodium, Na (mg/L)                  | Sensitive (S)                  | <115               |
|                                    | Moderately sensitive (MS)      | 230                |
|                                    | Moderately tolerant (MT)       | 460                |
|                                    | Tolerant (T)                   | 800                |
|                                    | Not suitable (N)               | >800               |
| Aluminium, Al (mg/L)               | Short-term trigger value (STV) | 20                 |
|                                    | Long-term trigger value (LTV)  | 5                  |
| Arsenic, As (mg/L)                 | Short-term trigger value (STV) | 2                  |
|                                    | Long-term trigger value (LTV)  | 0.1                |
| Beryllium, Be (mg/L)               | Short-term trigger value (STV) | 0.5                |
|                                    | Long-term trigger value (LTV)  | 0.1                |
| Boron, B (mg/L)                    | Short-term trigger value (STV) | Plant specific (P) |
|                                    | Long-term trigger value (LTV)  | 0.5                |
| Cadmium, Cd (mg/L)                 | Short-term trigger value (STV) | 0.05               |
|                                    | Long-term trigger value (LTV)  | 0.01               |
| Chromium, Cr (mg/L)                | Short-term trigger value (STV) | 1                  |
|                                    | Long-term trigger value (LTV)  | 0.1                |

| Analyte                                    | Parameter                          | Value  |
|--------------------------------------------|------------------------------------|--------|
| Cobalt, Co (mg/L)                          | Short-term trigger value (STV)     | 0.1    |
|                                            | Long-term trigger value (LTV)      | 0.05   |
| Copper, Cu (mg/L)                          | Short-term trigger value (STV)     | 5      |
|                                            | Long-term trigger value (LTV)      | 0.2    |
| Fluoride, F (mg/L)                         | Short-term trigger value (STV)     | 2      |
|                                            | Long-term trigger value (LTV)      | 1      |
| Iron, Fe (mg/L)                            | Short-term trigger value (STV)     | 10     |
|                                            | Long-term trigger value (LTV)      | 2      |
| Lead, Pb (mg/L)                            | Short-term trigger value (STV)     | 5      |
|                                            | Long-term trigger value (LTV)      | 2      |
| Lithium, Li (mg/L)                         | Short-term trigger value (STV)     | 2.5    |
|                                            | Long-term trigger value (LTV)      | 2.5    |
| Lithium, Li (citrus) (mg/L)                | Short-term trigger value (STV)     | 0.075  |
|                                            | Long-term trigger value (LTV)      | 0.075  |
| Manganese, Mn (mg/L)                       | Short-term trigger value (STV)     | 10     |
|                                            | Long-term trigger value (LTV)      | 0.2    |
| Mercury, Hg (mg/L)                         | Short-term trigger value (STV)     | 0.002  |
|                                            | Long-term trigger value (LTV)      | 0.002  |
| Molybdenum, Mo (mg/L)                      | Short-term trigger value (STV)     | 0.05   |
|                                            | Long-term trigger value (LTV)      | 0.01   |
| Nickel, Ni (mg/L)                          | Short-term trigger value (STV)     | 2      |
|                                            | Long-term trigger value (LTV)      | 0.2    |
| Selenium, Se (mg/L)                        | Short-term trigger value (STV)     | 0.05   |
|                                            | Long-term trigger value (LTV)      | 0.02   |
| Uranium, U (mg/L)                          | Short-term trigger value (STV)     | 0.1    |
|                                            | Long-term trigger value (LTV)      | 0.01   |
| Vanadium, V (mg/L)                         | Short-term trigger value (STV)     | 0.5    |
|                                            | Long-term trigger value (LTV)      | 0.1    |
| Zinc, Zn (mg/L)                            | Short-term trigger value (STV)     | 5      |
|                                            | Long-term trigger value (LTV)      | 2      |
| Nitrogen, N (as total nitrogen) (mg/L)     | Short-term trigger value (STV)     | 12–125 |
|                                            | Long-term trigger value (LTV)      | 5      |
| Phosphorus, P (as total phosphorus) (mg/L) | Short-term trigger value (STV)     | 0.8–12 |
|                                            | Long-term trigger value (LTV)      | 0.05   |
| pH                                         | Low (high corrosion potential)     | <5     |
|                                            | High (increased fouling potential) | >8.5   |

Table F2 Suitability of groundwater for irrigation based on ANZECC and ARMCANZ (2000) for bores 10WP33 to M1. Y(es) meets criteria; N(o) does not meet criteria (see Table F1 for analyte description)

[illegible]

[illegible]

| Analyte | 10WP33 | 10WP37 | 10WP39 | 11CS10RD | 11CS10RS | 11WP11RD | 11WP11RS | 11WP15R | 11WP43D | 11WP51D | 11WP51S | 11WP52D | 11WP56D | 11WP57 | CG4 | LIMESTONE | RN029660 | WP13 | WP5 | CG1 | CG2 | KC13 | KC3PB | KC14 | KC3 | KC3A | KCF1 | M1 |
|---------|--------|--------|--------|----------|----------|----------|----------|---------|---------|---------|---------|---------|---------|--------|-----|-----------|----------|------|-----|-----|-----|------|-------|------|-----|------|------|----|
| U LTV   | Y      | N      | Y      | N        | N        | Y        | Y        | N       | N       | Y       | Y       | Y       | Y       | N      | N   | Y         | Y        | Y    | N   | N   | Y   | N    | Y     | N    | Y   | Y    | N    | Y  |
| U STV   | Y      | Y      | Y      | Y        | N        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| V LTV   | Y      | Y      | Y      | Y        | N        | Y        | N        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| V STV   | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| Zn LTV  | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| Zn STV  | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| N LTV   | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | N         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| N STV   | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| P LTV   | N      | N      | N      | N        | N        | N        | N        | N       | N       | N       | N       | N       | N       | N      | N   | N         | N        | Y    | Y   | N   | N   | N    | Y     | N    | Y   | N    | Y    | Y  |
| P STV   | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| pH High | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | Y       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | Y     | Y    | Y   | Y    | Y    | Y  |
| pH Low  | Y      | Y      | Y      | Y        | Y        | Y        | Y        | Y       | Y       | Y       | Y       | Y       | N       | Y      | Y   | Y         | Y        | Y    | Y   | Y   | Y   | Y    | N     | Y    | N   | N    | Y    | Y  |

Table F3 Suitability of groundwater for irrigation based on ANZECC and ARMCANZ (2000) for bores 10WP31 to 11WP9RS. Y(es) meets criteria; N(o) does not meet criteria (see Table F1 for analyte description)

| Analyte | 10WP31 | 10WP32 | 10WP32PB | 10WP35N | 10WP35PB | 10WP35S | 10WP36N | 10WP36PB | 10WP36S | 10WP40 | 10WP41 | 10WP42 | 10WP44 | 10WP46 | 10WP47 | RN029659 | W2R | WP19 | 11WP16R | 11WP4R | 11WP50 | 11WP53D | 11WP53S | 11WP54D | 11WP54S | 11WP55 | 11WP56S | 11WP9RPB | 11WP9RS |    |
|---------|--------|--------|----------|---------|----------|---------|---------|----------|---------|--------|--------|--------|--------|--------|--------|----------|-----|------|---------|--------|--------|---------|---------|---------|---------|--------|---------|----------|---------|----|
| EC      | M      | H      | H        | H       | H        | H       | H       | H        | H       | E      | E      | V      | E      | E      | M      | V        | M   | E    | E       | H      | E      | E       | E       | E       | E       | E      | VH      | H        | H       | H  |
| SAR     | L      | M      | M        | M       | M        | M       | M       | H        | H       | VH     | VH     | M      | VH     | VH     | L      | VH       | L   | VH   | VH      | L      | VH     | VH      | VH      | VH      | VH      | VH     | VH      | L        | VH      | H  |
| CI      | S      | MT     | MS       | S       | MS       | S       | S       | S        | S       | N      | N      | MT     | N      | MS     | S      | N        | S   | N    | N       | MT     | N      | N       | N       | N       | N       | N      | T       | S        | MS      | MS |
| Na      | S      | MT     | MT       | MT      | MT       | MT      | MS      | MS       | MT      | N      | N      | MT     | N      | MT     | S      | T        | S   | N    | N       | MS     | N      | N       | N       | N       | N       | N      | N       | MS       | MT      | MT |
| Al LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       |    |
| Al STV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       |    |
| As LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       |    |
| As STV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       |    |
| Be LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       |    |
| Be STV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       |    |
| B LTV   | Y      | Y      | Y        | Y       | Y        | Y       | N       | N        | N       | N      | N      | Y      | Y      | Y      | Y      | Y        | Y   | N    | Y       | Y      | N      | N       | N       | N       | N       | N      | Y       | Y        | Y       | Y  |
| B STV   | P      | P      | P        | P       | P        | P       | P       | P        | P       | P      | P      | P      | P      | P      | P      | P        | P   | P    | P       | P      | P      | P       | P       | P       | P       | P      | P       | P        | P       | P  |
| Cd LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |
| Cd STV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |
| Cr LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |
| Cr STV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |
| Co LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |
| Co STV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |
| Cu LTV  | Y      | Y      | Y        | Y       | Y        | Y       | Y       | Y        | Y       | Y      | Y      | Y      | Y      | Y      | Y      | Y        | Y   | Y    | Y       | Y      | Y      | Y       | Y       | Y       | Y       | Y      | Y       | Y        | Y       | Y  |

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## Appendix G Salinity processes

### G1 Introduction

The groundwater salinity on the Weaber Plain ranges from 27mg/L TDS to 24 000mg/L TDS, with an average TDS around 3000mg/L. Using chloride isotope data, Tickell et al. (2007) showed that chloride has resulted from atmospheric disposition and has accumulated with time. Other ions that also resulted from atmospheric deposition would have likewise accumulated through time. Under the current Groundwater Management Plan (Strategen 2012a) for the Weaber Plain, it is proposed to pump this saline groundwater into the irrigation water supply and then reapply it to the irrigation areas. The resultant irrigation water will be considerably more saline than rainfall. Salts have accumulated in the landscape over time through introduction via rainfall. Therefore, what will be the longer term impacts of these management practices on the salinity of soil and groundwater on the Weaber Plain?

The chemical composition of groundwater on the Weaber Plain is also highly variable. The major ions, sodium, calcium, magnesium, sulfate and alkalinity species all show enrichment relative to chloride. However, potassium shows depletion relative to chloride. What are the major factors controlling the groundwater quality and chemistry? Through understanding these geochemical processes, the longer term risks of salinity caused by irrigation of agriculture on the Weaber Plain can be assessed.

### G2 Methodology

To identify the main geochemical processes controlling salinity and major ion concentrations, we investigated the statistical relationships between factors, such as soils and lithology, which were potentially controlling the processes. To increase the statistical robustness of the analysis, the chemistry dataset was expanded to include chemistry data from other areas of the ORIA with similar soils and hydrogeological conditions (Lillicrap et al. 2011). The additional chemistry data was screened for reliability and any bore that had a charge balance error of more than 10% was discarded. The major ion data was transformed by logging the data to obtain normal distributions. Any outliers were investigated and if the values could not be substantiated, such as concentrations of an analyte were significantly less than other related analytes for no obvious reasons, they were discarded. For each bore the major ion concentrations were divided by chloride to normalise the data ('normalised value'). This approach accounted for the influence of evapo-concentration and helped to identify other geochemical processes that were influencing the concentration of the ion.

The correlations between major ions and TDS were determined using GenStat 16th edition (VSN International Ltd). The effect of lithology and soil type on the major ions and TDS were also statistically analysed using a mixed model (Restricted Maximum Likelihood [REML]). Both the mean major ion concentrations and normalised values were used in the statistical analysis.

The rock type and soil for each bore used in the analysis was determined in a geographic information system by intersecting the bore location data with regional hydrogeology (O'Boy et al. 2001) and soil landscapes (Aldrick et al. 1990, Dixon 1996, Schoknecht & Grose 1996). The bores were assigned to a rock type based on

the dominant lithology. To aid statistical analysis, smaller groups with similar lithology were aggregated. The rock types used in the analysis were basalt, limestone, sandstone, sandstone/wackes/limestone and shale/siltstone. Bores in similar soil types were also aggregated, based on the presence of the tree, *Excoecaria parivifolia* (gutta percha) — an indicator of seasonal inundation — and the presence of gypsum in the subsoil in all of the units. The soil types used were Aquitaine/Cununurra gypsic clays, Cununurra clays, Keep clays and sand/silt.

To understand the causes of the correlations identified in the statistical analysis, and the evolution of groundwater types on the Weaber Plain, inverse modelling was undertaken in PHREEQCI (Parkhurst & Appelo 2013). Inverse modelling identifies a suite of possible chemical reactions that can cause the evolution of an initial solution to a final solution. The initial solution, used in all PHREEQCI models, simulated meteoric conditions and was based on DAFWA unpublished data and Hingston and Gailitis (1976). The final solutions used were the mean values of major and select minor ions (calcium, magnesium, sodium, potassium, chloride, bicarbonate, sulfate, bromide, silicon) for bores with chloride concentrations above 2000mg/L.

The phases, based on soil and geological records, used in the PHREEQCI inverse modelling were ion exchange reactions, biotite (ferromagnesian mica), plagioclase/albite (sodium plagioclase), halite (NaCl), sylvite (KCl), illite (clay with potassium bonding the interlayers), calcite ( $\text{CaCO}_3$ ), dolomite ( $\text{MgCa}(\text{CO}_3)_2$ ), gypsum ( $\text{CaSO}_4$ ) and evapotranspiration. Salinity processes in the soil and regolith were then derived from the results.

### G3 Results and discussion

Statistical analysis of groundwater chemistry for the ORIA showed that all the major ions, except for potassium, were highly correlated ( $r^2 \geq 0.67$ ) with each other and TDS (Table G1). From Figures G1 to G3, all the major ions increase with increasing chloride concentration. Tickell et al. (2007) showed chloride concentrations in groundwater had increased from evapotranspiration and therefore the other major ions would also have increased in concentration through evapotranspiration. The increase in concentrations of major ions due to evapotranspiration explains the correlations between the major ions evident in Table G1.

Except for sodium and potassium, all major ions at higher salinity concentrations — chloride concentration was used a surrogate for salinity — had chloride ratios greater than seawater, indicating enrichment relative to meteoric or seawater ratios (Figure G1). Therefore, there are additional sources besides atmospheric deposition. The sodium–chloride ratios at higher salinities were similar to seawater (>2000mg/L chloride) whereas at lower concentrations, sodium showed enrichment. Potassium–chloride ratios were less than the seawater ratio, indicating there was a depletion of potassium.

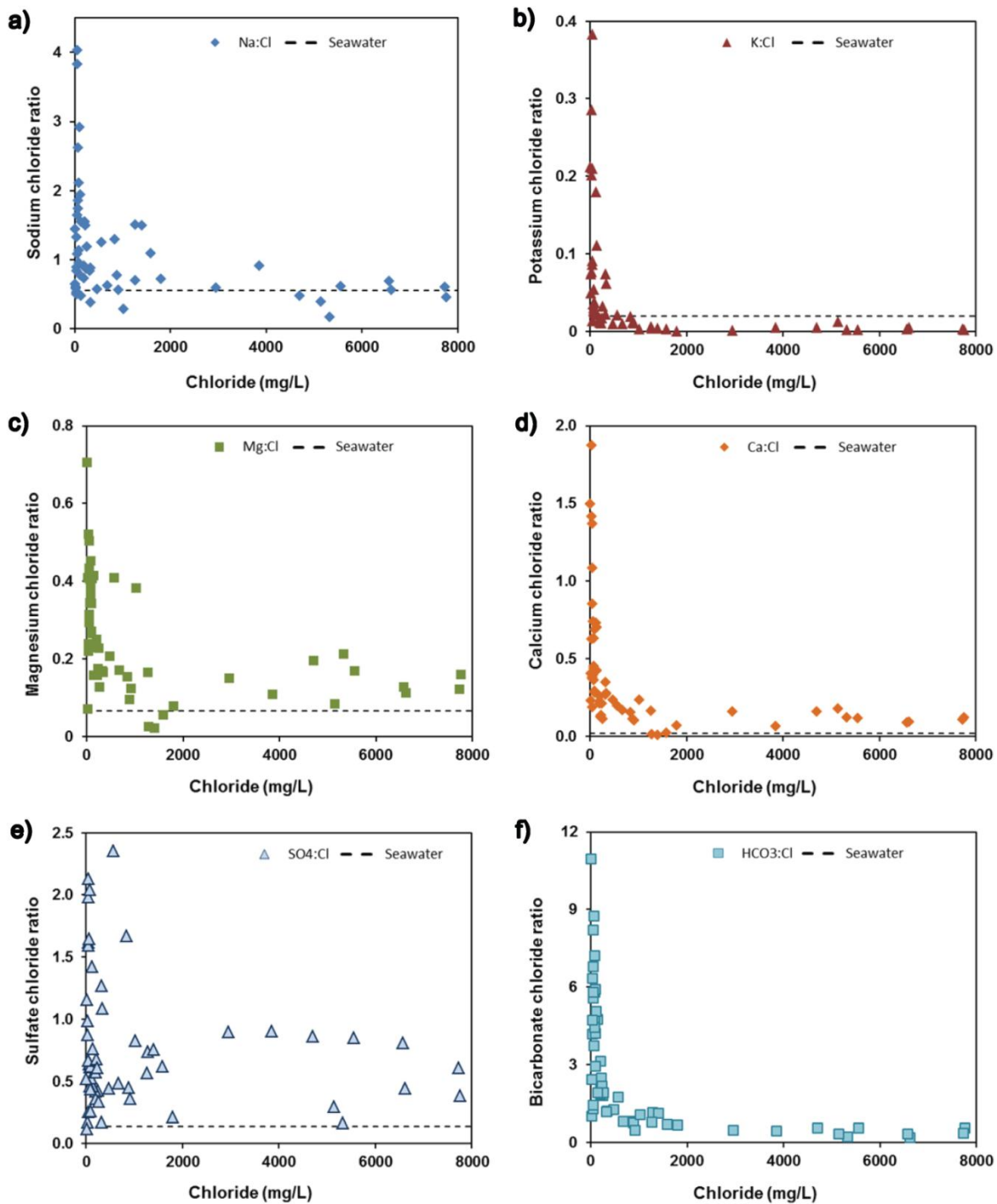


Figure G1 Major ion chloride ratios (in mg/L) plotted against chloride concentration. Chloride is a conservative ion that is a surrogate for salinity. The dashed black line is the seawater ratio, which is the expected chloride ratio for rainfall and atmospheric fallout

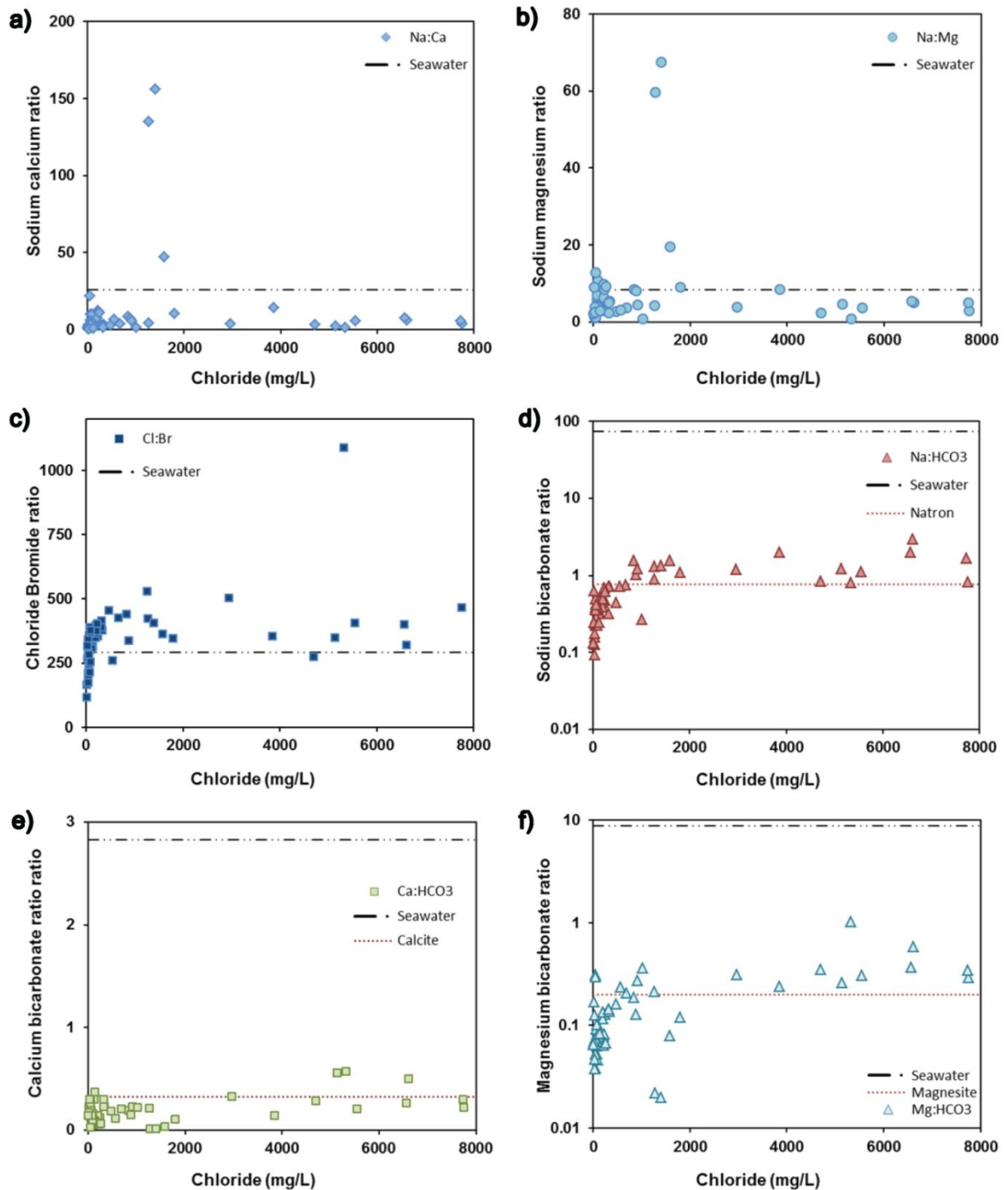


Figure G2 Ratio of major ions (in mg/L) plotted against chloride concentration. Chloride is a conservative ion that is a surrogate for salinity. The dashed black line is the seawater ratio, which is the expected ratio for rainfall and atmospheric fallout. The red dotted lines represent the ratios associated with the dissolution of common minerals: d) natron ( $\text{Na}_2\text{CO}_3$ ); e) calcite ( $\text{CaCO}_3$ ) and f) magnesite ( $\text{MgCO}_3$ )

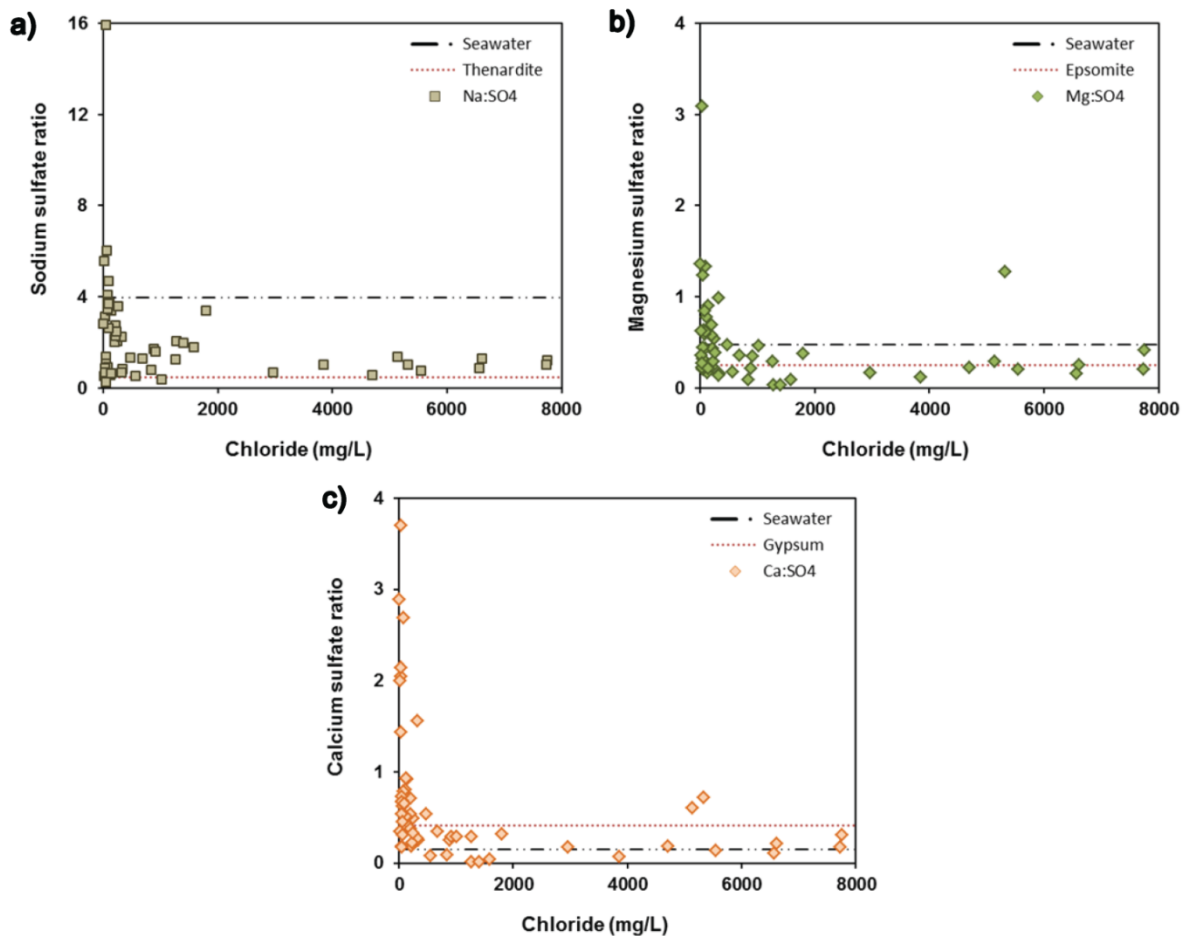


Figure G3 The ratio of major ions to sulfate (in mg/L) plotted against chloride concentration, a conservative ion that is a surrogate for salinity. The dashed black line is the seawater ratio, which is the expected ratio for rainfall and atmospheric fallout. The red dotted lines represent the ratios associated with the dissolution of common minerals: a) thenardite ( $\text{Na}_2\text{SO}_4$ ); b) epsomite ( $\text{MgSO}_4$ ); and c) gypsum ( $\text{CaSO}_4$ ).

The major ionic ratios (major ions plotted against other major ions) are shown in Figures G2 and G3. Bromide, another conservative halogen ion, shows depletion relative to chloride. Bromide is more likely to be sorbed than chloride (Davis et al. 1998). Conversely, calcium and magnesium show enrichment in concentrations relative to sodium.

The sodium:bicarbonate ratios at higher salinities (Figure G2) show enrichment of bicarbonate relative to sodium when compared to seawater, although these ratios are less than what would occur if sodium carbonate minerals, such as natron ( $\text{Na}_2\text{CO}_3$ ), had dissolved. The magnesium:bicarbonate ratios show a similar trend — bicarbonate is enriched relative to magnesium when compared to seawater, but less than what would occur if magnesium carbonate minerals, such as magnesite ( $\text{MgCO}_3$ ), had dissolved. On the other hand, the calcium:bicarbonate ratio shows an enrichment of bicarbonate relative to calcium when compared to seawater, but the ratios are similar to what would occur if the calcium carbonate minerals, such as calcite ( $\text{CaCO}_3$ ), had dissolved.

Both the calcium:chloride and sulfate:chloride ratios showed enrichment of calcium and sulfate, respectively, relative to seawater ratios. Though, the calcium:sulfate ratios for the Weaber groundwater were similar to seawater ratios (Figure G3). The calcium:sulfate ratios were less than the dissolution ratios of calcium sulfate minerals, such as gypsum ( $\text{CaSO}_4$ ). The magnesium:sulfate ratio, at higher salinities, shows enrichment of sulfate relative to magnesium when compared to seawater and has a similar ratio to the dissolution of magnesium sulfate minerals, such as epsomite ( $\text{MgSO}_4$ ). The sodium:sulfate ratios at higher salinities also show enrichment of sulfate relative to sodium when compared to seawater (Figure G3). Though sodium:sulfate ratios on the Weaber Plain were less than dissolution of sodium sulfate minerals, such as thenardite ( $\text{Na}_2\text{SO}_4$ ), at higher salinities, they were an average molar ratio of 4.0.

Table G1 Statistical correlations between major ions

|                                         | TDS<br>mg/L | SO <sub>4</sub><br>mg/L | Na<br>mg/L | Mg<br>mg/L | K<br>mg/L | HCO <sub>3</sub><br>mg/L | Cl<br>mg/L | Ca<br>mg/L |
|-----------------------------------------|-------------|-------------------------|------------|------------|-----------|--------------------------|------------|------------|
| Calcium (Ca)<br>mg/L                    | 0.87        | 0.80                    | 0.72       | 0.90       | 0.28      | 0.71                     | 0.91       |            |
| Chloride (Cl)<br>mg/L                   | 0.98        | 0.90                    | 0.92       | 0.91       | 0.24      | 0.75                     |            |            |
| Bicarbonate<br>(HCO <sub>3</sub> ) mg/L | 0.80        | 0.69                    | 0.76       | 0.67       | 0.22      |                          |            |            |
| Potassium<br>(K) mg/L                   | 0.27        | 0.26                    | 0.22       | 0.24       |           |                          |            |            |
| Magnesium<br>(Mg) mg/L                  | 0.89        | 0.87                    | 0.72       |            |           |                          |            |            |
| Sodium (Na)<br>mg/L                     | 0.95        | 0.89                    |            |            |           |                          |            |            |
| Sulfate (SO <sub>4</sub> )<br>mg/L      | 0.95        |                         |            |            |           |                          |            |            |
| TDS mg/L                                |             |                         |            |            |           |                          |            |            |

Comparing the variability in major ions of different rock types showed there were statistically significant variations ( $p < 0.05$ ) in bicarbonate, sulfate, calcium, magnesium and potassium (Table G2).

Table G2 The mean concentration and statistical correlations between factors influencing the strength of major ions

| Parameter                                       | TDS<br>mg/L | Cl<br>mg/L | HCO <sub>3</sub><br>mg/L | HCO <sub>3</sub><br>:Cl | SO <sub>4</sub><br>mg/L | SO <sub>4</sub> :Cl | Ca<br>mg/L | Ca:Cl | Mg<br>mg/L | Mg:Cl | Na<br>mg/L | Na:Cl | K<br>mg/L | K:Cl  |
|-------------------------------------------------|-------------|------------|--------------------------|-------------------------|-------------------------|---------------------|------------|-------|------------|-------|------------|-------|-----------|-------|
| <b>Lithology</b>                                |             |            |                          |                         |                         |                     |            |       |            |       |            |       |           |       |
| Basalt                                          | 2255        | 629        | 724                      | 5.64                    | 268                     | 0.52                | 61         | 0.37  | 76         | 0.27  | 564        | 2.12  | 7.6       | 0.07  |
| Limestone                                       | 1929        | 388        | 381                      | 1.97                    | 621                     | 1.51                | 112        | 0.42  | 84         | 0.30  | 384        | 0.97  | 19.3      | 0.06  |
| Sandstone                                       | 2561        | 762        | 465                      | 1.79                    | 504                     | 0.76                | 71         | 0.27  | 81         | 0.20  | 657        | 0.93  | 19.2      | 0.10  |
| Sandstone/wackes/limestone                      | 5887        | 1860       | 926                      | 2.59                    | 1423                    | 0.98                | 203        | 0.45  | 320        | 0.28  | 1255       | 1.01  | 30.6      | 0.10  |
| Shale/siltstone                                 | 3036        | 1024       | 658                      | 3.30                    | 571                     | 0.57                | 139        | 0.39  | 145        | 0.25  | 658        | 1.28  | 12.3      | 0.06  |
| p (logged values)                               | 0.053       | 0.102      | 0.010                    | 0.009                   | <0.001                  | <0.001              | <0.001     | 0.127 | 0.003      | 0.353 | 0.232      | 0.001 | <0.001    | 0.933 |
| <b>Soils</b>                                    |             |            |                          |                         |                         |                     |            |       |            |       |            |       |           |       |
| Aquitaine/Cununurra gypsic clays                | 3869        | 1266       | 673                      | 2.66                    | 854                     | 0.85                | 148        | 0.39  | 195        | 0.26  | 867        | 1.10  | 17.4      | 0.08  |
| Cununurra clays                                 | 2360        | 696        | 659                      | 4.75                    | 352                     | 0.52                | 87         | 0.36  | 82         | 0.27  | 553        | 1.82  | 10.4      | 0.06  |
| Keep clays                                      | 2168        | 538        | 748                      | 1.83                    | 154                     | 0.31                | 42         | 0.16  | 50         | 0.15  | 561        | 1.03  | 5.1       | 0.01  |
| Sand/silt                                       | 3043        | 846        | 672                      | 2.59                    | 761                     | 0.80                | 140        | 0.55  | 166        | 0.20  | 542        | 0.89  | 35.7      | 0.20  |
| p (logged values)                               | 0.315       | 0.174      | 0.697                    | 0.026                   | 0.019                   | 0.001               | 0.489      | 0.153 | 0.381      | 0.369 | 0.326      | 0.141 | 0.005     | 0.011 |
| <b>Interactions between lithology and soils</b> |             |            |                          |                         |                         |                     |            |       |            |       |            |       |           |       |
| p (logged values)                               | 0.845       | 0.678      | 0.986                    | 0.261                   | 0.611                   | 0.276               | 0.197      | 0.037 | 0.671      | 0.195 | 0.743      | 0.385 | 0.747     | 0.212 |

When the effects of evapotranspiration were removed by dividing the ion concentrations by chloride, only bicarbonate, sulfate and sodium remained statistically significant. This analysis supports the conclusion by Tickell et al. (2007) that one of the main controls on groundwater chemistry and individual ion concentrations is evapotranspiration.

The variations in bicarbonate, sulfate and sodium are explained by differences in local geology and mineralogy. Inverse modelling with PHREEQCI indicated that the variation in bicarbonate is because of the dissolution of limestone, with calcium carbonate causing enrichment relative to other rock types. Inverse modelling showed that differences in sodium concentrations are likely caused by the weathering of plagioclase found in the basalts of the Antrim Volcanics (Hem 1985). The difference in sulfate concentrations is likely caused by the weathering of sulfide ores in the Sorby Hills, which are located between the Weaber and Knox Creek plains.

The statistical analysis also found significant relationships ( $p < 0.05$ ) between soil type and bicarbonate, sulfate and potassium concentrations in groundwater (Table G2). The soils of the Weaber Plain have high potassium concentrations (Smolinski et al. 2011) and contain up to 20% illite (Dixon 1996). Inverse modelling indicated illite is the sink for potassium, leading to depletion in groundwater relative to seawater. The relationship between soil type and depletion of potassium in groundwater is most likely because of the sorbing of potassium from atmospheric deposition by soils.

There was also a statistically significant interaction ( $p < 0.05$ ) between soils and rock type for calcium. Inverse modelling showed that gypsum and carbonate minerals such as calcite are likely sources of sulfate and some of the bicarbonate enrichment in groundwater. The modelling also indicates that calcium concentrations in the groundwater are controlled by calcite solubility, which is reflected in Figure G2e. Some soils on the ORIA, particularly the Aquitaine family of soils, naturally accumulate salts, mainly the less soluble salts such as carbonates and gypsum (calcium sulfate), because drainage is restricted (Dixon 1996, Schoknecht & Grose 1996, Smolinski et al. 2011). Statistical and geochemical modelling indicates that gypsum and carbonates accumulate in the subsoil and are subsequently redissolved and flushed to groundwater through recharge, leading to enrichment of sulfate and bicarbonate in groundwater. George (1983) showed that Aquitaine soils experienced a reduction in subsoil salinity after the introduction of irrigation, further supporting that leakage occurs beneath these soils.

The results of the statistical and geochemical modelling provides complications for irrigation on the Weaber Plain because some soils naturally accumulate salts, particularly the Aquitaine soils with restricted subsoil drainage. Irrigation of Aquitaine and similar soils with water containing increased salt loads will lead to more rapid build-up of soil salinity. Therefore, until the data substantiates the use of mixtures, these types of soils should only be irrigated with unmixed scheme water to minimise the risk of salinity.

## Shortened forms

|                   |                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| CaCO <sub>3</sub> | calcium carbonate                                                                                                                 |
| EC                | electrical conductivity                                                                                                           |
| EPA               | Environmental Protection Authority                                                                                                |
| EPBC Act          | <i>Environmental Protection and Biodiversity Conservation Act 1999</i> , the main Australian Government environmental legislation |
| EPBC conditions   | Conditions imposed on developments by the Australian Government Minister for the Environment assessed under the EPBC Act          |
| GL                | gigalitre (1 billion litres)                                                                                                      |
| GMP               | groundwater management plan                                                                                                       |
| IRG               | Independent Review Group                                                                                                          |
| LOR               | limit of reporting (a substance is undetectable below this concentration)                                                         |
| LTV               | long-term trigger value                                                                                                           |
| Max.              | maximum value                                                                                                                     |
| mBGL              | metres below ground level                                                                                                         |
| mg/L              | milligrams per litre                                                                                                              |
| Min.              | minimum value                                                                                                                     |
| ML/d              | megalitres (1 million litres) per day                                                                                             |
| mS/m              | milliSiemens per metre (measurement of electrical conductivity)                                                                   |
| ORIA              | Ord River Irrigation Area                                                                                                         |
| ORP               | oxidation–reduction potential                                                                                                     |
| Redox             | reduction–oxidation reaction                                                                                                      |
| TDS               | total dissolved solids                                                                                                            |
| µg/L              | micrograms per litre                                                                                                              |

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